UNITED STATES DEPARTMENT OF THE INTERIOR
Bureau of Land Management
Southern Nevada District
Las Vegas Field Office
4701 N. Torrey Pines Drive
Las Vegas, Nevada 89130

DETERMINATION OF NEPA ADEQUACY WORKSHEET

DOI-BLM-NV-S010-2026-0080-DNA

New Proposed Action Title:

Townsite Data Center

Case File No: NVNV105844791 (N-099875)

Location of New Proposed Action:

Clark County, Nevada. Northern portion of the Eldorado Valley, 2.5 miles south of Railroad Pass and 0.75-mile west of U.S. Highway 95.

T. 23 S., R. 63 E., Sec. 14, Mt. Diablo Meridian

Applicant:

Townsite Solar 2 LLC, a wholly owned subsidiary of Townsite Solar 2 Parent, LLC, a joint venture between TS2 Investments, LLC and Arevon Energy Devco 1, LLC

A. DESCRIPTION OF THE NEW PROPOSED ACTION

(Note that the description below is based on information provided to the BLM in the Applicant's plan of development.)

Townsite Solar 2 LLC (Applicant) currently holds authorization to construct, operate, and maintain a 19 MW solar project plus a 35 MW Battery Storage System (BESS) pursuant to Bureau of Land Management (BLM) right of way (ROW) Grant N-99875 issued on September 22, 2023. The Applicant proposes to modify the ROW grant as outlined in the amended SF-299 submitted on November 4, 2025 and further updated on January 8, 2026 and in their updated plan of development (POD) dated May 2026.

The Applicant now proposes to construct, own, operate, maintain, and eventually decommission a data center on 81 acres of the existing 85-acre ROW area, with an additional 4 acres to be used for an 80 × 2,000–foot linear facilities segment (Project / new proposed action).

The Project consists of the data center building housing computer servers, storage, and network equipment to store, manage, process, and disseminate large amounts of data and digital information, and administrative facilities. Contiguous to the data center building will be surface parking, a 70 MW battery energy storage system (BESS), an electrical substation, and emergency backup generators fueled by biodiesel.

In addition, the linear facilities portion of the Project will contain a road to access the data center site (and temporary laydown area) during construction and operation, installation of a permanent 12- to 13.8-kilovolt (kV) electrical distribution powerline (distribution line), and an underground 34.5-kV to 230 kV power collection system used to interconnect the Project into a new substation proposed to support the expansion of the Southern Nevada Water Authority (SNWA) 230-kV transmission line (N-100382). The linear facilities will connect to the southeast corner of the data center and run due east for approximately 2,000 feet to the western border of the project being developed by the Applicant on municipal land owned by the City of Boulder City (BCNV Property). The Project will interconnect into the newly expanded SNWA 230-kV transmission line or into the existing Western Area Power Administration (WAPA) Mead 230-kV Substation from the existing Townsite Solar 1 facilities, both located on municipal land.

Site access will use two separate locations from U.S. Highway 95. The first point of site access is located approximately 1.24 miles south of the Interstate Highway 11 (I-11) / U.S. Highway 93 interchange with U.S. Highway 95, then west onto non-BLM lands using an unnamed road (the road to access Impact Sand and Gravel's Boulder Ranch Quarry (BRQ Pit)) for 0.25 mile, then north on Old U.S. Highway 95 (Old Hwy 95) for 0.35 mile to the proposed Project's linear facilities segment on BLM land. The second point of site access is a right in / right out only point of access located approximately 0.9 miles south of I-11 / U.S. Highway 93 interchange with U.S. Highway 95. If needed, the Applicant will obtain authorization from the governing jurisdictions for use of non-BLM roads for access.

To minimize potential environmental impacts, the proposed project area is slightly larger than required to allow for adjustments in the facility layout. Any land ultimately not needed for the Project will be relinquished back to the BLM at the appropriate time.

Below is a brief description of the Project's proposed elements:

- Geotechnical investigation work includes soil borings, soil resistivity testing, pile load testing, and access. Up to six exploratory test borings to depths of approximately 30 and 50 feet, two field soil resistivity tests (Wenner four-pin method) along two perpendicular traverses at each borehole, and pile load testing at four locations with a group of three piles installed at target depths of 7 and 13 feet. Underground Service Alert will locate and mark the horizontal and vertical locations of exiting utilities prior to boring. Access will be by existing trails and overland travel using a truck-mounted drill rig and conventional pickup truck(s) or four-wheel drive truck(s). All disturbance will be restricted to the Project Area.
- Construction Laydown area: Located adjacent to and outside of the southeast corner of the Project Area, measuring approximately 467 × 464 and totaling 5 acres. This acreage will initially be used as the laydown area and will subsequently be the location of a 70 MW BESS and switching station.
- Data center facility footprint:
    - Current GIS datasets provided by the Applicant show a total of four data center buildings, each with a rectangular footprint area of approximately 300,000 square feet. All four buildings are proposed for development within the footprint of the previously authorized solar facility analyzed in the existing NEPA document.
    - The exact dimensions are not fixed, as they depend on factors like server density, cooling systems, and overall building design. The data center building will not exceed 25 feet in height.

    - Permanent fencing will be installed around the perimeter of the data center. The perimeter fencing will include a 6- or 7-foot-tall chain-link fence with 1-foot-high barbed-wire security strands along the top. Installation of permanent Mojave desert tortoise (*Gopherus agassizii*) fencing will follow U.S. Fish and Wildlife Service (USFWS) guidelines and will be attached to the bottom of the perimeter fencing. Both will be installed concurrently.
    - One auxiliary voltage transformer, measuring 61 × 62 × 71 inches mounted on a concrete pad adjacent to and connected to the distribution line.
    - Installation of a BESS and appurtenant systems will provide up to 70 MW, 4- to 8-hour capacity system for the electric grid.
    - Installation of an electric switching station within the 5 acres to be initially used as an equipment laydown area during construction.
    - Installation of 167 MW of back-up biodiesel-fired generators.
    - Underground Power Collection Line alternating current (AC) collection system will transfer power from the DC collection system and PCSs and transfer the power through the underground Power Collection Line to the Linear Facilities segment where the electricity will interconnect into the grid.
- The Linear Facilities segment extending along the southern border of the project footprint to the east, measuring 80 feet wide × 2,000 feet long, ending at the western border of the proposed BCNV Property, on municipal land owned by the City of Boulder City, and will be used for the site road, auxiliary distribution line, underground Power Collection Line, and fiber optic cable:
    - Site road: The proposed Project will use the 30-foot-wide × 2,000-foot-long portion of the linear facilities segment for the main access to the data center facility, which will provide site access during construction and operation of the Project. The site road will be graded.
    - Auxiliary overhead distribution line: The remaining 50-foot-wide × 2,000-foot-long portion of the linear facilities segment will be used for an overhead 12- to 13.8-kV auxiliary distribution line to provide power during construction and operations. The Applicant intends to assign the 50 × 2,000-foot portion of the linear facilities segment to the owner of the distribution line at a later date.
    - Interconnection and communications: The 50-foot-wide × 2,000-foot-long portion of the linear facilities segment will also be shared for the construction and operation of the underground Power Collection Line and fiber optic cabling. The underground 34.5-230 kV collection line will deliver electricity into the on-site BESS or to the new SNWA substation in support of the newly expanded SNWA 230-kV transmission line. The Power Collection Line will require two pulling sites, up to 80 × 300 feet each, to be within a portion of the linear facilities segment (80 × 2,000-foot area) for an approximate two-month construction timeframe. Fiber optic cabling will be installed either underground with the Power Connection Line as a separate conduit or on overhead lines along the distribution line.
- Project security using a combination of perimeter fencing around the data center buildings, BESS and switching station, controlled access gates, on-site security patrols, lighting, electronic security systems and/or remote monitoring.

- Temporary fencing: Desert tortoise exclusion fencing (temporary) will be installed around the linear facilities segment (4 acres) to reduce the number of biological monitors required on-site. At least two desert tortoise gates will be installed within the linear facilities segment and one entrance gate to the data center facility (with permanent desert tortoise fencing attached). The desert tortoise fencing will be installed concurrently with the perimeter fencing.

Non-potable reclaimed water will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. If water from the City of Boulder City Wastewater Treatment Plant is not available, water would be obtained from commercial sources with existing water rights or availability within a reasonable distance of the project area. This water, if needed, would be trucked to the project area.

The Project will be designed in accordance with federal and industrial standards, including American Society of Mechanical Engineers standards, the National Electrical Code (National Fire Protection Association [NFPA] 2008), the International Building Code (International Code Council 2006), the International Energy Conservation Code (International Code Council 2006), the Uniform Plumbing Code (International Association of Plumbing and Mechanical Officials 2006), the Uniform Mechanical Code (International Association of Plumbing and Mechanical Officials 2006), NFPA standards, and Occupational Safety and Health Administration (OSHA) standards.

Construction is expected to take up to 18 months and will include mobilization, construction grading and site preparation, installation of drainage and erosion controls, data center and BESS construction.

B. LAND USE PLAN CONFORMANCE

LUP Name: Las Vegas Resource Management Plan Date Approved: October 5, 1998

The new proposed action is in conformance with the 1998 Las Vegas RMP. Specifically, it conforms with the following objective pertaining to right-of-way management.

> *Las Vegas Resource Management Plan Objective RW-1 (Rights-of-Way Management) - Meet public demand and reduce impacts to sensitive resources by providing an orderly system of development for transportation, including legal access to private inholdings, communications, flood control, major utility transmission lines, and related facilities.* (Las Vegas RMP, p. 19)

Data centers like the one in the new proposed action are a key component of modern-day cloud-computing infrastructure that are essential to the well-organized management and rapid access of information/data used by the whole of American society.

C. NEPA AND OTHER RELATED DOCUMENTS

NEPA environmental analysis of the new proposed action is covered by the following existing BLM NEPA document:

- Environmental Assessment – Townsite Solar 2 Project (September 2023)
  DOI-BLM-NV-S010-2023-0016-EA
  https://eplanning.blm.gov/Documents/?id=f37e7dd8-a7f2-f011-8406-001dd8008d46&spid=08622883-a8f2-f011-8407-001dd80c29f3#

D. NEPA ADEQUACY CRITERIA

1. Is the new proposed action a feature of, or substantially the same as an alternative analyzed in the existing environmental document(s)?

Yes, the new proposed action (data center) is essentially like the original 2023 proposed action (photovoltaic solar energy facility) analyzed in the existing 2023 Environmental Assessment. Both proposals are from LLC applicants operated by the same parent corporate entity. Both proposals will use the exact same location, same acreage, and same perimeter. Both are proposals for industrial uses that will operationalize cutting-edge technologies that are predominantly electrical and solid-state in nature. Both proposals are connected to external electrical infrastructure via the same proposed utility corridor. Both proposals will use the same access roads and construction laydown footprints. Both involve improvements/facilities that are visually geometric and less than 30 feet in height. Both propose grading the project site in order to accommodate development. Both proposals would require fencing of the site perimeter to ensure site security. Both proposals involve construction timeframes of 12-18 months using similar equipment and vehicles.

A description of the original 2023 proposed action is presented in the existing BLM NEPA document DOI-BLM-NV-S010-2023-0016-EA on pages 4 - 11.

2. Is the proposed action within the same analysis area, or if the location of the proposed action is different, are the geographic and resource conditions sufficiently similar to those analyzed in the existing environmental document(s)? If there are differences, can you explain why they are not substantial?

Yes, the new proposed action is within the exact same analysis area as that which was analyzed in the existing 2023 Environmental Assessment. Location, acreage, and perimeter shape and size are all the same.

A description of the analysis area for the original 2023 proposed action is presented in the existing BLM NEPA document DOI-BLM-NV-S010-2023-0016-EA in Appendix A – Table A-2 and Figures A-1 through A-4.

3. **Is the *range* of alternatives analyzed in the existing environmental document(s) appropriate with respect to the new proposed action, given current environmental concerns and resource values?**

Yes, the range of alternatives analyzed in the 2023 Environmental Assessment is appropriate with respect to the new proposed action. There are no known environmental concerns or resource values which would mandate the addition of new action alternatives beyond the alternatives that were analyzed in the existing 2023 Environmental Assessment: the original 2023 proposed action and the "no action" alternative.

Any action alternative [1] must satisfy the Purpose and Need statement (which – for applicant-driven proposals – typically is for BLM to respond to the application), [2] must be technically and economically feasible, [3] must be within the BLM's authority to implement, and [4] must address unresolved conflicts about the proposed action with respect to alternative uses of available resources. In the existing 2023 Environmental Assessment, the justifications for why certain potential action alternatives were considered but eliminated from detailed analysis remain relevant and applicable to the new proposed action.

A description of the potential action alternatives that were considered but eliminated from detailed analysis is presented in the existing BLM NEPA document (DOI-BLM-NV-S010-2023-0016-EA) on page 12.

In addition, it is important to note that the BLM ultimately approved (via its NEPA decision record) and authorized (via a ROW grant) the original 2023 proposed action. Consequently, this means that a potential denial of the new proposed action may result in the construction and operation of the original 2023 proposed action coming to fruition since it is already approved and authorized. In other words, a denial of the new proposed action does not mean that the project area in question would remain undeveloped.

4. **Is existing analysis valid considering any new information or circumstances relevant to the proposed action (such as, rangeland health standards assessment, recent endangered species listings, updated list of BLM sensitive species)? Can you reasonably conclude that new information and new circumstances do not warrant substantial change to the analysis of the new proposed action?**

Yes. The analysis provided in the existing 2023 Environmental Assessment remains valid with regards to the new proposed action. Because the 2023 Environmental Assessment is still less than three years old and there are no known new circumstances or information regarding the condition of resources present at the project location, the 2023 Environmental Assessment remains relevant to the new proposed action with regards to alternatives, affected environment, effects analysis, and mitigation measures. The project area is still surrounded by intensive land uses and heavy infrastructure including open-pit aggregate quarries, utility-scale solar photovoltaic farms, electric transmission corridors, a highway corridor, and an outdoor firearms shooting range.

5. **Are the environmental effects that would result from implementation of the new proposed action similar (both quantitatively and qualitatively) to those analyzed in the existing NEPA document?**

Yes. The effects analysis in the existing 2023 Environmental Assessment considered the original proposed action's potential effects on 14 distinct resources. Below is a concise resource-by-resource explanation as to why the existing analysis remains applicable to the new proposed action.

- **Air Quality (AQ) –** Because the new proposed action's location, size, and amount of ground disturbance remain the same as the original 2023 proposed action, and construction duration, construction equipment, and vehicle types and quantities are similar to the original 2023 proposed action, construction phase impacts from dust and equipment emissions are likely to remain similar as well, as are local AQ regulatory/permitting requirements.

  Operational phase on-site impacts on AQ are similar, since both projects do not directly generate large volumes of pollutant emissions, but may generate some emissions during periodic use of emergency backup generators.

  The potential fugitive emissions associated with the data center's air-cooling system are expected to be *de minimis* under Clean Air Act standards. Because the cooling compounds must be predominantly non-volatile / non-HFC (per 40 CFR Part 84) and the system is sealed, the Potential to Emit (PTE) falls below the threshold for Hazardous Air Pollutants (HAPs), thereby exempting the cooling infrastructure from major source air permitting requirements.

  Operational phase off-site pollutant emissions resulting from the generation of electricity for the new proposed action will be limited, since the new proposed action will receive its electrical power wholesale from the Western Area Power Administration's (WAPA's) Mead Substation. Mead Substation's physical inputs are dominated by the 2,074 MW Hoover Dam hydroelectric complex plus gigawatts of localized solar generation within the Eldorado Valley Energy Zone (such as the Copper Mountain and Townsite Solar projects). Consequently, during peak daylight hours, most of the power pooled and exported through the substation is sourced from non-carbon emitting renewable generation. With an estimated peak demand of up to 167 MW, the new proposed action represents less than 2% of the Mead Substation's total transport capacity.

  The environmental analysis in Section 3.2 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Cultural Resources –** Because the new proposed action's location, size, and amount of ground disturbance remain the same as the original 2023 proposed action, the potential for physical impacts to cultural resources remains the same as the original 2023 proposed action. Because the indirect APE has already been substantially impacted by surrounding land uses / development, the new proposed action's potential visual and atmospheric effects on cultural resources would be unchanged from the original 2023 proposed action.

  The environmental analysis in Section 3.3 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Socioeconomics –** Estimated workforce sizes for the new proposed action are slightly larger for both the construction phase and the operational phase in comparison to the original 2023 proposed action, thereby potentially resulting in a nominally increased benefit to the local labor market. According to the Applicant, the new proposed action will create 250-300 jobs during construction, and 40 full-time positions during the operational life of the project.

  The environmental analysis in Section 3.4 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Fish and Wildlife (excluding T&E species) –** Because the new proposed action's location, size, and amount of ground disturbance remain the same as the original 2023 proposed action, the potential for adverse impacts to wildlife habitat remains the same as the original 2023 proposed action. Although some specific types of potential impacts within the project area itself could be replaced by other impacts (e.g., "lake effect" on avian species would be no longer present, while new noise effects may occur), these differences in effects could be minor considering the surrounding lands and resources are already heavily disturbed with hundreds of acres of solar photovoltaic panels as well as noise-inducing mining quarry operations, high voltage powerline corridors, a highway corridor, and a firearms shooting range.

  The environmental analysis in Section 3.5 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Fuels and Fire Management –** The new proposed action's location remains the same as the original 2023 proposed action. The Existing 2023 Environmental assessment explains that the project area and immediate surrounding areas are open desert with minimal vegetation that do not support meaningful levels of vegetative fuel to carry wildland fire. Although a data center poses a larger fire risk than solar panel arrays, the project component with the greatest risk for fire – the battery energy storage system (BESS) – is common to both the original 2023 proposed action and the new proposed action. The existing 2023 Environmental Assessment analyzes the potential fire risk of an on-site BESS and prescribes mitigation measures, including development of a fire management plan, that remain valid with regards to the new proposed action.

  The environmental analysis in Section 3.6 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Meteorological Trends –** The analysis presented in the existing 2023 Environmental Assessment states that emissions known to impact meteorological trends "would be generated under the [original 2023] Proposed Action through the use of vehicles and off-road equipment during Project construction and decommissioning, with minimal emissions generated during O&M." This assessment holds true for the new proposed action, as the construction equipment and construction duration are largely similar.

  As noted above (in the Air Quality section), operational phase off-site emissions resulting from the electrical demand of the new proposed action will be limited, since it will receive its electrical power from WAPA's Mead Substation which has an energy generation mix that is majority renewable generation during solar production hours. With an estimated peak demand of up to 167 MW, the new proposed action represents less than 2% of the Mead Substation's total transport capacity.

Although the new proposed action poses a slight risk of emitting chemical pollutants that can contribute to changing meteorological trends (due to potential fugitive emissions of chemicals used for refrigerant purposes), these emissions are unquestionably negligible in comparison to the major sources of such emissions (generally speaking, the transportation sector and the electrical power generation sector). In addition, the cooling compounds must be predominantly non-volatile / non-HFC (per 40 CFR Part 84) and the system is sealed, further reducing the chance for emissions of HFC chemicals.

The 2023 EA states, "On a global scale, the GHG emission contribution of any single source on a subnational scale is dwarfed by the large number of comparable national and subnational contributors…. A single subnational contributor is very unlikely to influence global cumulative emissions. The [original 2023] Proposed Action alone is very unlikely to influence global cumulative emissions on a scale to make observable any increase or lessening of the social cost of all global emissions." This assessment remains true for the new proposed action as well.

The environmental analysis in Section 3.7 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Migratory Birds and Eagles –** The existing 2023 Environmental Assessment states that "Impacts to migratory birds (including golden eagles) are quantified as either displacement or removal of habitat and mortality. Due to a lack of suitable nesting habitat, impacts to golden eagles are limited to the amount of foraging habitat that would be permanently removed by the [original 2023] Proposed Action." Because the new proposed action will utilize the same location with the same acreage of disturbance as the original 2023 Proposed Action, this assessment holds true. The 2023 Environmental Assessment explains that impacts to nesting and foraging for golden eagles will be minimal, due to the small scale of the project and the availability of potential foraging habitat to the southeast and west of the project.

  Regarding migratory birds, the 2023 Environmental Assessment discusses that most impacts to these species would be caused by the loss of 85 acres of habitat (rather than the type of project proposed). Because the new proposed action will occur in the same location, this assessment remains accurate. The 2023 Environmental Assessment also explains that "Waterbird species may potentially mistake the solar panels for water features on which the birds can land…. These behaviors may lead to collisions with PV solar panels resulting in mortality, injury, or stranding of those species that require water to take off again". This unfortunate phenomenon is known as the "lake effect hypothesis". Switching the proposed action from a solar facility to a data center may avoid the potential for this particular adverse effect on waterbirds.

  The environmental analysis in Section 3.8 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Recreation / Travel / Wild and Scenic Rivers –** The existing 2023 EA demonstrates that potential impacts to these resources are mostly due to the original 2023 proposed action's location, access routes, construction duration and methods, and perimeter fencing needs. Since these project characteristics remain very similar for the new proposed action, it is reasonable to conclude that the effects would remain similar as well, thereby resulting in analysis in the 2023 EA that is adequate to cover the new proposed action.

The environmental analysis in Section 3.9 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Soils and Surface Hydrologic Conditions –** The existing 2023 Environmental Assessment demonstrates that potential impacts to these resources are mostly due to the original 2023 proposed action's location, construction duration and methods, and overall surface grading. Since these project characteristics remain very similar for the new proposed action, it is reasonable to conclude that the effects would remain similar as well, thereby resulting in analysis in the 2023 Environmental Assessment that is adequate to cover the new proposed action.

  The environmental analysis in Section 3.10 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Special Designations –** Because the new proposed action is in the same location as the original 2023 proposed action, the proximity to specially designated lands – specifically the Old Spanish National Historic Trail – does not change. Because the intensity of development and general type of development does not change (modern industrial use with relatively low-height improvements, in an area presently occupied with active mining operations, solar power generation facilities, and multiple high-voltage electrical transmission lines), the impacts to nearby specially designated lands is likely very similar to what is analyzed in the existing 2023 Environmental Assessment.

  The environmental analysis in Section 3.11 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Threatened, Endangered, or Candidate Animal Species –** Because the new proposed action is in the same location as the original 2023 proposed action, the list of T&E species which may have their habitat impacted does not change. Because the new proposed action's location, construction duration and methods, and intensity of development remain the same, it is reasonable to conclude that impacts, mitigation measures, and the outcome of interagency consultation would remain the same as well, thereby resulting in analysis in the 2023 EA that is adequate to cover the new proposed action.

  The environmental analysis in Section 3.12 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Vegetation –** Because the new proposed action is in the same location as the original 2023 proposed action, the type of vegetative communities and the list of botanical species which may be present in the project area (including special-status species) does not change. Because the new proposed action's location, construction duration and methods, and intensity of development (including grading) remain the same or similar, it is reasonable to conclude that impacts and mitigation measures would remain the same as well, thereby resulting in analysis in the 2023 EA that is adequate to cover the new proposed action.

  The environmental analysis in Section 3.13 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Visual Resources –** Although certain structures in the new proposed action would be different from the original 2023 proposed action, the general type of land use would remain the same (modern industrial use with relatively low-height structures and improvements) as would the location, intensity of use, size of the project area, and perimeter shape.

As the 2023 Environmental Assessment explains, the project area is currently designated as Visual Resource Inventory (VRI) Class IV. The new proposed action is unlikely to trigger a change in this VRI designation. The surrounding land uses which contribute to the VRI class would remain the same (active mining operations, solar power generation facilities, and multiple high-voltage electrical transmission lines).

The Visual Resource Management (VRM) designation for the project area is Class III. Because the new proposed action is similar to the original 2023 proposed action in size, location, and technologically-advanced industrial use, it is reasonable to conclude that the degree of visual impact felt from the new proposed action at the selected KOP's would remain weak, and that the new proposed action would remain in conformance with BLM's VRM objectives associated with the VRM Class III (2023 EA p. 58).

No longer including photovoltaic solar panels as part of the project area's development may improve the glare analysis for the new proposed action. VR mitigation measures specified in Appendix H of the exiting 2023 Environmental Assessment would remain applicable.

The environmental analysis in Section 3.14 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Wastes (Hazardous or Solid) –** Because the new proposed action will implement geotechnical testing and utilize construction equipment and methods similar to the original 2023 proposed action, the construction phase impacts will be similar as well. During the operational life of the new proposed action, limited quantities of hazardous materials will be used on site that are similar to the types of hazardous materials that were considered in the 2023 Environmental Assessment. Thus, potential impacts would be similar. Mitigation measures prescribed in the 2023 Environmental Assessment (including development of and adherence to a hazmat management plan, spill prevention plan, and an emergency response plan) would remain appropriate for and applicable to the new proposed action.

  Any liquid waste resulting from periodic maintenance of the closed-loop cooling system will be trapped and properly disposed of offsite through contractors approved to handle and dispose of the specific mixture in use.

  The environmental analysis in Section 3.15 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Effects of Past, Present, and Reasonably Foreseeable Future Actions –** The new proposed action is in the same location as the original 2023 proposed action with the same resource conditions and is similar in type and intensity of land use (modern high-technology electrically-focused industrial land use). Therefore, the anticipated impacts of the new proposed action when combined with the effects of past, present, and reasonably foreseeable future actions in the analysis area are sufficiently similar to the impacts that were analyzed and disclosed in the 2023 Environmental Assessment.

  The environmental analysis in Section 4 of the 2023 Environmental Assessment is appropriate with regard to the new proposed action.

- **Notes on water use:**

  The 2023 Environmental Assessment for the proposed solar project analyzed the Applicant's intended use of reclaimed effluent water discharged by the City of Boulder City Wastewater

Treatment Plant for construction phase dust suppression. The Applicant's 2026 POD for the new proposed action (data center) explains that this plan for sourcing dust suppression water remains unchanged. Although the volume of water estimated for dust suppression use is greater for the new proposed action (now projected at up to 500,000 gallons per day), this consumptive use will not adversely affect regional water supply volume. This is because – unlike other communities in southern Nevada – reclaimed effluent water discharged by the City of Boulder City Wastewater Treatment Plant is not recycled and returned to the Colorado River / Lake Mead system. Rather, Boulder City's reclaimed effluent water is sent to local evaporation ponds where this water is ultimately lost to the atmosphere. The 2026 POD further states that the Applicant does not intend to develop or purchase water rights (2026 POD p. 21).

During the operational life of the project, daily water usage estimates for the new proposed action are in the +/- 2,000 gallon-per-day range and are primarily for uses such as employee restroom and kitchen facilities (2026 POD p. 40). This estimate converts to approximately 2.24 acre-feet per year, which is consistent with the water usage estimate in the existing 2023 Environmental Assessment of 2-3 acre-feet per year (2023 EA p. 10). (Under the new proposed action, employee numbers would increase, but washing of solar panels would be eliminated.)

According to the Applicant's Plan of Development, approximately 90% of the potable water used onsite during the operational life of the project will be returned to Boulder City's wastewater treatment plant.

The Applicant proposes a design for the new proposed action that would rely on a closed-loop air-cooling system for the data center. Up to 400,000 gallons (1.23 acre-feet) of a purified water/propylene glycol mix would be required for an initial one-time filling of the closed loop system, but this type of cooling system does not rely upon constant evaporation (and an associated loss of water) as its means of heat dissipation. The purified water/propylene glycol mix will be purchased from a vendor of that mixture. The cooling system will not be filled with any municipal water, as it is not appropriate for this type of cooling system. The initial fill of the cooling system and any topping up of the system must be with the purchased purified water/propylene glycol mix. The first year, one-time consumption of 1.23 acre-feet would be 55% greater than the Applicant's estimated per year average water use of 2.24 acre-feet, but when considered across the proposed 30-year life of the project, the project remains well within the existing 2023 Environmental Assessment's water usage estimate of 2 to 3 acre-feet per year.

It is estimated that up to an additional 2,000 gallons (0.006 acre-feet) per year of purified water/propylene glycol mix could be needed for periodic maintenance activities of the cooling system and will be likewise sourced from a vendor of that mixture. Any waste from the system associated with periodic maintenance will be trapped and properly disposed of offsite through contractors approved to handle and dispose of the material.

6**. Are the public involvement and interagency review associated with existing NEPA document(s) adequate for the current proposed action?**

Yes. As expressed in Section 5 of the 2023 Environmental Assessment, the following steps were taken to ensure adequate public involvement and interagency review:

- The BLM notified adjacent ROW grant holders of the original 2023 proposed action before issuing a decision on the 2023 Environmental Assessment.
- A public comment period on the 2023 Environmental Assessment was held from August 3 through August 18, 2023. A total of 12 public submissions were received. Public comments and BLM responses were included in Appendix I of the final version of 2023 Environmental Assessment.
- BLM sent cooperating agency invitations for the 2023 Environmental Assessment via certified mail and electronic transmittal on February 24, 2023. Cooperating agencies included the Clark County Department of Aviation; Clark County Department of Environment and Sustainability, Division of Air Quality; EPA; and USFWS.
- On May 2, 2022, the BLM initiated government-to-government consultation with the following tribes: Chemehuevi Indian Tribe, Colorado River Indian Tribes, Fort Independence Indian Community of Paiute, Fort Mojave Indian Tribe, Hualapai Tribe, Kaibab Band of Paiute Indians, Las Vegas Paiute Tribe, Moapa Band of Paiutes, Paiute Indian Tribe of Utah, San Juan Southern Paiute Tribe, The Hopi Tribe, Twenty-Nine Palms Band of Mission Indians, and Utu Utu Gwaitu Paiute Tribe (Owens Valley Paiute Benton Reservation). The BLM also sent letters on August 21, 2022 to the Big Pine Paiute and Lone Pine Paiute Tribes.

  The BLM conducted government-to-government meetings with the Moapa Band of Paiutes regarding the original 2023 proposed action, as well as a preliminary field visit with tribal staff on February 18, 2022. Additionally, the BLM conducted informal coordination meetings with tribal staff from the Twenty-Nine Palms Band of Mission Indians. Moapa Band of Paiutes inquired about water resources, dust abatement measures, vegetation treatments, solar panels, and impacts to both cultural and wildlife resources. Both tribes expressed general interest in cultural resources, but no specific concerns were shared regarding impacts to traditional use areas, sacred sites, or Traditional Cultural Properties within the Project area. By request, the BLM provided the Moapa Band of Paiutes with a copy of the final draft of the Class III cultural report following completion of consultation with the Nevada State Historic Preservation Office (SHPO).

Because the new proposed action is in the same location as the original 2023 proposed action with the same resource conditions, and the new proposed action is similar in type and intensity of land use with similar anticipated impacts, the public involvement and interagency review associated with 2023 Environmental Assessment is adequate for the new proposed action.

E. CONCLUSION

Determination of NEPA Adequacy:

☑ Based on the review documented above, I conclude that this proposal conforms to the applicable land use plan and that the NEPA documentation fully covers the proposed action and constitutes BLM's compliance with the requirements of the NEPA.

No Significant Impacts

As documented in detail in Section D.3 of this DNA, because the environmental effects that would result from implementation of the new proposed action are similar (both quantitatively and qualitatively) to those effects analyzed and disclosed in the existing 2023 Environmental Assessment, the new proposed action will similarly result in no significant impacts.

Responsible Official: BRUCE SILLITOE Digitally signed by BRUCE SILLITOE Date: 2026.06.26 11:19:06 -07'00' Date: _____

ATTACHMENTS:

- Environmental Assessment for the Townsite Solar 2 Project (DOI-BLM-NV-S010-2023-0016-EA - September 2023), including the Biological Opinion on the Townsite Solar 2 Project (2022-0035189-S7-001 – October 2022)

- Townsite Data Center Plan of Development (May 2026)

U.S. Department of the Interior
Bureau of Land Management

Las Vegas Field Office

Environmental Assessment

Environmental Assessment

DOI-BLM-NV-S010-2023-0016-EA

Townsite Solar 2 Project

Case File Number NVN-099875

Applicant

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September 2023

TABLE OF CONTENTS

1 Introduction 1
1.1 Purpose and Need 2
1.2 Applicant's Project Objectives 2
1.3 Decision to be Made 2
1.4 Conformance Summary 2
2 Proposed Action and Alternatives 4
2.1 Introduction 4
2.2 Proposed Action 4
2.2.1 Project Overview 4
2.2.2 Project Elements 5
2.2.3 Construction of the Facilities 7
2.2.4 Laydown Area (Temporary) 8
2.2.5 Photovoltaic Solar and Ancillary Facility Construction 9
2.2.6 Operations and Maintenance 10
2.2.7 Decommissioning and Site Reclamation 10
2.2.8 Design Features and Standard Agency Requirements 10
2.3 No Action Alternative 11
2.4 Alternatives Considered but Eliminated from Detailed Analysis 12
2.4.1 Larger Project Boundary 12
2.4.2 Maintaining Vegetation, Soils, and Surface Water Flow within the Solar Field Area 12
3 Affected Environment and Environmental Effects 13
3.1 Resource Evaluation 13
3.2 Air Quality 13
3.2.1 Affected Environment 14
3.2.2 Environmental Effects of the No Action Alternative 15
3.2.3 Environmental Effects of the Proposed Action 15
3.2.4 Mitigation Measures 16
3.3 Cultural Resources 17
3.3.1 Affected Environment 17
3.3.2 Environmental Effects of the No Action Alternative 17
3.3.3 Environmental Effects of the Proposed Action 18
3.3.4 Mitigation Measures 18
3.4 Environmental Justice and Socioeconomics 18
3.4.1 Affected Environment 19
3.4.2 Environmental Effects of the No Action Alternative 20
3.4.3 Environmental Effects of the Proposed Action 21
3.4.4 Mitigation Measures 22
3.5 Fish and Wildlife (Excluding Federally Listed Species) 22
3.5.1 Affected Environment 22
3.5.2 Environmental Effects of the No Action Alternative 25
3.5.3 Environmental Effects of the Proposed Action 25
3.5.4 Mitigation Measures 28
3.6 Fuels and Fire Management 28
3.6.1 Affected Environment 28
3.6.2 Environmental Effects of the No Action Alternative 29
3.6.3 Environmental Effects of the Proposed Action 29

3.6.4 Mitigation Measures 31
3.7 Greenhouse Gas Emissions/Climate Change 31
3.7.1 Affected Environment 31
3.7.2 Environmental Effects of the No Action Alternative 34
3.7.3 Environmental Effects of the Proposed Action 34
3.7.4 Mitigation Measures 37
3.8 Migratory Birds and Eagles 37
3.8.1 Affected Environment 38
3.8.2 Environmental Effects of the No Action Alternative 38
3.8.3 Environmental Effects of the Proposed Action 38
3.8.4 Mitigation Measures 41
3.9 Recreation/Travel/Wild and Scenic Rivers 41
3.9.1 Affected Environment 41
3.9.2 Environmental Effects of the No Action Alternative 41
3.9.3 Environmental Effects of the Proposed Action 42
3.9.4 Mitigation Measures 42
3.10 Soils and Surface Hydrologic Conditions 42
3.10.1 Affected Environment 43
3.10.2 Environmental Effects of the No Action Alternative 44
3.10.3 Environmental Effects of the Proposed Action 44
3.10.4 Mitigation Measures 45
3.11 Special Designations 46
3.11.1 Affected Environment 46
3.11.2 Environmental Effects of the No Action Alternative 46
3.11.3 Environmental Effects of the Proposed Action 46
3.11.4 Mitigation Measures 47
3.12 Threatened, Endangered, or Candidate Animal Species 47
3.12.1 Affected Environment 47
3.12.2 Environmental Effects of the No Action Alternative 49
3.12.3 Environmental Effects of the Proposed Action 49
3.12.4 Mitigation Measures 50
3.13 Vegetation (Including Special-Status Plants, Cacti, and Invasive Plant Species and Noxious Weeds) 50
3.13.1 Affected Environment 51
3.13.2 Environmental Effects of the No Action Alternative 52
3.13.3 Environmental Effects of the Proposed Action 52
3.13.4 Mitigation Measures 55
3.14 Visual Resources 55
3.14.1 Affected Environment 56
3.14.2 Environmental Effects of the No Action Alternative 57
3.14.3 Environmental Effects of the Proposed Action 58
3.14.4 Mitigation Measures 60
3.15 Wastes (Hazardous or Solid) 60
3.15.1 Affected Environment 60
3.15.2 Environmental Effects of the No Action Alternative 60
3.15.3 Environmental Effects of the Proposed Action 60
3.15.4 Mitigation Measures 62
4 Cumulative Effects Analysis 63
4.1 Cumulative Effects Study Areas 63

4.2 Past and Present Actions 63
4.3 Reasonably Foreseeable Future Actions 64
4.4 Evaluation of Potential Cumulative Impacts 64
4.4.1 Air Quality 64
4.4.2 Cultural Resources 64
4.4.3 Environmental Justice and Socioeconomics 65
4.4.4 Fish and Wildlife (Excluding Federally Listed Species) 65
4.4.5 Fuels and Fire Management 66
4.4.6 Greenhouse Gas Emissions/Climate Change 66
4.4.7 Migratory Birds and Eagles 67
4.4.8 Recreation/Travel/Wild and Scenic Rivers 68
4.4.9 Special Designations 68
4.4.10 Soils and Hydrologic Conditions 68
4.4.11 Threatened, Endangered, and Candidate Species 69
4.4.12 Vegetation (including Special-Status Species, Cacti, and Invasive Species and Noxious Weeds) 69
4.4.13 Visual Resources 70
4.4.14 Wastes (Hazardous and Solid) 70

5 Consultation and Coordination 72
5.1 Scoping and Public Participation 72
5.2 Cooperating Agencies 72
5.3 Tribal Government-to-Government Consultation 72
5.4 NHPA Section 106 Consultation with Nevada SHPO and Advisory Council on Historic Preservation 73
5.5 ESA Section 7 Consultation with USFWS 73
5.6 Consultation with Others 73
5.7 List of Preparers and Reviewers 74

Appendices

Appendix A Acronyms, Tables, and Figures
Appendix B Literature Cited
Appendix C Applicant-Committed Environmental Protection Measures
Appendix D Bureau of Land Management Standard Stipulations and Mitigation Measures
Appendix E Project Biological Opinion
Appendix F Air Quality and Greenhouse Gas Support Data and Calculations
Appendix G Data Responses U.S. Fish and Wildlife Service and Nevada Department of Wildlife
Appendix H Visual Resources Mitigation Measures and Best Management Practices
Appendix I Public Review Comments and Responses

1 INTRODUCTION

Skylar Energy Resources, LLC (applicant), applied to the Bureau of Land Management (BLM) for a right-of-way (ROW) grant for a proposed solar energy development facility (N-99875) on 80 acres of BLM-managed public land and anticipates applying for a short-term ROW for a 5-acre temporary laydown area adjacent to and outside of the solar facility (Project). If approved, the ROW for the solar energy development facilities would be issued for an initial term of 30 years with the option to renew, and the ROW for the temporary laydown area would be issued for an initial 3-year term with an option to renew for an additional 3-year term.

Because the Project has a proposed generation of less than 20 megawatts (MW) (proposing up to 19 MW), it is not automatically governed by the decisions made in the *Final Programmatic Environmental Impact Statement for Solar Energy Development in Six Southwestern States* (Solar PEIS) (BLM and U.S. Department of Energy [DOE] 2012), including the application of the programmatic design features and variance process.

In accordance with 43 Code of Federal Regulations (CFR) 2804.25(e) and the BLM Southern Nevada District (SND) Instruction Memorandum (IM) NV-SNDO-2020-11 (Renewable Energy Priority), the BLM Las Vegas Field Office (LVFO) applied screening criteria to assess the priority for processing the application and determined the ROW application was a high priority for processing on September 17, 2021.

According to regulations, the BLM must evaluate the application for the Project in accordance with the criteria set forth in 43 CFR 2804.25(e)(2)(iii) and, in light of its findings, either deny the application or continue processing. As part of the application evaluation process, the BLM gathered information and input from the applicant; local, state, and federal agencies; tribal governments, BLM resource specialists, and the public. Virtual meetings were held to provide information about the proposed Project and gather input to inform the application evaluation, including one meeting for agencies and tribes, and one meeting for the public. Based on the information gathered, the BLM determined it was appropriate to continue processing the application and initiate the environmental review to further consider the application.

This Environmental Assessment (EA) has been prepared to analyze and disclose the environmental effects of approving a ROW for the proposed Townsite Solar 2 Project, which consists of a solar photovoltaic (PV solar) energy development facility (up to 19 megawatt alternating current [MWac]), a possible battery energy storage system (BESS) (up to 35 MW), a linear facilities segment with ancillary project components, and a short-term ROW for a temporary laydown area (Proposed Action). The Project is proposed on BLM-managed land situated approximately 5 miles southwest of downtown Boulder City in Clark County, Nevada (Figure A-1).

The BLM LVFO is preparing this EA in compliance with the National Environmental Policy Act (NEPA), as amended, to disclose impacts from the proposed Project and determine whether any significant effects could result from the analyzed actions. Following the requirements of NEPA (40 CFR 1501.5), as amended, this EA describes the potential impacts of the Proposed Action and a No Action Alternative. If the BLM determines the Project is not expected to have significant effects, a Finding of No Significant Impact will be issued, and a Decision Record will be prepared. If significant effects are anticipated, the BLM will prepare an Environmental Impact Statement or a Decision Record selecting the No Action Alternative.

1.1 Purpose and Need

In accordance with the Federal Land Policy and Management Act of 1976 (FLPMA), as amended, public lands are to be managed for multiple uses that consider the long-term needs of future generations for renewable and nonrenewable resources. The BLM is authorized to grant ROWs on public lands for systems of generation, transmission, and distribution of electrical energy (Section 501(a)(4)). Taking into account the BLM's multiple-use mandate, the BLM's purpose and need for this action is to respond to the FLPMA ROW application (for the solar facility and future short-term ROW request) submitted by the applicant under Title V of FLPMA (43 United States Code [USC] 1761) to construct, own, operate, and decommission a solar generation power plant and ancillary facilities (Proposed Action) on BLM land in Clark County, Nevada, in compliance with FLPMA, BLM ROW regulations (43 CFR 2800), the BLM NEPA Handbook (BLM 2008a), the U.S. Department of the Interior (DOI) NEPA regulations (43 CFR 46), and other applicable federal and state laws and policies.

1.2 Applicant's Project Objectives

The applicant's objective for the Project is to provide an efficient, economic, and reliable PV solar generating facility that would provide approximately 19 MW of solar-generated electricity. The Project would help meet the goals and requirements set forth by the State of Nevada for renewable energy resources. The Nevada Renewable Portfolio Standard (RPS) (Nevada Revised Statutes [NRS] 704.7821), first adopted by the State Legislature in 1997, mandates a minimum amount of electricity that electric utilities acquire from renewable energy sources. The Nevada RPS was revised and signed on April 22, 2019, known as Senate Bill 358, which states that by calendar years 2024 through 2026, no less than 34% of the total amount of electricity sold by electric providers to their retail customers in Nevada must be from renewable energy resources. NV Energy, the state's largest investor-owned utility and provider for the Las Vegas Valley, is working to acquire renewable energy from multiple generating facilities to meet, at a minimum, the mandated Nevada RPS target according to Senate Bill 358 of 20% of retail sales coming from renewable resources and gradually increasing in subsequent years until reaching 100% (zero carbon) by 2050. To help achieve this goal, the State of Nevada will "encourage and accelerate the development of new renewable energy projects" (Section 8 of Senate Bill 358).

1.3 Decision to be Made

The applicant submitted a ROW application for the proposed Project to the BLM LVFO on June 8, 2020.

The BLM will decide whether to approve the ROW application for the proposed Project, approve the ROW application for the proposed Project with modifications, or deny the ROW application for the proposed Project. The BLM may include any terms, conditions, and stipulations it determines to be in the public interest and may include modifying the proposed use or changing the route or location of the proposed facilities (43 CFR 2805.10(a)(1)). In the decision process, the BLM must consider how the BLM's resource management goals, objectives, opportunities, and/or conflicts relate to this nonfederal use of public lands.

1.4 Conformance Summary

The BLM lands in southern Nevada are managed under the Las Vegas Resource Management Plan (RMP) and Final Environmental Impact Statement (BLM 1998a). The ROW grant request is in conformance with the 1998 Las Vegas RMP, and the specific objectives and management directions that

allow for the actions proposed can be found in Appendix A of the 1998 Las Vegas RMP Record of Decision (BLM 1998a).

The Proposed Action would be consistent with other officially approved federal, state, and local plans, policies, and programs and with applicable federal regulations, policies, and laws. This EA incorporates by reference, and/or is tiered to, the following documents:

- The Final Programmatic Environmental Assessment (PEA) for the Eldorado Valley Transmission and Utility Corridor (DOI-BLM-NV-S010-2012-0024) Finding of No Significant Impact and Decision Record (BLM 2012a) streamlines the ROW permitting process for renewable energy projects that would require the construction of new facilities or interconnections to infrastructure within the BLM transportation and utility corridors within and near the Eldorado Valley Patent Area.
- The Project Biological Opinion (BO) File No. 2022-0035189-S7-001 is applicable because this Project will affect more than 20 acres of federal land administered by the BLM SND and is appended to Programmatic BO No. 08ENVS00-2019-F-0153 (U.S. Fish and Wildlife Service [USFWS] 2019, 2022d).
- The BLM's Final Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement (BLM 2007a) discusses the potential risks to human health and the environment posed by the proposed use of chemical herbicides on public lands within 17 western states, including Nevada. It also offers a cumulative impact analysis that accounts for the use of chemical herbicides in combination with other treatment techniques. The Record of Decision (BLM 2007b) lists the herbicides allowed for use on public lands and authorizes the continuous use of 14 herbicides, including those that contain the active ingredient glyphosate that may be applicable to maintenance activities during project operations.
- The Programmatic Agreement Among the Bureau of Land Management, the Advisory Council on Historic Preservation, and the National Conference of State Historic Preservation Officers Regarding the Manner in which the BLM Will Meet Its Responsibilities under the National Historic Preservation Act (BLM 2012b) was created to establish the process the BLM will use to comply with Section 106 of the National Historic Preservation Act of 1966 (NHPA) for solar energy development on BLM lands. The Proposed Action will be in accordance with the Programmatic Agreement's principles and procedures.
- The *Clark County Multiple Species Habitat Conservation Plan* (MSHCP) under Section 10(a) of the Endangered Species Act (ESA) permits "take" of threatened or endangered species resulting from otherwise lawful activities on nonfederal properties within Clark County (e.g., lands owned by the City of Boulder City). The MSHCP and resultant Section 10(a) Permit is also designed to allow the incidental take of "Covered Species" listed in the MSHCP (pp. 2–7) and constitutes a Special Purpose Permit under 50 CFR 21.27 for the take of ESA-listed birds (Regional Environmental Consultants, Inc. 2000). The 34.5-kilovolt (kV) collection system will interconnect into the new substation in support of the newly expanded Southern Nevada Water Authority (SNWA) 230-kV transmission line or to the existing Western Area Power Administration Mead (WAPA) 230-kV Substation from the existing Townsite Solar 1 Facility generation tie-in line; both substation options are on municipal land and are actions covered by the MSHCP.

Table A-1 in Appendix A lists the anticipated federal, state, and local permits or approvals that may be required for the Project analyzed in this EA. Note that this list may not be all-inclusive. Appendix B contains all literature cited.

2 PROPOSED ACTION AND ALTERNATIVES

2.1 Introduction

This section describes the Proposed Action as well as potential alternatives considered. Potential alternatives are in two categories: those carried forward for detailed analysis and those considered but not carried forward.

2.2 Proposed Action

The applicant has applied for a ROW grant (N-99875) for 80 acres of BLM-managed public land to construct, own, operate, maintain, and decommission an up-to 19-MWac PV solar generating facility. The Project consists of PV solar panels, a possible BESS, and an east–west-oriented linear facilities segment, approximately 2,000 feet long × approximately 80 feet wide. Within the linear facilities segment is a site road for access to the Project site, a 34.5-kV collector system, and an electrical distribution line. The Project will interconnect into the new substation in support of the newly expanded SNWA 230-kV transmission line or into the existing WAPA 230-kV Substation from the existing Townsite Solar 1 Facility generation tie-in line; both substations are located on municipal land. The Project may contain its own BESS or may share a BESS with the proposed Boulder City Nevada Facility (BCNV Facility). The BCNV Facility is a solar PV project proposed to be constructed on municipal land currently subject to a lease option agreement between Townsite Solar 2, LLC, and the City of Boulder City. The applicant also plans to request a short-term ROW grant for a 5-acre temporary laydown area on BLM-administered land located adjacent to and outside of the solar facility. For the purposes of this EA, the Project Area is defined as the 80-acre solar facility plus the 5-acre temporary laydown area, for a total of 85 acres.

Detailed locational information is included in Figure A-2, Table A-2, and Figure A-3 of Appendix A.

2.2.1 Project Overview

Prior to Project construction, geotechnical exploration investigations including soil borings, soil resistivity testing, and pile load tests would be conducted in the Project Area. The purpose of the geotechnical investigation is to provide information for preliminary design necessary for the notice to proceed for construction of the facilities. No reclamation is proposed for the geotechnical investigation as the testing locations would occur within areas proposed as part of the permanent disturbance associated with the Project. The applicant is planning to commence geotechnical exploration in Q4-2023.

The Project would be constructed using PV solar modules mounted on either a fixed-tilt mounting system and/or single-axis, horizontal tracker structures. Construction of the Project would take approximately 12 months and would commence following completion of the NEPA process and the obtainment of necessary permits. All Project staging and construction activities would occur within the 5-acre temporary laydown area, requested under a short-term ROW. The Project would operate year-round.

Primary access to the Project would be from U.S. 95 south 1.24 miles from the interchange of Interstate 11 (I-11)/U.S. 93 and U.S. 95, turning west onto non-BLM land using an unnamed road (that also provides access to Customers Team Members Community Crushing Boulder Ranch Quarry Pit [CTC BRQ Pit]) and then turning north onto Old U.S. Highway 95 for approximately 0.25 mile (past the existing Townsite Solar 1 Facility) to the linear facilities segment on BLM land.

2.2.2 Project Elements

Project elements would include geotechnical investigation, a laydown area (temporary), PV solar-generating facilities and solar modules (solar field), a possible BESS, linear facilities, and ancillary components. Each is summarized below; further details are provided in the Plan of Development (POD) (SWCA Environmental Consultants [SWCA] 2023). Project elements are included in Figure A-4 and Figure A-5, and the proposed interconnection pathways are included in Figure A-6.

- **Geotechnical Investigation within the Project Area:** Includes soil borings, soil resistivity testing, pile load testing, and access. Prior to testing, Underground Service Alert would locate and mark the horizontal and vertical locations of exiting utilities throughout the Project Area. A limited notice to proceed request will be made for the geotechnical work at the time the ROW is issued because it is a component of the overall ROW request for the Project.
- **Laydown area (temporary)**: Approximately 5 acres in size (467 × 464 feet), located outside and adjacent to the southeast corner of the solar facility that will undergo restoration at the end of construction. Mohave desert tortoise (*Gopherus agassizii*) (hereinafter, desert tortoise) temporary exclusion fencing would be installed around the 5-acre laydown area (and the 4-acre linear facilities segment with two gates to provide access). All fencing would be sequenced prior to construction and installed concurrently. A short-term ROW will be requested for this area at a later date. For this EA analysis, the Project Area includes the laydown area.
- **PV Solar-Generating Facilities and Solar Modules:** Installation of PV modules with the capacity to generate up to 19 MW of solar-generated electricity. High-efficiency solar panels mounted on fixed-tilt and/or single-axis, horizontal tracker mounting systems supported by driven steel posts into the ground (or other embedded foundation design) with steel table frames bolted to the driven posts connected to inverter/transformer equipment (power conversion station [PCS]) and a possible BESS installed on concrete footings. All drainage features and berms would be included within the solar facility area, as needed. Security perimeter fencing would include a 6- or 7-foot-tall chain-link fence with 1-foot-high barbed-wire security strand along the top with a controlled access gate at the entrance to the facility. Permanent desert tortoise exclusion fencing would be installed in accordance with USFWS guidelines and would be attached to the bottom of the perimeter fencing that surrounds the solar field (USFWS 2009). No on-site substation would be constructed as part of the Project. Rather, the Project would interconnect into a new substation (in support of the newly expanded SNWA 230-kV transmission line) or to the existing WAPA 230-kV Substation (from the existing Townsite Solar 1 230-kV generation tie-in line) from the proposed BCNV Facility using the interconnecting 34.5-kV underground collection line (see the linear facilities segment description below).
- **Battery Energy Storage System (BESS):** Installation of a possible BESS and appurtenances within the southeast corner of the solar facility would provide up to 35-MW, 4- to 8-hour capacity system for the electric grid or the Project would use the BESS on municipal land at the proposed BCNV Facility.
- **Linear Facilities Segment:** Extends along the southern border of the Project footprint to the east, measuring 80 feet wide × 2,000 feet long. Prior to construction, temporary desert tortoise fencing with two access gates would be installed and would remain until the start of operations. Figure A-5 depicts the components of the linear facilities segment in detail, with a brief description below:
    - **Site Road:** A newly constructed site road would provide access to the solar facility. This permanent road would be graded and run east–west in a 30-foot-wide × 2,000-foot-long portion of the linear facilities segment from the southeast corner of the solar field to the western border of the proposed BCNV Facility, on municipal land. The site road would

be at least 25 feet from the distribution power line's centerline and in the southern portion of the linear facilities segment.

- **Overhead Distribution Power Line:** The remaining 50-foot-wide × 2,000-foot-long area of the northern portion of the linear facilities segment will be used for an overhead 12- to 13.8-kV distribution line to provide auxiliary power during construction and operations. The distribution power line would be up to 2 miles in length, 2,000 feet crossing BLM-administered land and approximately 8,560 feet crossing municipal land owned by the City of Boulder City. (BLM does not have authority to approve Project components on municipal land; therefore, the portion of the distribution line planned for municipal land is not part of the BLM's Proposed Action and would require separate authorization from the City of Boulder City.) The poles would be installed in accordance with utility provider and Project BO guidelines, spaced an average of 300 feet apart and extending approximately 40 feet above the ground surface. The portion of the distribution line on BLM land (within the 50 × 2,000–foot portion of the linear facilities segment) would be assigned to the owner of the distribution line at a later date. To the extent applicable, all overhead electrical lines would be designed and installed in accordance with the Avian Power Line Interaction Committee's (APLIC's) guidelines for *Suggested Practices for Avian Protection on Power Lines* (APLIC 2006) and *Reducing Avian Collisions with Power Lines* (APLIC 2012). A fiber-optic communications cable could also be co-located on overhead lines along the distribution line (or buried underground) within the linear facilities segment.
- **Interconnection (buried) and Fiber Optic (overhead or buried):** The 50-foot-wide × 2,000-foot-long portion of the linear facilities segment would also be shared for the construction and operation of an interconnecting 34.5-kV underground collection line and fiber-optic cable. The 34.5-kV collection line would be installed between the 25-foot centerline for the distribution line and the access road, and would require two pulling sites, up to 80 × 300 feet each, to be located within a portion of the linear facilities segment during an approximate 2-month time frame. The 34.5-kV collector line is part of the 34.5-kV underground solar energy collection system that transports solar energy from the panels to the point of interconnection. There are two potential interconnection routes proposed. The 34.5-kV collector line within the linear facilities segment would deliver electricity to the proposed BCNV Facility (on municipal land), where the electricity would interconnect into the new substation in support of the newly expanded SNWA 230-kV transmission line; or the 34.5-kV collection line would interconnect into the existing WAPA 230-kV Substation from the existing Townsite Solar 1 Facility generation tie-in line (both on municipal land). Figure A-6 shows the proposed interconnection pathways, which are for illustration purposes only as they are not part of the Proposed Action. Fiber-optic cabling would be installed either underground in the same trench as the 34.5-kV collector line as a separate conduit or on an overhead line along the distribution line.

- Ancillary components:
  - An interior solar field access way would be constructed around the center of the solar field. The access way would be up to 20 feet wide × 6,000 feet long and would be graded and compacted to provide access to the solar equipment, as permitted by site topography.
  - Two meteorological stations using up to 30-foot-high steel lattice towers would be co-located within the east–west portions of the access way. The meteorological stations would be constructed at the time of solar facility construction.
  - Up to five inverter/transformer equipment stations (PCSs), approximately 20 × 10 × 8 feet in size, would be co-located within the east–west portions of the access way.

- One auxiliary voltage transformer, measuring 61 × 62 × 71 inches, mounted on a concrete pad adjacent to and connected to the distribution line.

Nonpotable reclaimed water would be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust-suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. Water transport on public roads would adhere to all federal and state transportation rules and regulations. The applicant does not intend to develop or purchase water rights.

The Project would be monitored 24 hours per day, 7 days per week during all phases through a combination of security patrols and electronic security systems.

2.2.3 Construction of the Facilities

2.2.3.1 OVERVIEW

Prior to construction, geotechnical investigations would be conducted in the Project Area. Once geotechnical exploration is complete, and a notice to proceed with construction is issued, construction is expected to take up to 12 months and would include the major phases of mobilization, grading and site preparation (disc and roll), installation of drainage and erosion controls, PV panel/tracker assembly, and solar field construction. Estimated temporary surface disturbance associated with geotechnical investigation will become part of the total permanent disturbance associated with the solar facility.

2.2.3.2 GEOTECHNICAL INVESTIGATION

All geotechnical disturbance would be restricted to the Project Area. The purpose of the geotechnical investigation is to obtain site-specific information for preliminary design, environmental analysis, and the permitting processes. No on-site staging areas will be needed during the geotechnical fieldwork.
No reclamation is proposed for the geotechnical investigation as the testing locations will become part of the permanent disturbance associated with the Project. Geotechnical work is part of the overall ROW request for the Project and would be authorized via a limited notice to proceed subsequent to the issuance of the BLM ROW grant for the Project itself. The applicant is planning to commence geotechnical exploration in Q4-2023.

2.2.3.3 GEOTECHNICAL BEST MANAGEMENT PRACTICES

As part of the geotechnical investigation, the following best management practices (BMPs) would be implemented:

- Hazardous material will not be drained onto the ground or into streams or drainage areas. Totally enclosed containment will be provided for all solid waste. All construction waste will be removed to a disposal facility authorized to accept such materials.
- Fire prevention practices, handling, storage, and disposal activities, including mandatory reporting requirements will be observed at all times.
- Boreholes will be backfilled with auger cuttings. No open borings will be left unattended, and active borings not completed will be covered overnight.
- Project activities will be designed to avoid or minimize impacts to sensitive plants and animal species. A BLM-approved monitor will conduct clearance surveys before ground disturbance

activities and will reroute equipment and vehicles to avoid impacts to sensitive plants and animals.

- The desert tortoise is the only federally protected species (federally threatened) with the potential to occur in the Project Area and will be considered in accordance with management policies set forth by the BLM and the USFWS. In addition to the proposed minimization measures required in the BO, the following BMPs will be followed in the interest of minimizing potential effects on desert tortoises.
    - A qualified desert tortoise monitor will be on-site during all geotechnical investigation activities.
    - An authorized desert tortoise biologist will conduct preconstruction clearance surveys, walk in front of all vehicles accessing the site, and monitor all construction activities in desert tortoise habitat. The biologist or monitor will clear all staging/parking (geotechnical work areas) and overland travel areas (access routes) before vehicles and/or equipment travel off road by walking in front of any vehicles accessing the site. There will be a minimum of one desert tortoise biologist for every vehicle or piece of equipment accessing the site.
    - Should a desert tortoise be found within or enter the area of activity, activities will immediately stop until such time as the animal has left the area of its own accord.
    - All potential desert tortoise burrows will be avoided by equipment. The authorized biologist will guide equipment around any potential desert tortoise burrows.
    - A litter-control policy will be implemented to minimize predation on tortoises by ravens, coyotes, or other predators drawn to the Project Area. This policy will include the removal of trash from the Project site at the end of each workday and proper disposal of trash in a designated solid waste disposal facility.
    - A Worker Environmental Awareness Program (WEAP) will be implemented for all workers who will be on the Project site prior to commencement of activities. Training materials and briefings will include discussion of ESA requirements and the consequences of noncompliance, identification and values of wildlife and natural plant communities, hazardous substance spill prevention and containment measures, and review of all design features of the proposed Project.
    - Workers, including maintenance personnel, will be instructed to check underneath all vehicles before moving them, as desert tortoises often take cover underneath parked vehicles.

2.2.4 Laydown Area (Temporary)

The 5-acre temporary laydown area would be located adjacent to and outside of the southeast corner of the Project footprint situated above the linear facilities segment. The laydown area would be for temporary installation of construction trailers, construction workforce parking, aboveground water tanks, materials receiving, and materials storage. The area would be cleared using traditional construction methods (disc and roll).

Temporary desert tortoise fencing would be installed around the laydown area and the linear facilities segment and would be connected to the perimeter fencing where it overlaps. If perimeter fencing installation is delayed, temporary desert tortoise fencing would be extended along the western boundary of the laydown area and the northwestern boundary of the linear facilities segment to ensure enclosure (see Figure A-5). Up to two gates would be installed to provide access to the linear facilities segment.

The site road located within the linear facilities segment would provide access to the laydown area. At the end of construction, the laydown area would undergo reclamation in accordance with the reclamation plan required by the BLM short-term ROW grant.

2.2.5 Photovoltaic Solar and Ancillary Facility Construction

The on-site construction workforce is expected to range from 100 to 150 workers and peak at 200 workers during any given time. Most construction staff and workers would commute daily to the jobsite from within Clark County, primarily from the Las Vegas Valley area. Construction would typically occur between 5:00 a.m. and 5:00 p.m. and may occur 7 days per week. Workers would utilize access from U.S. 95. The applicant would prepare a Traffic Management Plan for Project-related traffic, if needed.

It is anticipated that permitting and development of the Project would take place during 2022 and 2023 and construction would commence during the second quarter of 2024 and last up to 12 months. Construction would include the following activities, listed in approximate sequential order (some construction activities would occur simultaneously):

- Site preparation including surveying, staking, and geotechnical activities
- Site clearance surveys and desert tortoise exclusion (temporary and permanent) and perimeter fencing installation
- Site grading and installation of erosion control measures
- Construction of site road and interior way within the Project Area
- Trenching and installation of underground electrical system in solar field
- Assembling of array foundation/installation of solar array fields and ancillary facilities
- Construction of the electrical collection and transmission system
- Testing and commissioning
- Restoring temporary laydown area

Table A-4 provides a description of the on-site equipment expected to be used during Project construction, but actual equipment and durations may vary. During the Project construction period, construction-related activities, including dust control, would require an estimated 25 acre-feet of water.

Power for construction activities (and subsequent operation and maintenance [O&M]) would be provided by the newly constructed 12- to 13.8-kV distribution line for auxiliary power and would be sited within the linear facilities segment, which extends east onto municipal land owned by the City of Boulder City. Temporary construction power could be provided by generators prior to power line completion.

Primary wastes generated during construction (and O&M) would be solid and liquid wastes including scrap wood, steel, glass, plastic, paper and metals, empty hazardous waste containers, waste oil filters, oily rags and oil sorbent, spent lead acid batteries, waste oil and sanitary waste. The applicant would prepare a Spill Prevention and Emergency Response Plan, which would address waste and hazardous materials management, including BMPs related to spill response, transportation, and handling of materials and wastes. No hazardous materials would be stored on-site; instead, they would be brought on-site and off-site for use as needed.

All construction and operation contractors would be required to operate under a Health and Safety Program that meets industry standards. All site personnel would be required to go through a new hire

orientation and follow the WEAP, which would address Project-specific safety, health, and environmental concerns. In addition, the applicant would also prepare a Fire Management Plan and Lighting Management Plan that would be applicable during construction and operation of the Project.

2.2.6 Operations and Maintenance

O&M includes routine maintenance (i.e., potential solar panel washing); semiannual and annual inspections; repair and maintenance of the solar facility, BESS, and other project facilities; site security and management; and vegetation control. Up to three full-time equivalent (FTE) staff are anticipated to be required for O&M.

It is estimated that O&M activities would use up to 2 to 3 acre-feet of water per year. Water would be used for panel washing and for dust control. If needed, dust palliatives outlined in applicant-committed environmental protection measures (EPMs) (Appendix C), with prior approval from the BLM, may be used to supplement dust control measures.

O&M activities would likely require limited quantities of hazardous materials to be used and are listed in Table A-5.

2.2.7 Decommissioning and Site Reclamation

The Project has an anticipated operational life of 30 years, after which the applicant may choose to apply to the BLM for renewal of the ROW authorization or decommission the Project and remove the systems and their components and restore the Project Area. A Final Facility Decommissioning and Site Reclamation Plan would be created in accordance with BLM policy and objectives, and approved by the BLM, once the operational life of the Project has been reached.

2.2.8 Design Features and Standard Agency Requirements

Design features, including project components, applicant-committed EPMs, and management plans are features that are incorporated into a Proposed Action and alternatives with the intent to reduce potential Project impacts.

The analysis (see Section 3) assumes that the following design features, required plans below, applicant- committed EPMs (see Appendix C), and BLM standard stipulations (Appendix D) are adhered to during all phases of the Project. Mitigation measures identified and applied as part of the analysis are included in Section 3.

Conservation measures to be implemented as required by the Project BO (File No. 2022-0035189-S7-001) through Section 7 consultation with the USFWS would be used for mitigation to reduce potential adverse impacts to the desert tortoise from the Project activities (Appendix E) (USFWS 2022a).

2.2.8.1 REQUIRED PLANS

The BLM requires a POD for the development and implementation of the Project (SWCA 2023). The POD details the methods and procedures that would be used in construction, O&M, and decommissioning of the Project. Various management plans are prepared and implemented in accordance with BLM requirements and approval. The following is a list of the various management plans that may

be applicable to the Project. As appropriate, the BLM will include other agencies to review these plans prior to issuance of the notice to proceed. The list is not exhaustive, and plans may be added or removed:

- Bird and Bat Conservation Strategy Plan
- Cultural Resources Management and Mitigation Plan (includes Archeological Resources Monitoring and Discovery Plan, Native American Graves Protection and Repatriation Act Plan of Action/Unanticipated Burial Contingency Plan)
- Dust Abatement Plan
- Facility Decommissioning Plan and Site Reclamation Plan (at the conclusion of the life of the Project)
- Fencing Plan (Desert Tortoise Exclusion and Perimeter)
- Fire Management Plan
- Health and Safety Plan
- Insect and Indoor Rodent Control Plan
- Integrated Weed Management Plan
- Lighting Management Plan
- Raven Management Plan
- Restoration Plan (Temporary Laydown Area)
- Site Drainage Plan and Stormwater Pollution Prevention Plan (SWPPP)
- Spill Prevention and Emergency Response Plan
- Traffic Management Plan
- Worker Environmental Awareness Program (WEAP)

2.2.8.2 RESOURCE SURVEYS

The following resource surveys have been completed in support of the Project: pre-project desert tortoise 100% presence/absence surveys; cactus/yucca counts and vegetation surveys; and cultural resources class III inventory.

2.3 No Action Alternative

Under the No Action Alternative, the BLM would not approve a ROW for the Project. The PV solar facility would not be constructed, operated, maintained, or decommissioned. Therefore, there would be no ground disturbance or loss of habitat.

2.4 Alternatives Considered but Eliminated from Detailed Analysis

2.4.1 Larger Project Boundary

The original application requested 250 acres for development of the Project, but 165 acres were removed from the proposed Project due to overlap with an area important for community pit needs. An additional 5 acres were removed due to the presence of greater than 5% slope, to achieve Las Vegas RMP conformance for the Project. After the removal of all acres with such conflicts, the remaining 80 acres for the solar facility and the 5-acre laydown area (Project Area) are evaluated in this document.

2.4.2 Maintaining Vegetation, Soils, and Surface Water Flow within the Solar Field Area

An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared with other operational and proposed utility-scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography, including surface sediments composed of cobbles, gravel, and sand in combination with design constraints, will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required.

The proposed Project is also located within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and Project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.

3 AFFECTED ENVIRONMENT AND ENVIRONMENTAL EFFECTS

3.1 Resource Evaluation

As part of the prioritization process specified in 43 CFR 2804.35, BLM SND resource specialists reviewed the Proposed Action and assessed whether each resource was present or not present within the Project Area (BLM 2021a).

Table A-6 lists all resources/uses considered in the evaluation of the Proposed Action and the No Action Alternative. The resources/uses found that may be affected by this Project have been carried forward for detailed analysis and are discussed further in this section. The resources/uses that are not present or that are found to not be impacted by the Proposed Action because they would be completely mitigated with the implementation of applicant-committed EPMs (see Appendix C) and BLM standard stipulations (see Appendix D) are not carried forward for detailed analysis.

Impacts can either be short term (temporary) or long term (permanent). Short term generally refers to impacts during construction and/or decommissioning, up to 5 years. Long term generally refers to impacts lasting more than 5 years or for the life of the Project, which includes any impacts that may persist after decommissioning.

For the purposes of this EA, the Project or Project Area will be defined as the 80-acre solar facility plus the 5-acre temporary laydown area, for a total of 85 acres.

3.2 Air Quality

Air quality is characterized by ambient air quality standards and the factors that affect the National Ambient Air Quality Standards (NAAQS), including the location of pollutant sources, the type and magnitude of emissions, and the local atmospheric conditions. The NAAQS, set by the U.S. Environmental Protection Agency (EPA) under the federal Clean Air Act, specify the concentration and duration of exposure to criteria pollutants (carbon monoxide; lead; nitrogen oxides; ozone; particulate matter [$PM_{2.5}$ are fine inhalable particles with diameters that are generally 2.5 micrometers and smaller, and PM_{10} describes inhalable particles with diameters that are generally 10 micrometers and smaller]; and sulfur dioxide), which can be harmful to public health and the environment. Individual states can promulgate their standards for the criteria pollutants as long as they are at least as stringent as the federal standards (EPA 2022a).

Section 176 of the federal Clean Air Act requires federal agencies that fund, permit, or approve an activity to ensure that the activity complies with the applicable State Implementation Plan (SIP) adopted to eliminate or reduce air quality violations (42 USC 7506). In order to ensure that air pollutant emissions associated with federally approved or funded activities do not exceed emission budgets established in the applicable SIP and do not interfere with the state's ability to attain and maintain the federal NAAQS in areas working to attain or maintain the standard, the EPA passed federal conformity rules. The General Conformity rule applies to all projects that are not related to transportation. According to 40 CFR 51, Subpart W, a detailed determination of the General Conformity rule's applicability is required when federal actions or funding of non-transportation–related activities in nonattainment areas result in emissions that exceed de minimis threshold levels (EPA 2022b).

In Nevada, under the NRS for air pollution, each county in the state with a population equal to or greater than 100,000 people must establish a board of county commissioners to establish and implement an air pollution control program (NRS 445B.500). In 2001, the Clark County Board of County Commissioners established the Department of Air Quality to carry out the mandated program. There are 17 counties in the state of Nevada. All but two counties are overseen by the Nevada Division of Environmental Protection (NDEP) for the implementation of the Clean Air Act. Washoe and Clark Counties have a delegated authority by the Governor of the State of Nevada for the implementation of the Clean Air Act.
The Division of Air Quality under the Clark County Department of Environment and Sustainability is responsible for administering the air pollution control program for Clark County under the provisions of the Clark County Air Quality Regulations and the EPA-approved SIP for Clark County, Nevada (Clark County Air Quality Regulations Section 00 through Section 94 as adopted in 40 CFR 52, Subpart DD).

Due to the geographic size, topography, and high-density population of Clark County, hydrographic basins are used to delineate air quality management areas versus political boundaries such as counties (Clark County 2004). The Project lies within the northern portion of the Eldorado Valley, which is within the Eldorado Valley hydrographic basin (167) and represents the analysis area for air quality.

3.2.1 Affected Environment

The Basin and Range province, which is characterized by faulted mountain chains and flat, arid valleys, has a climate that is typical of the eastern Mojave Desert, where the Project is located. Daytime and nighttime temperatures are highly variable due to strong seasonal winds, and bimodal precipitation patterns in the late summer and winter are typical (Western Regional Climate Center [WRCC] 2012). Maximum daytime temperatures in the area in and around the Project range from 101.6 degrees Fahrenheit (°F) (38.7 degrees Celsius [°C]) in the summer to 38.6 °F (3.7°C) in the winter. According to the WRCC, the area receives 5.55 inches (14.10 cm) of annual average precipitation (WRCC 2012).

Two major transportation corridors are within the valley, I-11/U.S. 93 and U.S. 95. The annual average daily traffic volume along these corridors is 27,800 and 560 vehicles, respectively (Nevada Department of Transportation [NDOT] 2021). The closest city to the Project Area is Boulder City, Nevada.
The downtown area of Boulder City is approximately 5 miles northeast of the Project with a population of 14,879 residents (U.S. Census Bureau 2021a).

Air quality within the Eldorado Valley hydrographic basin (167) is considered in "attainment" or "unclassifiable" for carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter 10 microns in diameter or smaller (PM_{10}), and sulfur dioxide according to the 2008 NAAQS. As for the 1997 ozone NAQQS, the hydrographic basin is within an ozone maintenance area (Clark County 2021a) and is subject to a SIP (Clark County 2021b). The closest air monitoring station for ozone is more than 5 miles from the Project, within the Garret Junior High School in Boulder City, Nevada (Clark County 2022), and recent ozone data from this monitoring station are summarized in Appendix F.

Ozone, in most cases, is not directly released into the atmosphere but is instead produced by sunlight-induced chemical reactions between nitrogen oxides (NO_x) and volatile organic compounds (VOCs). NOx and VOCs are referred to as ozone precursor pollutants because they have the potential to engage in chemical processes that lead to the formation of ozone. A major contributing source of NO_x and VOCs is fossil fuel burning from gasoline and diesel-powered construction equipment and vehicles. VOCs also come from natural sources that include biogenic processes and wildfires (Montero-Montoya et al. 2018). Therefore, the Project's implementation will negatively impact air quality if proposed activities result in emissions that exceed the General Conformity de minimis levels for NO_x and VOCs.

Valley Fever, or Coccidioidomycosis, is a lung disease that is prevalent in the southwestern United States. The fungus *Coccidioides immitis*, which grows in soils with low rainfall, high summer temperatures, and moderate winter temperatures, causes Valley Fever. When the soil is disturbed by winds, construction, farming, or other activities, these fungal spores become airborne. Infection occurs when a spore is inhaled by a susceptible person or animal. Construction, agriculture, and archaeology workers are at a higher risk of exposure and disease because their jobs cause soil disturbance, which can lead to the presence of fungal spores. The Proposed Action, in Clark County, is in an area that may harbor the fungus that causes the disease Valley Fever. Cases of Valley Fever are highest in Arizona with over 100 cases per 100,000 people but is also present to a lesser degree in other states; the combined incidence in Nevada, New Mexico, and Utah in 2017 was 3.5 cases per 100,000 people. Clark County data show between 0 and 5.9 cases of Valley Fever per 100,000 people in 2017, the lowest bracket of incidence other than no incidence (Benedict et al. 2019). In 2020, 189 cases of Valley Fever in Nevada were reported (Centers for Disease Control and Prevention 2020). Therefore, while Clark County does have a risk of Valley Fever, it is a relatively low risk when compared to Nevada as a whole and other neighboring states.

3.2.2 Environmental Effects of the No Action Alternative

No air quality impacts associated with the Proposed Action would result under the No Action Alternative.

3.2.3 Environmental Effects of the Proposed Action

3.2.3.1 SHORT- AND LONG-TERM EFFECTS

The Project has the potential to disturb up to 85 acres, and air emissions generated from the Proposed Action could have short- and long-term adverse impacts to the existing air quality environment. Short-term increased air emission levels would occur as a result of the construction and decommissioning of the solar facility and ancillary components including the use of backup generators. These generators would be used for emergency backup service with limited operations for readiness testing, but not for prime power or peak shaving. The diesel generator engines would be maintained in accordance with the manufacturer's written emission-related instructions and would also be subject to the ongoing maintenance and inspection provisions of 40 CFR 63 Subpart ZZZZ. This construction also includes the geotechnical investigation, which consists of geotechnical soil borings, soil resistivity testing, and pile load testing. Minimal long-term direct and indirect emissions would occur during O&M of the solar facility as a result of the limited number of FTE staff and delivery vehicle emissions. Geotechnical investigation would contribute minimally to the short-term increased air emission levels, as these activities include a short duration of fieldwork (up to 12 days) and a small-scale disturbance (0.93 acre of the potential 85 acres).

For construction (and geotechnical investigation), O&M, and decommissioning, emission factors in grams per vehicle mile traveled for on-road vehicles were obtained from the EPA's Motor Vehicle Emission Simulator (MOVES3) model. Emissions from non-road construction equipment engines used during Project construction were estimated based on the anticipated types of non-road equipment and their associated levels of use. Emission factors in grams per hour were obtained using the most recent version of the EPA's MOVES3 model. Total disturbed acres, wind erosion, and paved and unpaved road fugitive dust emissions were calculated using EPA AP-42, a compilation of air pollutant emissions factors that has been published since 1972 as the primary compilation of the EPA's emissions factor information. Assumptions and complete air pollution emission calculation methodology and data are provided in Appendix F. Table A-7 shows the estimated annual Project emissions of ozone precursors (NO_x and VOCs) that are estimated to be generated by the Project. De minimis levels of NO_x and VOCs would not be exceeded as a result of the Proposed Action. Therefore, the Project is not subject to the federal General Conformity rule.

Fugitive dust—particles of soil that become airborne—can also impact air quality by causing hazy conditions that reduce visibility and increase particulate matter (PM_{10}), a criteria pollutant, in the air that can be inhaled by humans causing adverse health effects. Various types of activities cause fugitive dust, including excavation, grading, structure removal, and vehicle travel on unpaved surfaces. The amount of dust is dependent on the intensity of the activity, soil type and conditions, and wind speed. Fugitive dust air emissions would be greatest during the initial site grading and excavation and decommissioning activities but then would vary day to day depending on the work phase, level of activity, and prevailing weather conditions. Implementing dust control measures, i.e., BMPs such as using water to reduce airborne dust associated with surface disturbance, would greatly reduce dust emissions.

Construction- and decommissioning-related emissions also include exhaust from the operation of construction equipment, including diesel particulate matter (see Table A-4). The use of newer construction equipment with emissions controls and minimizing the times that the equipment is idling (BMPs) reduces construction equipment exhaust emissions. The transport and delivery of materials and construction workers commuting daily in their personal vehicles would also result in negligible increases in criteria pollutant emissions, which have been quantified in Appendix F. Decommissioning emissions would be similar to construction emissions for the purpose of the analysis. Long-term impacts to air quality are anticipated to be inconsequential; these operational emissions have been quantified in Appendix F.

The Proposed Action is in an area that may harbor the fungus that causes the disease Valley Fever. Construction could occur in an area that may be favorable to the growth of the "Valley Fever" vector, which is the fungus *Coccidioides immitis*. Project construction would disturb the soil and may cause the fungal spores to become airborne, potentially putting construction personnel, local communities, and wildlife at risk of contracting Valley Fever. Although most Valley Fever cases are very mild, and more than half of infected people either have no symptoms or experience flu-like symptoms and never seek medical attention, in extreme cases the disease can be fatal.

3.2.3.2 BENEFICIAL AND ADVERSE EFFECT

The Proposed Action would not contribute ozone precursors in an amount that would trigger federal General Conformity rules. Furthermore, solar energy technologies and power plants do not produce air pollution or greenhouse gases (GHGs) when operating, and as solar energy replaces or reduces the use of other energy sources, there may be positive, indirect effects on the environment (U.S. Energy Information Administration 2022). Surface disturbance of up to 85 acres during initial construction and decommissioning activities would contribute to particulate matter and ozone pollution and possible airborne Valley Fever fungal spores that could adversely impact air quality and the health of City of Boulder City residents. Implementation of BMPs (listed above) and applicant-committed EPMs listed in Appendix C would reduce the potential for effects on air quality and health concerns from airborne fungal spores. Additionally, there may be positive, indirect effects on the environment as the solar energy replaces or reduces the use of other energy sources (U.S. Energy Information Administration 2022).

3.2.4 Mitigation Measures

The Project will obtain and adhere to the Clark County Dust Control Operating Permit for Construction Activities and local government permit requirements. Backup generator(s) will be present, and the applicant will obtain a stationary source permit prior to construction, as required (Table A-1 in Appendix A). Implementation of the BMPs and the applicant-committed EPM #1 Air Quality (see Appendix C) would minimize the anticipated short-term, construction/decommissioning-related air quality impacts.

To mitigate the effects of fugitive dust, the applicant will adopt the following mitigation measure: a designated dust control monitor will be on-site to ensure dust control measures are implemented pursuant to the provisions of Section 94.8 of the Clark County Air Quality Regulations. No other mitigation measures are anticipated.

3.3 Cultural Resources

Cultural resources in Nevada are protected by federal and state laws, regulations, and statutes. Section 106 of the NHPA, as amended in 2000, requires government agencies to consider the effects of their actions on properties listed in or eligible for listing in the National Register of Historic Places (NRHP). Cultural resources refer to both human-made and natural physical features associated with human activity and, in most cases, are finite, unique, fragile, and nonrenewable. Cultural resources that meet the eligibility criteria for listing in the NRHP are considered "significant" historic properties and must be taken into consideration during the planning of federal projects.

3.3.1 Affected Environment

SWCA conducted a Class III cultural resources inventory of the entire Project Area (i.e., direct area of potential effects [APE]), as part of BLM compliance with Section 106 of the NHPA. The inventory included records searches, an intensive-level pedestrian survey of the direct APE, recordation and evaluation of cultural resource sites and isolates within the APE, and preparation of an inventory report (SWCA 2022a). In addition to the Class III inventory, a visual, auditory, atmospheric effects Class I analysis was conducted to determine potential atmospheric, auditory, vibrational, and visual effects on cultural resources within the indirect APE (i.e., 5-mile buffer of the Project Area).

The record search identified a single previously recorded site within the direct APE, consisting of a historic refuse scatter recommended ineligible for the NRHP. Also present within the direct APE was a single architectural resource consisting of the Old U.S. Highway 95/U.S. 95, which has been recommended eligible for the NRHP under Criterion A.

During the field inventory, SWCA recorded two newly identified historic sites, which were recommended ineligible for the NRHP. Old U.S. Highway 95/U.S. 95, which is eligible for the NRHP under Criterion A, was also observed in the direct APE during the survey. The previously recorded historic debris scatter had been destroyed and no longer exists.

The visual, auditory, atmospheric Class I records search identified a total of nine cultural resources that were identified as potentially sensitive to visual impacts from the Project. These resources include five transmission lines (Southern Sierras/Nevada-California Power Company 132kV Transmission Line, Los Angeles Department of Water and Power Boulder Transmission Line, Southern California Edison North Transmission Line- Hoover-Chino No. 1, Southern California Edison South Transmission Line-Hoover-Chino No. 2, and the Hoover Basic South Transmission Line), a branch of the Boulder City Branch Railroad, the Old U.S. Highway 95/U.S. 95 roadway, the historic Blue Quartz Mine, and an unnamed historic mining site. An alignment of the Old Spanish Trail was identified within the viewshed area according to the National Historic Trails database; however, the segment has not been formally recorded as an archaeological resource in this area and, therefore, was not analyzed for visual effects.

3.3.2 Environmental Effects of the No Action Alternative

The No Action Alternative would not result in impacts to cultural resources.

3.3.3 Environmental Effects of the Proposed Action

3.3.3.1 SHORT- AND LONG-TERM EFFECTS

Geotechnical investigation would not have any short- or long-term direct effects on cultural resources or visual effects associated with atmospheric, auditory, vibrational, and visual impacts would not be expected to result in an appreciable impact. Geotechnical investigation would occur over a short-time period (up to 12 days) and is temporary. The single historic property identified within the direct APE consists of a historic highway segment (i.e., Old U.S. Highway 95). The Project would utilize the historic road resource as an access route in a manner consistent with existing conditions.

The Proposed Action would not have any short- or long-term direct effects on cultural resources. No modifications or maintenance activities are proposed that would impact the eligibility status of Old U.S. Highway 95 as a historic road resource. The Project would utilize the historic road resource as an access route in a manner consistent with existing conditions. No modifications or maintenance activities are proposed that would impact the eligibility status of the road.

The Proposed Action would also not have any short- or long-term visual effects on cultural resources as a result of atmospheric, auditory, vibrational, and visual impacts. Key observation point (KOP) analysis indicated the visual setting within the indirect APE has been significantly compromised as a result of extensive development within the Eldorado Valley and surrounding areas. Consequently, the Project would not represent a significant impact to the eligibility status of any historic properties within the Project viewshed. Atmospheric, auditory, and vibrational effects from the Project would be restricted to construction and are therefore considered temporary and negligible. Therefore, no appreciable impact is expected.

3.3.3.2 BENEFICIAL AND ADVERSE EFFECT

The Project would have no adverse effects on cultural resources, either directly or indirectly, and it would not contribute to cumulative cultural resource impacts.

3.3.4 Mitigation Measures

No mitigation measures are recommended because there would be no impact to cultural resources as a result of the Project.

3.4 Environmental Justice and Socioeconomics

Executive Order (EO) 12898, “Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations,” was issued by President William J. Clinton in 1994 concerning environmental justice (EJ). The purpose of EO 12898 is to focus federal attention on the environmental and human health effects of federal actions on minority and low-income populations with the goal of achieving environmental protection for all communities. According to the EPA, EO 12898 directs federal agencies to identify and address the disproportionately high and adverse human health or environmental effects of their actions on minority and low-income populations, to the greatest extent practicable and permitted by law. The order also directs each agency to develop a strategy for implementing EJ. The order is also intended to promote nondiscrimination in federal programs that affect human health and the environment, as well as provide minority and low-income communities access to public information and public participation (EPA 1994).

3.4.1 Affected Environment

3.4.1.1 ENVIRONMENTAL JUSTICE

For this Project, the analysis area has been defined as census block groups (BGs) 320030057111, 32003005501, 32003005435, 32003005360, 32003005362, and 32003005358 in Clark County, Nevada (Table A-8) (BLM 2023; EPA 2023c; U.S. Census Bureau 2021b). These BGs were chosen because they overlap with the Project Area or are directly proximal to the Project Area and represent the communities most likely to be impacted by the Proposed Action. The reference area for this Project is the State of Nevada. EJ population data for the BGs within the analysis area and for the reference area are provided in Table A-8 and are visually represented in Supplemental Map 1 (see Table A-8).

Low-Income Community Analysis

A low-income community of concern is present if the population experiencing poverty in one or more analysis area geographies are near, at, or less than 200% of the federal poverty rate of the reference area or if the population of the community experiencing poverty is at or above 50%. The Nevada non-metropolitan reference threshold for the federal poverty rate is 31.2% (see Table A-8).

Based on a review of the BLM's Headwaters Economics database (BLM 2023) and EPA's EJ Screening tools (EPA 2023c), it is estimated that 25% of the analysis area population identified as low-income. The screening identified that BG 32003005360 has a low-income population that is greater than the reference population threshold, with 40.1% identifying as low income (see Table A-8) (BLM 2023). Therefore, a low-income EJ population is present in the analysis area.

Minority Community Analysis

A minority community of concern is present if the percentage of the population identified as belonging to a minority group in the analysis area is 1) equal to or greater than 50% of the population, or 2) meets the "meaningfully greater" threshold. Meaningfully greater is calculated by comparing the minority group population percentage with 110% of the reference area minority population. The State of Nevada threshold is 51.8% (see Table A-8); therefore, the meaningfully greater analysis threshold is 57%.

It is estimated that 19.4% of the analysis area identifies as belonging to a minority population group. However, no BGs in the analysis area have a minority population that is meaningfully greater than the reference population threshold, therefore, a minority EJ population is not present (see Table A-8) (BLM 2023).

American Indian Community Analysis

Tribal communities of concern are present if the percentage of the population identified as belonging to an indigenous community is equal to or greater than the reference population.

Based on a review of the BLM's Headwaters Economics database (BLM 2023) and EPA's EJ Screening tools (EPA 2023c), it is estimated that 1.1% of the analysis area identifies as belonging to a tribal population group. However, no BGs in the analysis area have an American Indian population greater than the reference population; therefore, an American Indian EJ population is not present in the analysis area (see Table A-8) (U.S. Census Bureau 2021b).

3.4.1.2 SOCIOECONOMICS

In 2017, the BLM contracted with Headwaters Economics to develop a BLM-specific version of its Economic Profile System tool. This new data tool is called the BLM Socioeconomic Profile tool and was released for use in 2018 to generate basic economic and demographic reports (in Microsoft Excel and PDF formats) for county-based geographic units that include field office, district office, county, certain special designation lands, and state boundaries (BLM 2023).

The Project Area is in Clark County, Nevada, and is less than 15 miles from Mohave County, Arizona. However, Mohave County is not included in this analysis because there are no populated communities in Mohave County within 25 miles of the Project Area. Therefore, for the analysis, Clark County has been used as the proxy for the Project Area or analysis area, and the State of Nevada is being used as the reference location for comparison. The data were obtained from the U.S. Department of Labor, Bureau of Labor Statistics, local area unemployment statistics, U.S. Department of Commerce, and U.S. Census Bureau, as compiled by Headwaters Economics for the BLM (BLM 2023).

Land Ownership

In Clark County, there is a total of 5,158,982 acres of land. Of that total, 86.2% of the land is federally managed, of which 51.0% is administered by the BLM. Private land holdings in Clark County accounts for 11.3% of land, Tribal land accounts for 1.5%, and other governmental entities (state, city, and county government) account for 1.0% (BLM 2023).

Population, Employment, and Income

In 2021, Clark County's population was 2,292,476, a 64.5% increase from its population of 1,393,909 in the year 2000 (BLM 2023). This population increase occurred at a faster rate than increases in the United States over the same period and can be attributed to population growth and migration to Clark County over the last 20 years (U.S. Census Bureau 2021c). Per capita income annually in Clark County has increased by 19.1%, from $52,859 to $62,938, between 2000 and 2021. The number of people employed in Clark County as of 2021 is 1,368,492; this is an increase of 60.4% since 2000. However, the number of people employed in Clark County is only 60% of the total residents of the county (BLM 2023). Therefore, it is reasonable to assume that the majority of those residing in Clark County are unemployed or employed elsewhere.

Jobs by Industry

According to the 2021 estimate, there are approximately 1,368,492 jobs in Clark County. Jobs in Clark County are generally service related, accounting for 82.6% of jobs in the county as of 2021. Non-service-related jobs and government jobs each constituted 8.7% of jobs in Clark County. The dominant industries in Clark County are accommodation and food service, retail trade, health care and social assistance, and administrative and waste services. The industries that grew the most between 2000 and 2021 are transportation and ware housing, and health care and social assistance. The only industries in Nye County that declined between 2000 and 2021 were farming and information (BLM 2023).

3.4.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, there would be no change in socioeconomic conditions.

3.4.3 Environmental Effects of the Proposed Action

3.4.3.1 SHORT- AND LONG-TERM EFFECTS

Environmental Justice

Disproportionate high and adverse effects are those effects that are greater than those experienced by other populations (Environmental Justice Interagency Working Group 2019). Although one EJ population is present within the analysis area, it is not anticipated that impacts from the Proposed Action would disproportionately affect this group. The closest communities to the Project Area where EJ populations would be present are Mission Hills and Boulder City, which are approximately 5 miles from the Project Area (see Table A-8 and Supplemental Map 2). No American Indian reservations are proximal to the Project Area; the closest are the Las Vegas Tribe of Paiute Indians and the Moapa Band of Paiute Indians, approximately 35 and 50 miles from the Project Area, respectively.

Previous development, such as renewable energy and transmission facilities, roads, and other land development, is already present near and the around the Project Area. A relatively small amount of additional land, 85 acres, would be impacted by the Proposed Action, and the impacts to EJ populations would be similar to the impacts of previous actions. The closest communities to the Proposed Action are likely to feel the effects of increased noise, fugitive dust, hazardous or solid waste, or other minor, localized impacts from constructing the Proposed Action.

The presence of the solar facility may reduce recreation access to local EJ communities, primarily during construction. However, access to public land for recreation would remain available during operation, so negligible long-term impact to recreation for EJ communities is anticipated. EJ communities recreating and living near the Project Area would also be aware of the visual impacts caused by the Proposed Action; however, the visual impacts would be similar to the existing development around the Project Area.

Other resources, such as cultural resources, biological resources, fuels and fire, and soils and surface hydrological conditions are expected to have negligible or minor impacts from the Proposed Action; however, EJ populations are not expected to feel the impacts to those resources.

Socioeconomics

Geotechnical investigation would not have an appreciable impact on the local economy due to the temporary and short-term nature of the work (up to 5 weeks). Employment participation would be limited to those individuals with the necessary skillset. Therefore, no impacts would occur to local services.

During construction and decommissioning (i.e., short term), the Project would have a direct positive effect on the local economy. Construction would typically require 100 to 150 laborers and supervisors over a 12-month period. Decommissioning would likely require less than half of the projected construction workforce. O&M staff (maximum of three FTE staff) would oversee, remotely monitor, and control the operation of the solar facility. It is anticipated that the majority of construction personnel would be from Clark County.

The addition of temporary jobs during construction and decommissioning may result in short-term beneficial impacts to income and employment in Clark County, as long as the individuals have the required experience/training required. The addition of temporary jobs or three permanent jobs during O&M activities is not anticipated to contribute to a significant population increase, as Clark County has almost 1 million labor force residents. Because the long-term employment opportunities are limited,

the Proposed Action is unlikely to have direct, indirect, or cumulative impacts to local social services or demographics, but may contribute to income and employment over the short term.

3.4.4 Mitigation Measures

No mitigation measures are anticipated.

3.5 Fish and Wildlife (Excluding Federally Listed Species)

Wildlife includes all fish and wildlife species that are not federally listed as threatened, endangered, or candidate species under the ESA. General wildlife species include those species that are common in terms of abundance and distribution across the landscape. Impacts to migratory birds and eagles are discussed in Section 3.8.

Some species of wildlife are considered special-status species by federal and state agencies largely because they are either scarce on a regional level, facing clearly defined threats, or in a position within the regional landscape to potentially become scarce. For the purposes of this EA, special-status species include BLM sensitive species, which are those species requiring special management consideration to avoid potential future listing under the ESA and have been identified in accordance with the BLM Manual 6480 – Special Status Species Management (BLM 2008b).

In the State of Nevada, certain wildlife species are classified as protected, threatened, or sensitive under NAC 503.103 and 503.104 or are considered species of conservation priority (SoCP) according to the Nevada Department of Wildlife's (NDOW's) 2012 Nevada Wildlife Action Plan (Wildlife Action Plan Team 2012). The Nevada Division of Natural Heritage (NDNH), a division of the Nevada Department of Conservation and Natural Resources, also maintains a database of known occurrences of federally listed, state-listed, and at-risk species. The USFWS also maintains a list of birds of conservation concern (BCCs) that warrant special attention and may be present in the analysis area. Species designated as threatened, endangered, or candidate under the ESA are discussed in Section 3.12.

The analysis area consists of the Project Area and an approximately 10-mile buffer surrounding the Project Area for raptor species (200,960 acres) and an approximately 4-mile buffer surrounding the Project Area for all other wildlife including special-status wildlife (37,342 acres total but 26,892 acres is undeveloped habitat) (Figure A-7). This area is intended to capture existing conditions and potential impacts to individuals, habitats, and movement corridors for species with larger home ranges. For other wildlife species with smaller home ranges (such as reptiles and small mammals), impacts would be limited to the Project Area and immediate vicinity.[1]

3.5.1 Affected Environment

3.5.1.1 GENERAL WILDLIFE

The analysis area supports a variety of desert-adapted wildlife species characteristic of the eastern Mojave Desert. Desert communities typically exhibit a low diversity of wildlife species due to the extreme environmental conditions (Farabee 2007). Wildlife diversity also varies according to topography, plant community, soil type, season, and proximity to water.

[1] Project vicinity includes the Project Area and a 4-mile buffer.

Wildlife species in the analysis area may include birds, insects, reptiles, rodents, and small mammals (including bats). Wildlife species that may use the Project Area and vicinity are listed in Table A-9. For a comprehensive list of potential wildlife species that may be present, please refer to the 1998 Las Vegas RMP and Final Environmental Impact Statement Volume II for the BLM SND (BLM 1998b).

3.5.1.2 SPECIAL-STATUS WILDLIFE SPECIES

Based on a review of the habitat requirements and historical and recent observations for special-status wildlife with a potential to occur, 29 of 86 species (including subspecies) have a "moderate" or "high" potential to occur within the analysis area (BLM 2017; NDNH 2022; NDOW 2022a). These species consist of seven avian species, eight mammal species, and 14 reptile species. Of the 29 species, 18 species have been identified as having a "high" potential to occur due to direct observation(s), widespread habitat for the species, or nesting/foraging habitat present within the analysis area. In total, 11 special-status species have "moderate" potential to occur because habitat is limited but the analysis area may be used for foraging, migration, or other temporary use. The other 57 special-status species were identified as having either "low" potential (unlikely to occur) or "no" potential because no suitable habitat is present to support the species and/or the species was not observed during surveys. Table A-10 lists the special-status wildlife species and identifies the potential for occurrence in the analysis area. Special-status plant species are discussed in Section 3.13.

There are no fish, mollusks, or amphibians present in the analysis area because there are no permanent water sources (aside from artificial ponds at the Boulder Creek Golf Club, approximately 3.5 miles from the Project Area).

NDNH provided information for observed special-status species within the analysis area. The only occurrence of a federally listed species in the Project vicinity was for the desert tortoise, which is discussed in Section 3.12. No other state-listed or at-risk species were recorded. However, several species were observed within the 4-mile analysis buffer, including the western burrowing owl (*Athene cunicularia hypugaea*), a BLM sensitive species and an NDOW SoCP; the banded Gila monster (*Heloderma suspectum cinctum*), a BLM sensitive species, Nevada protected species, and an NDOW SoCP; and three bat species (canyon bat [*Parastrellus hesperus*], Mexican free-tailed bat [*Tadarida brasiliensis*], and pallid bat [*Antrozous pallidus*]) that are BLM sensitive species and Nevada protected species, except for the Mexican free-tailed bat, which is also an NDOW SoCP (NDNH 2022). One western burrowing owl occurrence is documented on the valley floor near the northeastern portion of the 4-mile buffer. The eight occurrences of the banded Gila monster are limited to the mountainous areas in the northwestern and southwestern portions of the 4-mile buffer. Based on species-specific habitat requirements, both species have "high" potential for occurrence within the 4-mile buffer, but only the western burrowing owl has potential to occur within the Project Area. Occurrence data for the desert tortoise is discussed in Section 3.12.

Nevada SoCP are species identified by the NDOW as species in greatest need of conservation in Nevada (Wildlife Action Plan Team 2012). Of the 54 SoCP with potential to occur in the analysis area, 19 have "moderate" and "high" potential to occur (see Table A-10), and according to the NDOW, 14 of the 19 SoCP species have been observed in the vicinity of the Project Area (see NDOW data response letter in Appendix G).

Avian Species

The USFWS identified the following BCCs with the potential to occur in the analysis area: black-chinned sparrow (*Spizella atrogularis*), Clark's grebe (*Aechmophorus clarkii*), Costa's hummingbird (*Calypte costae*), golden eagle (*Aquila chrysaetos*), long-eared owl (*Asio otus*), marbled godwit

(*Limosa fedoa*), and willet (*Tringa semipalmata*) (see Appendix G) (USFWS 2022b). Habitat for the black-chinned sparrow is not present in the analysis area, and habitat for Clark's grebe, long-eared owl, marbled godwit, and the willet is unlikely (i.e., low potential for species to occur). Costa's hummingbird and the golden eagle are the only species that have potential to occur, but the potential is limited to foraging habitat or temporary use only (i.e., "moderate" potential) (Cornell Lab of Ornithology 2022; NatureServe Explorer 2022).

NDOW reported that various raptor species with varying special-status species designations (i.e., BCC by USFWS, BLM sensitive, Nevada protected, and/or NDOW SoCP) have distribution ranges that include the analysis area. These include bald eagle (*Haliaeetus leucocephalus*), barn owl (*Tyto alba*), Cooper's hawk (*Accipiter cooperii*), ferruginous hawk (*Buteo regalis*), flammulated owl (*Psiloscops flammeolus*), golden eagle, great horned owl (*Bubo virginianus*), long-eared owl, merlin (*Falco columbarius*), northern goshawk (*Accipiter gentilis*), northern harrier (*Circus hudsonius*), northern saw-whet owl (*Aegolius acadicus*), osprey (*Pandion haliaetus*), peregrine falcon (*Falco peregrinus*), red-tailed hawk (*Buteo jamaicensis*), rough-legged hawk (*Buteo lagopus*), sharp-shinned hawk (*Accipiter striatus*), short-eared owl (*Asio flammeus*), turkey vulture (*Cathartes aura*), western burrowing owl, and the western screech owl (*Megascops kennicottii*). Furthermore, the golden eagle (a BLM sensitive species, Nevada sensitive species, and a NDOW SoCP) is the only raptor species that has been directly observed in the Project vicinity (see Appendix G) (NDNH 2022; NDOW 2022a). The golden eagle is discussed in Section 3.8.

NDOW also reported that there are 52 known or suspected raptor nests that are within 10 miles of the Project Area. Of the 52 known nests, seven are eagle nests, three are eagle/buteo (birds of the hawk family) nests, 33 are falcon nests, five are buteo nests, three are corvid (birds of the crow family) nests, and one nest is considered as an unknown species (see Appendix G) (NDOW 2022a). Nesting and foraging habitat for various raptor species, including the ones identified above by NDOW, is present within 10 miles of the Project Area. Eagle nests are discussed in Section 3.8.

No special-status avian or other bird species nests were recorded within the Project Area during surveys (Allied Pacific Partners, LLC 2022).

Mammals

Larger mammals have the potential to occur mostly within the mountain ranges (i.e., Black Hill and River Mountains in the western and northern portions of the analysis area, including those protected by game laws in Nevada). All big game species are designated as "Protected" game mammals in the state of Nevada. The only big game mammal identified by NDOW with potential to occur in the analysis area is the desert (Nelson's) bighorn sheep (*Ovis canadensis nelsoni*) (see Appendix G) (NDOW 2022a).

Smaller mammals serve as the prey base for many large mammals and raptor species. These include mostly nocturnal species of rodents (such as brush mice [*Peromyscus boylii*], Merriam's kangaroo rats [*Dipodomys merriami*], and pocket mice [*Chaetodipus* sp.]) and bats. Habitat may be present within the analysis area for various small mammal species.

Bat species designated as BLM special-status species and/or NDOW SoCP were also evaluated for potential occurrence (see Table A-10). Bats may use rocky outcrops, cliffs with crevices, caves, and/or mines as roosting habitat; there are several active and closed mining claims within the analysis area (Nevada Division of Minerals 2022). Foraging habitat could also be present for all special-status bat species identified as having a "high" or "moderate" potential for occurrence (see Table A-10).

Reptiles

Out of all species, reptiles are most likely to occur within the analysis area. Of the 14 reptile species with "moderate" or "high" potential to occur, all are considered BLM sensitive species except for the desert night lizard (*Xantusia vigilis*), the spotted leaf-nosed snake (*Phyllorhynchus decurtatus*), and the western banded gecko (*Coleonyx variegatus*), which are SoCP by NDOW (see Table A-10; see Appendix G) (NDOW 2022a). The federally threatened desert tortoise was also identified by NDOW as having potential to occur in the analysis area and is discussed in Section 3.12.

3.5.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related impacts to general wildlife or special-status wildlife species would occur. The existing wildlife would continue to utilize the habitat at current levels.

3.5.3 Environmental Effects of the Proposed Action

Impacts to general wildlife and special-status wildlife are discussed above in terms of potential changes to habitat, habitat use, and/or behavior. Indicators used to analyze impacts to general wildlife and special-status wildlife include areas of existing habitat that would be disturbed and areas of existing habitat that would be precluded from use.

3.5.3.1 SHORT- AND LONG-TERM EFFECTS

General Wildlife

Geotechnical investigation would impact approximately 0.93 acre of potential wildlife habitat. The majority of the impacts are borehole-related, as vehicles would primarily use existing trails. Implementation of BMPs described in Section 2.2.3 and the presence of a biological monitor on-site would minimize any potential effects from geotechnical investigation. No interim reclamation is proposed because these sites will be permanently disturbed as part of the solar facility and are already included in the 80-acre footprint.

Project activities under the Proposed Action would have short- and long-term effects on wildlife habitat within the Project Area and adjacent habitat. Construction activities include removal of surface vegetation and grading as necessary; permanent removal of vegetation along the access road (within the linear facilities segment) and auxiliary roads; trenching for PV solar array cabling and collection systems; excavations for foundations associated with the solar array field and BESS (if needed); and other holes for various fencing and transmission line poles for the distribution line (if needed).

Even though construction activities are temporary and intermittent, the Proposed Action would remove up to 85 acres (0.3% of the undeveloped land within the analysis area) of wildlife habitat. The habitat removal within the solar field development area would last for the life of the Project (estimated up to 30 years), whereas the habitat within the 5-acre temporary laydown area may be unsuitable up to an additional 10 to 20 years after reclamation (Abella 2010; Chambers et al. 2013; Copeland et al. 2018; Lovich and Bainbridge 1999). It is anticipated that the loss of some wildlife and their habitat would result from the Proposed Action. However, wildlife use of the Project Area is likely already reduced due to significant existing disturbance (i.e., U.S. 93 and 95, mining/quarry sites, solar facility, and businesses) bounding the Project Area to the north, south, and west.

Potential impacts associated with construction and decommissioning could include habitat loss, habitat fragmentation, barriers to gene flow, direct mortality, increased noise, dust and dust-suppression effects,

light pollution effects, and increased fire risk as a result of the introduction and spread of invasive weed species (Abella 2010; Chambers et al. 2013; De Marco et al. 2014; Hernandez et al. 2014; Lovich and Bainbridge 1999; Lovich and Ennen 2011; Moore-O'Leary et al. 2017).

Habitat loss and degradation due to the construction, O&M, and decommissioning of the Proposed Action could lead to an increased susceptibility to the introduction and spread of noxious weeds and nonnative invasive plant species and increased erosion potential. During operations, vegetation would be allowed to regrow no higher than commercially reasonable to minimize fire risk and site maintenance requirements. Opportunities for invasive and noxious weed introductions would persist during operations due to vehicle/equipment and human presence in an already susceptible environment due to previous construction disturbance. Implementation of BLM standard stipulations and applicant-committed measures would minimize potential impacts to the Project Area and adjacent areas from noxious weeds and nonnative invasive plants and reduce erosion potential (see Appendices D and E). Long-term impacts associated with permanent removal, fragmentation, and degradation of habitat can also increase some species' vulnerability to predators (Moore-O'Leary et al. 2017).

Mortality of wildlife may occur, although mobile species would likely be able to avoid injury or mortality by leaving the area. The less mobile species, nocturnal species, or species that use burrows might be more susceptible to injury or mortality from Project activities. Vehicles and/or equipment could also injure or kill wildlife species along the access road to the Project or auxiliary roads from vehicle strikes. Wildlife would also be disrupted in the short term by noise, mostly during construction and decommissioning activities. The presence of humans, and dust and dust-suppression effects and light pollution effects, can also impact wildlife in the short-term.

Permanent security lighting associated with operational activities would be required at the Project entrance gate and associated infrastructure. However, the lighting on-site would be directed downward with shields to reduce light pollution and impacts to wildlife, such as bats and migrating birds. A Lighting Management Plan would be prepared by the applicant, approved by the BLM, and implemented throughout Project-related activities as required.

Once the Project is completed, reclamation of the 5-acre laydown area will begin. Because of the semiarid nature of the environment, reclamation efforts and natural vegetation reestablishment may take several seasons or decades (Abella 2010). The laydown area may still be unsuitable long term for most wildlife after reclamation efforts.

Following the end of project operations, decommissioning and reclamation (of the solar facility) would occur. Short-term impacts would be the same as those during construction. With the removal of the Project's facilities, wildlife could return to the Project Area. However, vegetation structure may not be representative of pre-Project conditions even with reclamation because desert ecosystems are known to take many years to recover from disturbance (Abella 2010).

Special-Status Wildlife

Avian Species

Geotechnical investigation would impact 0.93 acre of potential nesting and foraging habitat for special-status bird species (and foraging habitat for raptors). Nest abandonment and injury/mortality could occur to eggs and nestlings but would be minimized by the presence of a biological monitor during geotechnical investigation. Short-term displacement would occur during geotechnical work, but longer-term displacement would result at the start of Project construction. Long-term impacts include the permanent removal of habitat as the geotechnical sites would become part of the solar facility footprint, so no interim reclamation is recommended. Impacts to individual species may occur but long-term impacts to regional

special status avian population levels would not occur as the geotechnical work is temporary (up to 12 days) and will occur over a small-scale (0.93 acre) that is already planned to be incorporated into the solar facility footprint.

Potential short-term impacts to special-status bird species (and raptors) include the loss of nesting and foraging habitat, injury or mortality from construction/decommissioning activities, and displacement by noise from vehicles and equipment. Indirect effects from noise and vibration associated with construction activities and increased human noise and activity may also disturb avian behavior in the vicinity of the activities. Impacts to avian species associated with Project construction/decommissioning are not expected to result in a large reduction in avian population levels; however, displacement of some species may occur.

Long-term impacts include the permanent removal of usable habitat. There may be some small avian species that would reoccupy the Project Area, but foraging and nesting opportunities would be limited. During operations, the presence of solar panels in arrays can cause avian collisions with panels (i.e., "lake effect"). The introduction of structures (e.g., fencing and power poles) may provide new perching opportunities for raptors. At the end of Project operations, impacts to special-status avian species would be the same as those described for general wildlife. Due to the slow recovery of desert ecosystems, foraging and nesting habitat quality would likely vary throughout the Project Area based on success of restoration efforts.

No avian nests were observed within the Project Area (Allied Pacific Partners, LLC 2022); however, measures to minimize Project impacts would be applied (see BLM standard stipulations in Appendix D) should a nest be observed during the construction period.

Mammals

Desert bighorn sheep are unlikely to be disturbed by geotechnical investigation or construction (and decommissioning) and operation of the Project because their distribution is limited to the mountain ranges within the southwestern and southeastern portions of the analysis area (i.e., 4-mile radius). Therefore, no impacts to desert bighorn sheep are anticipated.

Nighttime lighting could create light pollution, which can disorient foraging bat species (Longcore and Rich 2004) during construction and O&M activities. To minimize potential impacts to foraging bats, nighttime lighting would be limited during construction and O&M and performed with temporary lighting, if needed. No geotechnical work will take place at night. There is no suitable roosting habitat present in the form of caves, mines, cliffs, bridges, or trees near perennial water sources.

The applicant would prepare a Lighting Management Plan for BLM approval that would contain several measures and BMPs that will minimize impacts. Therefore, it is unlikely that construction- or O&M-related lighting would have a measurable impact to bat populations in the analysis area.

Data associated with operational effects of solar panels on bats is very limited. Similar to birds, bats could mistake smooth, flat surfaces for water and attempt to land (Greif and Siemers 2010; Russo et al. 2012). Because solar panels are essentially smooth, flat surfaces, it is believed that this landing could result in injury or mortality. However, neither study was performed using solar panels as a proxy, so interpretation is limited. Also, glare and polarized light emitted by solar projects can serve as attractants for many insects (Horváth et al. 2009, 2010), which, in turn, could attract foraging bats (and birds). Because many of the bat species in the analysis area forage late in the day and into the evening when glare and polarized light emissions are relatively low, impacts to individual bats may occur, but impacts to the species on a regional level is not expected. Additionally, the Project's proposed PV panels are not expected to create glare as they would be nonreflective, dark in color, and absorb light.

Reptiles

Geotechnical investigation has the potential to injure or kill reptiles (including the desert tortoise) through equipment and vehicle strikes. A biological monitor would be on-site to survey work areas ahead of time and monitor equipment while geotechnical work was being performed. Implementation of BMPs described in Section 2.2.3 specific to the desert tortoise would also benefit other reptile species. In the short term, 0.93 acre would be disturbed. No interim reclamation is recommended because the geotechnical work areas would become part of the solar facility footprint, removing potential habitat permanently.

Short-term impacts to sensitive reptiles include injury and mortality from construction activities and vehicle strikes. Although vegetation within the perimeter fence would be allowed to grow no higher than commercially reasonable, it is unlikely to be of similar composition as existing vegetation. However, it may provide habitat for some reptile species (except for the desert tortoise). Sensitive reptiles would continue to be subject to vehicle strikes over the long term from vehicles associated with O&M activities and during Project decommissioning. The Proposed Action would result in the permanent loss of up to 85 acres of potential habitat for sensitive reptiles (see Table A-10).

3.5.3.2 BENEFICIAL AND ADVERSE EFFECT

The Proposed Action would remove up to 85 acres of potential habitat for general wildlife species and sensitive species, or 0.3% of the undeveloped land within the analysis area. Overall, impacts to special-status species would be similar to those for general wildlife and are not anticipated to lead to further decline of any special-status species or general wildlife species range wide. Any impacts to special-status species would be avoided and/or minimized through implementation of the applicant-committed EPMs (outlined in Appendix C) and BLM standard stipulations listed in Appendix D, including design features in Section 2.4.2, such as the Bird and Bat Conservation Strategy.

3.5.4 Mitigation Measures

No species-specific mitigation measures are anticipated as potential impacts to general wildlife and special-status species would be avoided/minimized through implementation of the design features BLM standard stipulations (Appendix D), and applicant-committed EPMs (Appendix C).

Access for all Project activities, including geotechnical investigations, would occur using the alignment proposed for the permanent access road (within the linear facilities segment). No access for any Project activity (including via overland travel) would be allowed outside of the Project Area analyzed within this document, without further NEPA compliance review.

3.6 Fuels and Fire Management

The analysis area for fuels and fire management is the Project Area and surrounding 0.50-mile area.

3.6.1 Affected Environment

3.6.1.1 FUELS

The Project is in the eastern Mojave Desert in an area of open desert with minimal vegetation and intermittently distributed solar and other energy infrastructure projects across the valley. The Project Area

and immediate surrounding areas do not support meaningful levels of wildland fuel to carry wildland fire (Nevada Division of Forestry 2023).

The BLM Las Vegas RMP classifies the Eldorado Valley area as Zone 2B, which consists of dry rural/wildland interface areas with a higher percentage of ephemeral/perennial plant communities that can periodically cause fuel loading of annual species and are vulnerable to more lightning-caused fires (BLM 1998a). Mediterranean grass (*Schismus* sp.) is an annual nonnative grass species that is present throughout the Project Area, mostly at the bases of perennial shrubs. No other invasive or noxious weeds are present (SWCA 2022b).

3.6.1.2 FIRE MANAGEMENT

The Project includes the installation of BESS and appurtenances within the solar facility footprint that would provide an up-to 35-MW, 4- to 8-hour capacity system for the electric grid. The increase in renewable energy projects with BESS capacity, an integral component of energy storage, in the Eldorado Valley has highlighted the potential for fire and explosion hazards associated with lithium-ion batteries, which are used in BESS. Due to the thermal instability of the battery materials, the fire risk is connected to electrolyte ignition and self-heating chemical reactions (University of Texas Fire Research Group 2017). Fire services are provided by the BLM SND for wildland fire protection to public lands in southern Nevada, and the Clark County Fire Department maintains first responder responsibility within areas of unincorporated Clark County. Fire protection for municipal lands within the Eldorado Valley is provided by the City of Boulder City Fire Department.

The Project is on BLM-administered public lands directly north and west of the existing Townsite Solar 1 facility, on lands owned by the City of Boulder City. Townsite Solar 1 has the only existing BESS in the vicinity of the Project (but outside of the 0.5-mile analysis area). The BESS is a 90-MW Tesla Megapack that is co-located within the southeast corner of the solar facility near the switching station and generation tie-in line that is approximately 2.8 miles northeast of the Proposed Action.

3.6.2 Environmental Effects of the No Action Alternative

No fire or fuels management impacts associated with the Proposed Action would result under the No Action Alternative.

3.6.3 Environmental Effects of the Proposed Action

3.6.3.1 SHORT- AND LONG-TERM EFFECTS

Fuels

Geotechnical investigation would disturb 0.93 acre and could temporarily increase the presence of Mediterranean grass (an annual nonnative grass species) on-site. Vehicles and equipment have the potential to introduce noxious weeds and nonnative invasive species into the Project Area. Based on the short duration of fieldwork (up to 12 days) and the small-scale disturbance (0.93 acre), geotechnical investigation is unlikely to contribute significantly to the risk of fire from vegetative fuels.

Surface disturbance of up to 85 acres during construction, operations, and decommissioning has both short- and long-term potential consequences, including an increase in the presence of Mediterranean grass on-site and, in the long term, a shift in the vegetation community structure. Construction equipment can also introduce noxious weeds and invasive species into the Project Area. Project operations are less likely to contribute to the introduction and spread of noxious weeds and invasive species due to the lack of

soil-disturbing activities. Implementation of the Project-specific Integrated Weed Management Plan and BLM standard stipulations would reduce potential impacts to nonsignificant levels (see Appendix D).

In the short term, surface-disturbing activities and construction-related traffic entering and exiting the laydown area can increase the introduction of various weed species and spread Mediterranean grass. Once construction is completed, the 5-acre temporary laydown area would be reclaimed. If present, noxious weeds would be treated, and nonnative invasive species would be reduced to pre-Project levels or less.

Fire Management

The Proposed Action (including geotechnical investigation) is unlikely to impact fire management activities performed by the BLM's Fire Prevention and Hazardous Reduction Program in the area due to a lack of combustible fuels present on-site. Additionally, the location and the size of the Project is unlikely to impact fire responsiveness from the City of Boulder City or Clark County Fire Departments, if responding to an incident in the vicinity.

Possible installation of an up-to 35-MW-capacity BESS on-site using lithium-ion batteries may lead to an increase in fire risk potential. From 2017 to 2019, numerous lithium-ion battery-powered BESS have been linked to high-profile fires across Korea and the United States (Cummings 2019; Spector 2019). The main factors that contribute to BESS fire incidents include thermal runaway, which is caused by lithium-ion battery defects or damage that generate excessive heat; failure of the BESS control system by one or more control components, which can cause overheating and fire; and the release of flammable gases from the heated battery cells (Cummings 2019; Spector 2019). To minimize potential fire risks associated with thermal runaway and control system component failure, multiple levels of safety design are incorporated at the software and hardware levels of the battery management system. To address the risk of flammable gas release, the inclusion of a fire suppression system can include combustible gas sensor systems, auto ventilation systems, smoke detectors, and dry pipe sprinkler systems running to each battery container. Additionally, the proposed BESS would meet National Fire Protection Association (NFPA) Standard 69: Standard on Explosion Prevention Systems (NFPA 2019) for the ventilation system and NFPA Standard 855: Standard for the Installation of Stationary Energy Storage Systems (NFPA 2022) for each BESS container, as well as Underwriters Laboratories 9540: the Standard for Energy Storage Systems and Equipment. These BESS design measures not only minimize the potential risk associated with fire but also provide containment so, in case of a fire, the risk of spreading outside of the container is unlikely. Additionally, the Project would also be designed in accordance with all federal and industrial standards including state and local codes. The implementation of a Project-specific Fire Management Plan would further minimize any potential risks introduced by the proposed on-site BESS.

3.6.3.2 BENEFICIAL AND ADVERSE EFFECTS

Geotechnical investigation and construction/decommissioning activities may increase the potential for introduction and spread of noxious weeds and invasive nonnative species that could increase the potential for wildfire fuels; however, implementation of EPMs (see Appendix C) and BLM standard stipulations (see Appendix D) would minimize potential impacts. O&M activities would likely have a net neutral impact to noxious weeds and invasive species through implementation of the Integrated Weed Management Plan, which would also reduce the risk of fire due to vegetative fuels.

Reclamation of the temporary laydown area after Project construction in conjunction with implementation of the Integrated Weed Management Plan would minimize potential introductions of noxious and nonnative weed species and would likely have a net neutral effect on wildfire fuels.

3.6.4 Mitigation Measures

Through the implementation of a Fire Management Plan, updated BESS components, including an updated fire suppression design, new software, and hardware design features, as well as meeting NFPA and Underwriters Laboratory certification requirements, the increased potential for fire risks associated with the proposed on-site BESS would be reduced. Therefore, no mitigation measures are proposed.

3.7 Greenhouse Gas Emissions/Climate Change

Characteristic meteorological conditions from two National Climatic Data Center stations near the analysis area in Boulder City and Searchlight, Nevada, are listed in Table A-11.

3.7.1 Affected Environment

GHGs include carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and several fluorinated species of gas. CO_2 is emitted primarily from the combustion of fossil fuels; CH_4 is emitted from the production and transport of coal, natural gas, and oil; and N_2O is emitted during agricultural and industrial activities. Fluorinated gases, which are synthetic, are emitted from a variety of industrial processes. CO_2 and other GHGs are naturally occurring gases in the atmosphere; their status as a pollutant is not related to their toxicity but instead is due to the added long-term impacts to climate because of their increased incremental levels in the Earth's atmosphere and their effect on the atmosphere's ability to retain heat that would otherwise be emitted to space. Section 3.0 of the *2021 BLM Specialist Report on Annual Greenhouse Gas Emissions and Climate Trends* (2021 BLM GHG report) (BLM 2022a) discusses these GHGs in more detail.

The global warming potential (GWP) of gases was developed to allow comparison of global warming impacts between different gases. The GWP of a gas depends on how well the gas absorbs energy and how long the gas stays in the atmosphere. It is a measure of the total energy that a gas absorbs over a particular period of time (usually 100 years) compared to CO_2, which has a GWP of 1. The larger the GWP, the more warming the gas causes. For example, CH_4 has a 100-year GWP estimated to be 29.8, meaning that CH_4 would cause 29.8 times as much warming as an equivalent mass of CO_2, over a 100-year time period (Intergovernmental Panel on Climate Change [IPCC] 2021). The GWP for N_2O is estimated to be 273. The term *carbon dioxide equivalent* (CO_2e) is used to describe different GHGs in a common unit. CO_2e is calculated with CO_2, CH_4, and N_2O multiplied by the high-end 100-year GWP values from the Intergovernmental Panel on Climate Change's (IPCC's) Sixth Assessment Report (IPCC 2021). Additional information about GWP can be found in the 2021 BLM GHG report (BLM 2022a).

Global, national, and state level GHG emissions can be found in the 2021 BLM GHG report (BLM 2022a: Chapter 6, Table 6-1 [Global and U.S.] and Table 6-3 [State]). However, the EPA's Inventory of the U.S. Greenhouse Gas Emissions and Sinks provides the most recent national and state level annual GHG emissions and are incorporated by reference (EPA 2022c). NDEP's Air Program prepares a GHG emissions inventory for the State of Nevada (NDEP 2022). Table A-12 lists the industry sector and total GHG emissions for the most recent reporting years (2021 for the United States and 2020 for Nevada). These data are useful to understand which large sources of anthropogenic emissions are contributing to GHG emissions at the national and state levels.

Also, triennially, the EPA publishes a summary of air emissions data, known as the National Emissions Inventory (NEI). The most recent NEI data available are from 2020. Table A-13 provides the 2020 GHG emissions for the United States, the State of Nevada, and Clark County (EPA 2023a).

Anticipated Emission and Mitigation Trends

Projected global and national emissions from the U.S. Energy Information Administration are presented in Sections 6.2 and 6.4, respectively, of the 2021 BLM GHG report (BLM 2022a). The U.S. Energy Information Administration provides long-term world energy and emissions projections in its International Energy Outlook report and an assessment of long-term domestic energy and emissions trends through 2050 in its Annual Energy Outlook report. There are various scenarios to forecast future energy needs and associated carbon emissions. However, the reference case, which reflects current trends and relationships among supply, demand, and prices in the future is a reasonable baseline case to compare with cases that include alternative assumptions about the future energy system. Globally, based on assumptions in the reference-case scenario, it is assumed that energy consumption would rise nearly 50% between 2020 and 2050. Under this scenario, it is forecasted that coal consumption would decrease while natural gas and renewable energy consumption would increase (BLM 2022a). Domestically, under the reference-case scenario, energy-related CO_2 is anticipated to decrease in the near term due to reduction in coal consumption and a rise in natural gas and renewable energy sources to meet demand. However, U.S. emissions from energy consumption are anticipated to increase from 2035 to 2050 due to population growth and associated increase in oil and natural gas consumption (BLM 2022a).

Under the 2015 Paris Agreement, countries agreed to cut GHG emissions with the goal of holding the increase in the global average temperature to well below 2 degrees Celsius (°C) above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C above pre-industrial levels to avoid some of the more dire consequences associated with climate change. Carbon budgets are an estimate of the amount of additional GHGs that could be emitted into the atmosphere over time to reach carbon neutrality while still limiting global temperatures to no more than 1.5°C or 2°C above pre-industrial levels. Carbon neutrality, or net zero emissions, means maintaining a balance between emitting and absorbing GHGs from the atmosphere. On a global scale, carbon neutrality would result in atmospheric concentrations of GHGs reaching an equilibrium, which could stabilize climate change and limit global warming. Additional technical information about carbon budget and carbon neutrality can be found in the 2021 BLM GHG report (BLM 2022a).

EO 14008, "Tackling the Climate Crisis at Home and Abroad" (January 27, 2021), directs the executive branch to establish policies or rules that put the United States on a path to achieve carbon neutrality, economywide, by no later than 2050. This goal is consistent with IPCC's recommendation to reduce net annual global CO_2 emissions between 2020 and 2030 in order to reach carbon neutrality by mid-century. Federal agencies are still in the process of developing policies that align with a goal of carbon neutrality by 2050. In the short term, the order has a stated goal of reducing economywide GHG emissions by 50% to 52% relative to 2005 emissions levels no later than 2030. In late 2021, the U.S. Department of State and the U.S. Executive Office of the President published a long-term strategy for achieving net-zero GHG emissions by 2050 (U.S. Department of State and the U.S. Executive Office of the President 2021).

Past and Present Climate Change Trends and Impacts

The 2021 BLM GHG report (BLM 2022a:Section 8.0) discusses climate science and trends, as well as past and present climate impacts. Some of this information is summarized in this section; however, additional technical information from the 2021 BLM GHG report is hereby incorporated by reference.

The annual average temperature over the contiguous United States has increased by 1.2ºF (0.7°C) over the last few decades and by 1.8°F (1°C) relative to the beginning of the last century. Additional increases in annual average temperature of about 2.5°F (1.4°C) are expected over the next few decades regardless of future emissions and increases ranging from 3°F to 12°F (1.6°C –6.6°C) are expected by the end of

century, depending on whether the world follows a higher or lower future emissions scenario, with proportionally greater changes in high temperature extremes (BLM 2022a).

The average annual temperature in Nevada has risen 2.8° over the past 120 years, from 49.0°F near the beginning of the twentieth century to 51.8°F over the past 10 years. Eight of the 10 warmest years since 1895 have occurred since 2000. The rate of warming is not uniform across the state. Urban areas, for example, are getting hotter faster relative to rural areas, which is most likely being driven by the heat island effect. For southern Nevada, including the rapidly growing Las Vegas metropolitan area, there have been trends demonstrating that urbanization has significantly altered the nighttime climate of the city and that the changes were most pronounced in the last 30 years (Miller 2011). Nevada is the driest state in the nation in terms of annual average precipitation (combined rain and snowfall). Average annual precipitation has decreased 2.2% between the beginning of the twentieth century and the past 10 years (BLM 2022a).

Another consideration is that climate strongly affects energy supply and demand. Climate change has contributed to an increase in average temperatures and unusually hot days and would continue to exacerbate projected increases in power demand throughout the analysis area and surrounding states (BLM 2022a; EPA 2022a). Western portions of the country are experiencing increases in drought conditions, limiting generation provided by hydropower. Drought conditions also impact thermal power generation: Low water availability combined with higher water temperatures limit the ability of water to provide cooling needed for thermal generators such as nuclear, coal, and natural gas and lowers the total generation available from those plants (DOE 2022).

Projected Climate Change Trends and Impacts

Projected climate change trends and impacts are described in detail in the 2021 BLM GHG report (BLM 2022a). Climate change is fundamentally a cumulative phenomenon, global in scope, and all GHGs contribute incrementally to climate change regardless of scale or origin. The multitude of interwoven natural systems and feedback mechanisms that contribute to climate variability over the entirety of the Earth further complicates analyses. Climate scientists provide analysis by modeling changes to these systems in response to a range of global emissions scenarios known as Representative Concentration Pathways (RCPs).

In preparation for the IPCC's Sixth Assessment Report (AR6), the climate science research community, economists, and energy systems modelers developed a new range of "pathways" that examine how global society, demographics, and economics might influence future climate impacts, vulnerabilities, adaptation, and mitigation over the next century. The scenarios are collectively known as the Shared Socioeconomic Pathways (SSPs). The SSPs explore how reductions in emissions would, or would not, be achieved and can therefore be thought of as potential mitigation alternatives. The RCPs and SSPs are meant to complement each other. For additional information on RCPs and SSPs, refer to the 2021 BLM GHG report, Sections 9.1 and 9.2 (BLM 2022a).

State climate change projections for Nevada summarized from the Fourth National Climate Assessment and National Oceanic and Atmospheric Administration's state climate summaries are described in Appendix F.

In addition to controlling or preventing emissions, strategies to offset emissions could be used to align BLM decision making with the goal of achieving net-zero emissions by 2050. There are various strategies available to the BLM for offsetting emissions: carbon sequestration, plugging orphaned and abandoned wells, energy substitutions, carbon capture, and compensatory mitigation. Energy substitution, in particular, is related to the current project type (BLM 2022a).

Overall, GHG emissions could be reduced if fossil fuel energy production is avoided by substituting the use of lower emitting forms of energy. The National Renewable Energy Laboratory reviewed and summarized life cycle assessment emissions of different electricity-generating technologies to reduce uncertainties around estimates of environmental impacts and increase the value of the life cycle assessments in decision making. As discussed in Section 10 of the 2021 BLM GHG Report, even renewable energy sources, referred to as carbon neutral, emit some GHGs during facility construction, maintenance, and decommissioning, as well as from transportation and distribution of energy. Over the past decade, the energy sector has seen a decrease in annual GHG emissions as energy production from coal has been replaced with energy production from natural gas and renewables. Over the short term, the trend of an increasing mix of natural gas and renewable energy production is expected to continue, eventually transitioning to mostly renewables in the long term (BLM 2022a).

3.7.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, the Project would not be constructed. There would be no GHG emissions that would contribute to climate change resulting from Project construction and operation.

The Project's estimated electricity generation rate for the 19 MW facility would equate to 49 million kilowatt hours per year (kWh/yr). Under the No Action Alternative, a range of potential CO_2e was calculated using the natural gas (low) and coal (high) emissions factors per kilowatt hour provided in the 2021 BLM GHG Report and the Project's proposed capacity. The range associated with the 46.3 million kWh/yr potential is 22,785 metric tons CO_2e/yr (natural gas) to 48,020 metric tons CO_2e/yr (coal). The benefits of the Project related to increased renewable energy availability within the analysis area and surrounding states would not occur, and benefits of the Project with respect to climate change adaptation would also be foregone. Given that climate strongly affects energy supply and demand, and that climate change has contributed to an increase in average temperatures and unusually hot days, climate change would continue to exacerbate projected increases in power demand throughout the analysis area and surrounding states (BLM 2022a; EPA 2022). Under the No Action Alternative, the Project, which would be an incremental contribution to supporting grid reliability and energy supply within the analysis area and surrounding states, would not occur.

3.7.3 Environmental Effects of the Proposed Action

3.7.3.1 SHORT- AND LONG-TERM EFFECTS

Greenhouse Gas Emissions and Climate Change Impacts

GHG emissions, primarily in the form of CO_2, N_2O, and CH_4, would be generated under the Proposed Action through the use of vehicles and off-road equipment during Project construction and decommissioning, with minimal emissions generated during O&M. This construction also includes the geotechnical investigation which consists of geotechnical soil borings, soil resistivity testing, and pile load testing.

Non-tree vegetation and soil disturbance could also result in an increase in GHG concentrations. Emissions as a result of soil disturbance would be short lived and return to background levels within several hours (IPCC 2006; Kessavalou et al. 1998). The small amount of carbon that would be stored in removed vegetation would be offset in time by the accumulation of carbon in soils. Also, with other larger contributors of GHG totals, there would not be a notable contribution from soil and vegetation disturbance compared to the overall Proposed Action GHG estimates. For these reasons, the temporary increase in GHG concentrations as a result of soil and vegetation disturbance are negligible and not quantified.

Direct GHG emissions resulting from the Proposed Action, including the geotechnical investigation, were calculated using the assumptions described in Appendix F. The Proposed Action could result in an estimated total of 4,150 metric tons of CO_2e emissions during the construction year through the use of vehicles and equipment during construction activities.

To provide context for these levels of emissions, the EPA mandatory reporting threshold for large emission sources of GHG is 25,000 metric tons of CO_2e emitted annually (74 FR 56260). Given the low level of CO_2e emission contributions and potential for indirect CO_2e benefits from the Proposed Action, the impacts of the Proposed Action on global CO_2e concentrations would be low. To put the estimated GHG emissions in a relatable context, potential emissions can be compared with other common activities that generate GHG emissions and with emissions at the state and national levels. The EPA GHG equivalency calculator can be used (https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator) to express the potential average year GHG emissions on a scale relatable to everyday life (EPA 2023b). For instance, the 4,150 metric tons of CO_2e emissions during the construction period are equivalent to 924 gasoline-fueled passenger vehicles driven for 1 year, or the emissions that could be avoided by operating 1.2 wind turbines as an alternative energy source or offset by the carbon sequestration of 4,949 acres of forest land. Given the low amount of GHG emissions and potential for indirect GHG benefits from the Proposed Action, impacts on global GHG concentrations would be minor.

GHG emissions associated with construction of the Project, including geotechnical investigation, and O&M is summarized in Table A-14 and detailed in Appendix F. GHGs expressed as CO_2e from the Project include CO_2, CH_4, and NO_2 occurring in Clark County. Impact intensity is assessed by comparing GHG emissions associated with the Project with the emission inventories of the impacted counties from the 2020 NEI. Table A-14 presents construction activity emission per year and the emissions from the construction of the Project as a percentage of the county's total emissions. Table A-14 also presents O&M activity emissions and the emissions from the O&M of the Project as a percentage of each county's total emissions.

As shown in Table A-14, GHGs emitted from Project operations would be less than 0.0002% of the respective county's total 2020 annual emissions, and GHGs emitted from Project construction would be less than 0.03% of the respective county's total 2020 annual emissions. Assumptions and complete GHG emission calculation methodology and data are provided in Appendix F.

Support for Federal Plans to Combat Climate Change

The proposed Project would provide PV solar generated energy to Nevada. This additional renewable resource would add to the diverse utility load profiles in the Desert Southwest/California. This Project would result in lowering the cost of Nevada RPS compliance for the Desert Southwest/California.

Within the Northern Grid Planning Area, the Project would contribute to meeting regional needs including the "Public Policy" and "Economic" regional needs by supporting the state-mandated RPS in Nevada (50% renewable by 2030) and by producing cost-effective renewable energy resources including support for the Energy Imbalance Market. The Energy Imbalance Market was launched by PacifiCorp and CAISO in 2014 to automatically balance demand every 5 minutes with the lowest-cost energy available across the combined grid (California, Oregon, Washington, Idaho, Utah, Wyoming, Arizona, Nevada, New Mexico, and Montana). Both increasing energy demand and grid reliability concerns are exacerbated by climate change as discussed in Section 3.3.3.1. GHG emissions during construction and O&M are presented in
Table A-14. The Project would comply with all applicable current federal plans, laws, and policies that are in place to combat climate change and contribute to meeting regional goals regarding renewable

resources. Ongoing routine maintenance and operations of the Project would result in minor amounts of ongoing GHG emissions, which would be outweighed by the Project's benefits.

In conclusion, since the GHG and climate change impacts associated with the Proposed Action would be low and would not conflict with current GHG and climate change plans, laws, policies, and goals, it is not anticipated that the Proposed Action construction and operational emissions would result in significant impacts to GHG concentrations and climate change. The overall Proposed Action impacts are expected to be beneficial with respect to climate change considerations. The Proposed Action would increase renewable energy supply, potentially offsetting electricity generation from fossil fuels. Additionally, the Proposed Action would be an incremental step toward adapting to the impacts of climate change to energy demand in the analysis area and surrounding states.

Table A-15 summarizes the construction, including geotechnical investigation, and O&M emissions over the life of the Project in terms of CO_2e based on 100-year GWP values. The emissions are presented in metric tons (MT) for construction and O&M. The construction and operation CO_2e emissions from the Proposed Action are a very small percentage of the potential offsets.

Social Cost of Greenhouse Gases

The "social cost of carbon," "social cost of nitrous oxide," and "social cost of methane," collectively referred to as the SC-GHG, are estimates of the monetized damages associated with incremental increases in GHG emissions in a given year. EO 13990 encourages federal agencies to "capture the full costs of GHG emissions as accurately as possible, including by taking global damages into account."

In accordance with President Biden's direction in EO 13990, this subsection provides estimates of the monetary value of changes in GHG emissions that could result from construction and operation of the Proposed Action. These numbers were monetized; however, they do not constitute a complete cost-benefit analysis, nor do the SC-GHG numbers present a direct comparison with other impacts analyzed in this document. For instance, the BLM's overall economic analysis for this Project does not monetize most of the major costs or benefits and does not include all revenue streams from the Proposed Action. SC-GHG is provided only as a useful measure of the benefits of GHG emissions reductions to inform agency decision making.

Given the data available to quantify the construction and operational emissions (see Table A-15) and therefore, to estimate the SC-GHG from these activities, this analysis includes an SC-GHG estimate for these Proposed Action impacts. The beneficial aspects of the Proposed Action, including potential replacement of fossil fuel generation with renewable energy generation, are quantified but include substantial uncertainties. Also discussed above, this SC-GHG analysis is not meant to be a complete cost-benefit analysis; instead, it is included for disclosure purposes and for context.

Table A-16 discloses the Social Cost of GHG, which is calculated in accordance with the IWG's Technical Support Document (IWG 2021). Detailed calculations are provided in Appendix F.

On a global scale, the GHG emission contribution of any single source on a subnational scale is dwarfed by the large number of comparable national and subnational contributors. The relative contribution of GHG emissions from construction and operation of the Proposed Action would vary depending on contemporaneous changes in other sources of GHG emissions. A single subnational contributor is very unlikely to influence global cumulative emissions. The Proposed Action alone is very unlikely to influence global cumulative emissions on a scale to make observable any increase or lessening of the social cost of all global emissions. This is especially true of this Proposed Action as the GHG emissions are relatively small. Because there would be no construction or operational emissions associated with the No Action Alternative, this alternative is also not analyzed in this section.

To reiterate, this SC-GHG analysis does not attempt to quantify any emission reductions that may be an indirect result of the Project (including fuel substitution effects), nor does it attempt to weigh the monetary cost of the damages of GHG emissions against the myriad of other benefits from the Project with respect to grid reliability or increased energy supply needed to support increased demand within the analysis area.

3.7.3.2 BENEFICIAL AND ADVERSE EFFECTS

In the future, renewable energy is anticipated to become a larger part of the U.S. energy mix and reducing energy-related carbon emissions. It has been estimated that with a 35% integration of wind and solar energy into the western U.S. electric grid, there would be an additional 25–45% reduction in carbon emissions. The U.S. Energy Information Administration International Energy Outlook expects renewable energy consumption to double between 2020 and 2050 and nearly equal liquid fuels consumption by 2050. Short-term increased GHG emission levels would occur as a result of the construction and decommissioning of the solar facility and ancillary components. This construction also includes the geotechnical investigation which consists of geotechnical soil borings, soil resistivity testing, and pile load testing. Minimal long-term direct and indirect GHG emissions would occur during O&M of the solar facility as a result of the limited number of FTE staff and delivery vehicle emissions. Geotechnical investigation would contribute minimally to the short-term increased GHG emission levels, as these activities include a short-duration of field work (up to 12 days). The applicant's objective for the Townsite Solar 2 Project is to provide an efficient, economic, and reliable PV solar generating facility that would provide approximately 19 MW of solar-generated electricity which serves an alternative to coal, natural gas or other non-renewable resources that would emit a larger amount of GHGs and have a greater impact on climate change. The Project would help meet the goals and requirements set forth by the State of Nevada for renewable energy resources. Therefore, adverse impacts to climate change and GHG emissions are not considered to be significant and there are beneficial impacts as solar projects provide renewable energy with reduced GHG emissions compared to alternatives like coal and natural gas.

3.7.4 Mitigation Measures

Implementation of the BMPs and the applicant-committed EPM #1 Air Quality (see Appendix C) would minimize the anticipated short-term, construction/decommissioning-related GHG impacts. No significant impacts are anticipated, and no additional mitigation measures are anticipated or proposed. The GHG impacts disclosed in the analysis would be irretrievable, (i.e., affecting GHG only until cessation of the activity) and would not affect GHG impacts over the long term.

3.8 Migratory Birds and Eagles

Birds, both migratory and most native-resident bird species including raptors, are protected under the Migratory Bird Treaty Act (MBTA), and their conservation by federal agencies is mandated by EO 13186 (Migratory Bird Conservation). Under the MBTA, it is unlawful by any means or in any manner to pursue, hunt, take, capture, kill, attempt to take, capture, or kill, [or] possess migratory birds or their nests or eggs at any time, unless permitted by regulation.

Bald and golden eagles are also protected under the MBTA and under the Bald and Golden Eagle Protection Act (Eagle Act) of 1940. The Eagle Act was initially passed to protect bald eagles—"the symbol of America freedom"—from extinction and later amended to include golden eagles to address their diminishing populations. The Eagle Act prohibits the "taking" of bald and golden eagles, including

their parts, nests, or eggs (16 USC 668-668d). Bald eagles are also protected in the state of Nevada and are further classified as endangered (NAC 503.050).

The analysis area for migratory birds consists of a 1-mile buffer around the Project Area, and bald and golden eagles were evaluated within a 2-mile buffer (USFWS 2020).

3.8.1 Affected Environment

The Project Area supports migratory birds that are characteristic of the eastern Mojave Desert, where in general, the scarcity of trees and permanent water limits the occurrence of many bird species. Furthermore, the Project Area may be undisturbed, but the Project Area is bounded to the north, south, and west by significant existing disturbance (i.e., U.S. 93 and 95, mining/quarry sites, solar facility, and businesses), which likely further restricts usage by various migratory bird species. The USFWS BCCs are discussed in Section 3.5.

Data obtained from NDOW indicated that the American kestrel (*Falco sparverius*), black-necked stilt (*Himantopus mexicanus*), black-tailed gnatcatcher (*Polioptila melanura*), and common raven (*Corvus corax*) were migratory birds observed in the Project vicinity.[2] Migratory bird species that have additional protection or management attention that have a "moderate" or "high" potential to occur are discussed in detail in Section 3.5.1.2. Migratory bird species known to occur within the analysis area that are also commonly found within the eastern Mojave Desert are included in Table A-9.

NDOW also reported that the distribution ranges of bald and golden eagles overlap the Project Area and 2-mile buffer, but only the golden eagle has been directly observed in the analysis area. A review of the NDOW raptor nesting database identified seven eagle nests and three eagle/buteo nests within 10 miles of the Project Area. An additional nest has an unknown species; however, eagle nests are very distinctive, so it is unlikely that the unknown nest belongs to an eagle (see Appendix G) (NDOW 2022a). Presently, golden eagle nesting habitat (i.e., cliffs, tall trees) is lacking within the Project Area, but foraging habitat may be present. The closest known nesting habitat for golden eagles is in the Sloan Canyon National Conservation Area, which is approximately 3.5 miles west. Also, there is no nesting (i.e., tall trees and cliffs when trees are lacking) or foraging habitats (i.e., open water bodies) present for bald eagles within the analysis area.

3.8.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, there would be no Project-related impacts to migratory birds or eagles.

3.8.3 Environmental Effects of the Proposed Action

Impacts to migratory birds (including golden eagles) are quantified as either displacement or removal of habitat and mortality. Due to a lack of suitable nesting habitat, impacts to golden eagles are limited to the amount of foraging habitat that would be permanently removed by the Proposed Action.

3.8.3.1 SHORT- AND LONG-TERM EFFECTS

Migratory Birds

Approximately 0.93 acre of potential nesting and foraging habitat for migratory birds and foraging habitat for eagles would be disturbed by geotechnical investigation. The presence of a biological monitor during

[2] Project vicinity includes the Project Area and a 4-mile buffer.

geotechnical investigation would reduce the possibility of nest abandonment and crushing of eggs/nestlings. Short-term displacement would occur during geotechnical work, but longer-term displacement would result at the start of Project construction. Because the geotechnical sites would become part of the solar facility footprint, long-term impacts include the permanent removal of habitat, so no interim reclamation is recommended. Individual species may be impacted, but there would be no long-term effects on regional migratory bird population levels because the geotechnical work is temporary (up to 12 days) and will occur over a small scale (0.93 acre) that is already planned to be incorporated into the solar facility footprint.

Construction activities would result in both short-term and long-term loss of bird nesting and foraging habitat for 85 acres. Ground-disturbing activities such as clearing and grading would reduce available cover, nesting and perching substrate, and foraging areas, and would result in displacement of individual birds. Because the Project Area is in an area characterized by existing human disturbance, the opportunity for individual birds to move into adjacent areas is limited. Bird behavior may also be indirectly affected by noise and vibrations associated with construction activities in the short term for those individual birds that may utilize the Project Area. Construction could result in injury or death of nesting birds and eggs. To prevent undue harm, habitat-altering Project activities would be scheduled outside of bird breeding seasons. If construction occurs during the migratory bird breeding season, an authorized biologist would clear construction zones ahead of crews to flag and monitor active nests. Work would be restricted to areas outside of the buffer zone until the nest is inactive. Over the long term, the presence of humans and associated noise and activity could indirectly deter use by some bird species too. Even though the Project may affect some individual migratory birds, the impacts are not expected to result in a reduction in migratory bird levels locally or regionally.

Operation of the Proposed Action may result in migratory bird mortalities resulting from collision and/or electrocution associated with the proposed solar facility and/or 12- to 13.8-kV distribution line. Collision with PV solar panels presents a direct mortality risk to migratory birds, which would potentially be greatest during the breeding season, when birds are more active. Collisions could be accentuated in insectivorous species attracted by high concentrations of insects drawn to the solar arrays (Horváth et al. 2009). To minimize potential collision risks of migrating birds who may be attracted to light sources at night (Longcore and Rich 2004), temporary lighting and limited steady lighting sources that would be directed downward to minimize backscatter would be used during nighttime maintenance activities. A Lighting Plan would be prepared by the applicant, approved by the BLM, and implemented throughout Project-related activities as required.

Waterbird species may potentially mistake the solar panels for water features on which the birds can land; this theory has been coined the “lake effect hypothesis” (Horváth et al. 2009). These behaviors may lead to collisions with PV solar panels resulting in mortality, injury, or stranding of those species that require water to take off again (e.g., grebes [*Podiceps* sp.] and loons [*Gavia* sp.]). Because bird fatality data for PV solar facilities have only recently become available, and are limited, science-based predictions of potential bird risk are limited as well.

No permanent water sources are present in the Project Area. The closest known water sources are the human-made ponds at Boulder Creek golf course approximately 3.5 miles north-northeast of the Project Area and Lake Mead National Recreational Area, which is more than 10 miles from the Project Area. The Eldorado Dry Lake Bed, more the 3 miles south of the Project Area, is a playa that can hold water if enough rain is received during monsoon and winter seasons (typically mid-June to the end of September and December through February). As such, the presence of water can be a draw for many migrating waterbird species from the nearby migratory corridor along the Colorado River, increasing the likelihood of these species to be present within the Project Area.

At the end of Project operations, impacts to migratory birds would be the same as those discussed for general wildlife and sensitive avian species. Based on the level of surface disturbance associated with construction of the Proposed Action and reclaimed vegetation structure, the quantity and quality of nesting and foraging habitat would likely vary within the Project Area because desert vegetation communities can years to recover from disturbance (Abella 2010).

Reclamation of the 5-acre temporary laydown following construction could provide habitat for some species. Reclamation efforts, however, may take years or even decades (Abella 2010). Use by migratory birds may be restricted as it is anticipated that some negative effects on vegetation density, cover, and composition will likely persist for up to 20 years after reclamation.

Golden Eagles

Surface disturbance caused by the geotechnical investigation would remove 0.93 acre of potential foraging habitat for the golden eagle in the short term. Because the work sites will be integrated into the footprint of the solar facility, no interim reclamation is proposed. Noise from the equipment and workers may displace some individual eagles. Due to the temporal and spatial scales of the geotechnical investigation, effects on individual golden eagles may occur, but not on the regional population.

Golden eagle nesting habitat is not present within the Project Area. As such, the Proposed Action is not anticipated to impact golden eagle nesting. The closest suitable nesting habitat is greater than 1 mile away to the west and northeast, in surrounding mountain ranges (i.e., Black Hill, McCullough Range, and River Mountains). Additionally, indirect impacts to potential nesting habitat from noise associated with construction activities is unlikely due to the temporary and intermittent nature of the construction and the distance from the Project to the closest potential nesting habitat.

The removal of approximately 85 acres of potential foraging habitat for golden eagles may impact some individuals locally. However, golden eagles are known to be sensitive to human activity and tend to avoid developed areas (USFWS 2022c). Due to the existing disturbance and human presence characteristic of the Eldorado Valley, quality foraging habitat is not likely present within the Project Area. Additionally, golden eagles are typically a wide-ranging species and tend to use a variety of habitats for foraging. This, in conjunction with potential foraging habitat (i.e., undeveloped land) that is available and to the southeast and directly west of the Project Area, suggests individual eagles can shift use to the undeveloped areas to optimize foraging success. Therefore, the impacts related to the loss of foraging habitat associated with the Proposed Action may impact some individual golden eagles, but it is not likely to impact the species on a regional level or the population as a whole.

3.8.3.2 BENEFICIAL AND ADVERSE EFFECTS

The Proposed Action would impact approximately 85 acres of nesting and foraging habitat for migratory birds and foraging habitat for golden eagles. No specific measures have been identified to minimize potential impacts to golden eagle foraging habitat. Adjacent habitat may be available for the species, though adjacent habitat is likely not of high value due to the existing disturbance and human presence characteristic of the Eldorado Valley.

To minimize potential impacts to migratory birds, the applicant would schedule surface-clearing activities outside of the migratory bird breeding and nesting season. In this region, migratory birds are typically most sensitive to disturbance during these times, generally between February 15 and August 31 (BLM 2019), whereas golden eagles would be subject to foraging pressures year-round as a resident species of southern Nevada (Cornell Lab of Ornithology 2022). If ground-clearing and other construction activities cannot be scheduled to avoid the breeding and nesting season, ground clearance surveys would be conducted by a qualified biologist to confirm the absence of nests or nesting activity before work

proceeded (on the ground and in burrows and vegetation). Active nests (containing eggs or young) would be avoided until they are no longer active, or the young birds have fledged. In the event a nest was observed, species-specific nesting bird buffers would be implemented upon approval from a BLM biologist. Implementation of applicant-committed EPMs and BLM standard stipulations would further minimize or avoid impacts to migratory birds (Appendices D and E).

3.8.4 Mitigation Measures

No mitigation measures are anticipated.

3.9 Recreation/Travel/Wild and Scenic Rivers

The analysis area for recreation and travel is limited to the Project Area.

3.9.1 Affected Environment

The Project is within the Nelson Hills/Eldorado Special Recreation Management Area (SRMA), which is managed specifically for competitive offroad vehicle events. Within the SRMA, off-highway vehicle usage is "limited to existing roads, trails, and dry washes" (BLM 1998a). In the vicinity of the Project Area, there are minimal casual recreation uses occurring in the area even though there are existing dirt roads and trails throughout the Project Area. The BLM LVFO has not issued any special recreation permits for events or commercial activities in the locale. Also, there are no developed recreation facilities or opportunities in this portion of the SRMA (BLM 2021a).

The Project Area is within the NDOW Game Management Hunt Unit 263 (NDOW 2022b). The principal focus of this unit is big game hunting for desert bighorn sheep, which are primarily found at higher elevations in the McCullough Range.

Access to the Project would be from either southbound I-11/U.S. 93 or northbound Boulder City Parkway/U.S. Route 93. The annual average daily traffic for Exit 15A from the I-11/U.S. 93, which can be used to access Old U.S. Highway 95 from the cities of Henderson and Las Vegas, Nevada, was 1,100 vehicles in 2021 (NDOT 2021). Residents of Boulder City can access Old U.S. Highway 95 from the Boulder City Parkway/U.S. 93, which has an annual average daily traffic of 420 vehicles (NDOT 2021).

Typical construction traffic would consist of trucks transporting construction equipment and materials to and from the site and management and construction employee vehicles during the construction and decommissioning periods. Most construction staff and workers would commute daily to the jobsite from within Clark County, primarily from the Las Vegas area. The on-site workforce is anticipated to be an average of 100 to 200 construction workers with a peak not expected to exceed 250 workers at any given time. Facility decommissioning would likely require less than half of the projected construction workforce, and O&M staff would require a maximum of three FTE staff. The applicant would be responsible for obtaining all necessary permits from NDOT.

The Project Area is not within any transmission corridors. Additionally, there are no wild or scenic rivers within 5 miles of the site (BLM 2021a).

3.9.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related effects on recreation and travel would occur.

3.9.3 Environmental Effects of the Proposed Action

3.9.3.1 SHORT- AND LONG-TERM EFFECTS

Geotechnical investigation is not likely to impact casual recreation or access.

At the start of construction, fencing would be installed. Eventually, the entire perimeter of the site would be enclosed in a 6- to 7-foot-high chain-link fence (with 1-foot-high barbed-wire security strands at the top), and public access would be restricted sitewide. Temporary desert tortoise fencing would be installed around the 5-acre temporary laydown area and the linear facilities segment before construction begins and removed at the end of construction.

The construction and operation of the Project would not affect hunting in the area because desert bighorn sheep are found at higher elevations in the adjacent mountains of the McCullough Range, which is west and more than 5 miles from the Project. Access to the hunt unit is not anticipated to be impacted by the Proposed Action because there are numerous access points to the hunt unit to the north and south of the Project as well as access to the west side of the mountain range.

Traffic on U.S. Route 95 could be temporarily affected during construction and decommissioning activities (short term), specifically during commuting hours of 5:00 am and 5:00 pm. An incremental long-term increase in traffic may also result from the operation and maintenance of the PV solar facility but the increase is expected to be negligible because up to three FTE positions are anticipated.

3.9.3.2 BENEFICIAL AND ADVERSE EFFECT

The installation of perimeter fencing during construction and operation and temporary desert tortoise fencing during construction may impact recreation access within and through the Project Area. However, given the area's lack of managed recreation opportunities and the fact that little known casual recreation use has been documented in the area (BLM 2021a), any potential access restrictions are expected to be negligible and would not contribute to cumulative recreational resource impacts.

3.9.4 Mitigation Measures

If the Proposed Action impacts traffic during construction (and decommissioning activities), the applicant would prepare and implement a Traffic Management Plan (i.e., design feature), if required by the BLM. The Traffic Management Plan would minimize potential short-term traffic impacts.

3.10 Soils and Surface Hydrologic Conditions

This section identifies the soil resources and hydrologic conditions within and adjacent to the Project that may be affected by construction, O&M, decommissioning, and reclamation activities of the Project and discusses the applicable regulations. Information in this section is largely based on information collected by the U.S. Geological Survey (USGS). Soil resource inventory data for the analysis area were gathered from the Natural Resources Conservation Service (NRCS) Soil Survey Geographic Database (NRCS 2022), which delineates unique soil map units. The analysis area for soils and hydrologic conditions is the Project Area with a 20-m (66-foot) buffer, which when combined measures approximately 100 acres.

3.10.1 Affected Environment

Soil

The Project is in Eldorado Valley, a southwest-trending, internally drained (i.e., closed) basin that consists of a series of alluvial fan lobes that form large, cone-shaped, sedimentary deposits confined by the McCullough Range to the west and the El Dorado Mountain to the east. Elevations in the Project Area range from approximately 1,800 to 1,840 feet above mean sea level (amsl).

There are three soil map units within the analysis area (Table A-17) (NRCS 2022). The soil map unit with the highest percentage of land cover within the analysis area is the Arizo very gravelly loamy sand at 57%. Caliza very cobbly loamy sand follows at 22%, and the Canutio-Akela complex is at 12%. The soils are not categorized as hydric soils (NRCS 2022). During resource surveys, the soils observed were representative of the NRCS soil profile for the area. Surface sediments consist primarily of cobbles, gravels, and sand.

The majority of the Project Area is susceptible to wind erosion, according to the wind erodibility index for the soil maps units, with the exception of the northwest region, which is represented by the Canutio-Akela complex.

NRCS provides a wind erodibility index for soils, which is a rough indication of the tons of soil per year estimated to be lost to wind erosion within each soil type. The wind erodibility index ranges from 0 for soils that are unlikely to be eroded by wind to 310 for soils most likely to be eroded by wind. The Arizo and Caliza soil units have a wind erodibility index of 86. The Canutio-Akela complex has a wind erodibility index of 48 (NRCS 2022).

The soil erodibility factor (K-factor) indicates the susceptibility of a soil to sheet and rill erosion by water. The estimates are based primarily on percentage of silt, sand, and organic matter and on soil structure and saturated hydraulic conductivity. K-factor values range from 0.02 to 0.69. Other factors being equal, the higher the value, the more susceptible the soil is to sheet and rill erosion by water. The Arizo soil unit and the Canutio-Akela complex have a K-factor of 0.1, and the Caliza unit has a K factor of 0.05, indicating low susceptibility to sheet and rill erosion (NRCS 2022).

Hydrologic soil groups are based on estimates of runoff potential. Soils are assigned to one of four groups according to the rate of water infiltration when the soils are not protected by vegetation, are thoroughly wet, and receive precipitation from long-duration storms. The soils in the analysis are in Group A. Group A soils have a high infiltration rate when thoroughly wet (low runoff potential). They consist chiefly of deep, well-drained to excessively drained sands or gravels. These soils have a high rate of water transmission (NRCS 2022).

In general, soil characteristics across all soil map units are slightly susceptible to sheet and rill erosion from water and are moderately susceptible to wind erosion, except for the Canutio-Akela complex, which is more resistant to wind erosion (NRCS 2022).

Biological soil crusts, or biocrusts, are a key component of desert ecosystems that are composed of living organisms, including cyanobacteria, green algae, lichens, microfungi, bryophytes, and other bacteria that form a crust on the soil surface (Rosentreter et al. 2007). These biocrusts are an important factor in soil stability, nutrient cycling, and water infiltration (Rosentreter et al. 2007). As such, SWCA recorded data on biocrust presence during the botanical survey. Biocrust was observed throughout the Project Area at low densities (i.e., discontiguous) and in locations where soils appeared to contain more silt and clay content and reduced sand content. Biocrust was also noticeably absent in washes and was recorded in

only the Arizo soil map unit (1.3% average cover). No biocrust was recorded within the other two soil map units as part of the botanical survey (SWCA 2022b).

No prime farmland or sensitive soil resources are within the analysis area.

Hydrologic Conditions

The existing landscape surrounding the Project has been modified over time by mining operations, major highways, transmission lines, and solar energy development, which has changed the drainage regime. Off-site surface water flows to the Project generally originate from culverts crossing under I-11/U.S. 93 or by overtopping it. A lesser source of storm flows is from the Black Hills of the McCullough Range in the northwest. These lesser flows also cross U.S. 95, with a portion directed south toward the Project.

On-site surface water is transported across the site within existing ephemeral drainages, only one of which is identified by the National Hydrography Dataset Plus. It is a north–south-trending drainage that crosses near the southeasternmost portion of the Project Area and access road (EPA 2020; Google Earth Pro 2022).

Soil units in the analysis area are rated as excessively drained to well drained. Water is drained readily to rapidly through the soil column, rather than by overland travel, due to coarse-textured sands and gravel. Depth to the saturation zone is more than 80 inches below surface where redoximorphic features (i.e., evidence of long-term soil saturation) are observed. The NRCS also rates the soil flooding frequency. The Arizo unit is rated with an occasional flood frequency, or a chance of flooding of 5% to 50% in any given year. The Caliza unit and Canutio-Akela complex have a nearly 0% chance of flooding in any given year; flooding occurs less than once every 500 years (NRCS 2022).

3.10.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related effects on soils or hydrologic conditions would occur. Impacts to soil and hydrologic conditions would continue under current conditions associated with sporadic recreational and wildlife use.

3.10.3 Environmental Effects of the Proposed Action

3.10.3.1 SHORT- AND LONG-TERM EFFECTS

Soil

Short-term soil impacts would occur from geotechnical work, specifically boring and pile testing locations due to soil replacement. Because the work sites would be integrated into the footprint of the solar facility, no interim reclamation is proposed. As a result, long-term soil impacts are expected from construction.

Long-term and permanent soil impacts (from construction, O&M, and decommissioning) would result from the Project. Construction techniques would restructure and extremely compact the soil. Although the existing soils have low susceptibilities to wind and water erosion, soils disturbed by grading and excavation are likely to become more susceptible to erosion once vegetation is removed. Water erosion could also lead to increased sedimentation into nearby drainages, and wind erosion could damage nearby plants. Surface-disturbing activities may indirectly contribute to the spread of noxious weeds and invasive plant species on-site. Long-term soil erosion impacts could result but would be minimized by maintaining appropriately designed stormwater management features associated with the Project.

O&M activities are not expected to have a significant impact to soil erosion because of the anticipated limited soil-disturbing activities. Water cleaning of the solar panels may be part of the Project's operations. Due to the infrequency of water applications, the small amount of water applied, and the flat topography of the Project Area, quantities and velocities of runoff sufficient to significantly erode soils are not expected.

Decommissioning activities have the potential to increase soil erosion and contribute to off-site sediment loads through the removal of solar panels and associated equipment. As such, short-term impacts are possible. However, the approval and implementation of the Facility Decommissioning Plan and Site Rehabilitation Plan would mitigate any potential short-term impacts.

After construction is completed, the temporary laydown area would be reclaimed. Surface soil would be leveled, and site drainage would be restored to pre-Project conditions, to the greatest extent possible. Even with reclamation, long-term negative effects on soil are possible.

Hydrologic Conditions

The geotechnical investigation is not anticipated to impact hydrological site conditions. Existing trails would be used to access the proposed geotechnical locations, except for a small portion of overland travel required to reach one of the sites. The geotechnical work is of short duration (up to 12 days), and the spatial extent is small (0.93 acre).

Surface water runoff patterns would likely be affected by the Proposed Action. The potential disturbance associated with the Proposed Action may lead to a short-term increase in erosion on-site and off-site, thereby increasing sediment loads in surface water runoff, altering the discharge and retention rates of water, and changes in the velocity of water moving through the Project Area. This could result in the degradation of surface water quality. However, any potential adverse impacts would be minimized through the implementation of a Site Drainage Plan and the BMPs outlined in the Project's SWPPP. Implementation of BMPs would ameliorate potential adverse impacts of surface water runoff from the Proposed Action.

3.10.3.2 BENEFICIAL AND ADVERSE EFFECT

Construction and decommissioning activities would result in short-term impacts to soil in the Project Area. However, such potential effects would be minimized through utilization of appropriate BMPs outlined in the SWPPP and other design features outlined in Section 2.4.2. Long-term and likely permanent soil impacts are anticipated, but off-site sediment transport would be controlled through maintained stormwater management features associated with the Project. During the Project's operational phase, erosion impacts are anticipated to be negligible.

The Proposed Action may impact hydrological conditions on-site due to construction and decommissioning activities and lead to an increased potential for on-site and off-site discharges of sediment-laden runoff. Implementation of BMPs outlined in the Project-specific SWPPP would minimize those effects.

3.10.4 Mitigation Measures

Implementation of a Site Drainage Plan, Project-specific SWPPP, and design features (see Section 2.4.2) would minimize potential short- and-long term soils impacts. Therefore, no mitigation measures related to soil or hydrologic connections are anticipated for the Proposed Action.

3.11 Special Designations

Special designation areas are lands that federal or state agencies manage for the protection and enhancement of specific resource values. These lands can include wilderness, wilderness study areas, areas of critical environmental concern (ACECs), national trails and parks, and other special management areas.

In December 2002, the Old Spanish Trail was designated as a National Historic Trail by Congress (BLM n.d.). The Old Spanish National Historic Trail (OSNHT) is jointly managed by the BLM and the National Park Service, working in partnership with other federal, state, and local government agencies, and private landowners who manage or own lands along the trail route.

3.11.1 Affected Environment

The OSNHT is a historical trade route that connected northern New Mexico to Los Angeles, California. The proposed solar field is approximately 0.3 mile west of the congressionally designated alignment of the OSNHT, and the alignment passes through the linear facilities segment. This portion of the OSNHT is known as Mojave Road. The landscape within and around the OSNHT in this area includes a broad and flat valley influenced by the surrounding mountains, U.S. 95, sand and gravel operations, existing solar facilities, and transmission lines. Additionally, the landscape character immediately surrounding the Project Area is characterized by desert landscapes with heavy development, including other solar array developments, transmission line corridors, substations, and industrial developments. The *Visual Resources Technical Report for the Townsite Phase 2 Solar Project Draft, Clark County, Nevada* corroborates this through images of the land surrounding the OSNHT and the Project Area (SWCA 2022c:Figures 9–11).

3.11.2 Environmental Effects of the No Action Alternative

The No Action Alternative would not result in impacts to special designations and the OSNHT.

3.11.3 Environmental Effects of the Proposed Action

3.11.3.1 SHORT- AND LONG-TERM EFFECTS

As described in Section 3.14 of the EA, while the visual setting adjacent to the OSNHT would be further modified by the Proposed Action, in context with impacts from existing projects and developments, the visual impact would be negligible. Therefore, because the Project would occur adjacent to other energy infrastructure projects, it would not result in discernible visual changes for any potential viewers from the trail and would not interfere with the trail's nature and purpose.

As identified in the OSNHT Comprehensive Administrative Strategy, the portion of the OSNHT within the Project's analysis area crosses BLM land and does not include any high potential historic sites, high potential historic route segments, or auto tour routes (National Park Service 2017). In addition, no trail traces or trail-related cultural resource sites have been identified, and there are no trail-specific recreational opportunities within the analysis area of the Project. Overall, there are no features of the OSNHT evident in this area today, and the Proposed Action would not affect the nature, purpose, and primary uses of the OSNHT Mojave Road segment.

3.11.3.2 BENEFICIAL AND ADVERSE EFFECTS

The presence of solar arrays and the Project's linear facilities would be noticeable to potential users of the OSNHT. However, the visual changes due to the proposed action are consistent with the existing development in the area and constitute a negligible impact to the OSNHT Mojave Road Segment.

3.11.4 Mitigation Measures

No mitigation measures are anticipated.

3.12 Threatened, Endangered, or Candidate Animal Species

The following section lists and describes any species that may be listed under the ESA as threatened, endangered, or candidate that has potential to occur within the Project Area. The purpose of the ESA is to protect species that are in danger of extinction because of low population levels and other environmental hazards. The Project Area is on federal land and therefore must undergo ESA Section 7 consultation with the USFWS if take of a listed species is anticipated.

Species identified for analysis were derived from 1) an online review of the USFWS's Information for Planning and Consultation (IPaC)-listed species for the Proposed Action area (Appendix G) (USFWS 2022b), 2) the listing of known occurrences of threatened, endangered, or candidate wildlife and plant species data maintained by the NDNH (2022), and known or potential occurrence of listed species within the vicinity (i.e., 4-mile buffer) of the Project according to the NDOW database (see Appendix G) (NDOW 2022a). Additionally, a presence/absence survey for the desert tortoise was performed by Allied Pacific Partners, LLC, in the fall of 2018 (Allied Pacific Partners, LLC 2022).

BLM sensitive species, species protected by the State of Nevada, species identified by NDOW as SoCP, known occurrences of state-listed and at-risk species by the NDNH, and bird species protected under the MBTA are discussed in Sections 3.5 and 3.8.

The analysis area consists of the Project Area and an approximately 4-mile buffer surrounding the Project Area (37,342 acres total, but 26,892 acres is undeveloped habitat). This area is intended to capture existing conditions and potential impacts to individuals, habitats, and movement corridors for species with large home ranges. For species with small home ranges (such as reptiles and small mammals), impacts would be limited to the Project Area and immediate vicinity.[3]

3.12.1 Affected Environment

Data obtained from the USFWS's IPaC online database indicates five ESA-listed species and candidates for listing have the potential to occur—the southwestern willow flycatcher (*Empidonax traillii extimus*), the endangered Ridgeway's rail (*Rallus obsoletus yumanensis*), the threatened yellow-billed cuckoo (*Coccyzus americanus*), the threatened Mojave desert tortoise, and the candidate monarch butterfly (*Danaus plexippus*)—and may be impacted by Project activities. The NDNH and NDOW reported that desert tortoise has been observed in the analysis area. Additionally, NDOW reported that the delisted brown pelican (*Pelecanus occidentalis*) was also seen in the analysis area.

The candidate monarch butterfly and the desert tortoise (federally threatened, BLM sensitive species, State of Nevada threatened species, and an SoCP by NDOW) have potential to occur within the analysis area. For the three avian species, no suitable habitat in the form of riparian habitat, freshwater marshes,

[3] Project vicinity includes the Project Area and a 4-mile buffer.

and springs is present. The closest water features to the Project Area are human-made ponds at Boulder Creek Golf Club, approximately 3.5 miles northeast. These ponds are unlikely to provide suitable habitat for the federally listed avian species, and no sightings have been reported (eBird 2021).

The monarch butterfly occurs in a variety of habitats but relies on nectar from flowering plants for food, shrubs and trees to overnight on, and milkweed (*Asclepias* spp.) plants for breeding, which serve as hostplants for the larval form of monarch butterflies (Greenstein et al. 2022). Habitat capable of supporting milkweed plant species may be present within the botanical survey area (i.e., Project Area with 66-foot buffer), even though no milkweed species were observed during surveys because existing drought conditions can cause plant species to become dormant. Various flowering plant species may also provide nectar for food and overnight habitat.

The Project is situated in the western portion of Eldorado Valley in the piedmont region of the McCullough Range and in the southeastern portion of the USFWS Eastern Mojave Recovery Unit as modified in the most recent recovery plan (USFWS 2011). The Project lies outside of any ACEC, BLM wilderness area, or designated critical habitat for desert tortoises and is within the Mojave Basin and Range ecoregion. The River Mountains ACEC is 3 miles northeast and across I-11/U.S. 93, the Boulder City Conservation Easement and Paiute-Eldorado desert tortoise critical habitat complex are 3 miles southeast, the Paiute-Eldorado Valley ACEC is 20 miles southwest, and the Sloan Canyon National Conservation Area is 5 miles west. The Project is bordered by the Townsite Solar 1 facility to the south, quarry operations to the west, and mining operations and a gun range to the north. The area immediately to the east of the Project site is open; however, U.S. 95, which has tortoise exclusion fencing, and the edge of the Townsite Solar 1 facility are less than 1 mile away.

Although the Solar PEIS does not apply to the Project because it is not a utility-scale solar development generating more than 20 MW, the majority of the Project Area is within Priority 1 desert tortoise connectivity habitat. However, the linkages to the north, east, and south are likely restricted because I-11/U.S. 93 and U.S. 95 (which includes desert tortoise fencing) act like barriers to the north and east and recent solar development to the south. The recent solar development in the south also restricts connectivity to the west into the Sloan Canyon National Conservation Area because desert tortoises likely have to travel south into Eldorado Valley before traversing west (USFWS 2022a). Additionally, the habitat in and around the Project is likely already isolated from other habitat due to existing disturbance (USFWS 2022a).

The USGS habitat model for desert tortoises has a habitat suitability value of 0.9 out of 1.0 within the Project Area (also the action area per USFWS Project BO [USFWS 2022a]), indicating that the area is likely to be suitable for tortoises (Nussear et al. 2009). The NRCS soil survey data show that the Project site is underlain by loamy sand soils with high drainage and low runoff (NRCS 2022). The Project Area also contains alluvial fans and valley wash features that is composed primarily of Sonora-Mojave Creosotebush-White Bursage Desert Scrub (SWCA 2022b).

Pre-project desert tortoise surveys were conducted by Allied Pacific Partners, LLC, in the fall of 2018 (October 27–30) for a broader survey effort on 458 acres that included the entire approximately 85-acre Project Area in accordance with the 2018 USFWS protocols in *Preparing for Any Action that May Occur within the Range of the Mojave Desert Tortoise (Gopherus agassizii*) (Allied Pacific Partners, LLC 2022; USFWS 2018). No desert tortoise or sign was observed in either the Project Area or within the broader survey area. Although neither desert tortoise nor sign were found within the Project Area during surveys, the area may be part of an individual's home range (USFWS 2022a). Thus, USFWS estimates that the 85-acre Project Area supports no more than one large adult desert tortoise (USFWS 2022d).

3.12.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related effects on desert tortoises would occur.

3.12.3 Environmental Effects of the Proposed Action

The desert tortoise is the only federally listed species with potential to occur within the analysis area. Impacts to desert tortoise are discussed in terms of potential changes to habitat, habitat use, and/or behavior.

3.12.3.1 SHORT- AND LONG-TERM EFFECTS

Approximately 0.93 acre of potential wildlife habitat would be impacted from geotechnical investigation over the short term. Impacts could consist of a loss of potential food sources and overnight habitat for the monarch butterfly and access-related vehicle strikes to the desert tortoise. Monarch butterfly mortality from equipment/vehicle strikes is unlikely because they have the capability to disperse if disturbed. To reduce potential vehicle strikes to the desert tortoise, a biological monitor would be on-site and would survey in advance of moving vehicles. Implementation of BMPs described in Section 2.2.3 would also reduce the potential effects to desert tortoise individuals on-site. Since these geotechnical sites would be permanently disturbed as part of the solar facility and are already included in the 80-acre footprint, no interim reclamation is proposed for the geotechnical investigation.

The Proposed Action's potential impacts on monarch butterflies are expected to be similar to those described above for geotechnical work. The potential impacts to desert tortoises during the installation of the perimeter fence and desert tortoise fencing (permanent and temporary) also include vehicle/equipment strikes that may cause injury or mortality. To prevent injury and death, a biological monitor would be present during fence installation. The biological monitor would locate desert tortoise(s) and, if necessary, direct the contractor to cease construction activities until the desert tortoise moves out of harm's way. If desert tortoise moving is required, the capturing, handling, and relocating of the desert tortoise could result in harassment and possibly injury or death.

The Project could result in habitat fragmentation; degradation of adjacent habitat from nonnative invasive species and noxious weed plant species; disturbance to desert tortoise individuals from human noise and activity; increased predation due to increases in litter, other human-caused attractive nuisances, and/or predators (such as ravens); and overhead lines serving as perching areas. Increased human presence in the area during and after construction could lead to death, harm, or harassment of individual desert tortoises or even illegal collection of a federally protected species due to encounters with construction workers.

Construction traffic entering/exiting could also increase the potential for tortoise/vehicle collisions in the short term and throughout the life of the Project. These activities could also result in increased noise levels, the introduction and spread of invasive species and noxious weeds, and increased soil erosion potential. Construction noise and vibrations could affect desert tortoises and their normal activity patterns in the short term.

In addition to the direct effects on the species itself, the Proposed Action would have a long-term impact to the desert tortoise by causing the loss of up to 85 acres of typical habitat for the species. Because reclamation efforts tend to take longer than expected, particularly during drought conditions, the temporary laydown area is included in the permanent impacts. As a result, the Proposed Action would likely contribute to the long-term displacement of resident tortoise from the area.

Construction activities that disturb the ground may increase the chance that invasive and noxious weed species will be introduced and spread. Construction vehicles and equipment are likely vectors. An Integrated Weed Management Plan would be implemented to reduce the possibility of noxious weed species being introduced and spread throughout the site, which could displace the native plant species on which desert tortoises forage.

During all phases of the Proposed Action, an increased human presence may result in increased human/desert tortoise interactions that may result in death. Indirectly, food opportunities for predators from site litter can also result in injury, death, or harassment. BMPs include a WEAP, speed limits on roadways, and a site litter control program to reduce or eliminate these effects.

Operational impacts to the desert tortoise would likely be reduced due to the presence of the perimeter fencing. The perimeter fencing would remain intact after construction, and desert tortoises would not be allowed to enter the Project Area. The presence of the perimeter fence and other structures, including the distribution line, would also provide increased perching opportunities for raptors and common ravens, which prey on desert tortoise. This could result in increased predation of the desert tortoise, particularly hatchings and juveniles.

Following the end of operations, decommissioning and reclamation would occur. The use of heavy equipment and other activities associated with decommissioning could result in impacts similar to those described above for construction.

3.12.3.2 BENEFICIAL AND ADVERSE EFFECT

Although no desert tortoise or sign was observed in the analysis area, the entire Project Area is within desert tortoise habitat. Therefore, desert tortoise may be affected by the Proposed Action.

3.12.4 Mitigation Measures

To reduce impacts to the desert tortoise, construction minimization measures outlined in the Project BO will be implemented (see Appendix E) (USFWS 2022a), to include the BLM's *Common Raven Management Plan for Energy Development within the BLM Southern Nevada District* (BLM 2018).

In accordance with the Project BO (File No. 2022-0035189-S7-001), to further offset the adverse effects from the Project, the applicant will pay a desert tortoise remuneration fee of $1,067 per acre to the BLM (subject to increase March 1, 2023), for a total of $90,695 ([80 acres + 5 acres laydown area] × $1,067). This fee will be used to support desert tortoise recovery actions (USFWS 2022a, 2023).

3.13 Vegetation (Including Special-Status Plants, Cacti, and Invasive Plant Species and Noxious Weeds)

This section identifies special-status species, weed species, and vegetation resources within and adjacent to the Project Area that would be affected by construction, O&M, and decommissioning of the Project. Surveys were performed in accordance with the botanical study plan prepared for the Project and are briefly described below.

Native plant community, special-status plant, cacti density estimates, and invasive species and noxious weed surveys were performed by SWCA from March 30 through April 2, 2022. Additionally, windshield surveys along existing roads within the surrounding 1-mile buffer were performed to identify weed

species that may be transported into the Project Area by vehicles or other dispersal vectors. A special- status plant reference population was also visited to time the survey window (SWCA 2022b).

The approximately 100-acre botanical survey area (Project Area plus 66-foot buffer) is intended to capture existing conditions and potential impacts to individuals, populations, and habitat from the Proposed Action, including indirect impacts associated with potential increased soil runoff and introductions of noxious and nonnative invasive weed species.

3.13.1 Affected Environment

3.13.1.1 VEGETATION COMMUNITY

The Project is located in the eastern Mojave Desert, in the northwestern portion of the Eldorado Valley, on relatively flat terrain with elevations ranging from 1,800 to 1,840 feet amsl. With the exception of some existing trails, the site is in relatively undisturbed habitat; however, the Project Area is practically surrounded by existing human disturbance, including I-11/U.S. 93, Old U.S. Highway 95, and U.S. 95; mining/quarry sites; solar facilities; and businesses.

According to the SWReGAP land cover classifications, the Project is completely within the Sonora- Mojave Creosotebush-White Bursage Desert Scrub vegetation community (Lowry et al. 2005). This land cover class ranges from sparse to moderately dense (2%–50%) cover and generally occurs below 4,000 feet amsl. In this vegetation community, creosote bush (*Larrea tridentata*) and white bursage (*Ambrosia dumosa*) are typically co-dominant, but a variety of shrub, sub-shrub, and cacti may be present to co-dominant (Lowry et al. 2005).

Botanical surveys noted that the vegetation present in the Project Area was drought stricken and water stressed with the exception of vegetation near water-concentrating areas. Few annual forbs were present within the Project Area, and many of the shrubs in the upland areas were brittle and dormant. The Eldorado Valley has been in a Severe (D2) to Exceptional (D4) Drought since the beginning of 2022 (National Drought Mitigation Center 2022), which likely explains site conditions. The Palmer Drought Severity Index for the previous year (2021) in this area was −2.47, indicating the presence of moderate drought conditions (Abatzoglou et al. 2017).

The observed plant species and composition is consistent with the vegetation community's general description. No species of mesquite (*Prosopis* sp.) were observed, but approximately 219 catclaw (*Senegalia greggii*) trees were recorded, mostly in drainages and along drainage margins (SWCA 2022b). The Project's plant species list is provided in Table A-18.

3.13.1.2 SPECIAL-STATUS PLANTS

Table A-19 lists the BLM special-status plant species with potential to occur in the area under the jurisdiction of the SND (BLM 2017). Through consultation with the BLM, the rosy two-tone beardtongue (*Penstemon bicolor* ssp. *roseus*) was the only special-status plant species with potential to occur in the botanical survey area (SWCA 2022b). A reference population was visited on March 30, 2022, prior to conducting protocol-level surveys, to ensure that a combination of floral, fruit, vegetative, or seed characteristics are visible for successful identification.

No rosy two-tone beardtongue plant species were observed during protocol-level surveys for special-status plant species even though potential habitat was observed in the form of gravely-sandy washes and wash margins. No incidental special-status plant species were observed within the Project Area.

Rosy two-tone beardtongue is a perennial species, and while drought conditions may affect germination, no past stems or other plant material was observed, further suggesting the absence of this species within the Project Area. Furthermore, visits to reference populations found this species present during the same time period (SWCA 2022b). Nonetheless, approximately 27 acres of potential habitat in the analysis area (approximately 19 acres in the Project Area) may exist along washes and margins (Figure A-8). Because the laydown area and a small portion of the linear facilities segment (easternmost end) were not surveyed during protocol-level surveys, all washes and margins within these areas are assumed to provide potential habitat for the species and are included in the 27 acres.

The 5-acre temporary laydown area is undisturbed and would be prepared using the disk and roll technique. Wash and wash margins would be left in place where possible to maintain site drainage. If needed, drainage and erosion controls would be installed to convey water flows or by spreading the flows to allow the water to leave the Project as sheet flow.

3.13.1.3 CACTI SPECIES

Seven species of cactus plants were observed: California barrel cactus (*Ferocactus cylindraceus*), beavertail pricklypear (*Opuntia basilaris*), buck-horn cholla (*Cylindropuntia acanthocarpa*), Wiggins'cholla (*Cylindropuntia echinocarpa*), branched pencil cholla (*Cylindropuntia ramosissima*), Engelmann's hedgehog cactus (*Echinocereus engelmannii*), and common fishhook cactus (*Mammillaria tetrancistra*). A total of 3,024 cacti were counted within the botanical survey area (Table A-20). Beavertail pricklypear was the most abundant cactus at 1,098 individuals observed. No yucca species were observed within the botanical survey area.

3.13.1.4 INVASIVE SPECIES AND NOXIOUS WEEDS

While Nevada Department of Agriculture (2021) noxious weeds species African mustard (*Brassica tournefortii*) and puncture vine (*Tribulus tribularis*) are known to occur within the Eldorado Valley, neither species nor any other noxious weeds were observed within the Project Area or along any public roads within the 1-mile buffer. Mediterranean grass, an annual nonnative grass species, was observed throughout the Project Area, mostly at the bases of perennial shrubs.

3.13.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related effects on vegetation, cacti, or noxious and nonnative weed species would occur.

3.13.3 Environmental Effects of the Proposed Action

Impacts to vegetation communities are indicated either by a direct physical removal of plant species or a change in vegetation community types that support plant species. Impacts to cactus species are quantified as either direct physical removal or a change in vegetation community type that supports the species. Noxious and nonnative invasive weed impacts are quantified as introduction or expansion of existing individuals and populations.

3.13.3.1 SHORT- AND LONG-TERM EFFECTS

Vegetation Community

Geotechnical work may impact vegetation in the short term; long term, the vegetation within the geotechnical sites would be permanently impacted because these sites would be integrated into the footprint of the solar facility. As a result, no interim reclamation is proposed.

Construction of the Proposed Action would result in short- and long-term impacts to the Sonora-Mojave Creosotebush-White Bursage Desert Scrub vegetation community. Most of the construction-related impacts would occur through grading/clearing including direct removal of plants. Soil disturbance and soil removal would also result from construction activities. As a result, construction impacts would alter the Sonora-Mojave Creosotebush-White Bursage Desert Scrub vegetation community throughout the Project Area. This would lead to long-term vegetation impacts, including permanently decreasing native plant cover and increasing soil disturbance. In the event that vegetation resprouts under the solar arrays, vegetation would be maintained.

Vegetation within the linear facilities segment would be removed during construction by clearing/grading and would be maintained during O&M of the Proposed Action. Vehicle use along the Project site road could indirectly lead to an increase in potential for noxious and nonnative weed species to be introduced and colonize the area. Other indirect impacts associated with the Proposed Action include edge effects on adjacent vegetation and increased bare ground resulting from construction activities, which can provide the opportunity for the introduction of noxious and nonnative weed species to colonize the Project Area. Additionally, bare ground can lead to an increase in wind and water erosion that has potential to damage plants and increase sedimentation into nearby drainages.

Following construction, to the extent feasible, the vegetation in the Project Area would be allowed to regrow and be maintained. A Site Rehabilitation Plan and Facility Decommissioning Plan would be developed and implemented at the end of the Project's operational capacity (approximately 30 years).

The applicant would avoid, to the maximum extent possible, all major washes within the Project Area, which coincides with the locations of catclaw, found within and along washes. Nonetheless, the Proposed Action has the potential to remove up to 219 catclaw plants.

Reclamation of the 5-acre temporary laydown area would occur once construction is complete. Because of the semiarid nature of the environment, reclamation efforts and natural vegetation reestablishment may take decades (Abella 2010). Long-term adverse impacts to vegetation density, cover, and composition would likely persist even after reclamation.

Special-Status Plants

The primary threat posed by geotechnical work to rosy two-tone beardtongue plants is crushing from geotechnical vehicles and equipment during borehole exploration. Existing trails would be used except for a small portion of overland travel required to access one of the borehole sites. To reduce potential crushing, an on-site biological monitor would survey ahead of vehicles/equipment to redirect them if necessary (per the BMPs in Section 2.2.3). However, it is unlikely that rosy two-tone beardtongue plants are present on-site as no live plants or remains were observed during surveys.

Construction would impact approximately 19 acres of potential rosy two-tone beardtongue habitat within the Project Area. No direct effects on individual plants are anticipated, but the Proposed Action would result in the long-term and permanent removal of potential habitat for the species within the Project Area. Construction and decommissioning activities could also have an indirect impact on approximately 8 acres

of potential habitat outside of the Project Area and within the analysis area from introduction of noxious and nonnative weed species, edge effects, and damage from water and wind erosion. O&M activities are not expected to further impact potential washes and margins beyond what was impacted under construction/decommissioning activities. Impacts to potential rosy two-tone beardtongue habitat (up to 27 acres) associated with the implementation of the Proposed Action would be permanent and localized.

The 5-acre laydown area would be restored at the end of construction and site drainage restored to natural conditions as much as possible. However, even with reclamation, potential rosy two-tone beardtongue habitat would likely suffer long-term, permanent impacts in this area.

Cactus Species

Geotechnical vehicles and equipment are the primary threat posed to cacti from crushing. To reduce the possibility of vehicle/equipment crushing, an on-site biological monitor would survey ahead of vehicles/equipment and, if necessary, redirect them (per the BMPs in Section 2.2.3).

Construction impacts to cacti would be the direct removal of cacti from the solar facility and linear facilities segment. Cacti would be either sold to the public or destroyed, with the appropriate cacti permit from the BLM, which requires fees for any cacti destroyed. Cacti within the temporary disturbance area (laydown area) are required to be salvaged (transplanted in nurseries until they can be transplanted back out into the restoration area). The cacti found on-site are relatively common throughout the eastern Mojave Desert, and removal would not contribute toward a downward trend across the species ranges.

Invasive Species and Noxious Weeds

No noxious weed populations were found on-site during vegetation surveys. However, because conditions were dry when surveys occurred, noxious weeds could be located on-site, and could also be introduced to the site by geotechnical and construction equipment. Geotechnical and construction/decommissioning activities have the potential to increase the presence of Mediterranean grass on-site, an annual nonnative species. However, implementation of the Integrated Weed Management Plan is expected to maintain or reduce Mediterranean grass cover. Given the level of soil disturbance proposed for the Project, there is potential for noxious weed and nonnative species to be introduced and successfully colonize the Project Area. This could affect the Project Area and surrounding landscape by modifying native plant assemblages, reducing biodiversity, increasing competition with native species, and potentially increasing fire hazards (Chambers, Bradley, et al. 2014; Chambers, Pyke, et al. 2014; Davies and Johnson 2011; Grodsky and Hernandez 2020; Lovich and Bainbridge 1999). Implementation of applicant-committed Project design features (see Appendix C) and BLM standard stipulations (see Appendix D) would reduce the risks associated with nonnative and noxious weed species being introduced or spread within the Project Area and adjacent habitat over time.

3.13.3.2 BENEFICIAL AND ADVERSE EFFECT

The Proposed Action would have short- and long-term impacts to up to 85 acres of Sonora-Mojave Creosotebush-White Bursage Desert Scrub vegetation. Even though no noxious weed species were found during baseline surveys, the soil surface disturbance associated with the Proposed Action increases the probability that an introduction could lead to successful colonization. Additionally, Mediterranean grass is already within the Project Area and has the ability to spread throughout the site during construction activities, as well as O&M activities but to a lesser degree. Implementation of applicant-committed EPMs (see Appendix C) and BLM standard stipulations (see Appendix D) would minimize impacts to the Project Area and adjacent habitat from noxious weeds and invasive species.

3.13.4 Mitigation Measures

The BLM's Forestry Program has a policy regarding the sale of vegetation resources, including cacti. The movement of cactus plants are governed by the Nevada Department of Conservation and Natural Resources, Division of Forestry, according to NRS 527. Cacti within temporary disturbance areas will require the applicant to obtain a permit from the BLM for destruction of those species; alternatively, the BLM could offer the cactus plants for sale to the general public.

3.14 Visual Resources

The term *visual resources* refers to the composite of basic terrain, geological and hydrological features, vegetative patterns, and built features that influence the visual appeal of a landscape. This section follows four steps to assess the existing visual environment and impacts to the visual environment from the Project: 1) defining an analysis area and Visual Resource Management (VRM) classifications within the analysis area; 2) describing existing visual resources (i.e., values) within the analysis area; 3) coordinating with the BLM to identify viewing locations (KOPs) from where the project may be viewed; and 4) completing contrast rating worksheets based on field observations incorporating environmental factors with supporting photographic simulations to assess Project conformance. The following analysis is based on information and data summarized from the Townsite Solar 2 Visual Technical Report (SWCA 2022c).

Analysis Area

For the purposes of this evaluation, the analysis area consists of the Project Area and a 5-mile buffer around the Project Area. Project impacts resulting from construction, operation, and decommissioning activities have the potential to affect visual resources both in and outside the Project Area. This section evaluates visual resources and associated Project effects throughout the geographic area where direct and indirect Project impacts are anticipated to occur and in the time frame in which effects would occur. The analysis area is where potential visual effects from Project development may be discerned by the casual observer and illustrates where in the surrounding landscape the Project would be theoretically visible. The Proposed Action, as described in Section 2, is completely within the analysis area.

Visual Resource Management

The BLM uses the VRM system to manage visual resources on BLM-administered lands (BLM 1984, 1986a, 1986b). This system provides a framework in which four VRM class objectives describe the different degrees of modification allowed to the basic elements of the landscape (i.e., line, form, color, and texture). The BLM uses RMPs to provide management direction and adapt to the changing resource and use demands, balanced with compliance with other federal, state, and local laws and policy. The Project Area is managed under the Las Vegas RMP (BLM 1998b). This analysis considers existing conditions and proposed project impacts to determine whether a project would conform with VRM Class III objectives.

Management Direction VS-1-b designates 1,579,800 acres of public land as VRM Class III for partial retention of the existing character of the landscape. In these areas, authorized actions may alter the existing landscape but not to the extent that they attract or focus attention of the casual viewer (BLM 1998b). Compliance with VRM objectives is assessed using BLM Form 8400-4 (Visual Contrast Rating Worksheet), as directed by BLM Manual 8431, *Visual Resource Contrast Rating*.

3.14.1 Affected Environment

The visual resources surrounding the Project Area are characterized by a flat valley surrounded by the distinctive jagged peaks of the Eldorado Mountains and Highland Range. Within this valley are multiple series of utility-scale solar array developments on either side of U.S. 95, which bisects the valley. Vegetation in the area is consistent with the eastern Mojave Desert including low bushes and occasional cacti growing out of the sandy soil, which is a dominant landscape feature in this area. Directly west of the Project Area is the Sloan Canyon National Conservation Area and the North McCullough Wilderness on the border of the analysis area. Boulder City lies to the northeast; the lush green grasses of the Boulder Creek Golf Club and Veterans Memorial Park are a stark contrast to the natural dry, sandy and silvery greens of the surrounding Mojave Desert landscape. South of the Project Area is the Eldorado Dry Lake Bed, which is populated with low shrubs and grasses. The landscape character immediately surrounding landscape scenery is characterized by desert landscapes with heavy development, including other solar array developments, transmission line corridors, substations, and industrial developments.

3.14.1.1 VISUAL RESOURCE INVENTORY

Existing visual conditions (i.e., values) relate to cultural resource modifications of the landscape that contribute to the overall visual character associated with a given area of land. These conditions can be documented through the BLM Visual Resource Inventory (VRI) process. The Project Area has been evaluated by the BLM as a VRI Class IV. The VRI objectives for this class are as follows:

- **VRI Class IV –** The objective of this class is to provide for management activities that require major modifications of the existing character of the landscape. The level of change to the characteristic landscape can be strong. These management activities may dominate the view and be the major focus of viewer attention. However, every attempt should be made to minimize the impact of these activities through careful location, minimal disturbance, and repeating the basic elements. The existing conditions for each category are described below.

The information collected provides descriptions and analysis of the landscape and viewer sensitivity in the Project Area, and the VRI class is broken down into three categories: scenic quality, visual sensitivity, and distance zones (BLM 2011). The existing conditions for each category are described below.

Scenic Quality

A scenic quality evaluation measures the visual appeal of a landscape. Public lands are rated as Class A (19 points or more), Class B (12–18 points), or Class C (11 points or less) based on the apparent scenic quality. Lands are reviewed and rated using seven key factors, and the total score determines the rating (BLM 2011).

The Project Area is contained within the Eldorado Valley Scenic Quality Rating Unit, which is rated as a Class C with a score of 9 points. The Project Area is a small, elongated, slightly bowl-shaped valley formed by the Eldorado Mountains and Highland Range populated with typical eastern Mojave Desert vegetation. U.S. 95, which bisects the valley, provides views of surrounding mountain ranges to vehicular travelers.

Visual Sensitivity

Visual sensitivity reflects attitudes and perceptions held by people regarding the landscape and, in general, reflects the public's level of sensitivity for noticeable change to the landscape. Visual sensitivity ratings range from high sensitivity (i.e., the public is highly aware of incremental changes in the subject landscape) to low sensitivity (i.e., little concern for changes in the subject landscape).

The project is contained within one Sensitivity Level Rating Unit (SLRU 027). Most land along this portion of U.S. 95 is privately owned, with residential and industrial development. Overall, the Project location has been given a moderate value for maintaining visual quality.

Distance Zones

The BLM defines distance zones as Foreground/Middleground (visibility from 3 to 5 miles), Background (visibility from 5 miles to a maximum of 15 miles based on atmospheric conditions), and Seldom Seen (landscape is not visible or is at distances greater than 15 miles).

The Project is contained within the Foreground/Middleground distance zone; Project elements introduced in this range would have increased proximity to viewers and would be in a more visually prominent distance zone.

3.14.1.2 SENSITIVE VIEWERS

Sensitive viewing platforms represent specific places, areas, and features that have visual importance relative to one's home, social, business, and recreational environments. KOPs represent points where the public would view the Project from a stationary location (e.g., residential area) or a linear (e.g., major roadway) location. KOPs are used to assess the level of change (contrast) introduced by a proposed project within a specific viewshed (BLM 1986b) and the sensitive viewers that would be affected. Sensitive viewing locations for this Project include the following:

- **Vehicular travel routes –** highways and roads used by origin/destination travelers, designated scenic or historic byways, and recreation destination roads (i.e., roads that provide access to designated recreation areas)

The selected KOPs represent the highest potentially impacted areas and viewers that may be affected by the Project. Other potential sensitive viewers (including residential and recreational) would feel impacts so minimal that it was not necessary to select KOPs to represent these groups. While representative KOPs were not selected for these viewer groups, there are residences and recreational opportunities surrounding the Project Area. However, any visual impacts felt from those areas would be similar to or less than the identified KOPs. The selected KOPs were identified through a combination of geographic information system (GIS) analysis, field observation, consultation with the applicant, and coordination with BLM SND staff and were selected because of their association with vehicular travelers along U.S. 95 and I-11/U.S. 93, two major vehicular routes in the area (Table A-21).

An analysis of visual dominance, scale, and contrast was used in determining to what degree the proposed alternatives would attract attention and in determining the relative change in character compared with the existing characteristic landscape and its inherent scenic quality. The amount of visual contrast created by a project is directly related to the amount of attention that is drawn to a project feature in the landscape. Potential changes in the viewshed from sensitive viewing locations (KOPs) were also evaluated and characterized. The analysis of visual impacts was used in the determination of conformance of the BLM VRM objectives where the Project would occur on BLM-administered lands. This process was used for each of the three selected KOPs.

3.14.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related effects on the visual resources in the area would occur.

3.14.3 Environmental Effects of the Proposed Action

Environmental effects of the Project were measured using the BLM contrast rating worksheets. This method allows for a level of objectivity and consistency in the process and reduces subjectivity associated with assessing landscape character and scenic quality impacts. Using the BLM's Visual Resource Contrast Rating system (BLM 1986b), the level of contrast for the Project was evaluated for each KOP to determine the degree to which the Project would affect the intrinsic visual character and the scenic quality of the landscape based on the level of contrast created between the Project and existing landscape. The degree of contrast for each landscape element was evaluated (e.g., land/water, vegetation, and structure) as none, weak, moderate, or strong (Table A-22).

Visual simulations, which provide a theoretical view of the Project, were prepared for each KOP. A digital rendering of the solar facility was superimposed on the KOP baseline photographs, illustrating a simulated view of the facility components. This information was used to aid in determining the level of visual contrast of the Project before and after construction.

3.14.3.1 SHORT- AND LONG-TERM EFFECTS

Short-term effects of the Proposed Action include visual impacts from geotechnical investigation including soil boring, soil resistivity testing, and pile load testing. Access for these activities would include overland travel using a truck-mounted drill rig and conventional pick-up truck(s) or four-wheel drive trucks(s). Other impacts would occur from fugitive dust, primarily caused by on-site vehicular traffic, grading, and hundreds of construction workers on-site over the construction period. Similar effects are expected to be perceived with decommissioning efforts. Reclamation of the area would begin once Project activities have ceased and equipment has been removed. Due to the semiarid nature of the environment, reclamation efforts and reestablishment of natural vegetation may require multiple seasons or decades, which will extend the duration of visual disturbance and impact. Temporary visual impacts like landscape scarring from decommissioning may be visible for some time but would be weak in contrast.

Long-term effects of the Proposed Action would include building or installing PV solar panels and a BESS facility, as well as an 80 × 2000–foot site access road on approximately 4 acres of previously undisturbed land. Impacts and resulting visual contrast would be weak because of continued introduction of form, line, and texture of elements into the landscape that are similar to elements in the immediate vicinity surrounding the Project and on municipal lands in the Eldorado Valley.

Several factors influence the degree of impact to observers on vehicular travel routes. These factors include the proximity of U.S. 95 and I-11/U.S. 93 to the Project Area, visual obstructions along the roadway that limit and obscure views of the Project Area, and the introduction of elements of form, line, color, and texture that are similar to existing utility-scale solar projects and activities in the surrounding landscape. With these factors, it is anticipated that the degree of visual impact felt from Project activities to vehicular travel routes and observers, as represented by the selected KOPs, would be weak. Project impacts from selected KOPs are summarized in Table A-23 and are further detailed in the Visual Technical Report (SWCA 2022c). It is anticipated that, based on the weak contrast and low level of visual impact when viewed from the three KOP locations, the Project, under the Proposed Action, would be in conformance with the BLM's VRM objectives associated with the VRM Class III.

There are several opportunities for recreation surrounding the Project, including the River Mountain Loop Trail (approximately 1.75 miles northeast of the Project), the Railroad Pass Hotel and Casino (approximately 1.8 miles northeast of the Project), and the Boulder Creek Golf Course (approximately 3 miles east of the Project). These recreational areas would have varying obstructed views of the Project

Area due to topography, highways, and the interstate roadway system. With the introduction of elements of form, line, color, and texture that are similar to existing utility-scale solar projects and activities in the surrounding landscape, we anticipate that the degree of visual impact to recreational users would be similar to or less than those from vehicular KOPs (weak visual impact).

Additionally, a branch of the OSNHT intersects with the linear facilities segment, approximately 0.3 miles west of the solar array area. While the setting adjacent to the congressionally designated alignment would be further modified by the Project, in context with impacts from existing projects and developments, visual impact would be negligible to weak. Since the Project would occur adjacent to other energy infrastructure projects, the Project would not result in any discernible changes and would not interfere with the trail's nature and purpose.

3.14.3.2 GLARE ANALYSIS

An analysis of glare was conducted using GlareGauge (also known as Solar Glare Hazard Analysis Tool) to evaluate potential glare from solar panels and other Project components. The analysis focused on potential glare effects on observation points (OPs). Aircraft landing and approach were considered, including the Harry Reid International Airport (in Las Vegas approximately 14 miles from the Project Area), Henderson Executive Airport (approximately 12 miles from the Project Area), and Boulder City Municipal Airport (approximately 6 miles from the Project Area). This glare analysis used a total of six OPs; OPs 1, 2, and 3 were set at the KOP locations, and OPs 4, 5, and 6 were points selected to represent residential areas on the western border of Boulder City approaching the 5-mile extent of the analysis area. OPs used a height of 4 feet (an average height of passenger cars, trucks, and diesel trucks). Using Project parameters provided by the client, the glare analysis will indicate a quantified assessment of when and where glare would occur throughout the year for a prescribed solar installation and the potential effects on the human eye. The sources of potential glint and glare for the Project is the proposed PV panels; however, PV panel surfaces are designed specifically not to reflect light, thus reducing the potential for glint and glare. For the purposes of this EA, glint is defined as a bright, momentary flash of light; glare is defined as a more continuous and sustained presence of light that may appear to "sparkle" from sensitive viewing locations. There are three types or degrees of glare. "Green" glare is glare with a low potential to cause an afterimage (flash blindness) when observed prior to a typical blink response time. "Yellow" glare is glare with potential to cause an afterimage (flash blindness) when observed prior to a typical blink response time and create a temporary after-image. "Red" glare is glare with potential to cause permanent eye damage. For this analysis, it was assumed that the solar project would operate 365 days per year, during daylight hours. Glare analyses do not account for physical obstructions between reflectors and receptors. This includes buildings, tree cover, and geographic obstructions. Impacts for the glare analysis were analyzed and summarized by their representative type of sensitive viewers.

The analysis concluded that the Project would not produce glare over the course of a typical year for any of the glare receptor locations. Regarding the nearest airports, 16 potential flight paths were considered at three nearby airport locations. During takeoff and landing procedures, airborne viewers (e.g., pilots) would be elevated in relation to the Project Area. The results of the glare analysis conducted for this Project indicate that airborne viewers would not experience glare within the four 2-mile-long takeoff and landing flight path segments analyzed by ForgeSolar, with parameters established to meet Federal Aviation Administration (FAA) requirements. SWCA reviewed the Project Area and confirmed the airports and their associated flight paths.

3.14.3.3 BENEFICIAL AND ADVERSE EFFECT

The Proposed Action would adversely impact up to 85 acres with the introduction of Project components into a previously undeveloped piece of public land and alter the existing landscape character. Removal of

vegetation and grading for installation of Project components would leave a visually altered landscape, continuing the increasing degree of visual impacts from utility-scale developments in the area. These impacts would adversely affect the existing visual character of the landscape.

3.14.4 Mitigation Measures

Implementation of mitigation measures, which are listed in Appendix H, and BMPs identified by the BLM (2013) would lessen visual impacts.

3.15 Wastes (Hazardous or Solid)

Hazardous and toxic materials or substances are generally defined as materials or substances that pose a risk (through either physical or chemical reactions) to human health or the environment. Applicable federal laws would include the following: the Resource Conservation and Recovery Act of 1976; Hazardous and Solid Waste Amendments; Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA [aka Superfund]); and the Superfund Amendments and Reauthorization Act of 1986. Release of a reportable quantity of a hazardous substance into the environment within a 24- hour period must be reported to the National Response Center in accordance with regulations issued under Section 102 of CERCLA, as amended (40 CFR 302). The BLM must be informed whenever a reportable quantity is released on public property.

The use and production of hazardous materials by the Project are also subject to the laws and regulations governing hazardous materials and waste in the State of Nevada. NAC 445A.347 mandates the immediate reporting of a release of a reportable quantity of a hazardous substance to the NDEP.

To assess potential risks associated with interactions with solid and hazardous wastes, the analysis area is the Project Area plus a 66-foot buffer.

3.15.1 Affected Environment

A review of aerial imagery identified two existing potential sources for hazardous materials in the analysis area: Quarry 187 sand and gravel pit adjacent and directly west and southwest of the Project Area and the Henderson Community sand and gravel pit 0.25-mile northeast of the Project Area. Primary access to both mineral pits is from Old U.S. Highway 95, which would be the same access road used for the Project. No specific health and safety issues have been identified for the sand and gravel pits, and it is assumed that the facilities are operated in compliance with all applicable laws and regulations regarding sand and gravel pits. No other potential sources of hazardous materials were identified within the analysis area, as the area is primarily undeveloped.

3.15.2 Environmental Effects of the No Action Alternative

Under the No Action Alternative, no Project-related solid waste and hazardous materials impacts would occur.

3.15.3 Environmental Effects of the Proposed Action

Impacts are quantified based on whether the Project would result in the acute or chronic exposure of the public or occupational workers to hazardous materials through the generation of hazardous materials and generation of solid wastes.

3.15.3.1 SHORT- AND LONG-TERM EFFECTS

A comprehensive analysis of hazardous materials and environmental exposure from PV solar panels, batteries, BESS, semiconductors, transformers, and inverters were completed and included in the Draft Solar PEIS (BLM and DOE 2012). Therefore, environmental effects will focus on impacts associated with geotechnical, construction, O&M, and decommissioning activities associated with the Proposed Action.

Leaks from equipment and vehicles used for geotechnical work, including lubricating oils, hydraulic fluids, brake fluids, glycol-based coolants, fuel (including both gasoline and diesel), and dielectric fluids, pose a risk to workers if direct exposure occurs. However, implementing BMPs for proper equipment and vehicle maintenance, including frequent inspections in accordance with manufacturer specifications, reduces the risk of leaks and spills.

During construction/decommissioning, short-term impacts could result due to the increased presence and use of hazardous materials during construction of the solar facility and ancillary components. In addition, an increase in construction vehicle traffic and equipment would increase the possibility of a release of operating fluids including lubricating oils, hydraulic fluids, brake fluids, glycol-based coolants, fuel (including both gasoline and diesel), and dielectric fluids that could pose a risk to workers should direct exposure occur. As such, a short-term impact is possible. However, BMP implementation of proper maintenance of equipment and vehicles, including frequent inspections in accordance with manufacturer specifications, reduces the risk of leaks and spills. Design features include Project-specific plans outlined in Section 2.4.2, in combination with BLM standard stipulations (see Appendix D), would ensure that industry standard BMPs and protocols are implemented during all stages of Project activities to help minimize the occurrence of spills and leaks and potential exposure to workers. Implementation of these practices would serve to ensure any adverse impact is further minimized.

Additional measures to minimize public exposure include restricting public access through a combination of measures, including installing chain-link fencing with barbed wire around the entire Project Area, installation of an electronic security system and lighting, controlled access gates, and on-site security patrols. Implementation of these measures would reduce the risk of exposure during construction (short term) and O&M activities (long term).

Primary solid wastes anticipated to be generated during construction (short term) would include solid and liquid wastes such as scrap wood, steel, plastic, paper, metals, and sanitary waste. No significant short-term impacts from solid wastes are anticipated. All solid waste would be disposed of by a local waste disposal company in accordance with federal and state laws.

During O&M operations, limited quantities of hazardous material would be used and would be brought on site as needed. No hazardous material would be stored on-site. It is anticipated that one or more propane aboveground storage tanks with secondary containment would be installed to service backup generators. In addition, routine maintenance activities include panel washing and routine equipment repairs. Panel washing would not increase the risk of harmful exposure of hazardous materials. Routine equipment repairs would follow manufacturer specifications, and removal of broken solar panels, spent lead acid–based or lithium-ion batteries, and failed transformers, all of which contain toxic material, would include disposal of or recycling in accordance with federal and state laws. As part of the purchase agreement, the modules will be recycled with the module manufacturer. To minimize potential short-term impacts associated with hazardous material exposure, the applicant would prepare a Hazardous Materials and Waste Management Plan, as well as a Spill Prevention and Emergency Response Plan, which would address waste and hazardous materials management, including BMPs related to storage, spill response, transportation, and handling of materials and wastes. In the event of a release, handling would occur in

accordance with the approved Hazardous Materials and Solid Waste Management Plan and would also include a list of required notifications in the unlikely event of a major spill or risk to the public from potentially hazardous materials.

Solid waste materials typical of O&M activities would include defective or broken electrical materials, empty containers, refuse generated by workers and small office operations, and other miscellaneous solid wastes. O&M activities are not expected to result in significant amounts of solid waste, and disposal would follow manufacturers' recommendations.

Decommissioning activities and reclamation of the temporary laydown area would have the potential in the short term to lead to an increase in the generation of solid waste and the potential for spills and leaks from construction equipment. Implementation of the Project-specific Decommissioning and Site Reclamation Plan and other existing plans (i.e., design features) listed in Section 2.4.2 would further minimize the potential for impacts. Disposal would be managed in accordance with applicable federal and state laws.

3.15.3.2 BENEFICIAL AND ADVERSE EFFECT

The Proposed Action is not anticipated to result in a significant impact through the generation of hazardous materials or the production of solid waste, increase the exposure of workers to hazardous substances, increase the presence of hazardous materials in the environment, or place substantial restrictions on property use due to hazardous waste, materials, or site remediation.

3.15.4 Mitigation Measures

Implementation of BLM standard stipulations and design features outlined in Section 2.4.2 (i.e., Hazardous Materials and Waste Management Plan, Spill Prevention and Emergency Response Plan, Emergency Response Plan, etc.) would reduce potential impacts to negligible levels and would not result in environmental degradation cumulatively. As a result, no mitigation measures are anticipated.

4 CUMULATIVE EFFECTS ANALYSIS

The Council on Environmental Quality regulations defines cumulative impacts as "the impact on the environment which results from the incremental impact of the action when added to other past, present, and reasonably foreseeable future actions (RFFAs) regardless of what agency (federal or nonfederal) or person undertakes such other actions" (40 CFR 1508.1(g)(3)). Federal agencies have the responsibility of determining how and the extent to which cumulative impacts are assessed in NEPA documents and documenting that effort.

In the context of this EA, cumulative impacts are the total of all past, present, and RFFAs brought on primarily by renewable energy, mining, utility infrastructure, and public uses. The cumulative analysis in this EA is used to assess the significance of the Project's contributions to the cumulative environment.

4.1 Cumulative Effects Study Areas

Each resource/use under consideration has a distinct cumulative effects study area (CESA) for the Proposed Action and the No Action Alternative. Figure A-9 depicts the geographic area of the CESAs by resource. In terms of time frame, the cumulative effects analysis is considered over a 30-year period. Based on the anticipated lifespan of the Project facilities and key components, the Proposed Action has a life expectancy of 30 years.

The geotechnical investigation impacts (0.93 acre) are already factored into the 80-acre solar facility footprint. Therefore, the cumulative analysis for the Proposed Action already accounts for this disturbance.

4.2 Past and Present Actions

Much of the land in the Eldorado Valley is owned by the City of Boulder City. Past and present actions within 5 miles of the Proposed Action include renewable energy and transmission facilities, mining, roads, and land development for rural and urban business and residential areas. Solar generation facilities near the Proposed Action include, but are not limited to, Boulder Solar (I and III), Copper Mountain Solar (I, II and III), El Dorado Solar, Nevada Solar One, Techren Solar (I, II, III and IV), and Townsite Solar 1. To facilitate energy transmission, the Mead, Marketplace, McCullough, and Eldorado substations are present. Also nearby are businesses and residential areas including Pro Gun Vegas, Safari and Expedition, Blastco Sandblasting & Powder Coating, Goodfellow Corporation, Railroad Pass Casino, the City of Henderson (southeasternmost portion of city), Boulder City (westernmost portion of city), Boulder Creek Golf Club, Boulder City Municipal Airport, and Veterans Memorial Park. I-11/U.S. 93, U.S. 95, and Old U.S. Highway 95 are the major transportation corridors present, and active quarries/mining are present north and southwest of the Project Area. Several roadway improvements are currently in the planning stage or proposed for construction, including I-11/U.S. 93 at Wagonwheel Drive Interchange, the Boulder Highway Flyover to I-11/U.S. 93, and the College Drive Interchange at I-11/U.S. 93, by the City of Henderson. Additionally, the City of Boulder City also plans to make improvements along Industrial Road with pedestrian safety upgrades and American with Disabilities Act upgrades along Georgia Avenue (Regional Transportation Commission of Southern Nevada 2022). The compiled acreage associated with past actions utilized the USGS National Land Cover Dataset (USGS 2016).

4.3 Reasonably Foreseeable Future Actions

Table A-24 provides a summary of the RFFAs and associated surface disturbance within 5 miles of the Proposed Action.

4.4 Evaluation of Potential Cumulative Impacts

4.4.1 Air Quality

The analysis area for cumulative air quality impacts includes the Eldorado Valley hydrographic basin (approximately 339,200 acres in size [State of Nevada 2023]) and RFFAs within 5 miles of the Proposed Action. Past and present actions, including existing land development, have contributed to the existing air quality conditions within the analysis area (see Table A-24).

4.4.1.1 PROPOSED ACTION

In general, cumulative impacts to air quality from Project components under the Proposed Action, when considered with existing uses and RFFAs, could occur where construction associated with land development occurs. RFFA that have been identified with potential to affect air quality include projects associated with county land and BLM-administered land and future development of municipal owned land parcels for solar power generation and energy storage facilities on municipal lands in Eldorado Valley, including the Boulder Flats Solar Project on 1,100 acres and the Boulder Flats Solar Transmission Line on 145 acres (see Table A-24).

Renewable energy projects being built on municipal land in Eldorado Valley could have detrimental short-term cumulative effects on air quality from fugitive dust and ozone precursors (NO_x and VOCs) emissions. This could be exacerbated with the addition of other infrastructure and private projects within the analysis area (see Table A-24). Future projects, as stated in Section 5.4.9 of the Eldorado Valley Transmission and Utility Corridor PEA, may have overlapping construction schedules, contributing to temporary increases in ozone precursors and the production of fugitive dust (i.e., particulate matter) during construction (BLM 2012a). Upcoming projects from the SNWA and Southwest Gas could potentially overlap the construction of the Project. The removal of vegetation causes soil disturbance, which in turn would increase the amount of particulate matter in the Eldorado Valley's air. Implementation of EPMs (see Appendix C) and BLM standard stipulations (see Appendix D) would reduce impacts to temporary regional air quality.

Operation of the Project and future renewable energy initiatives would have a beneficial effect on air quality because of the reduction in emissions when compared with other types of energy-generating facilities.

4.4.2 Cultural Resources

According to Section 5.4.6 of the Eldorado Valley Transmission and Utility Corridor PEA, development of the Eldorado Valley is likely to have cumulative effects on cultural resources (BLM 2012a). As a result, the Project Area's 5-mile radius is the CESA for cumulative effects to cultural resources.

4.4.2.1 PROPOSED ACTION

Based on the analysis in the resource evaluation section (Section 3.3), the Proposed Action would not have any direct, indirect, or cumulative effects on cultural resources. Consequently, the Proposed Action would not contribute cumulatively to cultural resource impacts in the Eldorado Valley.

4.4.3 Environmental Justice and Socioeconomics

The CESA for socioeconomics and EJ is limited to Clark County, Nevada.

4.4.3.1 PROPOSED ACTION

Environmental Justice

Past and present actions that could have impacted and may be currently impacting EJ populations include renewable energy and transmission facilities, mining, and ROW construction and maintenance. There are EJ communities within 3 miles of the Project Area, and others distributed throughout Clark County.

These communities are likely to notice increased noise, fugitive dust, waste, or other localized impacts from construction of the actions in closest proximity to them. Other potential impacts from past actions, present actions, and RFFAs may be attributed to dispersed recreation access restrictions. EJ communities recreating and living near the RFFAs would also be aware of the visual impacts; however, the visual impacts would be of similar nature to existing development within Clark County.

Socioeconomics

The cumulative impacts of past, present, and RFFAs on socioeconomics in Clark County would likely be beneficial. According to the POD, the Project would require 100 to 150 workers during construction, approximately half of that workforce during decommissioning activities, and fewer workers for geotechnical investigation. The addition of the RFFAs listed in Table A-24 would contribute to the total workforce but not all at the same time. Regardless, the creation of construction jobs would be beneficial and likely contribute cumulatively to local populations and socioeconomic local conditions in the short term.

During operation, the three FTE staff required for the Proposed Action combined with RFFAs listed in Table A-24, estimated to be less than 20 FTE positions, would not measurably contribute cumulatively to socioeconomic conditions in Clark County.

4.4.4 Fish and Wildlife (Excluding Federally Listed Species)

The geographic scope for cumulative impacts to wildlife including special-status species is a 2-mile area (7,982 acres of undisturbed habitat of the total 10,699 acres) surrounding the Project and a 3-mile area (17,255 acres of undisturbed habitat of the total 22,012 acres) for nesting raptors. Cumulative impacts to federally listed species are analyzed in Section 4.3.10.

4.4.4.1 PROPOSED ACTION

Past actions have had an impact on 2,717 acres of the CESA's 10,699-acre special-status species habitat, or 25% of the area, and 4,757 acres of the CESA's 22,012-acre raptor habitat, or 22% of the analysis area. In conjunction with the Proposed Action, up to an additional 85 acres (80 acres permanently impacted from the solar facility footprint and 5 acres from the temporary laydown area, which is unlikely to return

to pre-disturbance conditions even with reclamation over the 30-year period) would be impacted. RFFAs listed in Table A-24 would continue to impact wildlife species including special-status species through displacement, loss of habitat, and injury or death during construction activities. The development of the projects listed in Table A-24 would result in the long-term disturbance of up to an additional 152 acres of the 2-mile CESA and 214 acres of the 3-mile CESA for raptors. This would account for additional surface disturbance of approximately 3% in the 2-mile CESA and 2% in the 3-mile CESA. When considered with the cumulative impacts of 6,670 acres of land disturbance analyzed in Section 5.4.4. of the Eldorado Valley Transmission and Utility Corridor PEA (BLM 2012a), the Proposed Action and RFFAs would have an incremental cumulative adverse effect on general wildlife, special-status species, and nesting raptors habitat in the analysis area. Furthermore, the RFFAs would likely be subject to either the mitigation measures outlined in the Clark County MSHCP for county/municipal lands or BLM standard stipulations on federal lands, which would further reduce cumulative impacts to these species.

4.4.5 Fuels and Fire Management

The analysis area for cumulative impacts to fuels and fire management is a 0.5-mile area surrounding the Project Area.

4.4.5.1 PROPOSED ACTION

The cumulative impact of past and current actions and RFFAs that would result in an increase in fire fuels in the analysis area is associated with the introduction and spread of noxious weeds and nonnative invasive species. The presence of noxious weeds and invasive species present with the CESA is unknown. Mediterranean grass, an invasive plant, is already known to be prevalent throughout the Project Area, so it is likely that the species is also present throughout the CESA as well. The geotechnical investigation and construction/decommissioning activities of the Proposed Action has the potential to lead to an increase in Mediterranean grass and other species that could support fire. Implementation of the Integrated Weed Management Plan and Project-specific EPMs (see Appendix C) would minimize potential introductions of noxious weed species and expansion of current invasive species on-site. Therefore, it is unlikely that the Proposed Action would result in an impact that would be cumulatively considerable.

The possible inclusion of an on-site BESS as part of the Project has the potential to lead to an increase in fire risk potential in the CESA. Currently, there are no other BESS within CESA analysis area. The nearest BESS is at Townsite Solar 1, which is on nonfederal land owned by the City of Boulder City and is greater than 2 miles east of the Project's proposed BESS location. The proposed BCNV Facility, a RFFA, includes a BESS but the proposed location is more than 1 mile northeast of the BESS location for this Project on land owned by the City of Boulder City. Therefore, the analysis is limited to the BESS (up to a 35 MW) proposed for the Project.

The BESS proposed for the Project has industry-imposed levels of protection to minimize the fire risk potential associated with the BESS. These include updated software and hardware measures including the combustible gas sensor systems, as well as meeting the certifications set by the NFPA and the Underwriters Laboratories 9540 standards. The application of these protection levels would offset any potential impact that is cumulatively considerable during operations.

4.4.6 Greenhouse Gas Emissions/Climate Change

The analysis area for cumulative GHG impacts includes the Eldorado Valley hydrographic basin (approximately 339,200 acres in size [State of Nevada 2023]) and RFFAs within 5 miles of the Proposed Action. Past and present actions, including existing land development, have contributed to the existing GHG conditions within the analysis area (see Table A-24). However, as discussed, climate

change is fundamentally a cumulative phenomenon, global in scope, and all GHGs contribute incrementally to climate change regardless of scale or origin.

4.4.6.1 PROPOSED ACTION

Section 0 discussed projected climate change trends and impacts such as increased temperatures and droughts in Nevada. On a global scale, the GHG emission contribution of any single source on a subnational scale (such as the Proposed Action or RFFAs) would be dwarfed by the large number of comparable national and subnational contributors. The relative contribution of GHG emissions from construction, including geotechnical investigation, and operation of the Proposed Action would vary depending on contemporaneous changes in other sources of GHG emissions. A single subnational contributor is very unlikely to influence global cumulative emissions.

The overall impacts from RFFAs that support agency goals to combat climate change by increasing renewable energy supply, potentially offsetting electricity generation from fossil fuels, are expected to be beneficial with respect to climate change considerations. The Project would add incrementally to these beneficial impacts.

4.4.7 Migratory Birds and Eagles

The geographic scope of cumulative impacts to migratory birds and golden eagles is a 1-mile area (2,091 acres of undisturbed habitat of the total 3,403 acres) and a 2-mile area (7,982 acres of undisturbed habitat of the total 10,699 acres), respectively, that surrounds the Project.

4.4.7.1 PROPOSED ACTION

The Project's additional land disturbance of up to 85 acres would leave 2,006 acres and 7,897 acres of habitat in each CESA analysis area. Additional land disturbance would total 151 acres when combined with the RFFAs listed in Table A-24 but would be limited to the 1-mile CESA for migratory birds because no additional RFFAs have been identified for the 2-mile CESA. The combination of past and present actions, as well as RFFAs, would affect approximately 45% of the 1-mile analysis area for migratory birds or a 6% increase in disturbance. No additional disturbance would occur within the 2-mile CESA.

As a result of past, present, and RFFAs, minor impacts to migratory bird nesting and foraging habitat within 1 mile of the Project are possible. Adverse impacts include those associated with construction, which can lead to injury or even death, and cumulative impacts to numerous species from habitat loss. When considered with cumulative impacts of 6,670 acres of land disturbance analyzed in Section 5.4.3 of the Eldorado Valley Transmission and Utility Corridor PEA (BLM 2012a), the additional reduction in available habitat for migratory birds may cumulatively impact some species more than others. Implementation of BLM standard stipulations and Project EPMs should reduce the contribution of the RFFAs to cumulative impacts on federal lands. Furthermore, nonfederal private lands within the CESA are subject to the Clark County MSHCP requirements and would further reduce the contribution of these future projects to cumulative impacts on migratory birds.

The Proposed Action would impact up to 85 acres, or 1% of the golden eagle CESA. RFFAs would contribute an additional 72 acres of land disturbance in the golden eagle CESA, which results in an incremental impact of approximately 1%. When considered with the Proposed Action, effects on availability of foraging habitat may occur, but impacts to the regional population within the CESA are not anticipated. Cumulative effects to individual eagles utilizing this area could occur, especially when considering the 6,670 acres of land analyzed in Section 5.4.3 of the Eldorado Valley Transmission and

Utility Corridor PEA (BLM 2012a). Despite this, the concentration of renewable development in the analysis area has the potential to affect less total land disturbance due to the consolidation of related infrastructure, which indirectly benefits foraging golden eagles. As a result, small-scale cumulative effects on individual golden eagles are possible.

4.4.8 Recreation/Travel/Wild and Scenic Rivers

The analysis area for recreation and travel is limited to the Project Area.

4.4.8.1 PROPOSED ACTION

Based on the analysis in the resource evaluation section (Section 3.9), the Proposed Action would have negligible effects on travel and recreational resources. Implementation of a Travel Management Plan, if needed, would reduce the Projects contribution to cumulative effects in the Eldorado Valley.

4.4.9 Special Designations

The geographic scope of cumulative impacts to the OSNHT is a 5-mile radius around the Project Area and access road due to the visual character of the landscape.

4.4.9.1 PROPOSED ACTION

The Proposed Action is anticipated to have a negligible impact to the purpose and nature of the OSNHT trail. This is because there are no formally recorded archeological resources, trail traces, trail-related cultural resource sites, and trail-specific recreational opportunities in the area. While the trail alignment would be additionally disturbed by this project, the extent of existing development on or surrounding the trail alignment makes the disturbance of the Proposed Action negligible. Opportunities to experience the trail as a historical and experiential resource are minimal to non-existent in the analysis area. The cumulative impact of reasonably foreseeable future actions associated with cumulative projects as described in Table A-24 would contribute approximately an additional 1,477 acres of disturbance (some projects currently do not have acreages associated with them) in addition to the 85 acres of the Proposed Action. Given the small scale of development in the Proposed Action, the amount of current development, and foreseeable projects, the degree of cumulative impacts to special designations (OSNHT) would be negligible, and the Proposed Action would not result in any discernable changes to the OSNHT or interfere with the trail's nature and purpose.

4.4.10 Soils and Hydrologic Conditions

The geographic scope of cumulative impacts to soils and hydrologic conditions is a 200-m area (or 206 acres) around the Project Area and access road due to the presence of similar soils and surface hydrology.

4.4.10.1 PROPOSED ACTION

Quantifiable past actions in the CESA account for approximately 7 acres of disturbance (or 3% of the CESA). The Proposed Action would impact up to an additional 85 acres or approximately 43% of the CESA, which results in an incremental impact from the Proposed Action of approximately 40%. Because quantifiable data for all activities within the CESA are limited, this calculation is a conservative analysis of the potential incremental impact of the Proposed Action. Project-related impacts to erosion and off-site drainage would be localized and minimized due to implementation of the SWPPP and adherence to the Nevada Construction Stormwater General Permit conditions and future site reclamation. Therefore, based

on the above analysis and findings, incremental impacts to soils and hydrologic conditions as a result of the Proposed Action, and RFFAs (listed in Table A-24) with the implementation of the Project design features and adherence to permit conditions, are expected to have minor contributions to cumulative effects for both resources.

The Proposed Action would disturb up to 85 acres, which could disrupt hydrologic connections on-site and off-site. The applicant has committed to avoiding major drainages on-site to the maximum extent possible. If drainages are impacted from the Proposed Action, the applicant will construct drainage features and berms to convey overland surface water to minimize water ponding, soil erosion, and off-site sediment transport. Implementation of a SWPPP and adherence to the Nevada Construction Stormwater General Permit conditions would further minimize potential impacts to hydrologic connections.

4.4.11 Threatened, Endangered, and Candidate Species

The analysis area for cumulative impacts to threatened, endangered, and candidate species is approximately 1 mile (2,091 acres of undisturbed habitat of the total 3,403 acres).

4.4.11.1 PROPOSED ACTION

Approximately 1,312 acres (or 39% of the CESA) have been previously disturbed by past operations. The Project would provide up to an additional 85 acres of land disturbance, resulting in a 2% incremental impact due to the Proposed Action.

The RFFAs propose affecting an additional 152 acres (see Table A-24), accounting for an additional 5% or 46% of the CESA. As a result, there would be long-term permanent habitat removal and adverse effects on the desert tortoise, such as individual injury and death during construction activities. Furthermore, new perches for common ravens would be created from the Proposed Action (distribution line and perimeter fence) and RFFAs project elements such as transmission and distribution lines and fencing. However, Section 7 consultation would be necessary for each of the RFFAs on federal lands and compliance with the Clark County MSCHP and incidental take permits on nonfederal lands, thereby reducing their ability to contribute to overall impacts during operation.

4.4.12 Vegetation (including Special-Status Species, Cacti, and Invasive Species and Noxious Weeds)

The analysis area for cumulative impacts to vegetation (including special-status species, cacti, and invasive species and noxious weeds) is a 200-m radius of the Project Area.

4.4.12.1 PROPOSED ACTION

Past projects within 200 m of the Project Area and access road would contribute incrementally to Sonora-Mojave Creosotebush-White Bursage Desert Scrub vegetation and cacti (mostly beavertail pricklypear) disturbance and removal. Within 200 m of the Project, which represents the CESA, there are approximately 282 acres, of which 84 acres have already been disturbed by previous and ongoing activities. The Proposed Action will disturb up to 85 acres, with 80 acres permanently impacted from the solar facility footprint and 5 acres from the temporary laydown area, which is unlikely to return to pre-disturbance conditions even with reclamation over the 30-year period. The additional 85 acres would account for 60% of the CESA, which is an incremental increase of 30%. No additional disturbance to the CESA would occur from RFFAs (see Table A-24). Therefore, based on the above analysis and findings, minor impacts to Sonora-Mojave Creosotebush-White Bursage Desert Scrub community and cacti species would result from the Proposed Action. However, implementation of BLM standard stipulations and

EPMs, including revegetation of areas after construction, reclamation of the area after decommissioning activities, and a potential BLM sale of public-salvaged cacti from permanent disturbance areas, would result in a reduced contribution to cumulative effects in the CESA.

Although no rosy two-tone beardtongue plants were observed, potential habitat is present. The Proposed Action would negatively affect the ability of the species to occur in the CESA and permanently remove 27 acres of suitable habitat. Approximately 3.5 miles west of the Project, in the Sloan Canyon National Conservation Area, are known occurrences. Therefore, while the Proposed Action would gradually contribute cumulatively to the removal of the species' potential habitat, it would not result in a trend toward federal listing of the plant species.

Land disturbance associated with the Proposed Action could provide opportunities for invasive species and noxious weeds to colonize the area over time or lead to an increase in cover of Mediterranean grass, which is within the Project Area. An increase in cover of Mediterranean grass could impact vegetation community structure and also lead to an increase in wildfire potential in the Project Area and adjacent lands. Following the EPMs (see Appendix C) and BLM standard stipulations (see Appendix D) would reduce those effects.

4.4.13 Visual Resources

The analysis area for impacts to visual resources includes projects within 5 miles of the Proposed Action. In context with the impacts from existing projects and developments within 5 miles (detailed in Section 4.2), the development of this Project would have a negligible impact to the visual resources in the analysis area, including impacts to the OSNHT's visual resources.

4.4.13.1 PROPOSED ACTION

Reasonably foreseeable future actions associated with cumulative projects as described in Table A-24 would contribute approximately an additional 1,477 acres (some projects currently do not have acreages associated to the projects) of disturbance and will cause additional impacts to visual resources. With the completion of this Project, an additional 85 acres would be visually impacted, adding to the cumulative visual impact of the existing solar array areas, increasing the visibility and scale of surrounding utility developments, and resulting in an increased degree of impact that could be felt by casual observers. With the small scale of the development, the amount of development currently existing, and foreseeable projects, the degree of cumulative impacts to visual resources would be weak to negligible.

4.4.14 Wastes (Hazardous and Solid)

The analysis area for impacts to waste (hazardous or solid) includes the Project Area and access road boundary.

4.4.14.1 PROPOSED ACTION

The construction of the Proposed Action has the potential to contribute to waste in the short term and increase the potential for spills to occur. However, BMP and design feature implementation in conjunction with BLM standard stipulations (see Appendix D) would minimize potential adverse effects. The potential for waste and spills during O&M operations would be further reduced due to the limited quantities of hazardous material being used on-site and waste generated. Therefore, the cumulative contribution to waste is negligible.

In conjunction with the RFFAs listed in Table A-24, which are mostly construction-related projects that do not require the use of large quantities of hazardous materials, the cumulative impact contribution is much less. According to Section 5.4.15 of the Eldorado Valley Transmission and Utility Corridor PEA, projects that include the construction of gas pipelines or energy generation plants other than solar projects, tend to use hazardous materials and have a higher potential for accidental spills or releases of hazardous materials (BLM 2012a). None of these types of projects are proposed currently but potential exists for future construction. Implementation of BMPs listed in Section 4 of the Eldorado Valley Transmission and Utility Corridor PEA would reduce these impacts, which would reduce the Project's contribution to cumulative effects (BLM 2012a).

5 CONSULTATION AND COORDINATION

5.1 Scoping and Public Participation

The BLM resource specialists participated in internal scoping discussions to exchange information about the Project and to identify resources and issues to address during the preparation of this EA. Comments and input provided by the BLM resource specialists are summarized and analyzed in Section 3.

The BLM notified adjacent ROW grant holders of the pending request from the applicant for a new ROW request to construct, operate, maintain, and eventually decommission/recommission the proposed solar facility and BESS.

The public comment period was held from August 3 through August 18, 2023. A total of 12 public submissions were received. Comments and responses are included in Appendix I of this EA. Comments focused primarily on the following topics: air quality, desert tortoise, the OSNHT, land use, water resources, and construction practices. The content of resource-specific comments focused on complying with permit requirements, suggestions for a proper and thorough analysis, and proposed additional mitigation measures. Minor text and figure changes to the EA have resulted from some of these comments. Final edits and review of the document also resulted in minor editorial corrections.

5.2 Cooperating Agencies

Cooperating agency invites were sent by certified mail and transmitted electronically on February 24, 2023. Cooperating agencies include the Clark County Department of Aviation; Clark County Department of Environment and Sustainability, Division of Air Quality; EPA; and USFWS.

5.3 Tribal Government-to-Government Consultation

On May 2, 2022, the BLM invited the following tribes to government-to-government consultation: Chemehuevi Indian Tribe, Colorado River Indian Tribes, Fort Independence Indian Community of Paiute, Fort Mojave Indian Tribe, Hualapai Tribe, Kaibab Band of Paiute Indians, Las Vegas Paiute Tribe, Moapa Band of Paiutes, Paiute Indian Tribe of Utah, San Juan Southern Paiute Tribe, The Hopi Tribe, Twenty-Nine Palms Band of Mission Indians, and Utu Utu Gwaitu Paiute Tribe (Owens Valley Paiute Benton Reservation). The BLM also sent letters on August 21, 2022, to the Big Pine Paiute and Lone Pine Paiute Tribes.

The BLM has conducted government-to-government meetings with the Moapa Band of Paiutes about the Project, as well as a preliminary field visit with tribal staff on February 18, 2022. Additionally, the BLM has conducted informal coordination meetings with tribal staff from the Twenty-Nine Palms Band of Mission Indians. Moapa Band of Paiutes has inquired about water resources, dust abatement measures, vegetation treatments, solar panels, and impacts to both cultural and wildlife resources. Both have expressed general interest in cultural resources, but no specific concerns have been shared regarding impacts to traditional use areas, sacred sites, or Traditional Cultural Properties within the Project Area. By request, the BLM will be providing the Moapa Band of Paiutes with a copy of the final draft of the Class III cultural report after Nevada State Historic Preservation Office (SHPO) consultation is complete.

5.4 NHPA Section 106 Consultation with Nevada SHPO and Advisory Council on Historic Preservation

The BLM is required to consider the effects of its actions on historic properties and provide the Advisory Council on Historic Preservation (ACHP) an opportunity to comment, in compliance with Section 106 of the NHPA (54 USC 306108) and its implementing regulations at 36 CFR 800. Consulting parties in this process include the ACHP (should they choose to participate), the SHPO, Indian tribes, the project proponent, state and local governments, other parties, and the public. In May and June 2022, the BLM initiated NHPA Section 106 compliance process for the Proposed Action by inviting the ACHP, SHPO, and Indian tribes to consult on the undertaking. The ACHP declined to participate in a letter dated July 6, 2022. The BLM requested concurrence from the SHPO on the Agency's determinations of eligibility and finding of no historic properties affected by letter dated October 3, 2022. Consultation on the determinations of eligibility and finding of effect is ongoing.

5.5 ESA Section 7 Consultation with USFWS

The USFWS has jurisdiction over threatened and endangered species listed under the ESA (16 USC 1531 et seq.). Formal consultation with the USFWS under Section 7 of the ESA is required for any federal action that may adversely affect a federally listed species.

Formal consultation with the USFWS under Section 7 of the ESA for the Project was initiated in July 2022 and concluded with the October 18, 2022, issuance of a BO related to potential impacts to the federally threatened Mojave desert tortoise (File No. 2022-0035189-S7-001). The BO identified conservation (minimization) measures to reduce impacts to this species (USFWS 2022a).

5.6 Consultation with Others

On April 12, 2023, the BLM held an interdisciplinary team kickoff meeting with BLM resource specialists and cooperating agencies.

The following federal, state, and local agencies were contacted for information, consulted with, or provided an opportunity to consult during preparation of the EA:

- Clark County Department of Environment and Sustainability, Division of Air Quality
- Harry Reid International Airport
- NDOW
- NDNH
- SHPO
- NV Energy
- EPA
- USFWS

5.7 List of Preparers and Reviewers

BLM

Beth Ransel	SND BLM Energy & Infrastructure Team Supervisor
Jessica Headen	SND BLM Energy & Infrastructure Team Project Manager
Matthew Klein	NEPA Coordinator
Frankin Giles	Air Quality; Climate Change
Dagmar Galvan	Cultural Resources
Matthew Fockler	Environmental Justice; Socioeconomics
Sean McEldery	Fuels and Fire Management
Kelsey Bynum	Lands and Realty
Janyne Pringle; Mark Slaughter	Migratory Birds; Special Status Species; Federally Listed Species; General Wildlife
Kenneth Kendrick	Recreation; Travel; Wild and Scenic Rivers
Lara Kobelt	Vegetation; Cacti; Noxious Weeds, Invasive and Non-native Species
Ernest Johnson	Soils; Surface Hydrology
Stephen Leslie	Special Designations
Ernest Johnson	Groundwater Resources; Floodplains; Wetland and Riparian Zones
Stephen Leslie	Visual Resources
Lisa Christianson	Wastes, Hazardous and Solid

SWCA Environmental Consultants, Inc.

Laura Megill	Project Manager and EA Author
Josh Moss	GIS Data Management and Figure Production
Patty Hardesty	Technical Review and Senior Advisor
Erin Wielenga	Air Quality and Climate Change
Mary Ann Vicari	Cultural Resources
Sophie Butler	Environmental Justice & Socioeconomic Resources
Garet Openshaw	Visual Resources

APPENDIX A

Acronyms, Tables, and Figures

LIST OF ACRONYMS AND ABBREVIATIONS

ACEC	Area of Critical Environmental Concern
ACHP	Advisory Council on Historic Preservation
amsl	above mean sea level
APE	area of potential effects
BCC	Bird of Conservation Concern
BCNV	Boulder City Nevada Solar Project (proposed)
BESS	battery energy storage system
BG	block group
BLM	Bureau of Land Management
BMP	best management practice
BO	Biological Opinion
BRQ	Boulder Ranch Quarry
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act of 1980
CESA	cumulative effects study area
CFR	Code of Federal Regulations
CH_4	methane
CO_2	carbon dioxide
CO_2e	carbon dioxide equivalent
CTC	Customers Team Members Community
DOE	U.S. Department of Energy
DOI	U.S. Department of the Interior
EA	Environmental Assessment
Eagle Act	Bald and Golden Eagle Protection Act
EIS	Environmental Impact Statement
EJ	environmental justice
EO	Executive Order
EPM	environmental protection measure
ESA	Endangered Species Act
FAA	Federal Aviation Administration
FEMA	Federal Emergency Management Agency
FLPMA	Federal Land Policy and Management Act
FTE	full-time equivalent
GHG	greenhouse gas
GWP	global warming potential

I-11	Interstate 11
IPaC	Information for Planning and Consultation
IPCC	Intergovernmental Panel on Climate Change
KOP	key observation point
kV	kilovolt
LVFO	Las Vegas Field Office (BLM)
MBTA	Migratory Bird Treaty Act
MSHCP	Multiple Species Habitat Conservation Plan
MW	megawatt
N_2O	nitrous oxide
NAAQS	National Ambient Air Quality Standards
NAC	Nevada Administrative Code
NDNH	Nevada Division of Natural Heritage
NDOT	Nevada Department of Transportation
NDOW	Nevada Department of Wildlife
NEPA	National Environmental Policy Act
NFPA	National Fire Protection Association
NHPA	National Historic Preservation Act of 1966
NO_x	nitrogen oxides
NRCS	Natural Resources Conservation Service
NRHP	National Register of Historic Places
NRS	Nevada Revised Statutes
O&M	operation and maintenance
OP	observation point
OSNHT	Old Spanish National Historic Trail
PCS	power conversion station
PEA	Programmatic Environmental Assessment
PEIS	Programmatic Environmental Impact Statement
PL	Public Law
$PM_{2.5}$	particulate matter equal to or less than 2.5 microns in diameter
PM_{10}	particulate matter equal to or less than 10 microns in diameter
POD	plan of development
Project	Townsite Solar 2 Project
PUP	pesticide use proposal
PV	photovoltaic
RFFA	reasonably foreseeable future action

RMP	Resource Management Plan
RPS	Renewable Portfolio Standard
ROW	right-of-way
SHPO	State Historic Preservation Office
SIP	State Implementation Plan
SND	Southern Nevada District
SoCP	species of conservation priority
SRMA	Special Recreation Management Area
SWCA	SWCA Environmental Consultants
SWPPP	stormwater pollution prevention plan
SWReGAP	Southwest Regional Gap Analysis Project
USC	United States Code
USFWS	U.S. Fish and Wildlife Service
USGS	U.S. Geological Survey
VOC	volatile organic compound
VRI	Visual Resource Inventory
VRM	Visual Resource Management
WEAP	Worker Environmental Awareness Program
WRCC	Western Regional Climate Center

Table A-1. Conformance Summary – Potential Permits and Authorizations

Authorization	Agency Or Authority	Statutory Reference	Permit or Authorization Trigger
Federal			
ROW for land under federal management	BLM	FLPMA (Public Law [PL] 94-579; 43 USC 1761–1771; 43 CFR 2800)	Federal land
NEPA compliance to grant ROW	BLM	NEPA (PL 91-190, 42 USC 4321−4347, January 1, 1970, as amended by PL 94-52, July 3, 1975; PL 94-83, August 9, 1975; and PL 97-258, 4(b), September 13, 1982	Federal land, federal permit, (i.e., federal nexus)
ESA	USFWS	ESA (PL 93-205, as amended by PL 100-478 [16 USC 1531 et seq.])	Section 7a(2) consultation: consultation requirements are triggered by the potential for adverse effects on a listed species or species' habitat
NHPA compliance	Nevada State Historic Preservation Office (SHPO)	NHPA Section 106 (PL 89-665; 16 USC 470 et seq.)	Federal nexus
National Trails System Act	BLM	16 USC 1241 et seq.	Proximity to Old Spanish National Historic Trail
Federal Noxious Weed Act	Federal land management agencies	7 USC 2801 and 2814	In the event an EA or EIS is required under NEPA to implement plant control agreements, federal agencies shall complete such assessments within 1 year
State			
NHPA Section 106 Determination of Effect Concurrence	Nevada SHPO	16 USC 470 et seq., NRS 383	Federal nexus or if human remains are discovered
Desert tortoise and Gila monster handling permit	Nevada Department of Wildlife (NDOW)	Nevada Administrative Code (NAC) 503.093	Handling of protected wildlife
Native cacti and yucca commercial salvaging and transportation permit	Nevada Division of Forestry	NRS 527.050–527.110	Removal or possession of six or more protected cacti in any single calendar day or less than six for 7 or more consecutive calendar days
Incidental Take Permit	NDOW	NRS 503.584–503.589	Capture, removal, or destruction of a fully protected species
Energy Planning and Conservation Fund and the Fund for the Recovery of Costs	NDOW	NRS 701.600–NRS 701.640 and NAC 701.800–701.820	Requires the owners/applicants of all proposed energy projects (of applicable size) to file a notice and provide an initial fee to NDOW for evaluation of the project
ROW occupancy permit	Nevada Department of Transportation (NDOT)	NRS 408.423, 408.210, NAC 408	Construction within an NDOT ROW
Over legal size/load permit	NDOT	NRS 484.437−775, NAC 484.300−580	An exceedance of 80,000 pounds of gross weight; 8 feet, 6 inches in width; 14 feet in height; 10 feet of front or rear overhang; or 70 feet in length

Authorization	Agency Or Authority	Statutory Reference	Permit or Authorization Trigger
Uniform Permit (for Transportation of Hazardous Materials)	Nevada Department of Public Safety	NAC 459.979	Transportation of hazardous materials in a vehicle on a public highway
Local*			
Dust operating control permit	Clark County Division of Air Quality	Clark County Air Quality Regulations. Clean Air Act of 1977 and amendments (NRS 445B.100, 40 CFR Subpart C, 42 USC 7408–7409)	Construction activities that impact more than 0.25 acre
Stationary source permit	Clark County Division of Air Quality	Clark County Air Quality Regulations. Clean Air Act of 1977 and amendments (NRS 445B.225)	Any source located in Clark County that has a potential to emit a regulated pollutant including mechanical equipment (e.g., backup generators, boilers, cooling towers).

* Various permits may also be required from local government(s) with jurisdiction over the Project Area and authority over certain components/actions of the Project.

Table A-2. Legal Description of Proposed Action

Mount Diablo Meridian, Clark County, Nevada		
Townsite Solar 2 Project		
Solar Facility Area		
Township 23 South, Range 63 East	Section 14	S½NE¼NW¼SW¼, NW¼NW¼SW¼, S½NW¼SW¼, SW¼SW¼, S½SE¼SW¼, and Lot 7
Laydown Area* (Temporary)		
Township 23 South, Range 63 East	Section 14	SW¼SE¼SW¼

* Temporary laydown yard (5 acres) will be requested under a separate short-term-use ROW grant, at a later date.

Table A-3. Disturbance

Facility Type	Acres of Disturbance	Notes
Permanent Disturbance		
Solar facility	76*	19-MWac PV solar facility with ancillary components including PCSs and MET stations.
BESS (optional)	(4)	Storage and inverter are within the solar facility footprint.

Facility Type	Acres of Disturbance	Notes
Linear facilities segment (access road, distribution line, and interconnection cabling)	4	The linear facilities segment is approximately 4 acres in size, measures 80 feet wide × 2,000 feet long, and would commence from the southeast corner of the solar array field and extend eastward along the southern border of S½SE¼NW¼ of Section 14, T23S, R63E of BLM land to end at the proposed BCNV Facility, on municipal land. The newly constructed site road, measuring 30 feet wide × 2,000 feet long, would be co-located within the linear facilities segment (in the southern portion closest to the Townsite Solar 1 Facility) and would be used to access the Project during construction and operations. The site road would be at least 25 feet from the distribution power line's centerline and would be graded and covered with gravel base. The remaining 50-foot-wide × 2,000-foot-long portion of the linear facilities segment will be used for an overhead 12-kV to 13.8-kV auxiliary distribution line to provide power to the Project Area during construction and operations. The applicant intends to assign the 50 × 2,000–foot portion of the linear facilities segment to the owner of the distribution line at a later date. The construction and operation of a 34.5-kV interconnecting underground line and a fiber optic cable installed either underground in a separate trench (adjacent to the -kV underground solar energy collection line) or on overhead lines along the distribution line would also be co-located in the linear facilities segment between the site road and the centerline of the distribution line (25-foot offset).
Drainage features	(TBD)	The drainage features are included within the solar array footprint.
Berms	(TBD)	The berms are within the solar array footprint.
Project interconnection	N/A	The Project will interconnect into through the proposed BCNV Facility on municipal land owned by the City of Boulder City to the new substation from the newly expanded SNWA 230-kV transmission line; or it would interconnect through the proposed BCNV Facility into the WAPA's Mead 230-kV Substation via the Townsite Solar 1 Facility (on municipal land). Regardless of the interconnection pathway selected, no new surface disturbance is expected.
Collection line construction	(1)	Two pulling sites, up to 80 × 300 feet, would be located within a portion of the linear facilities segment (80 × 2,000–foot area) with an approximate 2-month construction time frame.
Fencing (permanent)	(0.5)	Security perimeter fencing and an entrance gate will be installed. Permanent desert tortoise fencing, up to 7,917 feet long × 3 feet (surface clearance area for installation) will be attached to the bottom of the security perimeter fence and solar facility entrance gate.
Interior access way	(3)	The access way will be graded and compacted with site materials, up to 20 feet wide × 6,000 feet long, anticipated to be placed around the center of the solar field to provide vehicle access to solar equipment.
Total	**80**[†]	
Temporary Disturbance for Geotechnical Exploration (within solar facility footprint)		
Borehole/soil resistivity	(0.92)	Up to four boreholes at depths of approximately 30 and 50 feet. Geotechnical boring sites would be approximately 100 × 100 feet (to account for space for drilling and support vehicles), for a total disturbance of 0.23 acre per site. At each borehole, two soil resistivity surveys would be conducted along two perpendicular traverses. Resistivity testing would occur entirely within the disturbance area for geotechnical boring sites.
Pile Load Testing	(0.01)	At four locations, a group of three piles (6 × 9 feet or 54 square feet) will be installed using a pile installation rig for a total disturbance of 0.01 acre. Target depth will be 7 and 13 feet. Pile load testing would occur at the geotechnical boring sites.
Site access	(0)	Access would be by existing trails and overland travel. All disturbance will be restricted to the Project Area.
Total	**(0.93)**	
Temporary Disturbance (granted through a short-term use ROW, outside of the solar facility)		
Laydown area	5	The applicant will submit a temporary short-term ROW application for a construction laydown area to be located adjacent to and outside of the southeast corner of the Project Area. The area would measure approximately 467 feet wide × 464 feet long, would be graded as needed, and would undergo reclamation after construction is complete.

Facility Type	Acres of Disturbance	Notes
Fencing (temporary)	(0.4)	Temporary desert tortoise exclusion fencing up to 6,068 × 3 feet (surface clearance area for installation), will be installed within and around the laydown area and linear facilities segment before surface disturbance and removed after construction is completed. No additional area is required. Two gates would be installed to access the linear facilities segment from the east and west sides of Old U.S. Highway 95.
Total Laydown Area	**5**†	
Total Project Area	**85**‡	

() Parentheses indicate the acreage is within the solar facility footprint and is part of the 76-acre total, or within the linear facilities segment and is part of the 4-acre total, or within the temporary laydown area and is part of the 5-acre total.

N/A = not applicable

TBD = to be determined

* The solar array field is within the Project footprint and measures approximately 1,300 feet wide × 1,700 feet long.

† All acreages listed under permanent disturbance are included within the requested 80-acre solar facility and the acres listed under temporary disturbance (5 acres) will be included within the requested short-term ROW.

‡ The Project Area includes the 80-acre solar facility plus the 5-acre temporary laydown area (to be requested at a later date as a short-term ROW).

Table A-4. Estimated On-Site Equipment for Project Construction Activities

Equipment Description	Daily Quantity	Horsepower	Fuel Type	Equivalent Full-Load Operating Time (hours/day)	Vehicle Miles Travelled per Day on Unpaved Surface
Site Preparation					
Rough terrain forklift	4	75	Diesel	1.7	10
Delivery/work trucks	6	200	Diesel	2	5
Site Preparation – Solar Arrays					
Truck, pick-up (survey crew)	4	180	Gas	1.7	5
Grader	6	200	Diesel	6.8	20
Backhoe/front loader	4	120	Diesel	3.4	20
Tractor/disc	6	210	Diesel	6.8	40
Scraper	8	265	Diesel	3.4	30
Compactor	4	120	Diesel	1.7	10
Water truck	4	175	Diesel	6.8	N/A
Site Preparation – Roads					
Grader	6	200	Diesel	6.8	20
Backhoe/front loader	2	120	Diesel	6.8	10
Compactor	4	120	Diesel	6.8	20
Water truck	4	175	Diesel	6.8	N/A
Dump truck	8	235	Diesel	2.7	10
Site Preparation – Fencing					
Rough terrain forklift	4	75	Diesel	1.7	10
Delivery/work trucks	6	200	Diesel	1	5
Site Preparation – Post Installation					
Delivery/work trucks	4	200	Diesel	1	5
Post machine	10	45	Diesel	8.1	1

Equipment Description	Daily Quantity	Horsepower	Fuel Type	Equivalent Full-Load Operating Time (hours/day)	Vehicle Miles Travelled per Day on Unpaved Surface
Rough terrain forklift	4	75	Diesel	6.8	10
Site Preparation – Install Support Structures					
Rough terrain forklift	4	75	Diesel	6.8	10
Delivery/work trucks	4	200	Diesel	1	5
Site Preparation – Install Inverters, Switchgear, and Support Equipment					
Crane	4	125	Diesel	4.5	1
Backhoe/front loader	4	120	Diesel	6.8	10
Delivery/work trucks	4	200	Diesel	1	5
Site Preparation – Install Direct Current (DC) and Alternating Current (AC) Electrical					
Backhoe/front loader	8	120	Diesel	6.8	10
Crawling trencher	4	100	Diesel	4.1	1
Mini-excavator	8	42	Diesel	6.8	10

Table A-5. Hazardous Materials that May Be Used during Operation

Hazardous Material	Storage Description; Capacity	Storage Practices and Special Handling Precautions
Mineral insulating oil (or biodegradable oil)	Carbon steel transformers; total on-site inventory of 80,000 gallons.	Used only in transformers within PCSs; secondary containment for each transformer would be managed in accordance with the Spill Response and Emergency Response Plan.
Batteries, lead acid based and/or lithium ion	Battery-based emergency back-up power at each PCS (which include both inverter and transformer equipment) plus BESS.	Enough cooling capacity to maintain ambient temperatures appropriate for the selected battery would be provided.
Propane	Generator-based emergency back-up power at each of the PCS shelters (or one centralized generator); tanks at PCS will be sized between 20 and 100 gallons (or 1,000 gallons if a single centralized tank).	Would be managed in accordance with the Spill Response and Emergency Response Plan.
Herbicide Roundup® (glyphosate) or equivalent; pesticide	Brought on-site by licensed contractor; used immediately and as approved by the BLM.	No mixing will occur on-site, and no herbicides/pesticides will be stored on-site.

Table A-6. Resources Considered in the Evaluation of the Proposed Action and Alternatives

Resource	May be Impacted (carried forward for analysis)	Present and Not Impacted or Not Present	Rationale for Not Impacted
Air Quality	X		Carried forward for analysis, see Section 3.2.

Resource	May be Impacted (carried forward for analysis)	Present and Not Impacted or Not Present	Rationale for Not Impacted
Areas of Critical Environmental Concern (ACECs)		X	The River Mountains ACEC is approximately 2.5 miles to the north-northeast of the Project (BLM 2021b). Given the distance and topography of the area, no impacts to this ACEC are expected. The Project is also north of the Piute-Eldorado ACEC (BLM 2021b); however, the Project is nearly surrounded by existing human disturbances, including I-11/U.S. 93 and U.S. 95/Veterans Memorial Highway, mining/quarries, and existing solar facilities; therefore, the addition of this Project in this area is not expected to result in impacts to this ACEC. Therefore, no adverse impacts to the ACECs would result from the Proposed Action.
Conservation Lands or National Monument		X	Sloan Canyon National Conservation Area is approximately 3.5 miles west of the Project. Given the distance and the topography of the area, no impacts are expected (BLM 1998b). No other conservation lands or national monuments are present within 5 miles of the Proposed Action.
Cultural Resources	X		Carried forward for analysis, see Section 3.3.
Environmental Justice and Socioeconomics	X		Carried forward for analysis, see Section 3.4.
Fish and Wildlife Excluding Federally Listed Species	X		Carried forward for analysis, see Section 3.5.
Floodplains		X	The Project Area is not within a floodplain, and the area is subject to minimal flooding (Zone X) according to the Federal Emergency Management Agency (FEMA) Flood Map Service Center (FEMA 2011). Therefore, no impacts to floodplains would result from the Proposed Action.
Forestry	X		This resource is present in the form of cactus plant species with approximately 3,024 plants. This resource is discussed under Section 3.13, Vegetation.
Fuels and Fire Management	X		Carried forward for analysis, see Section 3.6.
Geology/Mineral Resources		X	The Project Area is entirely within a mineral materials community pit (BLM Community Pit, N-48712) in the W½ of Section 14, T23S, R63E. The Project is also adjacent to an existing disturbed mineral site, N-96925, that has roads, pipelines, and underground utilities within the legal description (BLM 2021a). The Proposed Action is not expected to impact uses associated with existing mineral resource claims. Implementation of BLM standard stipulations (see Appendix D) would be followed and would minimize any potential temporary impacts to mineral resources from excess production of mineral materials during construction activities. Therefore, geology and mineral resources would not be impacted to a degree that detailed analysis is required.
Greenhouse Gas/Climate Change	X		Carried forward for analysis, see Section 3.7.
Hydrologic Conditions	X		Carried forward for analysis, see Section 3.10.
Invasive Species/Noxious Weeds	X		Carried forward for analysis, see Section 3.13.

Resource	May be Impacted (carried forward for analysis)	Present and Not Impacted or Not Present	Rationale for Not Impacted
Lands and Realty		X	The Proposed Action is within or adjacent to existing ROWs and the BLM Henderson Community Pit (case file N-48712). Additionally, the Project is adjacent to a previously disturbed mineral site (N-96925) and there are roads, pipelines, and underground utilities that run within the Project Area. No land or access conflicts were identified (BLM 2021a).
Livestock Grazing		X	The Project Area is not in an active grazing allotment; therefore, no impact to livestock grazing would result from the Proposed Action. The Project is within the McCullough Mountain allotment; however, the grazing rights have been purchased by Clark County to protect the desert tortoise under the Clark County Multiple Species Habitat Conservation Plan (MSHCP) (BLM 2021a). Given the multiple physical barriers (i.e., material pits and mining, solar facilities, and U.S. 95), desert tortoise habitat would not be impacted. Further information regarding the desert tortoise is provided in Section 3.10. Therefore, the Proposed Action would not impact livestock grazing
Migratory Birds	X		Carried forward for analysis, see Section 3.8.
Native American Concerns		X	Consultation between the BLM and the appropriate tribal entities has occurred, and no major concerns have been expressed regarding the Project. Therefore, no additional review is needed at this time.
Old Spanish National Historic Trail		X	The Old Spanish National Historic Trail (OSNHT) is a historical trade route that connected northern New Mexico to Los Angeles, California (BLM n.d.). The proposed solar field is located approximately 0.3 mile west of the congressionally designated alignment of the OSNHT, and the alignment runs directly though the linear facilities segment. This portion of the OSNHT is known as the Mojave Road. The segment of the OSNHT within the Project's analysis area is located in a broad and flat valley influenced by the surrounding mountains, U.S. 95, sand and gravel operations, existing solar facilities, and transmission lines. As identified in the OSNHT Comprehensive Administrative Strategy (National Park Service 2017), the portion of the OSNHT within the Project's analysis area crosses private land, does not include any high potential historic sites, high potential historic route segments, or auto tour routes. In addition, no trail traces or trail-related cultural resource sites have been identified, and there are no trail-specific recreational opportunities within the analysis area of the Project. There are no features of the OSNHT evident in this area today and the Proposed Action would not affect the nature, purpose, and primary uses of the OSNHT Mojave Road segment. While the visual setting adjacent to the OSNHT would be further modified by the Project, in context with impacts from existing projects and developments, visual impact would be negligible to weak. Since the Project would occur adjacent to other energy infrastructure projects, the Project would not result in any discernible changes and would not interfere with the trail's nature and purpose. For further information, including maps and photos, refer to Section 3.3 of the Townsite Solar 2 Project Visual Resources Technical Report (SWCA 2023).

Resource	May be Impacted (carried forward for analysis)	Present and Not Impacted or Not Present	Rationale for Not Impacted
Paleontological Resources		X	The potential fossil yield classification for the Project Area and 1-mile buffer is “Class 1” (very low) and “Unknown,” with an “Unknown” designation for the Project Area (BLM 2022b). Although the Project Area is designated with an unknown classification, quaternary alluvial deposits covering the valley floor of Boulder City, including the Project Area, are known to have little or no paleontological potential (California Public Utilities Commission and BLM 2010). Therefore, paleontological resources would not likely be impacted by the Project operations as the Project Area is unlikely to host supporting geological structures for vertebrate paleontological resources. If unanticipated discovery occurs, the BLM would be contacted immediately. Therefore, potential adverse impacts to paleontological resources from the Proposed Action are unlikely and no further evaluation is required.
Recreation/Travel/ Wild and Scenic Rivers	X		Carried forward for analysis, see Section 3.9.
Soils	X		Carried forward for analysis, see Section 3.10.
Threatened Endangered or Candidate Animal Species	X		Carried forward for analysis, see Section 3.12.
Threatened Endangered or Candidate Plant Species		X	The Project Area is not within any designated critical habitat or in known habitat for Threatened or Endangered plant species.
Transmission Corridors		X	The Project is not within any transmission corridors (BLM 2021a); therefore, no further evaluation is required.
Vegetation (and Special Status Plant Species)	X		Carried forward for analysis, see Section 3.13.
Visual Resources	X		Carried forward for analysis, see Section 3.14.
Wastes (hazardous or solid)	X		Carried forward for analysis, see Section 3.15.

Resource	May be Impacted (carried forward for analysis)	Present and Not Impacted or Not Present	Rationale for Not Impacted
Water Resources		X	The Project is not within 3 miles of natural surface water, spring, riparian area, or wetland (U.S. Environmental Protection Act [EPA] 2020). The Basin and Range Basin-Fill Aquifer system would not be impacted by Project activities because excavations would not go below 300 feet below ground surface to encounter groundwater levels (U.S. Geological Survey [USGS] 2022). Additionally, the Proposed Action has the potential to impact precipitation infiltration due to the presence of solar panels and O&M facilities. However, groundwater recharge within the valley floor from precipitation is minimal to nonexistent (Harrill and Prudic 1998); therefore, the Project is not likely to impact groundwater recharge. The applicant would use non potable reclaimed water from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. Water transport on public roads would adhere to all federal and state transportation rules and regulations. The applicant does not intend to develop or purchase water rights. The Project is located in Nevada Hydrographic Area number 167 – Eldorado Valley. The perennial yield of the Eldorado Valley is 500-acre-feet per year, and there are 2,256 acre feet per year of committed water rights, making the basin over allocated. This is a designated ground water basin, meaning the permitted groundwater rights are approaching or exceeding the estimated average annual recharge and the water resources are being depleted or require additional administration (Nevada Division of Water Resources 2023). No active groundwater wells are within 5 miles of the Project (USGS 2022). No groundwater withdrawal is anticipated. Therefore, no impacts to water resources would result from the Proposed Action.
Wetlands and Riparian Areas		X	No wetlands or riparian habitats are present in the Project Area (USFWS 2022d). There are human-made ponds associated with the Boulder Creek Golf Club approximately 3.5 miles to the north-northeast of the Project Area (Google Earth Pro 2022). The Proposed Action would not impact the human-made ponds. There is also a catchment, the Eldorado Dry Lake Bed, more than 3 miles south of the Project (EPA 2020; USFWS 2022d). Surface water runoff from the Project Area has the potential to drain into the dry lake bed during high rainfall events. The Proposed Action would manage surface water runoff under the site-specific SWPPP, which would reduce potential impacts to negligible levels.
Wild Horse and Burros		X	There are no herd management units within 5 miles of the Project Area (BLM 1998b); therefore, no impacts to wild horses or burros would result from the Proposed Action.
Wilderness/Wilderness Study Areas/Lands with Wilderness Characteristics		X	There are no wilderness lands or lands with wilderness characteristics within 5 miles of the Project Area (BLM 2021a); therefore, no further evaluation is required.

Table A-7. Estimated Annual Project Emissions

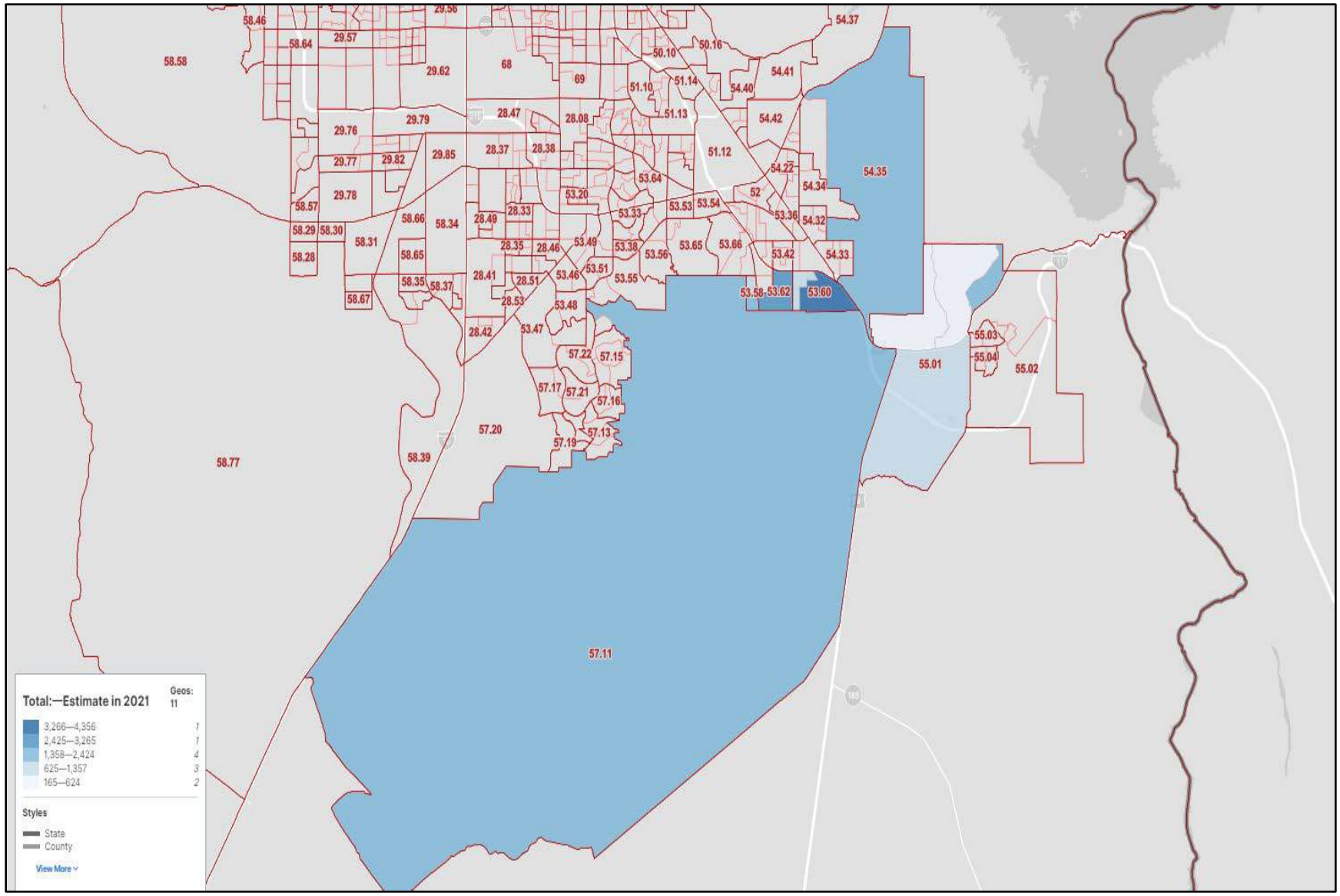
Source	NO_x (tons/year)	VOCs (tons/year)
Annual construction emissions	12.0	1.3
Annual operation emissions	0.09	0.02
Annual decommissioning emissions	12.0	1.3
General conformity levels	50	50
Exceeds de minimis levels?	No	No

Table A-8. Environmental Justice Populations

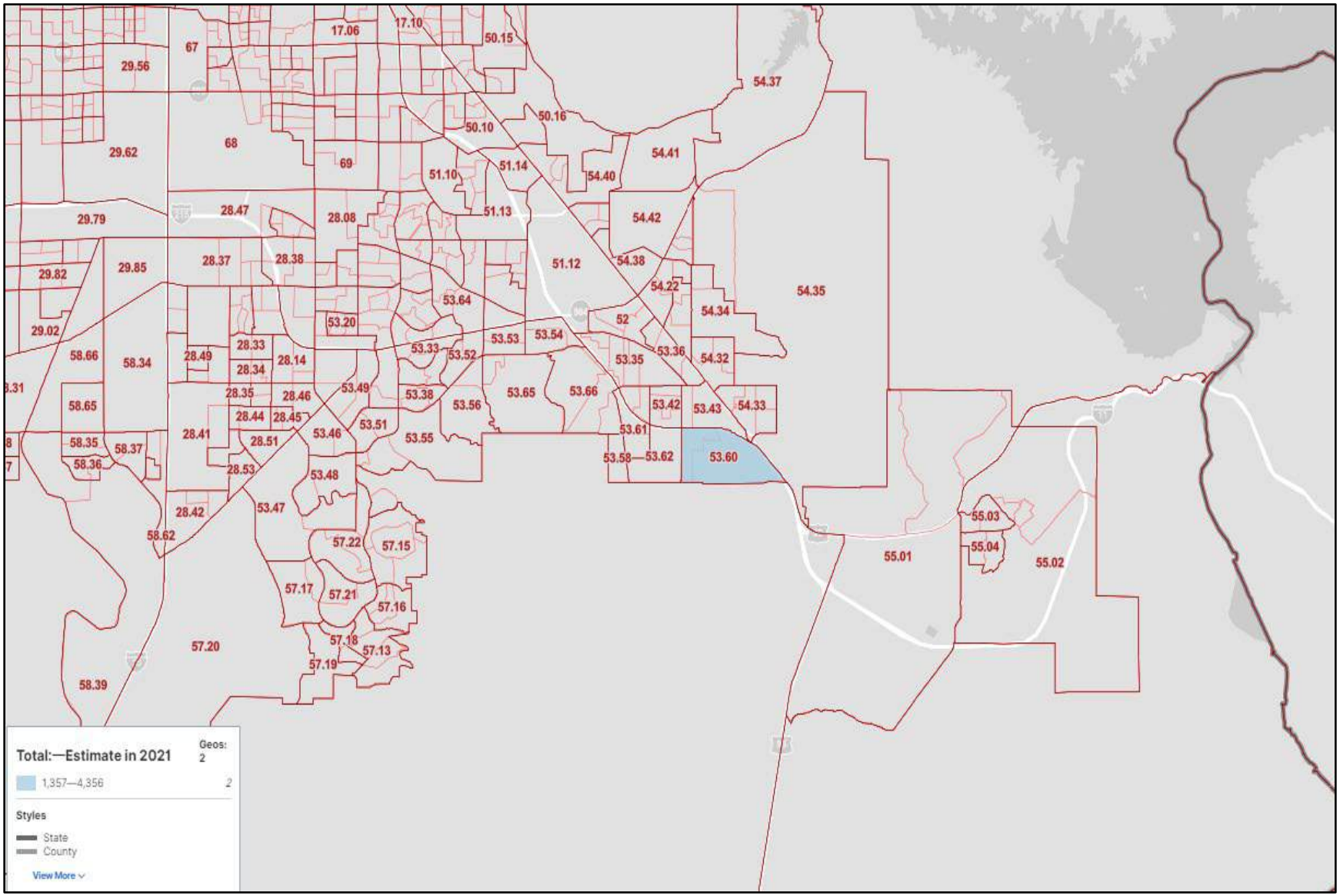
Population	Low Income	Minority	American Indian
Block Group 320030057111*	5.6%	45%	1.5%
Block Group 32003005501*	32.7%	10.7%	1.4%
Block Group 32003005435*	29.8%	40%	1%
Block Group 32003005360*	40.1%	33.9%	1%
Block Group 32003005362*	5%	36%	1.3%
Block Group 32003005358*	18.5%	19.4%	0.3%
State of Nevada	31.2%	51.8%	2.5%

Sources: EPA (2023c), U.S. Census Bureau (2021b)

* The analysis area includes all Block Groups combined.



Supplemental Map 1. Block groups (see Table A-8).



Supplemental Map 2. Low-income block group (see Table A-8).

Table A-9. General Wildlife Species with Potential to Occur in the Analysis Area

Common Name	Scientific Name
Birds	
American avocet	*Recurvirostra americana*
American kestrel	*Falco sparverius*
Barn owl	*Tyto alba*
Black-necked stilt	*Himantopus mexicanus*
Black-tailed gnatcatcher	*Polioptila melanura*
Common raven	*Corvus corax*
Cooper's hawk	*Accipiter cooperii*
Great-horned owl	*Bubo virginianus*
Long-eared owl	*Asio otus*
Merlin	*Falco columbarius*
Northern harrier	*Circus hudsonius*
Northern saw-whet owl	*Aegolius acadicus*
Osprey	*Pandion haliaetus*
Red-tailed hawk	*Buteo jamaicensis*
Rough-legged hawk	*Buteo lagopus*
Sharp-shinned hawk	*Accipiter striatus*
Turkey vulture	*Cathartes aura*
Western screech-owl	*Megascops kennicottii*
Reptiles	
Common kingsnake	*Lampropeltis getula*
Desert banded gecko	*Coleonyx variegatus variegatus*
Great Basin gophersnake	*Pituophis catenifer* ssp. *deserticola*
Great Basin whiptail	*Aspidoscelis tigris tigris*
Long-nosed snake	*Rhinocheilus lecontei*
Mojave patch-nosed snake	*Salvadora hexalepis mojavensis*
Mojave rattlesnake	*Crotalus scutulatus*
Nevada shovel-nosed snake	*Chionactis occipitalis talpina*
Nevada side-blotched lizard	*Uta stansburiana nevadensis*
Northern desert night snake	*Hypsiglena chlorophaea deserticola*
Northern sagebrush lizard	*Sceloporus graciosus graciosus*
Red racer	*Coluber flagellum piceus*
Southwestern speckled rattlesnake	*Crotalus mitchelli pyrrhus*
Spotted leaf-nosed snake	*Phyllorhynchus decurtatus*
Tiger whiptail	*Aspidoscelis tigris*
Variable groundsnake	*Sonora semiannualata semiannualata*
Western fence lizard	*Sceloporus occidentalis*
Western long-tailed brush lizard	*Urosaurus graciosus*
Yellow-backed spiny lizard	*Sceloporus uniformis*

Common Name	Scientific Name
Zebra-tailed lizard	*Callisaurus draconoides*
Mammals	
Desert woodrat	*Neotoma lepida*
Long-tailed pocket mouse	*Chaetodipus formosus*
Merriam kangaroo rat	*Dipodomys merriami*
Western harvest mouse	*Reithrodontomys megalotis*

Table A-10. Special-Status Wildlife Species with Potential to Occur in Analysis Area

Common Name	Scientific Name	Species Status	Potential for Occurrence
Amphibians			
Arizona toad	*Anaxyrus microscaphus*	BLM (S); SoCP; S2	None
Relict leopard frog	*Lithobates onca*	BLM (S); NV (P); NDOW (SoCP); S1; CS	None
Insects			
Aegialian scarab beetle	*Aegialia knighti*	BLM (S); S1	None
Ash Meadows naucorid	*Ambrysus amargosus*	BLM (S); S1	None
Big Dune aphodius scarab	*Aphodius sp. 1*	BLM (S)	None
Big Dune miloderes weevil	*Miloderes sp. 1*	BLM (S)	None
Devils Hole warm spring riffle beetle	*Stenelmis calida calida*	BLM (S); S1	None
Giuliani's dune scarab	*Pseudocotalpa giulianii*	BLM (S); S1	None
Large aegialian scarab	*Aegialia magnifica*	BLM (S); S1	None
MacNeill sooty wing skipper	*Hesperopsis gracielae*	BLM (S); S1	None
Moapa Warm Spring riffle beetle	*Stenelmis moapa*	BLM (S); S1	None
Mojave gypsum bee	*Andrena balsamorhizae*	BLM (S); S1	L
Mojave poppy bee	*Perdita meconis*	BLM (S); S1	L
Monarch butterfly	*Danaus plexippus plexippus*	USFWS (C); BLM (S); S3	L
Northern Mojave blue	*Euphilotes mojave virginensis*	BLM (S); S1	L
Reptiles			
Banded Gila monster	*Heloderma suspectum cinctum*	BLM (S); NV (P); NDOW (SoCP); S2	H
Common chuckwalla	*Sauromalus ater*	BLM (S); NDOW (SoCP); S3	H
Desert glossy snake	*Arizona elegans eburnata*	BLM (S); S4	H
Desert horned lizard (and northern subspecies)	*Phrynosoma platyrhinos*	BLM (S); NDOW (SoCP); S4	H
Desert iguana (and northern subspecies)	*Dipsosaurus dorsalis*	BLM (S); NDOW (SoCP); S3; CS	H
Desert night lizard	*Xantusia vigilis*	NDOW (SoCP)	H
Desert rosy boa	*Lichanura trivirgata*	BLM (S); NDOW (SoCP); SU	None
Great Basin collared lizard	*Crotaphytus bicinctores*	BLM (S); NDOW (SoCP); S4; CS	M
Long-nosed leopard lizard	*Gambelia wislizenii*	BLM (S); NDOW (SoCP); S4	H

Common Name	Scientific Name	Species Status	Potential for Occurrence
Mojave Desert tortoise	*Gopherus agassizii*	USFWS (T); BLM (S); NV (T); NDOW (SoCP); S2S3; CS	H
Ring-necked snake	*Diadophis punctatus*	BLM (S); NDOW (SoCP); S3	None
Sidewinder	*Crotalus cerastes*	BLM (S); NDOW (SoCP); S4; CS	H
Spotted leaf-nosed snake	*Phyllorhynchus decurtatus*	NDOW (SoCP)	M
Western banded gecko	*Coleonyx variegatus*	NDOW (SoCP); CS	H
Western shovel-nosed snake (including Mojave and Nevada subspecies)	*Chionactis occipitalis (Chionactis occipitalis occipitalis, Chionactis occipitalis talpina)*	BLM (S); NDOW (SoCP); S4	H
Western long-tailed brush lizard	*Urosaurus graciosus*	NDOW (SoCP)	M
Mammals			
Allen's big-eared bat	*Idionycteris phyllotis*	BLM (S); NV (P); NDOW (SoCP); S1	None
Ash Meadows montane vole	*Microtus montanus nevadensis*	BLM (S); NV (S); SH	None
Big brown bat	*Eptesicus fuscus*	BLM (S); NV (P); S3S4	M
Big free-tailed bat	*Nyctinomops macrotis*	BLM (S); NV (P); NDOW (SoCP); S1	L
Brazilian (Mexican free-tailed bat	*Tadarida brasiliensis*	BLM (S); NV (P); NDOW (SoCP); S4	M
California leaf-nosed bat	*Macrotus californicus*	BLM (S); NV (S); NDOW (SoCP); S2	L
California myotis	*Myotis californicus*	BLM (S); NV (P); S3S4	M
Canyon bat	*Pipistrellus hesperus*	BLM (S); NV (P); S3S4	H
Cave myotis	*Myotis velifer*	BLM (S); NV (P); NDOW (SoCP); S1	None
Desert bighorn sheep	*Ovis canadensis* ssp. *nelsoni*	BLM (S); NV (G); NDOW; S4	M
Fringed myotis	*Myotis thysanodes*	BLM (S); NV (P); NDOW (SoCP); S2	L
Hoary bat	*Lasiurus cinereus*	BLM (S); NV (P); NDOW (SoCP); S2S3	L
Little brown bat	*Myotis lucifugus*	BLM (S); NV (P); NDOW (SoCP); S2S3	None
Long-eared myotis	*Myotis evotis*	BLM (S); NV (P); NDOW (SoCP); S3; CS	None
Long-legged myotis	*Myotis volans*	BLM (S); NV (P); S3S4, CS	L
Pale kangaroo mouse	*Microdipodops pallidus*	BLM (S); NV (S); NDOW (SoCP); S2	None
Pallid bat	*Antrozous pallidus*	BLM (S); NV (P); S3	H
Pocket gopher	*Thomomys bottae*	BLM (S); NDOW (SoCP); SNR	None
Silver-haired bat	*Lasionycteris noctivagans*	BLM (S); NV (P); NDOW (SoCP); S3; CS	L
Spotted bat	*Euderma maculatum*	BLM (S); NV (T); NDOW (SoCP); S2	L
Townsend's big-eared bat	*Corynorhinus townsendii*	BLM (S); NV (S); NDOW (SoCP); S2	L
Western red bat	*Lasiurus blossevillii*	BLM (S); NV (S); NDOW (SoCP); S2	None
Western small-footed myotis	*Myotis ciliolabrum*	BLM (S); NV (P); NDOW (SoCP); S3S4	L
Yuma myotis	*Myotis yumanensis*	BLM (S); NV (P); S3	M
Birds			
American avocet	*Recurvirostra americana*	NDOW (SoCP), S4B	L
American kestrel	*Falco sparverius*	S3	H

Common Name	Scientific Name	Species Status	Potential for Occurrence
Bald eagle	*Haliaeetus leucocephalus*	BLM (S); NV (S); NDOW (SoCP); S2B, S4N	L
Bendire's thrasher	*Toxostoma bendirei*	BLM (S); NDOW (SoCP); S1B	L
Black-chinned sparrow	*Spizella atrogularis*	USFWS (BCC); NDOW (SoCP); S2B	None
Brewer's sparrow	*Spizella breweri*	BLM (S); NV (S); NDOW (SoCP); S3B	L
Clark's grebe	*Aechmophorus clarkii*	USFWS (BCC); S4B	L
Costa's Hummingbird	*Calypte costae*	USFWS (BCC); S3B	M
Crissal thrasher	*Toxostoma crissale*	BLM (S); S3	L
Ferruginous hawk	*Buteo regalis*	BLM (S); NDOW (SoCP); S3B, S4N	L
Flammulated owl	*Psiloscops flammeolus*	BLM (S); SoCP; S3	None
Golden eagle	*Aquila chrysaetos*	BLM (S); NV (S); NDOW (SoCP); S4	M
Great Basin willow flycatcher	*Empidonax trailii adastus*	BLM (S); NDOW (SoCP); S1S2B	None
Greater sage-grouse	*Centrocercus urophasianus*	BLM (S); NV (G); NDOW (SoCP); S3	None
Greater sandhill crane	*Antione canadensis tabida*	BLM (S); NDOW (SoCP); S2B,S2M	L
LeConte's thrasher	*Toxostoma lecontei*	BLM (S); NDOW (SoCP); S2	H
Loggerhead shrike	*Lanius ludovicianus*	BLM (S); NV (S); NDOW (SoCP); S3	H
Long-eared owl	*Asio otus*	USFWS (BCC); S3	L
Marbled godwit	*Limosa fedoa*	USFWS (BCC); S3M	L
Peregrine falcon	*Falco peregrinus*	BLM (S); NV (S); NDOW (SoCP); S3; CS	L
Phainopepla	*Phainopepla nitens*	BLM (S); SoCP; S3; CS	H
Pinyon jay	*Gymnorhinus cyanocephalus*	BLM (S); NDOW (SoCP); S3	L
Ridgeway's rail (Yuma clapper rail)	*Rallus obsoletus yumanensis*	USFWS (E); BLM (S); NV (E); NDOW (SoCP); S1B	None
Sage thrasher	*Oreoscoptes montanus*	BLM (S); NV (S); NDOW (SoCP); S4B	L
Southwestern willow flycatcher	*Empidonax trailii extimus*	USFWS (E); BLM (S); NV (E); NDOW (SoCP); S1B; CS	None
Swainson's hawk	*Buteo swainsoni*	BLM (S); NDOW (SoCP); S3B	L
Western burrowing owl	*Athene cunicularia hypugaea*	BLM (S); NDOW (SoCP); S3B	H
Western snowy plover	*Charadrius nivosus nivosus*	BLM (S); NDOW (SoCP); S3B	None
Western yellow-billed cuckoo (yellow-bulled cuckoo)	*Coccyzus americanus occidentalis*	BLM (S); NV (T); NDOW (SoCP); S1B; CS	None
Willet	*Tringa semipalmata*	USFWS (BCC); S3B	L

USFWS Status Definitions:

E = Endangered. Is a species in danger of extinction throughout all or a significant portion of its range.

T = Threatened. Is one that is likely to become endangered in the foreseeable future throughout all or a significant portion of its range.

C – Candidate. Is one for which we have on file sufficient information on biological vulnerability and threats to support a proposal for listing as endangered or threatened, but for which preparation and publication of a proposal is precluded by higher priority listing actions.

BCC = Bird of conservation concern. Are species, subspecies, and populations of all migratory nongame birds that, without additional conservation actions, are likely to become candidates for listing.

BLM Status Definitions:

S = Sensitive. Species are listed as Sensitive if there is information that the species has recently undergone, is undergoing, or is predicted to undergo a downward trend such that viability of the species is at risk, or because the species depends on ecological refugia or specialized or unique habitats on BLM-administered lands, and there is evidence that such areas are threatened with alteration such that the viability of the species is at risk.

Nevada (NV) Status Definitions (NAC 503.030 - 503.080):

P = Protected. Protected from "take" under Nevada state law. Protected species may be further classified as endangered, threatened, or sensitive.

E = Endangered. Endangered species are those in imminent jeopardy of extinction.

G = Game mammal. Are wild mammals further classified as game mammals.

T = Threatened. Threatened species are those in imminent jeopardy of becoming endangered.

S = Sensitive. The population or distribution of the wildlife is in a significant decline, the population is threatened because of disease or predation or ecological or human causes, and/or the primary habitat of the wildlife is deteriorating.

NDNH Definitions:

S1 = Species is critically imperiled and at very high risk of extirpation in the jurisdiction due to very restricted range, very few populations or occurrences, very steep declines, severe threats, or other factors.

S2 = Species is imperiled and at high risk of extirpation in the jurisdiction due to restricted range, few populations or occurrences, steep declines, severe threats, or other factors.

S3 = Species is vulnerable and at moderate risk of extirpation in the jurisdiction due to a fairly restricted range, relatively few populations or occurrences, recent and widespread declines, threats, or other factors.

S4 = Species is apparently secure and at a fairly low risk of extirpation in the jurisdiction due to an extensive range and/or many populations or occurrences, but with possible cause for some concern as a result of local recent declines, threats, or other factors.

B = Conservation status applies to the breeding population of the species.

M = Migrant species occurring regularly on migration at particular staging areas or concentration spots where the species might warrant conservation attention. Conservation status refers to the aggregating transient population of the species in the state.

N = Conservation status applies to the nonbreeding population of the species.

SH = Species possibly extirpated because known from only historical records but still some hope of rediscovery. There is evidence that the species or ecosystem may no longer be present in the jurisdiction, but not enough to state this with certainty.

SNR = Species is unranked and conservation status not yet assessed.

S#S# = Range rank used to indicate any range of uncertainty about the status of the species or ecosystem.

Clark County Multiple Species Habitat Conservation Plan

CS – Covered Species

Potential for Occurrence Definitions:

None = Does not occur.

L = Low, unlikely to occur

M = Medium, habitat is limited, may be used for foraging or passing through.

H = High, observed on-site or habitat is widespread, likely nests and forages on-site.

Table A-11. Average Annual Meteorological Conditions in the Analysis Area

Meteorological Parameter	Boulder City, Nevada	Searchlight, Nevada
Minimum daily temperature (°F)	39.5	36.6
Maximum daily temperature (°F)	107.1	97.5
Total precipitation (inches)	6.5	7.8
Snowfall (inches)	0	0.5

Source: National Climatic Data Center (2023) for the latest 30-year period (1991–2020)

Table A-12. 2005, 2020, and 2021 Greenhouse Gas Emissions by Sector

Sector	2005 U.S. GHG Emissions (MMT CO_2e)	2021 U.S. GHG Emissions (MMT CO_2e)	2020 Nevada GHG Emissions (MMT CO_2e)
Transportation	N/A	N/A	13.76
Energy	6,351.5	5,196.6	13.26
Industry	356.1	376.4	7.52
Agriculture	577.7	598.1	1.86
Waste	192.1	169.2	1.99
Residential and commercial	N/A	N/A	4.69
Land use, land use change, and forestry	−781.1	−754.2	−5.75

Sector	2005 U.S. GHG Emissions (MMT CO_2e)	2021 U.S. GHG Emissions (MMT CO_2e)	2020 Nevada GHG Emissions (MMT CO_2e)
Total (gross)	7,477.4	6,340.2	43.08
Total (net)	6,696.3	5,586.0	37.34

Sources: EPA (2023a); NDEP (2022)

Note: MMT = million metric tons; N/A = not available, GWP values have been applied

Table A-13. National Emissions Inventory 2020 Emissions Data for the United States, Nevada, and Clark County

Pollutant	United States (tons)	Nevada (tons)	Clark County (tons)
CO_2e	4,567,587,840	34,566,069	21,579,446
Carbon dioxide (CO_2)	4,378,761,025	33,915,829	21,244,787
Methane (CH_4)	5,304,516	19,211	9826
Nitrous oxide (N_2O)	108,357	285	153

Source: EPA (2023a)

Note: Values are in metric tons; N/A = not available

Table A-14. Proposed Action Construction and Operation and Maintenance Greenhouse Gas Emissions

County	GHG	2023 Construction (tons)	O&M (tons)
Clark County	CO_2	4,127	25.4
	CH_4	0.07	0.001
	N_2O	0.08	0.0001
	CO_2e	4,150	25.4
Clark County NEI (2020)	CO_2e	21,579,446	
Percentage of NEI	CO_2e	0.02%	0.0001%

Source: EPA (2023a)

Note: Values are in metric tons; N/A = not available

Table A-15. Proposed Action Summary for Potential Greenhouse Gas Emissions from Construction and Operation of the Project

Parameter Description	Proposed Action
Number of years in operation (years)	30
Annual O&M emissions (MT CO_2e/year)	25.4
Total life-of Project O&M emissions (MT CO_2e)	762
Project total construction emissions (MT CO_2e) 2023	4,150
Total life-of Project CO_2e emissions (MT CO_2e)	**4,937**
Emissions Displacement Potential per Year – Natural Gas*	**22,785 MT/yr**
Emissions Displacement Potential per Year – Coal*	**48,020 MT/yr**

Note: Assumes GHG emissions global warming potentials published by IPCC (IPCC 2021).

*Estimation of emissions avoided due to displacement of fossil fuel powered electricity generation

Table A-16. Social Cost of Greenhouse Gases from the Proposed Action (in 2020 dollars)

Social Cost Metric	5% Discount Rate – Average	3% Discount Rate – Average	2.5% Discount Rate – Average	3% Discount Rate – 95th Percentile
SC-CO_2	$74,358	$257,441	$382,399	$770,902
SC-CH_4	$69	$158	$208	$418
SC-N_2O	$525	$1,627	$2,379	$4,282
Total	**$74,751**	**$259,226**	**$384,986**	**$775,602**

Table A-17. Soil Map Units within the Analysis Area

Soil Map Unit Name	Ecological Sites	Wind Erodibility Index*	K Factor	Hydrologic Soil Group	Acres†	Percentage of Analysis Area
Arizo very gravelly loamy sand, flooded, 0 to 4 percent slopes	R030XB028NV	86	0.1	A	46	57
Caliza very cobbly loamy sand, 4 to 8 percent slopes	R030XB005NV	86	0.05	A	22	28
Canutio-Akela complex, 2 to 15 percent slopes	R030XB038NV and R030XB010NV	48	0.1	A	12	15
Total	**–**	**–**			**100**	**100**

***** The wind erodibility index is a numerical value indicating the susceptibility of soil to wind erosion and is represented in tons per acre per year that can be expected to be lost to wind erosion. The rating ranges from zero to 310.

† Acres are rounded up to the nearest whole number.

Source: NRCS (2022)

Table A-18. Plant Species List for the Analysis Area

Scientific Name	Common Name
Ambrosia dumosa	white bursage
Bebbia juncea	sweetbush
Chamaesyce micromera	Sonoran sandmat
Chorizanthe rigida	devil's spineflower
Cylindropuntia acanthocarpa	buck-horn cholla
Cylindropuntia echinocarpa	Wiggins' cholla
Cylindropuntia ramosissima	branched pencil cholla
Dalea mollissima	soft prairie clover
Echinocereus engelmannii	Engelmann's hedgehog cactus
Ephedra nevadensis	Nevada ephedra
Eriogonum fasciculatum	Eastern Mojave buckwheat
Eriogonum inflatum	desert trumpet
Eriogonum species	buckwheat
Ferocactus cylindraceus	California barrel cactus
Hymenoclea salsola	burrobush
Krameria erecta	littleleaf ratany
Larrea tridentata	creosote bush
Lotus strigosus	strigose bird's-foot trefoil

Scientific Name	Common Name
Mammillaria tetrancistra	common fishhook cactus
Opuntia basilaris	beavertail pricklypear
Porophyllum gracile	slender poreleaf
Schismus sp.	Mediterranean grass
Senegalia greggii	catclaw acacia
Stephanomeria pauciflora	brownplume wirelettuce

Table A-19. BLM Special-Status Plant Species with Potential to Occur in Analysis Area

Common Name*	Scientific Name	Species Status	Potential for Occurrence†
Alkali mariposa lily	*Calochortus striatus*	BLM (S);	None; no habitat present
Amargosa niterwort	*Nitrophila mohavensis*	BLM (S); NV (CE); S1	None; no habitat present
Antelope Canyon goldenbush	*Ericameria cervina*	BLM (S); S1	None; no habitat present
Beatley scorpionflower	*Phacelia beatleyae*	BLM (S); S2	None; no habitat present
Beaver Dam breadroot	*Pediomelum castoreum*	BLM (S); S2	None; no habitat present
Black woolypod	*Astragalus funereus*	BLM (S); S2	None; no habitat present
Blue Diamond cholla‡	*Cylindropuntia multigeniculata*	BLM (S); NV (CE); S2	None; not observed
Bullfrog Hills sweetpea	*Lathyrus hitchcockianus*	BLM (S); S2	None; no habitat present
Clokey buckwheat	*Eriogonum heermannii* var. *clokeyi*	BLM (S); S2	None; no habitat present
Darin buckwheat	*Eriogonum concinnum*	BLM (S); S3	None; no habitat present
Death Valley beardtongue	*Penstemon fruticiformis* ssp. *amargosae*	BLM (S); S2	None; no habitat present
Death Valley sage	*Salvia funerea*	BLM (S); S1	None; not observed
Gilman's milkvetch	*Astragalus gilmanii*	BLM (S); S1	None; no habitat present
Gold Butte moss	*Didymodon nevadensis*	BLM (S); S2	None; no habitat present
Halfring milkvetch	*Astragalus mohavensis* var. *hemigyrus*	BLM (S); S3	None; not observed
Jaeger beardtongue	*Penstemon thompsoniae* ssp. *jaegeri*	BLM (S); S3	None; no habitat present
Jaeger ivesia	*Ivesia jaegeri*	BLM (S); S2	None; no habitat present
Las Vegas bearpoppy	*Arctomecon californica*	BLM (S); S2	None; not observed
Las Vegas buckwheat	*Eriogonum corymbosum* var. *nilesii*	BLM (S); S2	None; not observed
Mojave thistle (Virgin River thistle)	*Cirsium mohavense (*or *C. virginense)*	BLM (S); SNR	None; no habitat present
Mokiak milkvetch	*Astragalus mokiacensis*	BLM (S); S1S2	None; no habitat present
Nevada willowherb	*Epilobium nevadense*	BLM (S); S2	None; no habitat present
Pahrump silverscale	*Atriplex argentea* var. *longitrichoma*	BLM (S); S1	None; no habitat present
Pahrump Valley buckwheat	*Eriogonum bifurcatum*	BLM (S); S2	None; no habitat present
Pahute Mesa beardtongue	*Penstemon pahutensis*	BLM (S); S3	None; no habitat present
Parish phacelia	*Phacelia parishii*	BLM (S); S3	None; no habitat present
Polished blazingstar	*Mentzelia polita*	BLM (S); S2	None; no habitat present
Red Rock Canyon aster	*Ionactis caelestis*	BLM (S); S1	None; no habitat present

Common Name*	Scientific Name	Species Status	Potential for Occurrence†
Rock purpusia	*Ivesia arizonica* var. *saxosa*	BLM (S); S2	None; no habitat present
Rosy twotone beardtongue	*Penstemon bicolor* ssp. *roseus*	BLM (S); S3	None; not observed but potential habitat is present within and along wash margins
Rough angelica	*Angelica scabrida*	BLM (S); S2	None; no habitat present
Rough dwarf greasebush	*Glossopetalon pungens* var. *pungens*	BLM (S); S2	None; no habitat present
Scrub lotus	*Lotus argyraeus* var. *multicaulis*	BLM (S); S2	None; no habitat present
Sheep fleabane	*Erigeron ovinus*	BLM (S); S2	None; no habitat present
Silverleaf sunray	*Enceliopsis argophylla*	BLM (S); S2	None; no habitat present
Smooth dwarf greasebush	*Glossopetalon pungens* var. *glabrum*	BLM (S); S2	None; no habitat present
Spring-loving centaury	*Centaurium namophilum*	USFWS (T); BLM (S); NV (CE); S1	None; no habitat present
Spring Mountains milkvetch	*Astragalus remotus*	BLM (S); S2	None; no habitat present
Sticky buckwheat	*Eriogonum viscidulum*	BLM (S); NV (CE); S2	None; no habitat present
Sticky ringstem	*Anulocaulis leiosolenus var. leiosolenus*	BLM (S); S2	None; not observed
Straw milkvetch	*Astragalus lentiginosus* var. *stramineus*	BLM (S); S1	None; no habitat present
Stream stippleback lichen (Silverskin lichen)	*Dermatocarpon luridum*	BLM (S); S1	None; no habitat present
Threecorner milkvetch	*Astragalus geyeri* var. triquetrus	BLM (S); NV (CE); S2S3	None; no habitat present
Torrey milkvetch (one-leaflet)	*Astragalus calycosus* var. *monophyllidus*	BLM (S); S3	None; no habitat present
White bearpoppy	*Arctomecon merriamii*	BLM (S); S3	None; not observed
White-margined beardtongue	*Penstemon albomarginatus*	BLM (S); S2	None; no habitat present
Yellow two-toned beardtongue	*Penstemon bicolor* var. *bicolor*	BLM (S); S2	None; no habitat present

Notes:

* No Ash Meadow plant species (five) have potential to occur because the Project is outside the Ash Meadows.

† Plant species potential to occur determined from botanical field surveys (SWCA 2022b).

‡ Plant species protected as cactus, yucca, or Christmas tree in State of Nevada.

BLM Status Definitions:

S = Sensitive. Species are listed as Sensitive if there is information that the species has recently undergone, is undergoing, or is predicted to undergo a downward trend such that viability of the species is at risk, or because the species depends on ecological refugia or specialized or unique habitats on BLM-administered lands, and there is evidence that such areas are threatened with alteration such that the viability of the species is at risk.

Nevada (NV) Status Definitions (NAC 527.050):

CE – Critically endangered species or subspecies of native plants declared to be threatened with extinction and placed on the list of fully protected species of native flora by the State Forester pursuant to Nevada Revised Statutes 527.270.

NDNH Definitions:

S1 = Species is critically imperiled and at very high risk of extirpation in the jurisdiction due to very restricted range, very few populations or occurrences, very steep declines, severe threats, or other factors.

S2 = Species is imperiled and at high risk of extirpation in the jurisdiction due to restricted range, few populations or occurrences, steep declines, severe threats, or other factors.

S3 = Species is vulnerable and at moderate risk of extirpation in the jurisdiction due to a fairly restricted range, relatively few populations or occurrences, recent and widespread declines, threats, or other factors.

SNR = Species is unranked and conservation status not yet assessed.

S#S# = Range rank used to indicate any range of uncertainty about the status of the species or ecosystem.

Potential for Occurrence Definitions:

None; no habitat present – Plant species desktop analysis identified no habitat for the plant species.

None; not observed – Plant species was not observed during botanical surveys.

Table A-20. Cacti Estimates within the Botanical Survey Area

Scientific Name	Common Name	Count
Cylindropuntia acanthocarpa	buck-horn cholla	915
Cylindropuntia echinocarpa	Wiggins' cholla	56
Cylindropuntia ramosissima	branched pencil cholla	176
Echinocereus engelmannii	Engelmann's hedgehog cactus	29
Ferocactus cylindraceus	California barrel cactus	742
Mammillaria tetrancistra	common fishhook cactus	8
Opuntia basilaris	beavertail pricklypear	1,098
Total		**3,024**

Table A-21. Selected KOP Locations and Sensitive Viewer Types

KOP Number	Name	Sensitive Viewer Group	Approximate Distance to Nearest Mine feature (miles) Distance Zone
1	U.S. Route 95 northbound	Vehicular Travel Route	0.92 Foreground
2	U.S. Route 95 southbound	Vehicular Travel Route	0.73 Foreground
3	I-11/U.S. Route 93 eastbound	Vehicular Travel Route	1.03 Foreground

Table A-22. Criteria for Degree of Contrast

Degree of Contrast	Criteria
None	The element contrast is not visible or perceived.
Weak	The element contrast can be seen but does not attract attention.
Moderate	The element contrast begins to attract attention and begins to dominate the characteristic landscape.
Strong	The element contrast demands attention, cannot be overlooked, and is dominant in the landscape.

Source: BLM (1986b)

Table A-23. KOP Impact Summary

KOP Number	Name Location Sensitive Viewers Overall Level of Contrast Impact Magnitude	Summary of Impacts
1	U.S. 95 Northbound Approximately 1 mile southeast of the existing Townsite solar fields on U.S. 95 Vehicular Travel Route Weak Low	Views of the Project components from this KOP would be predominantly backdropped against mountains and be in proximity to other solar arrays. Project components may be present but would repeat form, line, color, texture or scale common in the landscape and would be visually subordinate within the visual setting of the existing landscape character. The landscape would appear intact with only minor modifications visible, if any. Project elements introduced would have similar form, line, color, texture, or scale common in the landscape, reducing their visibility and assisting in blending Project components in the landscape. The overall impacts from Project elements into the existing landscape would be low.

KOP Number	Name Location Sensitive Viewers Overall Level of Contrast Impact Magnitude	Summary of Impacts
2	U.S. 95 Southbound Directly east approximately 0.75 mile of the Project Area on U.S. 95 Vehicular Travel Route Weak Low	Views of the Project components from this KOP would be predominantly backdropped against mountains and be in proximity to other solar arrays. Project components may be present but would repeat form, line, color, texture or scale common in the landscape and would be visually subordinate within the visual setting of the existing landscape character. The landscape would appear intact with only minor modifications visible, if any. Project elements introduced would have similar form, line, color, texture, or scale common in the landscape, reducing their visibility and assisting in blending Project components in the landscape. The overall impacts from Project elements into the existing landscape would be low.
3	I-11/U.S. Route 93 Eastbound Approximately 1 mile northeast of Project Area on I-11/U.S. Route 93 Vehicular Travel Route Weak Low	Views of the Project components from this KOP would be predominantly backdropped against mountains and be in proximity to other solar arrays. Topographical variation in this area would provide additional screening of the Project. Project components may be present but would repeat form, line, color, texture or scale common in the landscape and would be visually subordinate within the visual setting of the existing landscape character. The landscape would appear intact with only minor modifications visible, if any. Project elements introduced would have similar form, line, color, texture, or scale common in the landscape, reducing their visibility and assisting in blending Project components in the landscape. The overall impacts from Project elements into the existing landscape would be low.

Table A-24. Summary of Reasonably Foreseeable Future Actions

Proposed Projects	Applicant	Potential Surface Disturbance
Within 1 mile of Project		
Mineral material sale	BLM	80 acres (existing)
Expanded SNWA 230-kV transmission line to support the Boulder Flats Solar Project	Southern Nevada Water Authority	145 acres, 10 miles long*
Transmission line	NV Energy	0.217 acres, 3,081 feet long by width change to 30-feet wide from 10-feet wide
Road and drainage	Quarry 187 LLC	6.25 acres
Within 2 miles of Project		
None	None	None
Within 3 miles of Project		
Road and Drainage	Southwest Gas Corporation	62.61 acres
Within 4 miles of Project		
Boulder City Municipal Airport	City of Boulder City	Various improvements (no acreage provided)
Wastewater treatment plant	City of Boulder City	Basin concrete liner (no acreage provided)
Reclamation project (short term)	Las Vegas Valley Water Authority	1.31 acres (reclamation)
Within 5 miles of Project		
Residential development	City of Boulder City	45 acres
Boulder Flats Solar Project access road and water pipeline	ibV Energy Partners	24.86 acres, 2.4 miles long × 54 feet wide

Proposed Projects	Applicant	Potential Surface Disturbance
Boulder Flats Solar Project temporary water pipeline	ibV Energy Partners	12.21 acres, 26,576 feet long × 20 feet wide
State Railroad Museum expansion	City of Boulder City	Expansion on existing site (no acreage provided)
Boulder Flats solar facility	ibV Energy Partners	1,100 acres

Notes:

* All disturbance associated with the 230-kV line is accounted for within 1 mile of the Project for CESA calculation purposes.

Source: BLM (2022c) and City of Boulder City (2022).

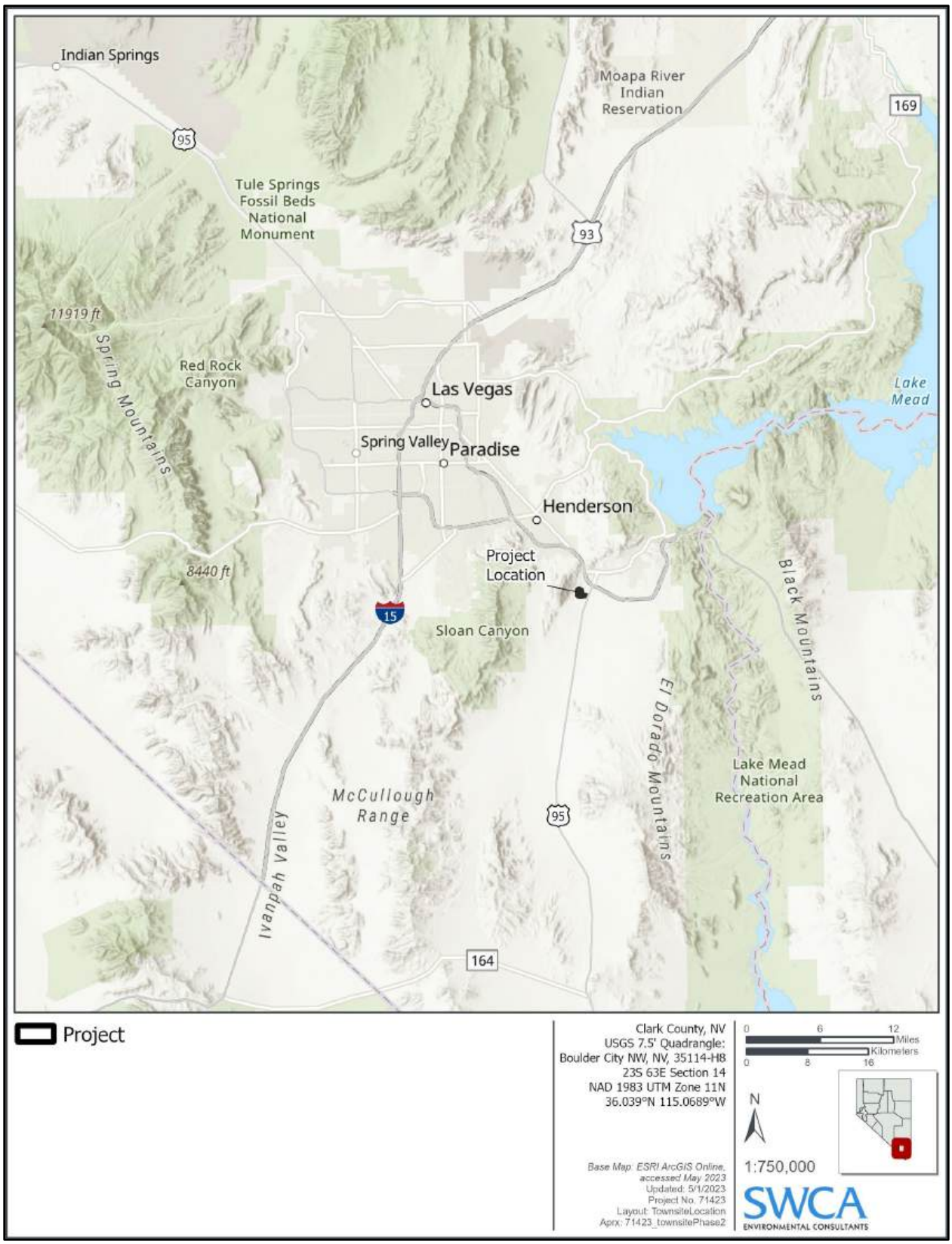


Figure A-1. Project location.

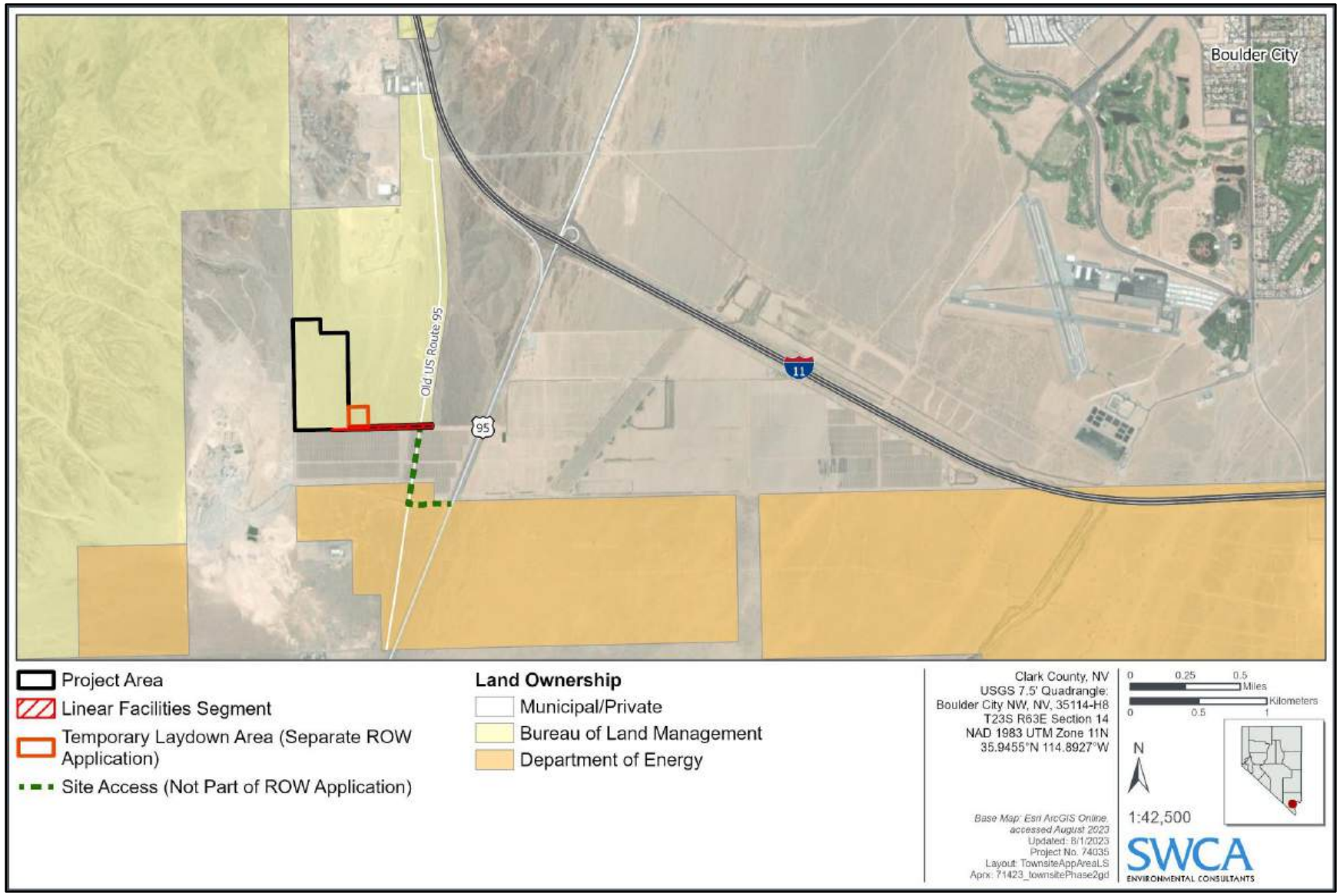


Figure A-2. Project area and land ownership.

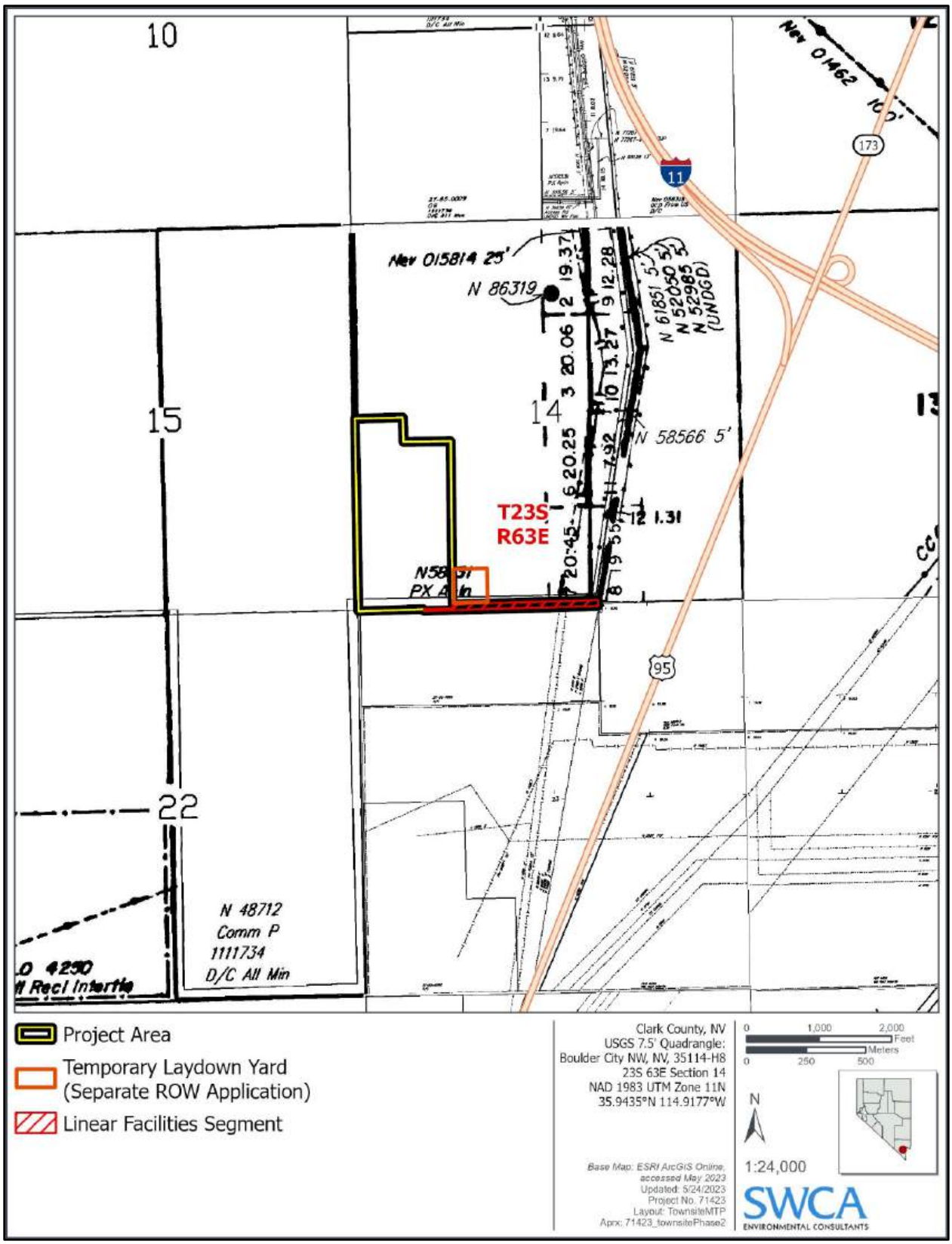


Figure A-3. Master title plat.

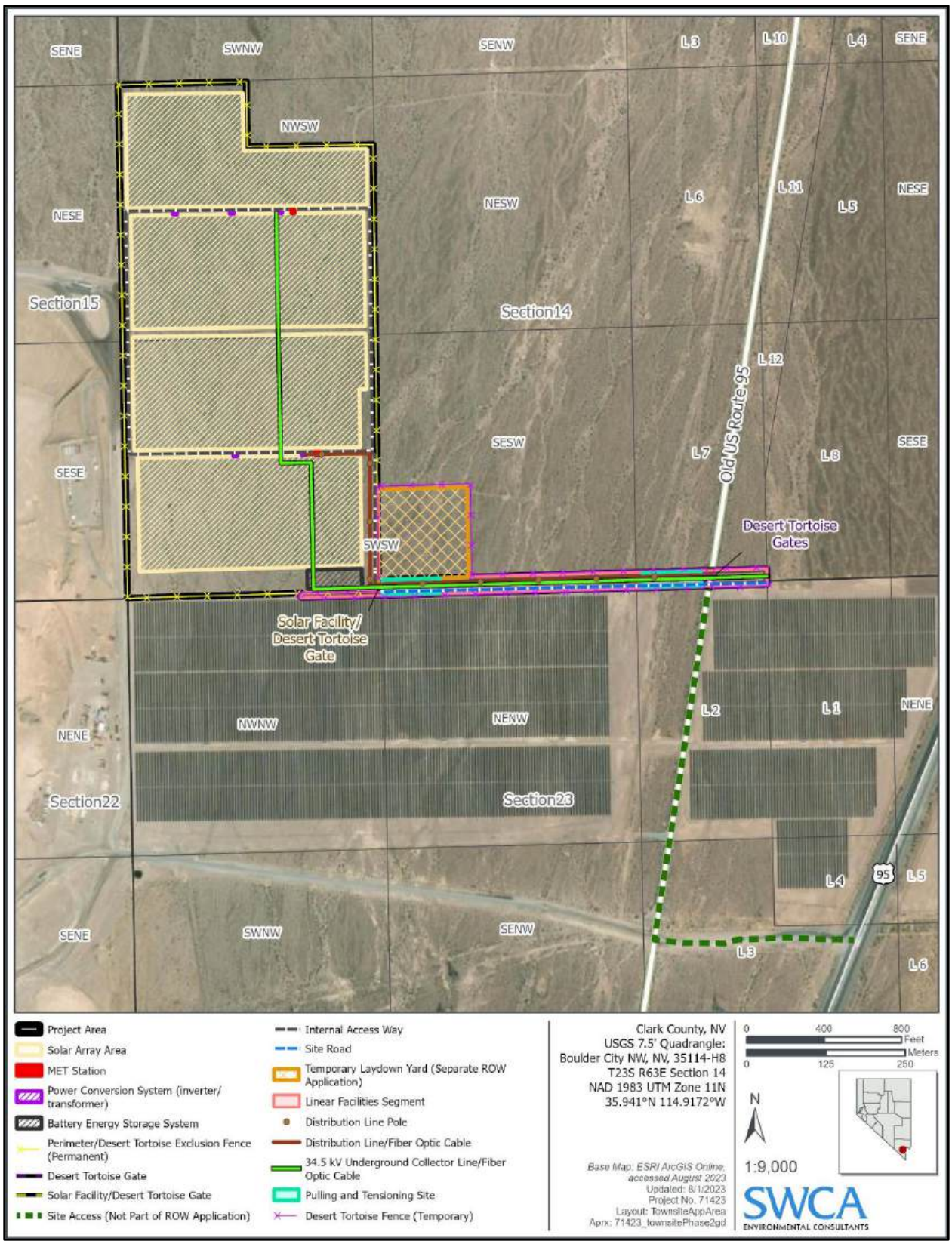


Figure A-4. Project elements.

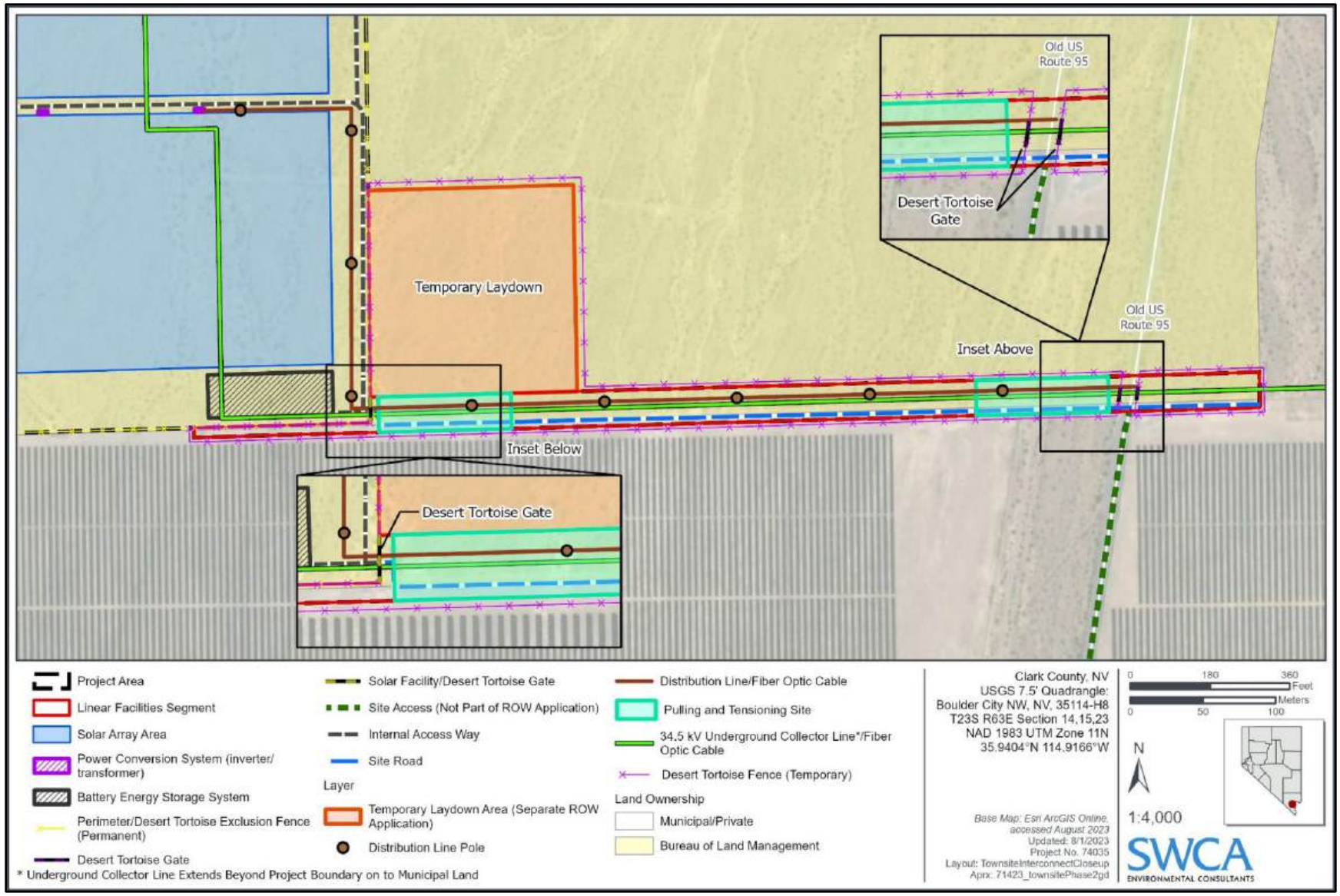


Figure A-5. Detailed view of the linear facilities segment.

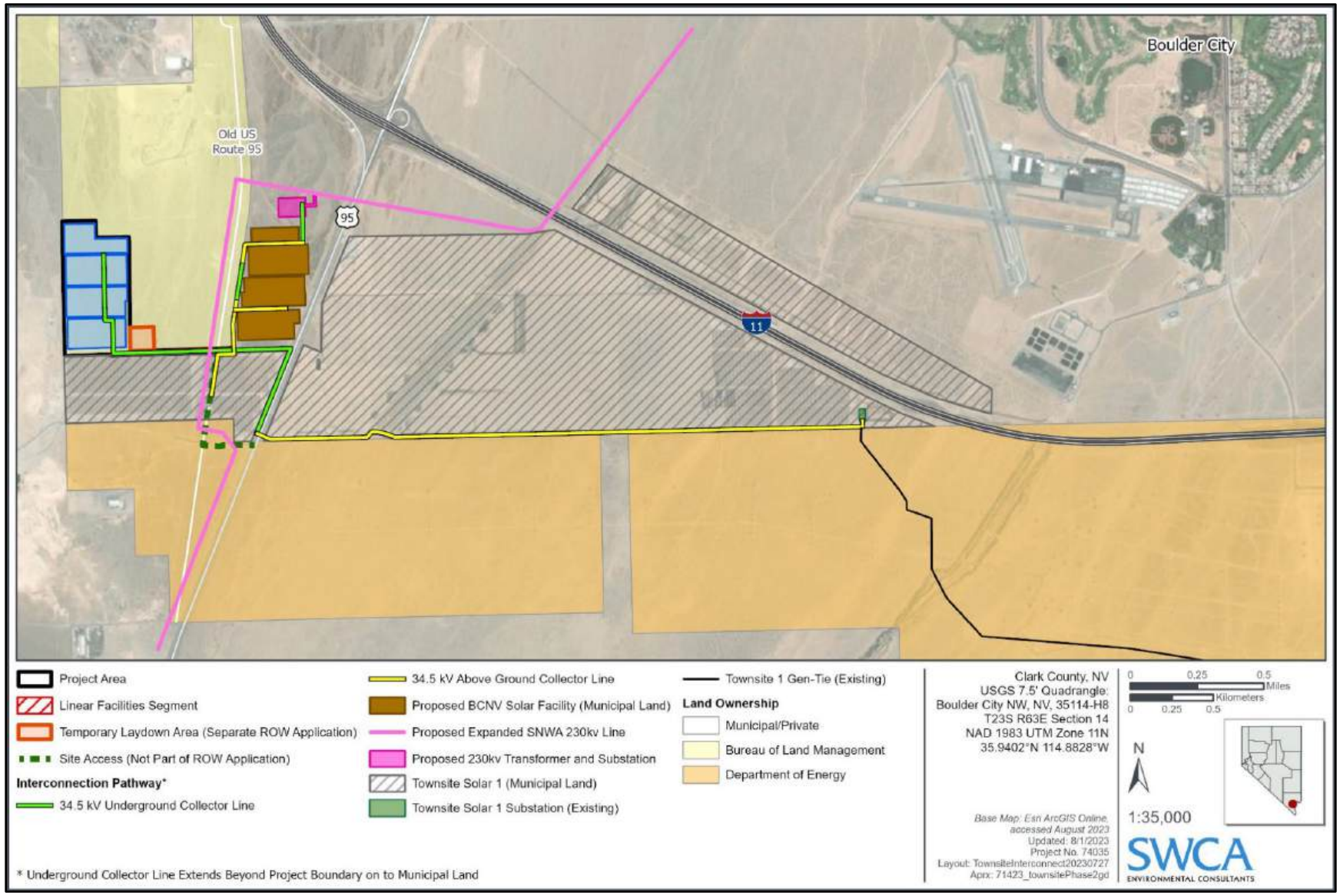


Figure A-6. Interconnection pathways (for illustration purposes only, not part of the Proposed Action).

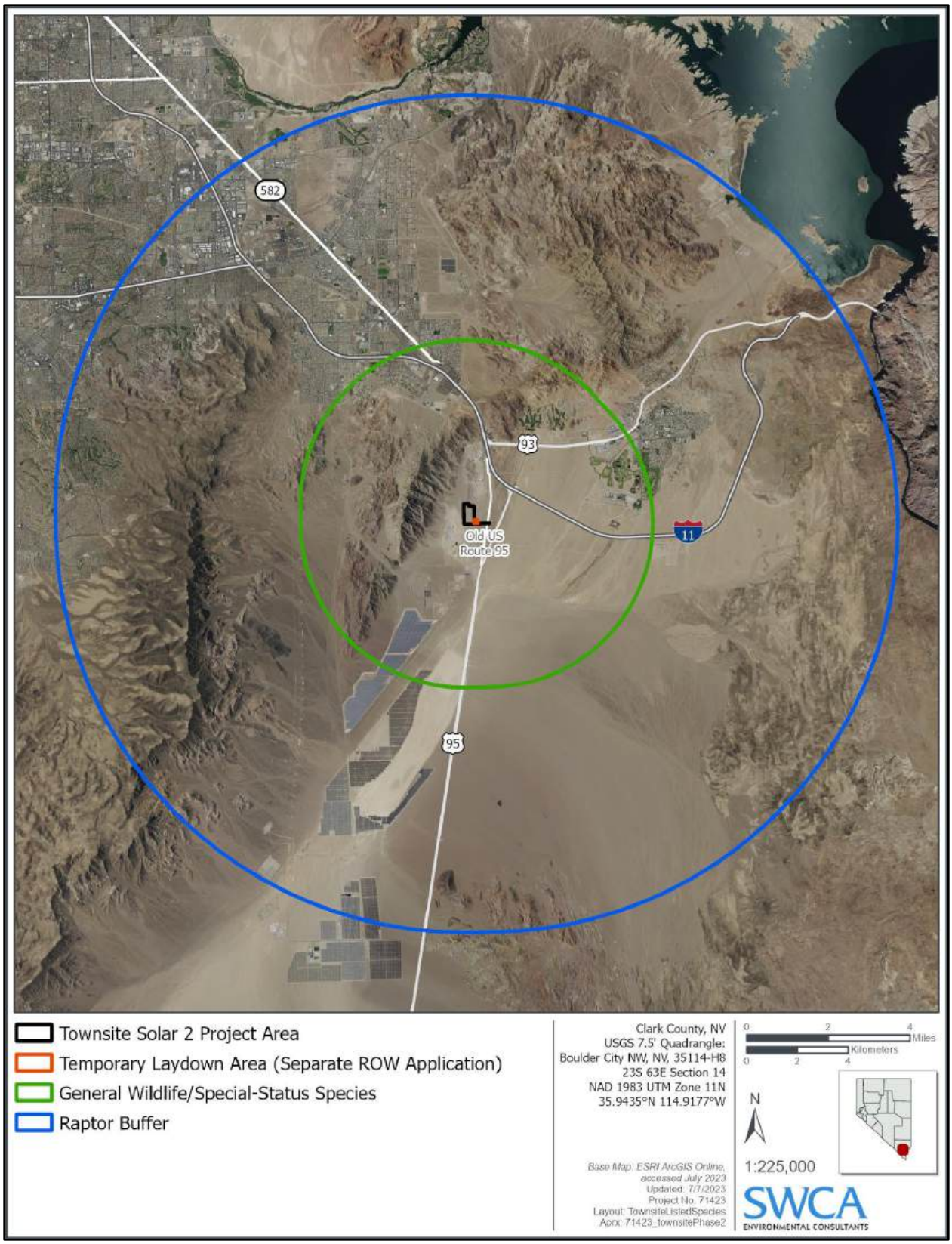


Figure A-7. Analysis area buffers for general wildlife, special-status species, and raptors.

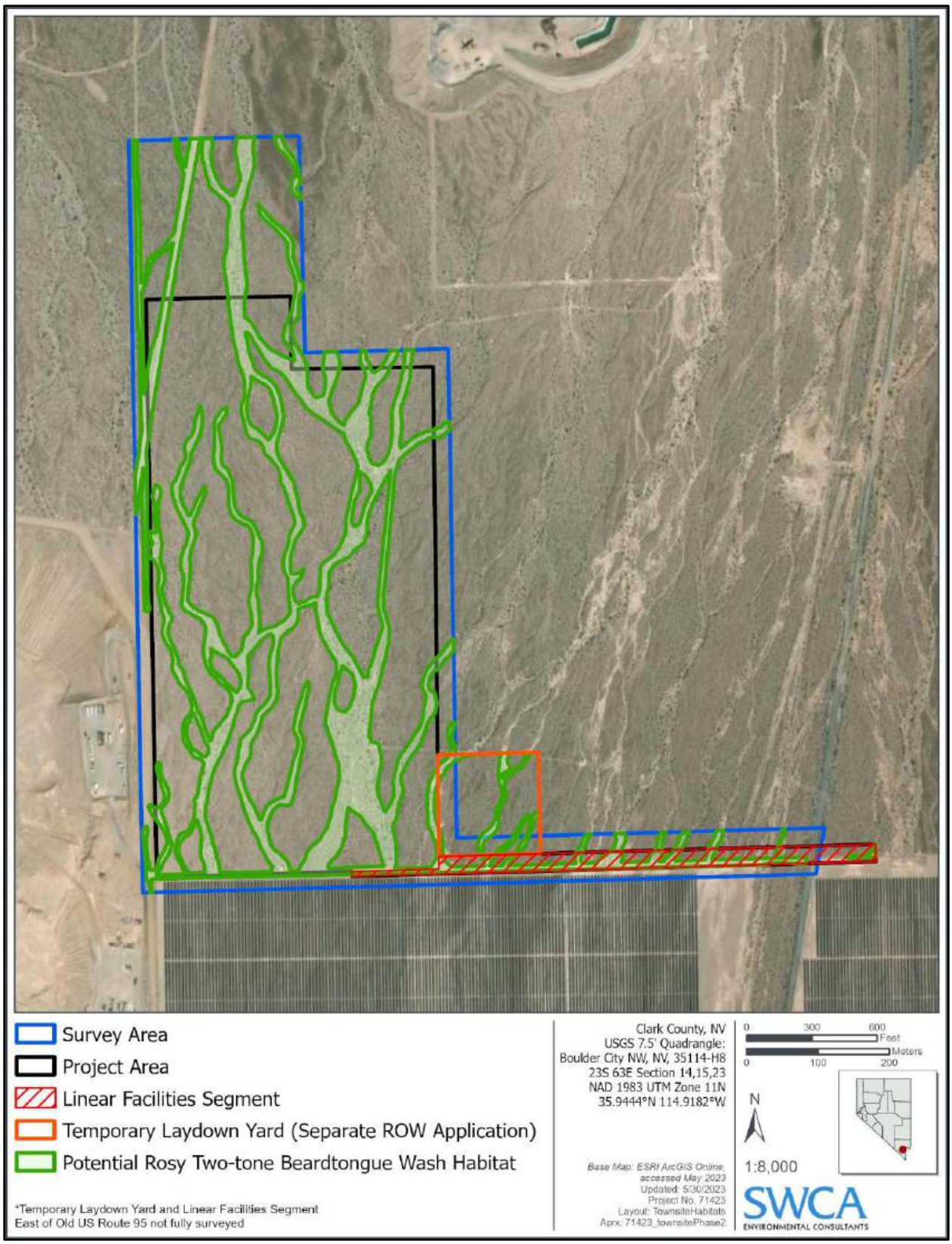


Figure A-8. Potential habitat for rosy two-tone beardtongue.

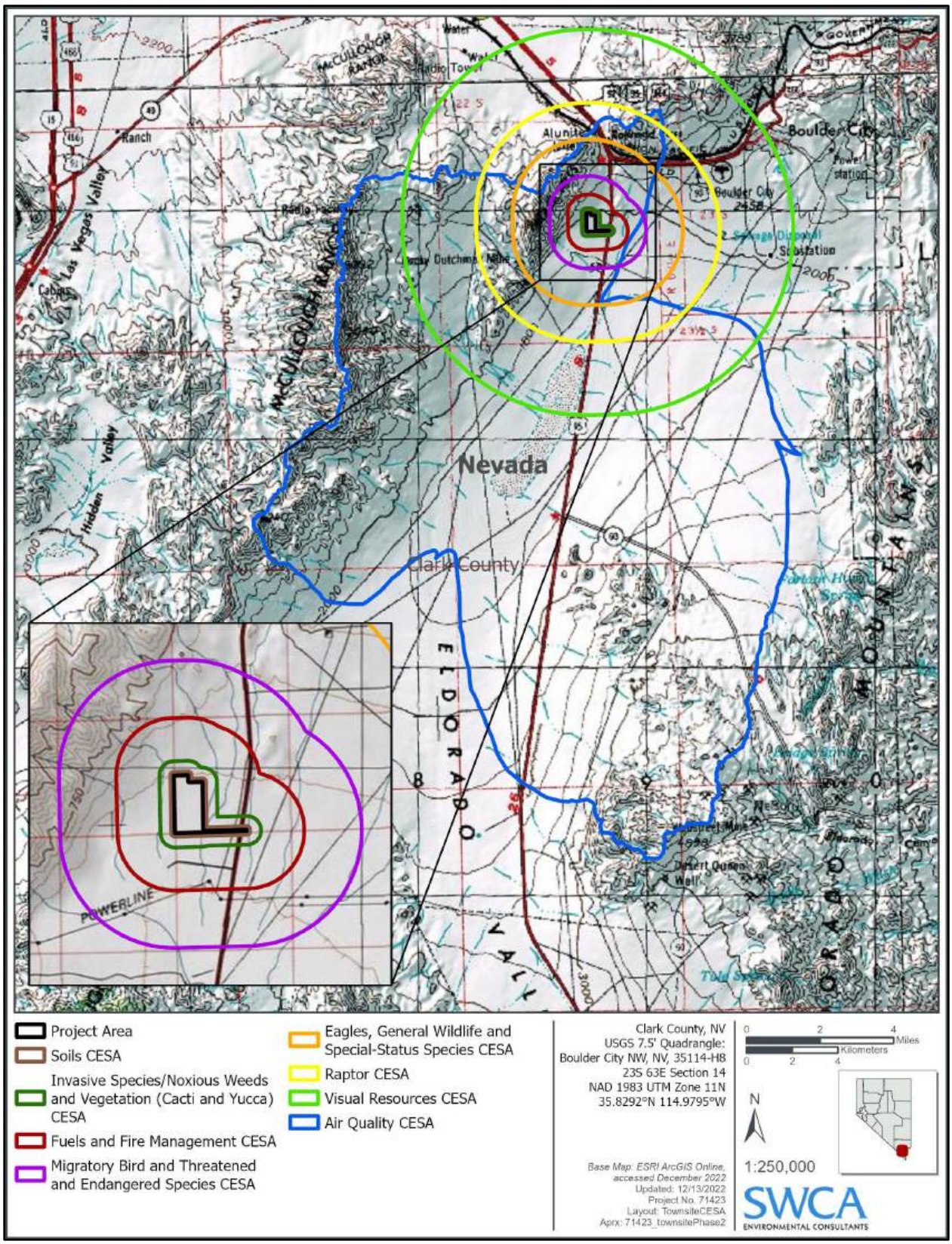


Figure A-9. CESAs.

APPENDIX B

Literature Cited

Literature Cited

Abatzoglou, J.T., D.J. McEvoy, and K.T. Redmond. 2017. The west wide drought tracker: drought monitoring at fine spatial scales. *Bulletin of the American Meteorological Society* 98(9):1815–1820.

Abella, S.R. 2010. Disturbance and plant succession in the Mojave and Sonoran Deserts of the American Southwest. *International Journal of Environmental Research and Public Health* 7(4):1248–1284. Available at: http://dx.doi.org/10.3390/ijerph7041248. Accessed July 29, 2022.

Allied Pacific Partners, LLC. 2022. *Townsite Solar Phase 2 Solar Project Desert Tortoise Pre-Project Survey Report*. Submitted to Bureau of Land Management Southern Nevada District Office. February 11, 2022.

Avian Power Line Interaction Committee (APLIC). 2006. *Suggested Practices for Avian Protection On Power Lines: The State of the Art in 2006*. Edison Electric Institute, Avian Power Line Interaction Committee, and the California Energy Commission. Washington, D.C. and Sacramento, CA. Accessed August 10, 2022.

———. 2012. *Reducing Avian Collisions with Power Lines: State of the Art in 2012.* Edison Electric Institute and Avian Power Line Interaction Committee. Washington, D.C. Accessed August 10, 2022.

Benedict K., O.Z. McCotter, S. Brady, K. Komatsu, G.L. Sondermeyer, Cooksey, A. Nguyen, S. Jain, D.J. Vugia, B.R. Jackson. 2019. Surveillance for Coccidioidomycosis — United States, 2011–2017. Available at: https://www.cdc.gov/mmwr/volumes/68/ss/ss6807a1.htm?s_cid=ss6807a1_w#F6_down. Accessed: July 7, 2023.

Bureau of Land Management (BLM). 1984. Manual 8400 – *Visual Resource Management*. April 5. Available at: https://www.blm.gov/policy/manuals. Accessed November 2022.

———. 1986a. Manual H-8410-1 – *Visual Resource Inventory*. January 17. Available at: https://www.blm.gov/policy/handbooks. Accessed November 2022.

———. 1986b. Manual H-8431 – *Visual Resource Contrast Rating*. Washington, D.C.: U.S. Department of the Interior, BLM, Washington Office. Available at: https://www.blm.gov/policy/handbooks. Accessed November 2022.

———. 1998a. *Record of Decision for the Approved Las Vegas Resource Management Plan and Environmental Impact Statement: Appendix A.* Las Vegas, Nevada: Bureau of Land Management, Las Vegas Field Office. May 1998. Available at: https://eplanning.blm.gov/eplanning-ui/project/78155/570. Accessed on June 27, 2022.

———. 1998b. *Proposed Las Vegas Resource Management Plan and Final Environmental Impact Statement: Volume II: Appendices, Glossary, References, and Maps.* Las Vegas, Nevada: Bureau of Land Management, Las Vegas Field Office. May 1998. Available at: https://eplanning.blm.gov/public_projects/lup/78155/128953/156905/1998_LV_RMP_Volume_1.pdf. Accessed on June 27, 2022.

———. 2007a. Final Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement. DOI-BLM-WO-WO2100_2007_0002-EIS. Available at: https://eplanning.blm.gov/eplanning-ui/project/70300/570.

———. 2007b. Record of Decision for the Final Vegetation Treatments Using Herbicides on Bureau of Land Management Lands in 17 Western States Programmatic Environmental Impact Statement.

———. 2008a. H-1790-1 – National Environmental Policy Act. Available at: https://www.blm.gov/policy/handbooks. Accessed August 5, 2022.

———. 2008.b. H-6840 – Special-Status Species Management. Available at: https://www.blm.gov/policy/manuals. Accessed August 5, 2022.

———. 2011. BLM National Data Viewer. Available at: https://blm-egis.maps.arcgis.com/apps/webappviewer/index.html?id=6f0da4c7931440a8a80bfe20eddd7550. Accessed November 2022.

———. 2012a. *Final Programmatic Environmental Assessment (PEA) for the Eldorado Valley Transmission and Utility Corridor Finding of No Significant Impact and Decision Record*. DOI-BLM-NV-S010-2012-0024. Available at: https://eplanning.blm.gov/eplanning-ui/project/33155/510. Accessed November 16, 2022.

———. 2012b. *Programmatic Agreement Among the Bureau of Land Management, the Advisory Council on Historic Preservation, and the National Conference of State Historic Preservation Officers Regarding the Manner in Which the BLM Will Meet Its Responsibilities Under the National Historic Preservation Act.* Available at: https://www.achp.gov/digital-library-section-106-landing/blm-national-pa-compliance-section-106. Accessed August 6, 2022.

———. 2013. *Best Management Practices for Reducing Visual Impacts of Renewable Energy Facilities on BLM-Administered Lands*. First Edition: Cheyenne, Wyoming.

———. 2017. Final BLM Nevada Sensitive and Special-Status Species List. Nevada State Office, Reno, Nevada. Available at: https://www.blm.gov/policy/nv-im-2018-003.

———. 2018. *Common Raven Management Plan for Energy Development within the BLM Southern Nevada District.*

———. 2019. *Southern Nevada Nesting Bird Management Plan*. Prepared by the Bureau of Land Management, Southern Nevada District Office.

———. 2021a. *Townsite Phase II Solar Project ROW N-99875 Priority Determination*. BLM Southern Nevada District Office, Las Vegas Field Office. September 17, 2021.

———. 2021b. BLM National Area of Critical Environmental Concern Designated Polygons GIS Layer. Available at: https://gis.blm.gov/arcgis/rest/services. Accessed June 27, 2022.

———. 2022a. *2021 BLM Specialist Report on Annual Greenhouse Gas Emissions and Climate Trends*. Available at: https://www.blm.gov/content/ghg/2021/.

———. 2022b. BLM National PFYC Potential Fossil Yield Classification Geographic Information System Geologic Formations Coverage GIS Layer. U.S. Department of the Interior, Bureau of Land Management (BLM), Paleo Program. Spatial Reference 102100 (3857). Available at: https://gis.blm.gov/arcgis/rest/services/geophysical/BLM_Natl_PFYC/MapServer. Accessed July 20, 2022.

———. 2022c. Townsite Solar 2 EA-Inquiry for Cumulative Effects Analysis-Follow-up. Electronic Correspondence to Jessica Headen, BLM Project Manager for Townsite Solar 2 Project. October 21, 2022.

———. 2023. BLM Socioeconomic Profile Reports. Available at: headwaterseconomics.org/tools/blm-profiles/. Accessed July 2023.

———. n.d. Old Spanish National Historic Trail. Available at: https://www.blm.gov/visit/old-spanish-national-historic-trail. Accessed August 30, 2023.

Bureau of Land Management (BLM) and U.S. Department of Energy (DOE). 2012. *Final Programmatic Environmental Impact Statement (PEIS) for Solar Energy Development in Six Southwestern States*. FES 12-24, DOE/EIS-0403. Washington, D.C.: Bureau of Land Management and U.S. Department of Energy. July 2012. Available at: https://solareis.anl.gov/index.cfm. Accessed May 17, 2022.

California Public Utilities Commission and Bureau of Land Management (BLM). 2010. *Final Environmental Impact Report/Environmental Impact Statement for the Eldorado-Ivanpah Transmission Project.* Application No. 09-05-027. Available at: https://ia.cpuc.ca.gov/environment/info/ene/ivanpah/Ivanpah_FEIR-EIS.html. Accessed on August 5, 2022.

Centers for Disease Control and Prevention. 2020. Nationally Notifiable Infectious Diseases and Conditions, United States: Annual Tables. Table 2e. Annual Reported cases of notifiable diseases, by region and reporting area, United States, U.S. Territories, and Non-U.S. Residents, 2020. Available at: https://wonder.cdc.gov/nndss/static/2020/annual/2020-table2e.html. Accessed July 7, 2023.

Chambers, J.C., M.L. Brooks, B.K. Pendleton, and C.B. Raish. 2013. *The Southern Nevada Agency Partnership Science and Research Synthesis: Science to Support Land Management in Southern Nevada*. General Technical Report RMRS-GTR-303. Fort Collins, Colorado: U.S. Forest Service, Rocky Mountain Research Station.

Chambers, J.C., B.A. Bradley, C.S. Brown, C. D'Antonio, M.J. Germino, M.J., J.B. Grace, and D.A. Pyke. 2014. Resilience to Stress and Disturbance, and Resistance to Bromus Tectorum L. Invasion in Cold Desert Shrublands of Western North America. *Ecosystems* 17(2):360–375. doi:10.1007/s10021-013-9725-5.

Chambers J.C., D.A. Pyke, J.D. Maetas, M. Pellant, C.S. Boyd, S.B. Campbell, S. Espinosa, D.W. Havlina, K.E. Mayer, and A. Wuenschel. 2014. *Using Resistance and Resilience Concepts to Reduce Impacts of Invasive Annual Grasses and Altered Fire Regimes on the Sagebrush Ecosystem and Greater Sage-Grouse: A Strategic Multi-Scale Approach.* Gen. Tech. Rep. RMRS-GTR-326. Fort Collins, Colorado: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station. 73 pp.

City of Boulder City. 2022. City of Boulder City-Request for Reasonably Foreseeable Projects 10-11-22. Electronic Correspondence from Brok Armantrout, City of Boulder City Revenue Contracts & Real Estate Manager. October 19, 2022.

Clark County. 2004. Nevada Air Quality Designations Boundary Recommendations for the 8-Hour Ozone NAAQS for Clark County, Nevada. Available at: https://archive.epa.gov/airquality/ozonepollution/designations/web/pdf/boundary.pdf. Accessed January 3, 2023.

———. 2021a. State Implementation Plans Relayed Documents. Department of Environment and Sustainability. Division of Air Quality. Available at: https://www.clarkcountynv.gov/government/departments/environment_and_sustainability/division_of_air_quality/planning/state_implementation_plans.php Accessed August 29, 2022.

———. 2021b. Second Maintenance Plan for the 1997 8-hour Ozone NAAQS. Clark County, Nevada. Available at: Clark County, NV (clarkcountynv.gov). Accessed October 28, 2022.

———. 2022. Clark County Air Quality Data. Division of Air Quality, Monitoring Department. Available at: https://desaqmonitoring.clarkcountynv.gov/AgilaireWeb/. Accessed September 8, 2022.

Copeland, S.M, S.M. Munson, D.S. Pilliod, J.L. Welty, J.B. Bradford, B.J. Butterfield. 2018. Long-term trends in restoration and associated land treatments in the southwestern United States. *Restoration Ecology* 26(2):311–322.

Cornell Lab of Ornithology. 2022. All About Birds. Cornell Lab of Ornithology, Ithaca, New York. Available at: https://www.allaboutbirds.org. Accessed on July 5, 2022.

Cummings, S. 2019. South Korea Identifies Top 4 Causes for ESS Fires. Li-Ion Tamer. June 14, 2019. Available at: https://liiontamer.com/south-korea-identifies-top-4-causes-that-led-to-ess-fires/ (liiontamer.com).

Davies, K.W., and D.D. Johnson. 2011. Are we "missing the boat" on preventing the spread of invasive plants in rangelands? *Invasive Plant Science and Management* 4:166–171.

De Marco A., I. Petrosillo, T. Semeraro, M.R. Pasimeni, R. Aretano, and G. Zurlini. 2014. The contribution of utility-scale solar energy to the global climate regulation and its effects on local ecosystem services. *Global Ecology and Conservation* 2:324–337.

eBird. 2021. eBird: An Online Database of Bird Distribution and Abundance [Web Application]. eBird, Cornell Lab of Ornithology, Ithaca, New York. Available at: http://www.ebird.org. Accessed August 26, 2022.EJ Interagency Working Group. 2019. Community Guide to Environmental Justice and NEPA Methods. March 2019. Available at: https://www.energy.gov/sites/default/files/2019/05/f63/NEPA%20Community%20Guide%202019.pdf. Accessed July 2023.

Environmental Justice Interagency Working Group. 2019. Community Guide to Environmental Justice and NEPA Methods. Federal Interagency Working Group on Environmental Justice & NEPA Committee.

Farabee, M.J. 2007. Community and Ecosystem Dynamics. On-Line Biology Book. Revised Edition. Available at: https://www2.estrellamountain. Available at: edu/faculty/farabee/biobk/biobookcommecosys.html. Accessed July 7, 2022.

Federal Emergency Management Agency (FEMA). 2011. Flood Insurance Rate Map, No. 32003C2975F. Project Site Vicinity. FEMA Flood Map Service Center. Available at: https://msc.fema.gov/portal/home. Accessed July 20, 2022.

Google Earth Pro. 2022. *Townsite Solar 2 Project Area in Clark County, Nevada*. Latitude 35.940393°, Longitude -114.916603°, Eye Altitude 8,458 ASL, Version 7.3.4.8642. Available at: Earth Versions – Google Earth. Accessed July 20, 2022.

Greenstein L, C. Steele, and C.M. Taylor. 2022. Host plant specificity of the monarch butterfly *Danaus Plexippus*: A systematic review and meta-analysis. *PLOS ONE* 17(6): e0269701. Available at: https://doi.org/10.1371/journal.pone.0269701. Accessed June 14, 2022.

Greif, S., and B.M. Siemers. 2010. Innate recognition of water bodies in echolocating bats. *Nature Communications* (1):107. Available at: https://doi.org/10.1038/ncomms1110.

Grodsky, S.M., and R.R. Hernandez. 2020. Reduced ecosystem services of desert plants from ground-mounted solar energy development. *Nature Sustainability* 3:1036–1043.

Harrill, J.R., and D.E. Prudic. 1998. *Aquifers Systems in the Great Basin Region of Nevada, Utah, and Adjacent States – Summary Report*. Regional Aquifer-System Analysis – Greta Basin, Nevada – Utah. U.S. Geological Survey Professional Paper. 1409-A. Available at: https://pubs.usgs.gov/pp/1409a/report.pdf. Accessed July 28, 2022.

Hernandez, R.R., S.B. Easter, M.L. Murphy-Mariscal, F.T. Maestre, M. Tavassoli, E.B. Allen, C.W., and S. Ravi. 2014. Environmental impacts of utility-scale solar energy. *Renewable and Sustainable Energy Reviews* 29:766–779.

Hováth, G., M. Blahó, A. Egri, G. Kriska, I. Seres, and B. Robertson. 2010. Reducing the maladaptive attractiveness of solar panels to polarotactic insects. *Conservation Biology* 24:1644–1653.

Hováth, G., G. Kriska, P. Malik, and B. Robertson. 2009. Polarized light pollution: a new kind of ecological photopollution. *Frontiers in Ecology and the Environment* 7:317–325.

Intergovernmental Panel on Climate Change Climate Change (IPCC). 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Available at: https://www.ipcc.ch/report/2006-ipcc-guidelines-for-national-greenhouse-gas-inventories/.

———. 2021. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge and New York: Cambridge University Press. Available at https://www.ercevolution.energy/ipcc-sixth-assessment-report/.

Interagency Working Group (IWG). 2021. Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide, Interim Estimates under Executive Order 13990. Available at: https://costofcarbon.org/resources/entry/2021-IWG-tsd.

Kessavalou A., A.R. Mosier, and J.W. Doran. 1998. Fluxes of carbon dioxide, nitrous oxide, and methane in grass sod and winter wheat-fallow tillage management. *Journal of Environmental Quality* 27:1094–1104.

Longcore, T., and C. Rich. 2004. Ecological Light Pollution. *The Ecological Society of America Frontiers in Ecology and the Environment*. Available at: https://doi.org/10.1890/1540-9295(2004)002[0191:ELP]2.0.CO;2. Accessed July 29, 2022.

Lovich, J.E., and D. Bainbridge. 1999. Anthropogenic degradation of the southern California desert ecosystem and prospects for natural recovery and restoration. *Environmental Management* 24:309–326.

Lovich, J.E., and J.R. Ennen. 2011. Wildlife conservation and solar energy development in the Desert Southwest, United States. *BioScience* 61(12):982–992.

Lowry, J.H., Jr., R.D. Ramsey, K. Boykin, D. Bradford, P. Comer, S. Falzarano, W. Kepner, J. Kirby, L. Langs, J. Prior-Magee, G. Manis, L. O'Brien, T. Sajwaj, K.A. Thomas, W. Rieth, S. Schrader, D. Schrupp, K. Schulz, B. Thompson, C. Velasquez, C. Wallace, E. Waller, and B. Wolk. 2005. Southwest Regional Gap Analysis Project: Final Report on Land Cover Mapping Methods, RS/GIS Laboratory, Utah State University, Logan, Utah. Available at: https://swregap.org/data/landcover/. Accessed May 2022.

Miller, James. 2011. Urban and regional temperature trends in Las Vegas and southern Nevada. *Journal of the Arizona-Nevada Academy of Science* 43(1, 2011):27–39. Available at: https://www.jstor.org/stable/41510543.

Montero-Montoya, R., R. López-Vargas, and O. Arellano-Aguilar. 2018. Volatile organic compounds in air: sources, distribution, exposure and associated illnesses in children. *Annals of Global Health* 84(2):225–238. DOI: http://doi.org/10.29024/aogh.910

Moore-O'Leary, K.A., R.R. Hernandez, D.S. Johnston, S.R. Abella, K.E. Tanner, A.C. Swanson, J. Kreitler, and J.E. Lovich. 2017. Sustainability of utility-scale solar energy–critical ecological concepts. *Frontiers in Ecology and the Environment* 15(7):385–394.

National Climactic Data Center. 2023. Normal Monthly Station Details. Available at: https://www.ncei.noaa.gov/access/us-climate-normals/.

National Drought Mitigation Center. 2022. U.S. Drought Monitor. University of Nebraska-Lincoln. Available at: https://droughtmonitor.unl.edu/CurrentMap.aspx. Accessed September 8, 2022.

National Fire Protection Association (NFPA). 2019. NFPA 69: Standard on Explosion Prevention Systems. Available at: https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=69.

———. 2022. NFPA 855: Standard for the Installation of Stationary Energy Storage Systems. Available at: https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=855.

National Park Service. 2017. Old Spanish National Historic Trail Comprehensive Administrative Strategy. Available at: https://parkplanning.nps.gov/document.cfm?documentID=83540. Accessed November 2022.

Natural Resources Conservation Service (NRCS). 2022. Custom Soil Resource Report for Las Vegas Valley Area, Nevada, Part of Clark County. Townsite Solar 2 Project. Web Soil Survey Online Database. Available at: https://websoilsurvey.sc.egov.usda.gov/App/HomePage.htm. Accessed July 20, 2022.

NatureServe Explorer. 2022. Online Guide to Species and Ecosystems. Available at: https://explorer.natureserve.org/. Accessed August 24, 2022.

Nevada Department of Agriculture. 2021. Nevada Noxious Weed List by Category. Available At: https://agri.nv.gov/Plant/Noxious_Weeds/Noxious_Weed_List/. Last updated February 2, 2021.

Nevada Department of Wildlife (NDOW). 2022a. NDOW Letter Response to Data Request. Accessed May 17, 2022

———. 2022b. Hunt Unit Map. Available at: https://www.ndow.org/wp-content/uploads/2022/03/NDOW_HuntUnits_2022_ArchD_topo.pdf. Accessed July 29, 2022.

Nevada Division of Environmental Protection (NDEP). 2020. *Nevada Statewide Greenhouse Gas Emissions Inventory and Projections, 1990-2040, 2020 Supplemental Report*. Carson City, Nevada: Nevada Division of Environmental Protection.

———. 2022. *Nevada Statewide Greenhouse Gas Emissions Inventory and Projections, 1990-2042*. Available at: https://ndep.nv.gov/uploads/air-pollutants-docs/ghg_report_2022.pdf.

Nevada Division of Forestry. 2023. Natural Resources and Fire Information Portal – Public Viewer. Available at: https://nevadaresourcesandwildfireinfo.com/Map/Public/#whats-your-risk. Accessed April 20, 2023.

Nevada Division of Minerals. 2022. Nevada Mining Claim Listings. Open Data Site. Available at: https://data-ndom.opendata.arcgis.com/pages/mining-claims. Accessed on July 29, 2022.

Nevada Division of Natural Heritage (NDNH). 2022. Nevada Natural Heritage Program Website. Carson City: Nevada Department of Conservation and Natural Resources. Available at: http://heritage.nv.gov/.

Nevada Department of Transportation (NDOT). 2021. Traffic Related Information Access (TRINA) Application. Available at: https://www.dot.nv.gov/doing-business/about-ndot/ndot-divisions/planning/traffic-information. Accessed August 5, 2022.

Nevada Division of Water Resources. 2023. Hydrographic Basin Summary - 167 Eldorado Valley. Available at: water.nv.gov/UndergroundActive.aspx. Accessed September 8, 2023.

Nussear, K.E., Esque, T.C., Inman, R.D., Gass, Leila, Thomas, K.A., Wallace, C.S.A., Blainey, J.B., Miller, D.M., and Webb, R.H. 2009. Modeling habitat of the desert tortoise (Gopherus agassizii) in the Mojave and parts of the Sonoran Deserts of California, Nevada, Utah, and Arizona: U.S. Geological Survey Open-File Report 2009-1102, 18 p.

Regional Environmental Consultants, Inc. 2000. *Final Clark County Nevada Multiple Species Habitat Conservation Plan and Environmental Impact Statement*. Prepared for Clark County Administrative Services. September 2000. Available at: https://www.clarkcountynv.gov/government/departments/environment_and_sustainability/desertconservation_program/current_mshcp.php.

Regional Transportation Commission of Southern Nevada. 2022. Seeing Orange Construction Projects - Online Database. Available at: https://gis.rtcsnv.com/seeingorange/. Accessed October 11, 2022.

Rosentreter, R., M. Bowker, and J. Belnap. 2007. *A Field Guide to Biological Soil Crusts of Western U.S. Drylands*. U.S. Government Printing Office, Denver, Colorado. January 2007.

Russo, D., L. Cistrone, and G. Jones. 2012. Sensory Ecology of Water Detection by Bats: A Field Experiment. *PLoS ONE.* Available at: https://doi.org/10.1371/journal.pone.0048144.

Spector, J. 2019. *The Arizona Battery Explosion is Changing Conventional Wisdom on Safety*. Greentech Media. October 10, 2019. Available at: https://www.greentechmedia.com/articles/read/arizona-battery-explosion-conventional-wisdom-safety.

State of Nevada. 2023. Division of Water Resources – Hydrographic Regions and Basins. Available at: http://water.nv.gov/hydrographicregions.aspx?region=Clark. Accessed April 27, 2023.

SWCA Environmental Consultants, Inc. (SWCA). 2022a. *Cultural Resources Inventory for the Townsite Phase 2 Solar Project, Clark County, Nevada.* Submitted to the Bureau of Land Management Southern Nevada District Office, Las Vegas Field Office. BLM Report Number 5-2914. Accessed May 2022.

———. 2022b. *Botanical Resource Survey Report for the Townsite Solar Project, Clark County, Nevada.* Submitted to the Bureau of Land Management Southern Nevada District Office, Las Vegas Field Office. Accessed August 2022.

———. 2022c. *Visual Resources Technical Report for the Townsite Phase 2 Solar Project Draft, Clark County, Nevada.* Las Vegas: SWCA Environmental Consultants.

———. 2023. *Plan of Development – Townsite Solar 2 Project*. Revision 5. Prepared for Bureau of Land Management, Las Vegas Field Office. Las Vegas: SWCA Environmental Consultants.

University of Texas Fire Research Group. 2017. Battery Fires. Available at: https://www.utfireresearch.com/battery-fires. Austin Campus. Accessed December 3, 2022.

U.S. Census Bureau. 2021a. Quick Facts for Nevada, Clark County, and Boulder City, Nevada. Available at: https://www.census.gov/quickfacts/bouldercitycitynevada. Accessed November 10, 2022.

———. 2021b. Census Data Mapping. Available at: https://data.census.gov/map?layer=VT_2021_040_00_PP_D1&loc=43.3751,-113.1138,z2.6270. Accessed March 2023.

———. 2021c. Historical Population Change Data (1910–2020). Available at: https://www.census.gov/data/tables/time-series/dec/popchange-data-text.html. Accessed October 25, 2022.

U.S. Department of Energy (DOE). 2022. Building a Better Grid: Addressing Climate Change and Bolstering Electric Grid Security Through Planning & Innovation. Available at: https://www.energy.gov/policy/articles/building-better-grid-addressing-climate-change-and-bolstering-electric-grid.

U.S. Department of State and the U.S. Executive Office of the President. 2021. The Long-Term Strategy of the United States: Pathways to Net-Zero Greenhouse Gas Emissions by 2050. Available at: https://www.whitehouse.gov/wp-content/uploads/2021/10/US-Long-Term-Strategy.pdf.

U.S. Energy Information Administration. 2022. *Solar Energy and the Environment*. Last Updated February 25, 2022. Available at: https://www.eia.gov/energyexplained/solar/solar-energy-and the-environment.php. Accessed September 8, 2022.

U.S. Environmental Protection Agency (EPA). 1994. Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations (Executive Order 12898). Available at: https://www.archives.gov/files/federal-register/executive-orders/pdf/12898.pdf. Accessed July 2023.

———. 2020. Viewing WATERS Data using Google Earth KMZ File. WATERSKMZ v1.10. Updated January 19, 2020. Used to View Surface Water Features and Impaired Waters. Available at: https://www.epa.gov/waterdata/viewing-waters-data-using-google-earth. Accessed July 20, 2022.

———. 2022a. NAAQS Table. Available at: https://www.epa.gov/criteria-air-pollutants/naaqs-table. Accessed November 10, 2022.

———. 2022b. Basic Information about the General Conformity Rule. Available at: https://www.epa.gov/general-conformity/basic-information-about-general-conformity-rule. Accessed November 11, 2022.

———. 2022c. *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2021*. Available at: https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks-1990-2021.

———. 2023a. 2020 National Emissions Inventory (NEI) Data. Available at: https://www.epa.gov/air-emissions-inventories/2020-nei-supporting-data-and-summaries.

———. 2023b. Greenhouse Gas Equivalencies Calculator. Available at: https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator#results.

———. 2023c. Environmental Justice Screening and Mapping Tool (Version 2.12). Available at: https://ejscreen.epa.gov/mapper/. Accessed July 2023.

U.S. Fish and Wildlife Service (USFWS). 2009. *Desert Tortoise (Mojave Population) Field Manual (Gopherus agassizii)*. Region 8, Sacramento, California. Available online at: https://www.fws.gov/sites/default/files/documents/Desert-Tortoise-Field-Manual.pdf.

———. 2011. Revised Recovery Plan for the Mojave Desert Tortoise (*Gopherus agassizii*). Region 8, Pacific Southwest Region. Sacramento, California. Pp. 246. May 6, 2011.

———. 2018. *Preparing for Any Action that May Occur within the Range of the Mojave Desert Tortoise (Gopherus agassizii).* Version October 26, 2018. Available at: https://fws.gov/media/mojave-desert-tortoise-pre-project-survey-protocol-2019.pdf.

———. 2019. *Programmatic Biological Opinion for Bureau of Land Management Activities Adversely Affecting 17 Listed Species and Critical Habitat.* Southern Nevada Fish and Wildlife Office. File No. 08ENVSOO-2019-F-0153.

———. 2020. Eagle Surveys Memorandum. Washington, D.C. April 21, 2020.

———. 2022a. Biological Opinion on the Townsite Solar 2 Project, Clark County, Nevada. Southern Nevada Fish and Wildlife Office. File No. 2022-0035189-S7-001. Accessed October 18, 2022.

———. 2022b. Information for Planning and Consultation (IPaC) Online Screening Tool. Available at: https://ecos.fws.gov/ipac/. Accessed May 16, 2022.

———. 2022c. *Golden Eagle*. Search for a Species Online Database Search. Available at: https://www.fws.gov/species/golden-eagle-aquila-chrysaetos. Accessed August 5, 2022.

———. 2022d. National Wetlands Inventory Website. U.S. Department of the Interior, Fish and Wildlife Service, Washington, D.C. Available at: https://www.fws.gov/program/national-wetlands-inventory. Accessed August 5, 2022.

———. 2023. Adjustment for 2023 Mojave Desert Tortoise Remuneration Fees Collected under Endangered Species Act Section 7 Biological Opinion. Southern Nevada Fish and Wildlife Office. January 20, 2023.

U.S. Geological Survey (USGS). 2016. National Land Cover Database (NLCD). Available at: https://www.usgs.gov/node/279743. Accessed April 27, 2023.

———. 2022. National Water Information System: Mapper. Available at: https://maps.waterdata.usgs.gov/mapper/index.html. Accessed July 15, 2022.

Western Regional Climate Center (WRCC). 2012. *Boulder City, Nevada. Period of Record General Climate Summary*. Available at: https://wrcc.dri.edu/cgi-bin/cliMAIN.pl?nv1071. Accessed August 29, 2022.

Wildlife Acton Plan Team. 2012. Nevada Wildlife Action Plan. Nevada Department of Wildlife, Reno. Available at: https://www.ndow.org/blog/2012-nevada-wildlife-action-plan/#. Accessed April 27, 2023.

APPENDIX C

Applicant-Committed Environmental Protection Measures

Applicant-Committed Environmental Protection Measures

The following are applicant-proposed EPMs. These measures are subject to change based on the findings of technical analyses and the analysis during the NEPA process.

As part of the construction, O&M, and decommissioning activities, the applicant has identified the following EPMs to be implemented:

1. Air Quality

- Water would be applied to the ground during the construction and use of the Project Area, access roads, and other disturbed areas as necessary to control dust.
- A Dust Control Operating Permit from Clark County Department of Environment and Sustainability, Division of Air Quality would be obtained prior to construction, and requisite dust control measures and BMPs would be implemented during the Proposed Project.
- Appropriate water erosion and dust-control measures would be implemented to prevent an increased dust and sediment load to ephemeral washes around the construction site and to comply with Clark County dust control requirements. Dust during construction would be controlled and minimized by applying water and/or BLM-preapproved palliatives.
- The applicant would employ BMPs to protect the soil surface by covering or binding soil particles. The Proposed Project would incorporate erosion-control measures required by regulatory agency permits and contract documents as well as other measures selected by the contractor. Project specific BMPs would be designed by the contractor and included in the Project SWPPP. BLM has allowed the use of several dust palliatives on other projects within the Southern Nevada District. If dust palliatives are used in place of water for the Proposed Project, the total amount of water needed during construction would be reduced. The applicant may opt to use such palliatives, as authorized by the BLM for the Proposed Project. The soil binder/dust palliatives that are proposed for the Project, and which BLM previously has allowed are:
    - Road Bond 1000
    - For roads and heavy traffic areas: Soil Cement
    - For nontraffic areas on finer soils; Formulated Soil Binder FSB 1000
    - For nontraffic areas on sandier/rocker soils: Plas-Tex

2. Cultural Resources

- Cultural resources are defined as buildings, sites, structures, or objects, each of which has historical, architectural, archaeological, cultural, and/or scientific importance. Numerous laws, regulations, and statues, on both the federal and State levels, seek to protect and target the management of cultural resources.
- In consultation with BLM and with SHPO concurrence, any areas which contain cultural resources of significance or whose eligibility for inclusion on the NRHP is unevaluated, would be avoided, mitigated, or “treated” and recorded as appropriate.

3. Fire Protection

- All federal, State, and county laws, ordinances, rules, and regulations that pertain to prevention, pre-suppression, and suppression of fire would be strictly adhered to.

4. Invasive Species and Noxious Weed Management

- Noxious weeds within the construction area are to be addressed by the initiation of mitigation measures in consultation with the BLM noxious weed management specialists.
- All contractor vehicles and equipment would arrive at the work site clean and weed free.
- Prior to being allowed access to vehicles and equipment in the ROW or ancillary facilities, an inspector would ensure that vehicles and equipment are free of soil and debris capable of transporting noxious weed seeds, roots, or rhizomes.
- The linear facilities segment and ancillary facilities would be inspected for noxious weeds prior to vegetation clearing. Any infestations would be recorded for reference to inform post-construction monitoring.
- The contractor would implement the reclamation of disturbed lands following construction as outlined in the Reclamation Plan.
- Continuing revegetation efforts would ensure adequate vegetative cover to prevent the invasion of noxious weeds.

5. Reclamation

- For temporary disturbance areas that require clearing and/or grading work, restoration and reclamation procedures will follow the Restoration Plan and will be based on site-specific requirements and techniques commonly employed at the time the area is to be reclaimed and/or based on the Restoration Plan and would include regrading, top soiling, and revegetating all disturbed areas.
- After construction is complete, disturbed work areas would be graded to the approximate original contour, and the area would be revegetated with BLM-approved seed mixtures per the Restoration Plan. Most postconstruction work would entail scarifying soils to reduce compaction and reseeding.
- A Facility Decommissioning and Site Reclamation Plan will be prepared in accordance with BLM policy and objectives at the time and approved by the BLM, once the operational life of the Proposed Project has been reached.

6. Desert Tortoise

- The applicant would implement the minimization measures and compensatory mitigation requirements included in the Project BO, File No. 2022-0035189-S7-001 (USFWS 2022a) as described in Appendix E.

APPENDIX D

Bureau of Land Management
Standard Stipulations

Bureau of Land Management

The following stipulations must be implemented unless they are not applicable to the Proposed Action. The standard stipulations that include "if applicable, if used, or if constructed" are to be implemented if the Proposed Action includes that activity or design.

1. *General Resource Stipulations*
    a. The Holder shall comply with all applicable local, state, and federal laws and regulations for the protection of resources and the environment, to include but not limited to air, cultural, hazmat, soil, vegetation, water, wildlife.
    b. As part of project reclamation, the Holder will be responsible for ensuring that any boreholes, wells, or other openings in the ground are backfilled and properly covered, according to the Nevada Regulatory Statues.
    c. The Holder shall remove from public land and properly dispose of any and all trash, litter, debris, waste, excess materials, including flagging and signs, or other substances and materials resulting from the use under this authorization. All trash and food items shall be promptly contained within closed, raven-proof containers.
2. *Threatened, Endangered or Candidate Animal Species*
    a. Compliance with the special stipulations below will help to ensure desert tortoises are not impacted:
        i. A speed limit of 25 miles per hour shall be required for all vehicles travelling on existing roads.
        ii. Should a desert tortoise enter the area of activity, all activity shall cease until such a time the animal leaves the area of its own accord.
    b. All drivers must check underneath vehicles and equipment before moving to ensure no tortoise has taken cover underneath parked vehicles.
    c. The Holder will comply with the terms and conditions of the Project BO File No. 2022-0035189-S7-001 as appended to Programmatic BO No. 08ENVS00-2019-F-0153 for this Project. The BO is on file at the Bureau of Land Management, Southern Nevada District Office. The terms and conditions are listed in Appendix E.
    d. The Holder, upon completion of the Proposed Action, must submit Appendix G found in the terms and conditions of the BO (attached in Appendix E). Please forward Appendix G to the BLM, Attn: Wildlife Biologist, 4701 N. Torrey Pines Drive, Las Vegas, Nevada, 89130. Failure to abide by the terms and conditions of the grant and BO, could result in temporary suspension of all activities within your ROW area per 43 CFR 2807.16 and 43 CFR 2807.17.
3. *Cultural and Paleontological Resources*
    a. Any cultural and/or paleontological resources (historic or prehistoric site or object) discovered by the Holder, or any person working on his behalf on public or federal lands shall be immediately reported to the Authorized Officer. Holder shall immediately suspend all operations in the immediate area of such discovery until written authorization to proceed is issued by the Authorized Officer. The Holder will make every effort to protect the site from further impacts, including looting, erosion, or other human or natural damage. In some cases, this may delay activity at the site until the discovery may be recovered, or the project is modified to avoid impacting the find.
    b. An evaluation of the discovery will be made by the Authorized Officer to determine appropriate actions to prevent the loss of significant cultural or paleontological values.

Any decision regarding suitable mitigation measures will be made by the Authorized Officer after consulting with the Holder. The Holder will be responsible for the cost of evaluation. Holder shall be responsible for the resultant mitigation costs.

4. *Hazardous Materials*

   a. If hazardous materials/substances are used or present within the authorized area, the Holder shall immediately notify the Authorized Officer of any release (leaks, spills, etc.) of hazardous substances, toxic substances, or hazardous waste. As required by law, Holder shall have responsibility for and shall take all action(s) necessary to respond to and fully remediate releases (leaks, spills, etc.) within the authorized area. A copy of any report required or requested by any federal, state, or local government agency as a result of a reportable release or spill of any hazardous substances shall be furnished to the Authorized Officer concurrent with the filing of the reports to the involved federal, state, or local government agency.

5. *Survey Monuments*

   a. Holder shall protect all survey monuments found within the authorization area. Survey monuments include, but are not limited to, General Land Office and Bureau of Land Management Cadastral Survey Corners, reference corners, witness points, U.S. Coast and Geodetic Survey benchmarks and triangulation stations, military control monuments, and recognizable civil (both public and private) survey monuments. If any of the above are to be disturbed during operations, the Holder shall secure the services of a Professional Land Surveyor or Bureau cadastral surveyor to perpetuate the disturbed monuments and references using surveying procedures found in the Manual of Instructions for the Survey of the Public Lands of the United States and Nevada Revised Statutes, Chapter 329, Perpetuation of Corners. The Holder shall record such survey in the appropriate county and send a copy to the Authorized Officer. If the Bureau cadastral surveyors or other federal surveyors are used to restore the disturbed survey monuments, the Holder shall be responsible for the survey cost.

6. *Fire and Fuels Management*

   a. Compliance with fire restrictions is mandatory while fire restrictions are in effect (43 CFR 9212). Fire restrictions are generally enacted May through October. Fire restriction orders are available for review at BLM district offices and on the BLM website.

   b. The use of standard fire prevention measures should be practiced at all times (43 CFR 2805.12). Conditions that support wildfires can occur any time of the year in Southern Nevada.

   c. The Holder shall immediately report fires to 911 or (702) 631-2350 and make all accommodations to allow immediate safe entry of firefighting apparatus and personnel.

   d. An Origin and Cause Investigation will be carried out on any human caused fire by BLM law enforcement or their designated representative. To minimize disturbance of potential evidence at the fire scene, the applicant/proponent shall properly handle and preserve evidence in coordination with the BLM. The BLM shall pursue cost recovery for all costs and damages incurred from human-caused fires on BLM lands when the responsible party(s) has been identified and evidence of legal liability or intent exists. Legal liability includes, but is not limited to, negligence and strict liability (including statutory and contractual liability), products liability, etc.

7. Vegetation
    a. Restoration – for any habitat-disturbing activities (ground disturbance outside of maintenance in roads, permanently-disturbed areas, etc.):
    b. For temporary habitat disturbance, the Holder is responsible for filling out BLM's Restoration Plan Template, which describes how temporary disturbance will be restored, to BLM for approval prior to issuance of authorization of habitat-disturbing activities.
    c. For permanent habitat disturbance, the Holder is responsible for developing a decommissioning plan, following BLM's Restoration Plan Template, to be approved by BLM prior to issuance of authorization of habitat-disturbing activities, which describes how permanent disturbance will be restored.
    d. Disturbances will not be released from restoration requirements until the standards are met as described in BLM's Restoration Plan Template. Limit disturbance as much as possible to reduce restoration timelines. Weed Management stipulations must be followed.
    e. Any use of seed or native plant materials will be approved by BLM in advance. Plant materials must originate from the appropriate Seed Transfer Zone and have appropriate seed tags, evidence of permits, and be weed-free.

Cacti, Yucca, and Succulent Species:

    a. For temporary habitat disturbance, cacti, yucca, or other succulent species that cannot be avoided must be salvaged by a BLM-approved contractor, stored appropriately during construction, and then planted back into the disturbed area in natural patterns and densities after construction. Survival standard is 80%. Coordinate with BLM.
    b. For permanent habitat disturbance, cacti, yucca, or other succulent species in permanent disturbance areas must be salvaged by a BLM-approved contractor and moved to a BLM storage facility; in some circumstances when there are too many plants to be used, BLM may issue a forestry permit for destruction of these species. Coordinate with BLM.

8. *Nonnative and Invasive Species and Noxious Weeds*
    a. Weed Management Plan - is required for any ground-disturbing activities greater than 1 acre:
    b. The Holder is responsible for filling out BLM's Weed Management Plan Template, to be approved by BLM prior to issuance of authorization of initial habitat-disturbing activities, which describes how weed treatment will be conducted.
    c. The Holder is, for the lifetime of the responsibility for the disturbance or ROW, responsible for:
        i. Surveying for, and treating, all noxious weeds within the disturbed area during biologically appropriate times and before the noxious weeds have gone to seed.
        ii. Surveying for, and treating, non-native weeds within the disturbed area during biologically appropriate times and before the non-native weeds have gone to seed. Non-native weeds that were common in the project area prior to disturbance must be kept at levels (cover and density) less than or equal to pre- disturbance. Non-native weeds that were not common or non-existent in the project area prior to disturbance must all be treated (this also applies to new introductions that spread off the disturbed area/ROW).
        iii. Monitoring for, and reporting to BLM, non-native and noxious weeds occurrence, spread, and treatment (providing treatment data).

d. Any new detections of non-native or noxious weeds shall be reported to the SND Weed Management Specialist immediately (702-515-5000) to determine best course for treatment.

e. The use of pesticide treatment requires the Holder to coordinate with the BLM SND weed management specialist (702-515-5000) and prepare, submit, obtain, and maintain a pesticide use proposal (PUP) to utilize pesticides for project activities. The proponent shall submit a new PUP 6 months prior to their current PUP's expiration date.

f. To reduce the accidental spread of non-native and noxious weeds, the Holder and any contractors shall avoid or minimize all types of travel through state-listed noxious weed-infested areas that can be carried to the project area. To minimize the threat of spreading noxious weeds, project-related equipment (i.e., undercarriages and wheel wells) should be cleaned of all mud, dirt, and plant parts before moving into relatively weed-free areas or out of relatively weed-infested areas. Project workers shall inspect, remove, and dispose of weed seeds and plant parts found on their clothing and personal equipment, bag the product, and dispose of it in a dumpster. If you have questions, consult with the BLM SND noxious weed coordinator.

g. During construction and maintenance activities the Holder shall:

   i. Review the annual weed inventory prior to any ground disturbance.

   ii. Limit the size of any vegetation and/or ground disturbance to the absolute minimum necessary to perform the activity safely and as designed.

   iii. Begin activities in weed-free areas whenever feasible before operating in weed-infested areas.

   iv. Locate equipment storage, machine and vehicle parking or any other area needed for the temporary placement of people, machinery and supplies in areas that are relatively weed-free.

   v. Avoid or minimize all types of travel through weed-infested areas or restrict major activities to periods of time when the spread of seed or plant parts are least likely.

h. If landscaping is part of the project design, the Holder will ensure that landscaping does not contain non-native species or state-listed noxious weeds, such as fountaingrass (*Pennisetum setaceum*).

9. *Mineral Resources*

a. If construction activities produce excess mineral materials from within the boundaries of the Proposed Action, the mineral materials must be used within the boundaries of the Proposed Action or stockpiled within the boundaries of the Proposed Action for future disposal by the BLM.

b. If construction activities require that excess mineral materials be exported from within the boundaries of the Proposed Action as they are generated, then written authorization, a mineral material sales contract, a free-use permit, etc. must be obtained from the BLM by the Holder prior to exporting the excess mineral materials from within the boundaries of the Proposed Action.

c. If mineral materials are to be stockpiled on-site for a future disposal, specific BLM use authorization in the form of a written authorization, mineral material sales contract, free-use permit, etc. must be obtained from the BLM prior to exporting the excess mineral materials from within the boundaries of the Proposed Action.

10. *Migratory Birds*

    a. Projects that require ground disturbance or actions that could affect nesting birds, should try to be scheduled outside of the bird breeding season. The breeding season in the SND generally occurs from February 15 to August 31. If a project cannot be scheduled outside of those dates, a qualified biologist may be required to conduct a survey for nesting birds. If nesting birds are found, methods to reduce project impacts to nesting birds will be developed in coordination with the BLM.

    b. Any infrastructure for projects will be designed and constructed in a manner that does not allow open pipes that birds or other wildlife could be trapped in. This includes fencing, gates, or other materials with open holes. All open pipes will be capped or secured so that wildlife cannot access.

    c. If lighting is installed on buildings or required by the FAA, lighting on buildings should be down-shielded and those structures/towers required by FAA to have lighting installed, should have flashing lights with the minimum intensity required by the FAA to prevent migratory bird collisions.

    d. If project involves power lines and/or power line posts, the Holder shall follow Avian Power Line Interaction Committee (APLIC) guidelines, *Suggested Practices for Avian Protection on Power Lines* (2006) and *Reducing Avian Collisions with Power Lines* (2012) to reduce this risk through facility design and comply with MBTA and other federal wildlife laws, due to potential for electrocution, collision, and nesting/perching by migratory birds on overhead power lines.

    e. If guy wires are used on structures (including power line posts and communication towers) they must be marked with bird diverters, so they are visible to prevent injury/mortality to birds through collision.

11. *Fish and Wildlife, Excluding Federally Listed Species*

    a. If artificial water sources are used, ensure that they have a properly installed and designed escape ramp to allow wildlife to flee in the event of accidental entrapping.

    b. Project supplies or equipment where wildlife could temporarily hide will be inspected prior to moving them to reduce the potential for injury to wildlife. Supplies and equipment that cannot be inspected, or from which wildlife cannot escape or be removed, will be covered or otherwise made secure from wildlife intrusion or entrapment at the end of each work day.

    c. If any Gila monsters are encountered during project construction, they must be reported immediately to the Nevada Division of Wildlife at (702) 486-5127.

12. *Wild Horse and Burro*

    a. If wild horses and/or burros are encountered in or near the authorized area do not feed, harass, or otherwise interact with the animal. Report sick or injured animals, or violations to animals to the BLM immediately.

13. *Recreation*

    a. Unless expressly stated, a land use authorization does not create an exclusive right of use of an area by the holder. The holder shall not interfere with other valid uses of the federal land by other users, such as casual recreationists.

APPENDIX E

Project Biological Opinion

United States Department of the Interior

FISH AND WILDLIFE SERVICE
Southern Nevada Fish and Wildlife Office
4701 North Torrey Pines Drive
Las Vegas, Nevada 89130

IN REPLY REFER TO:
2022-0035189-S7-001

October 18, 2022
Sent electronically

Memorandum

To: Assistant Field Manager
Las Vegas Field Office
Bureau of Land Management
Las Vegas, Nevada

From: Field Supervisor
Southern Nevada Fish and Wildlife Office
Las Vegas, Nevada

GLEN KNOWLES
Digitally signed by GLEN KNOWLES
Date: 2022.10.18 14:06:31 -07'00'

Subject: Biological Opinion on the Townsite Solar 2 Project, Clark County, Nevada

This document transmits the U.S. Fish and Wildlife Service's (Service) biological opinion based on our review of the proposed Townsite Solar 2 project and its effects on the federally threatened Mojave desert tortoise (*Gopherus agassizii*), in accordance with section 7 of the Endangered Species Act of 1973, as amended (Act) (16 U.S.C. 1531 et seq.). We received your July 5, 2022 request for formal consultation on July 7, 2022.

We have based this biological opinion on information that accompanied your July 5, 2022 request for consultation, including the biological assessment prepared for the Bureau of Land Management (BLM) by Allied Pacific Partners, LLC (BLM 2022); correspondence between the Service and the BLM; interagency section 7 consultation regulations in 50 CFR Part 402; scientific publications, articles, and reports; and our files.

cc: Supervisory Biologist, Habitat, Nevada Department of Wildlife, Las Vegas, Nevada

BIOLOGICAL OPINION

On July 5, 2022, the U.S. District Court of the Northern District Court of California vacated the 2019 regulations implementing section 7 of the Endangered Species Act (ESA). On September 21, 2022, the Ninth Circuit Court of Appeals granted a request to stay the U.S. District Court of Northern California's July 5, 2022, order that vacated the 2019 ESA regulations. As a result, the 2019 regulations are again in effect, and the Service has relied upon the 2019 regulations in rendering this biological opinion. However, because the outcome of the legal challenges to 2019 ESA Regulations is still unknown, we considered whether our substantive analyses and conclusions in this consultation would have been different if the pre-2019 regulations were applied. Our analysis included the prior definition of "effects of the action," among other prior terms and provisions. We considered all the “direct and indirect effects” and the “interrelated and interdependent activities” when determining the “effects of the action.” As a result, we determined the substantive analysis and conclusions would have been the same, irrespective of which regulations applied.

Consultation History

On April 24, 2022, Allied Pacific Partners, LLC (APP) obtained an official species list from the Service’s Information and Planning for Consultation website (Project Code: BWDSSKZD45FA7HMTRUVHJ2J5A4). The biological assessment for the Townsite Solar 2 project, provided to the Service on July 7, 2022, lists these species, their status, their proximity to the proposed project areas, and the subsequent need for consultation on each.

On July 7, 2022, the Service received BLM’s biological assessment and request to initiate formal consultation for the Townsite Solar 2 Project to address potential adverse effects to the desert tortoise, at which time formal consultation was initiated.

On July 29, 2022, the BLM met with the Service to discuss remuneration fees for the permanent and temporary disturbance areas, raven control, decommissioning and management plans, and translocation. Both agencies agreed that if desert tortoises are found during project activities, they would need to be translocated away from the site due to the small area of remaining habitat in the area. The Paiute-Eldorado Valley Area of Critical Environmental Concern (ACEC) was discussed as a possible recipient site for translocated tortoises.

DESCRIPTION OF THE PROPOSED ACTION

Action Area

The Townsite Solar 2 project would be located within Eldorado Valley, approximately 5 miles southwest of Boulder City, Nevada, and 2 miles south of Interstate 11 (Figure 1). The action area consists of approximately 85 acres of undeveloped land managed by the BLM, with some remnants of old roads and tracks present. Surrounding the action area is the Townsite Solar 1 facility (located on Boulder City-owned land, completed in 2022) to the south, quarry operations to the west, and a gun range and mining operations to the north.



Figure 1. Location of the proposed Townsite Solar 2 Project (BLM 2022).

Proposed Action

Skylar Energy Resources, LLC (Applicant) proposes to construct, operate, maintain, and decommission the Townsite Solar 2 project, a 19-megawatt (MW) photovoltaic solar generating facility (Project). The Project would include three components: (1) a solar field with photovoltaic (PV) solar panels and a 35 MW/140 MWh battery energy storage system (BESS), (2) a temporary laydown area, and (3) an 80-foot right-of-way (ROW) corridor with access roadway and transmission lines (Figure 2). The Project would be located on approximately 85 acres of BLM-managed land and would use the gen-tie transmission infrastructure at the adjacent Townsite Solar 1 facility on Boulder City land to interconnect into the Western Area Power Administration (WAPA) Mead 230 kV substation. All Project components would occur within a 458-acre area that was surveyed for desert tortoises.

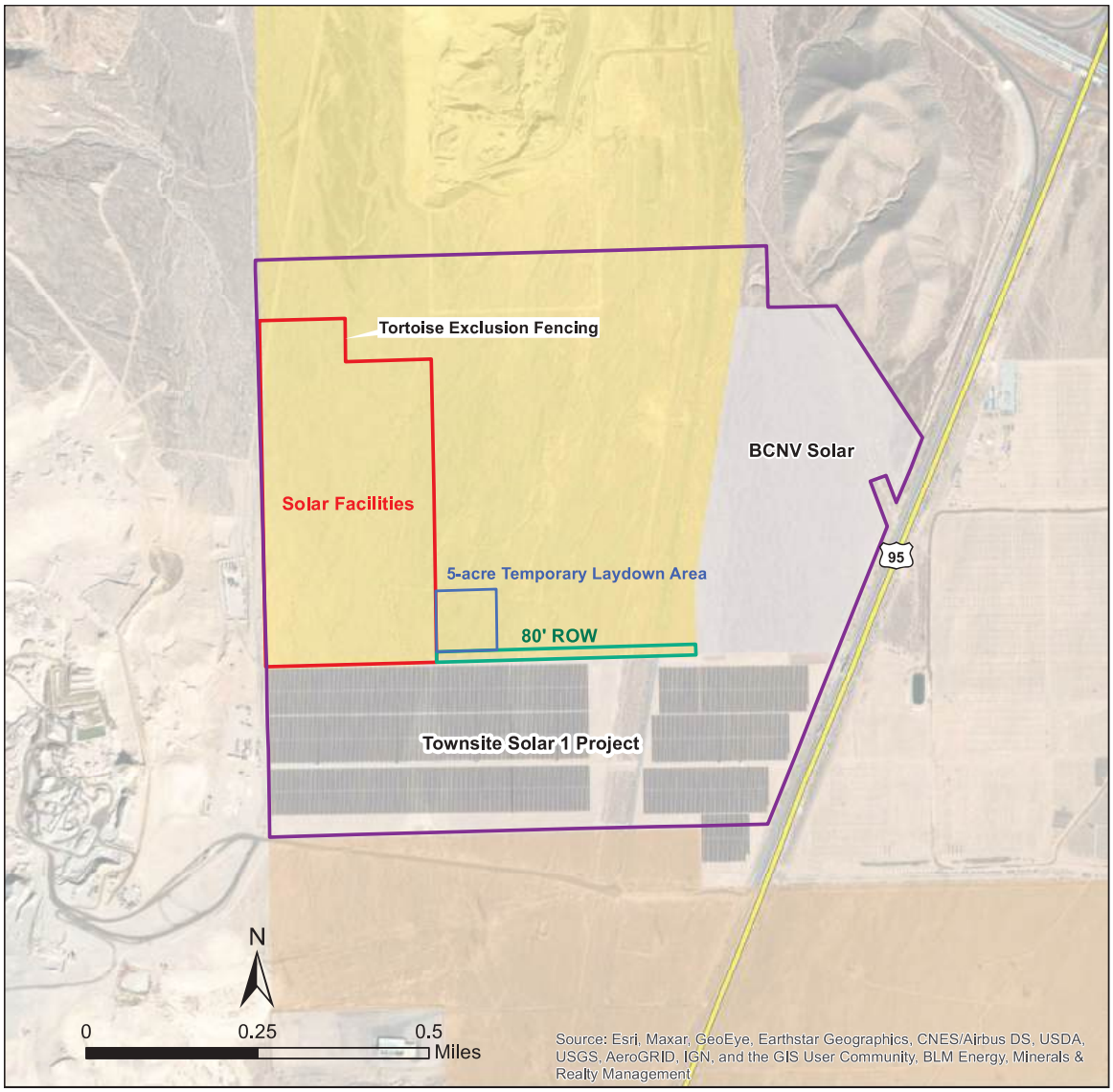


Figure 2. Proposed project overview map (BLM 2022).

Project Components

The solar field component of the Project would occur on 76 acres of the action area and would be enclosed within permanent desert tortoise exclusion fencing (Table 1). The following facilities would be located within this boundary:

- Solar array blocks consisting of PV modules mounted on fixed-tilt mounting systems and/or single-axis, horizontal tracker mounting systems supported by driven steel posts or other embedded foundation design
- Meteorological stations, and if tracker technology is utilized, up to five meteorological towers (steel lattice), approximately 30 feet high, mounted on concrete foundations would be installed around the perimeter of the solar field
- Lithium-ion BESS
- Interior access ways and a perimeter road
- Direct current collection system and Power Conversion Stations (PCSs) to collect power from the array blocks
- Underground 34.5 kV alternating current collection system to convey electricity from the PCSs to the switching station located on non-federal land at the Townsite Solar 1 facility
- Project security using a combination of perimeter security fencing, controlled access gates, onsite security patrols, minimal/motion-sensor lighting, electronic security systems and/or remote monitoring
- Drainage control structures
- Fiber optic communications cable installed underground or on overhead lines along the Distribution Line or the gen-tie transmission line

The temporary laydown area and 80-foot ROW corridor components of the Project would occur on approximately 9 acres of the Project area and would be outside of the permanent desert tortoise exclusion fencing (Table 1). The following elements would be located within these boundaries:

- An approximately 5-acre temporary construction mobilization and laydown area near the southeastern border of the proposed Project for temporary installation of construction trailers, construction workforce parking, above ground water tanks, materials receiving, and materials storage (graded/compacted earth);
- An 80-foot-wide by 2,000-foot-long linear ROW, approximately 4 acres in size, in which the following elements would be constructed:

- 30-foot-wide access roadway (graded and covered with a gravel base) for access during construction and operations
- 50-foot ROW for auxiliary power distribution line (25 feet on each side of the road offset)
- A 34.5 kV interconnecting cable (overhead and buried within the 50-foot roadway offset)
- Fiber optic line (overhead within the 50-foot roadway offset).

Table 1. Proposed acres of new disturbance for the Project components.

Project Component	Tortoise Exclusion Fencing	Temporary Disturbance (acres)	Permanent Disturbance (acres)
Solar Field	Permanent	0	76
Laydown Area	None or Temporary	5	0
80-Foot ROW Corridor	None or Temporary	0	4
Total (acres)		5	80

The following Townsite Solar 1 facilities were completed at the end of 2021. These facilities would be shared with the proposed Project:

- An approximately 190,000 square-foot substation
- A 3-mile 230 kV gen-tie line owned by Townsite Solar 1 connecting the substation to WAPA's Mead 230 kV substation and a 20-foot-wide road along the Townsite gen-tie used for periodic maintenance.
- An existing water pipeline that would deliver reclaimed water from the Boulder City Wastewater Treatment Plant to be used for dust control during construction and operations and maintenance (O&M). Additional water, if needed, would be obtained from existing commercial sources.
- Site access from an existing access road. The Applicant will not be building or improving any roads outside the solar field other than the ROW corridor access road.

The Project will occur over several phases including construction, O&M, and decommissioning. We provide a summary of these phases below, and a more detailed description of Project activities can be found in the BA and draft Plan of Development (BLM 2022).

Project Construction

Construction is anticipated to commence in the first quarter of 2023 and continue for 9-12 months. The Project onsite construction workforce is anticipated to be an average of 200-400 construction workers, with a peak not expected to exceed 500 workers at any given time. The

onsite construction workforce would consist of laborers, craftsmen, supervisory personnel, support personnel, and construction management personnel. All vehicles would access the Project site using existing paved roads (i.e., US-93, I-11, and US-95). Construction generally would occur between 5:00 am and 5:00 pm and may occur seven days per week. Additional hours may be necessary to make up schedule deficiencies, or to complete critical construction activities. For example, during hot weather, it may be necessary to start work earlier (e.g., 3:00 am) to avoid work during high ambient temperatures. Additionally, construction activities would require some night activity for installation, service or electrical connection, inspection, and testing activities. Lighting during these nighttime activities would be minimal and lights would be focused downward to minimize impacts on wildlife.

Prior to construction, the limits of construction disturbance areas would be determined by surveying and staking. Permanent desert tortoise exclusion fencing would be installed around the perimeter of the solar field (76 acres). Temporary tortoise exclusion fencing may be installed in some areas around the laydown area and 80-foot ROW corridor to reduce the number of biological monitors required onsite (9 acres). Once installed, authorized biologists would conduct clearance surveys to remove and translocate any tortoises within the Project area.

Within the solar field, areas would be graded and existing vegetation would be worked into the underlying surface soils using the disk and roll technique. In areas where the terrain is not suitable for disk and roll, conventional cut and fill grading would be used to prepare the relevant area. Vegetation would be permanently cleared from roadways, access ways, and areas where concrete foundations are used for the inverter equipment, substations, and O&M facilities. A 10-foot-wide fire break would be established around the outside of the perimeter fence and maintained clear of vegetation. All earthwork required to install drainage control detention basins, access roads, and foundations for Project-related buildings would be balanced on site. Trenching would be required for placement of collector lines and would vary between 2 to 3 feet wide and 2 to 3 feet deep, depending on the number of conductors and voltage of equipment needed to comply with applicable electrical codes.

Concrete would be poured in place for equipment and building foundations, fence footing, and miscellaneous small pads. Aggregate material would be used for the trench backfill, parking lot, and substation area (and if determined necessary, for the perimeter road and access roads), and rip-rap material would be required for erosion control. The Applicant would determine a source for these materials that would be presented to the BLM for review and approval. An estimated 25 acre-feet of water would be required over the proposed Project construction period for dust control. Water for the Project would be obtained from commercial water sources with existing water rights, including Boulder City.

Solar equipment would be installed on steel posts to which steel table frames would be attached. The solar PV modules would be mounted on fixed-tilt mounting systems and/or single-axis horizontal tracker mounting systems. Mounted PV modules, inverters, and transformers would be combined to form array blocks of 1.25 to 2.5 MW. Solar field assembly would require trucks, small cranes, tractors, and forklifts.

Outside the permanent desert tortoise exclusion fencing, the temporary laydown area would be

mowed and compacted. Along the ROW corridor, the area for roadway construction would be graded and overlayed with gravel. Trenching would be required to install underground powerlines. Overhead powerlines would be installed with concrete foundations and wood, metal, or concrete poles. At the end of construction, temporary desert tortoise fencing, if it had been installed, would be removed. Temporary disturbance areas would be reclaimed in accordance with the Restoration Plan developed with the BLM.

Project Operation and Maintenance

Operation of the facility is expected to start the fourth quarter of 2023 and to continue for 30 years or longer. Operation would require a workforce of up to 2-3 full time-equivalent positions. This workforce would include administrative and management personnel, operators, and security and maintenance personnel. Employees would be based at an O&M building at a nearby solar facility or at an office in the surrounding area.

A solar PV project uses no process water, gas, or fuels for the power generation process. Proposed water consumption during O&M would require up to 2-3 acre-feet per year for panel washing and dust control. The maintenance protocol is mainly routine inspections. During the first year of operation, inspections would occur more frequently (quarterly) to address settling and electrical termination torque. Subsequently to year one, inspections would be performed less frequently. Inspections would include visual inspections of all major facilities including, but not limited to, transmission lines, roadways, vegetation, and backup power. At designated intervals, approximately every 10 to 15 years, major equipment maintenance would be performed. Inspections of permanent desert tortoise exclusion fencing would occur in accordance with Service (2009) regulations and requirements.

Operations and maintenance would require the use of vehicles and equipment including crane trucks for minor equipment maintenance. Additional maintenance equipment would include forklifts, manlifts, and chemical application equipment for weed abatement and BLM-approved soil stabilizer treatment in the reclamation area. Pickup trucks would be in daily use on the site. No heavy equipment would be used during normal plant operation.

Project Decommissioning

The Project would operate at a minimum for the duration of the 30-year ROW with the BLM. At the end of the Project's planned service life, operations would either cease or be renewed. If the Project were to be decommissioned, a final Decommissioning and Reclamation Plan would be developed, and all structures and equipment would be removed from the site. Revegetation with native species would be completed as needed.

Proposed Minimization Measures

The following minimization measures would be implemented as part of the Project to avoid, minimize, and offset potential adverse effects to the desert tortoise. These measures include general best management practices as well as strategies specific to desert tortoise conservation. The Applicant proposed these measures and the Service and BLM participated in their

development.

Construction Minimization Measures

The following measures will be implemented to reduce effects of the proposed action on desert tortoise and other species during construction activities:

1. **Worker Education and Awareness Plan (WEAP).** A WEAP will be presented to all personnel onsite during construction, O&M, and decommissioning before being permitted to work in the project area. This program will address (1) the types of construction activities that may affect the desert tortoise, (2) the required desert tortoise protective measures, (3) desert tortoise biology and threats, (4) legal protections and penalties, and (5) reporting requirements.

2. **Authorized desert tortoise biologist.** The Applicant will employ Authorized Desert Tortoise Biologists (ADTBs) and desert tortoise monitors to ensure proper implementation of the conservation measures outlined in this biological opinion. Prior to starting construction, ADTBs will submit documentation of authorization from the Service and approval from the Nevada Department of Wildlife. Potential ADTBs will submit their statement of qualifications to the Service for approval. Guidance on ADTB qualifications is outlined in the *Desert Tortoise (Mojave Population) Field Manual* (Service 2009). Desert tortoise monitors will be approved by the ADTB.

   All ADTBs will have the right and responsibility to halt any activities that are in violation of the conservation measures intended to protect the desert tortoise. ADTBs will report to the Applicant, the BLM, and the Service concurrently regarding all compliance issues and take of desert tortoises.

3. **Construction area flagging.** Boundaries of the proposed action area will be demarcated as appropriate so that all disturbance activities stay within the BLM ROW areas. A biological monitor will escort all survey crews on site prior to construction. In cases where construction vehicles are required to go off existing roads, a biological monitor (on foot) will precede the vehicles.

4. **Desert tortoise exclusion fencing.** Permanent Service-approved desert tortoise exclusion fencing will be installed around the 76-acre solar field (affixed to security fencing) (Service 2009). The Applicant may install temporary tortoise fencing in areas outside the solar field (i.e., the temporary laydown area and 80-foot ROW corridor) to reduce the number of biologists required on site during construction. ADTBs, assisted by biological monitors, will monitor installation of security fencing with affixed permanent or temporary desert tortoise exclusion fencing. Fence specifications will be consistent with those approved by the Service (Service 2009).

   Desert tortoise exclusion fencing will be inspected daily and 24 hours following storm events while active construction is taking place within its boundaries. All fence breaches will be repaired within 24 hours.

5. **Biological monitoring.** Under supervision of an ADTB, biological monitors will be present during all construction operations outside the desert tortoise exclusion fencing (on approximately 8.67 acres). Desert tortoise monitors will provide oversight to ensure proper implementation of conservation measures. Typically, this includes escorting vehicles along the ROW, clearing the area of desert tortoises ahead of construction equipment, and checking under parked vehicles before they are operated. Temporary Service-approved desert tortoise exclusion fencing may be used to reduce the number of ADTBs and biological monitors in these areas.

6. **Desert tortoise clearance surveys.** After installation of the exclusion fencing and prior to any construction activities, desert tortoise monitors, supervised by ADTBs, will conduct clearance surveys to locate and remove all desert tortoises from within fenced Project areas. Clearance survey methods will meet the standards established by the Service (Service 2009). These methods currently require 100 percent coverage of the Project area, with a focus on locating all desert tortoises above and below ground. An ADTB will also excavate burrows potentially containing desert tortoises located in the Project area to locate and remove all desert tortoises and desert tortoise eggs. No surface-disturbing activities shall begin until two consecutive surveys find no live tortoises. In areas where a live tortoise is found, surveys will be repeated until the two-pass standard is met. For areas not enclosed by desert tortoise exclusion fencing, an ADTB will oversee pre-construction clearance surveys to detect and remove tortoises from these areas within 24 hours prior to onset of surface disturbance.

7. **Desert tortoise translocation.** If tortoises are found in or around the action area, the Applicant will contact the BLM and the Service immediately for disposition. Short-distance relocations will not be possible because of the small area of remaining habitat outside the Project perimeter and its isolation from other suitable desert tortoise habitat. If tortoises are found, an ADTB will perform a health assessment and draw blood to determine whether the individuals suffer from upper respiratory tract disease. Depending on current weather conditions, time of year, and the health assessment of the individual(s), the BLM and the Service may decide to either translocate desert tortoises immediately to the Paiute-Eldorado Valley ACEC or transfer and hold the individual(s) at the Desert Tortoise Center until conditions improve and they can be released.

8. **Trenching.** If trenches are needed in the area outside the desert tortoise exclusion fencing, trenches will have tortoise escape ramps built to Service standards and placed at least every 250 feet. An ADTB or biological monitor will inspect the trench each morning prior to commencement of construction activities to ensure no tortoises are trapped inside the trench.

9. **Trash and litter control.** Trash and food items will be disposed of properly in predator-proof containers with resealing lids. Trash will be emptied and removed from the project area periodically. Trash removal reduces the attractiveness of the area to opportunistic predators such as ravens, coyotes, and foxes.

10. **Raven monitoring and management.** The Applicant will implement the *Common*

Raven Management Plan for Energy Development Within the BLM Southern Nevada District (BLM 2018), which includes information on common raven biology, avoidance and minimization measures, and monitoring and reporting. The common raven (*Corvus corax*) is a subsidized predator of desert tortoises and other sensitive species, and this plan details a program to monitor raven presence, implement raven control measures as needed, and avoid any Project-related increases in common raven numbers during the Project.

11. **Vehicle access and speed limits.** During construction, site access will be limited to established and Project roadways. No cross-country travel will occur. A maximum speed limit of 25 miles per hour will be maintained while traveling on unpaved access roads. During the desert tortoise active seasons (i.e., April 1 – May 31 and September 1 – October 31), a maximum speed limit of 15 miles per hour will be maintained on roads outside of the permanent desert tortoise exclusion fencing. Speed limit and caution signs will be installed along access and service roads.

12. **Integrated Weed Management Plan.** Prior to construction, an Integrated Weed Management Plan will be developed in consultation with the BLM and will follow the Las Vegas Resource Management Plan (BLM 1998), Noxious Weed Plan (BLM 2006), and the Vegetation Treatments on BLM Lands Programmatic Environmental Impact Statement (PEIS; BLM 2007). The plan will include measures to reduce the propagation and spread of designated noxious weeds, undesirable plants, and invasive non-native species. Measures in the plan will include, but are not limited to, the following:

    - All contractor vehicles and equipment will be required to arrive at the work site clean and weed free. An inspector will ensure that vehicles are free of soil and debris capable of transporting noxious weeds, seeds, roots, or rhizomes.

    - The ROW corridor and ancillary facilities will be inspected for noxious weeds prior to vegetation clearing. Any infestations will be recorded for reference in clearing the ROW and ancillary features for construction and post-construction monitoring.

    - Infested areas will be cleared last to avoid distributing seeds throughout the ROW.

13. **Management Plans.** Prior to construction, the Applicant will work with the BLM to develop and implement management plans for Project activities. Management plans may include, but are not limited to, the following: Bird and Bat Conservation Strategy, Dust Control Plan, Lighting Management Plan, Hazardous Materials and Waste Management Plan, Spill Prevention and Emergency Response Plan, and a Fire Management Plan.

14. **Avian Power Line Protection.** All overhead electrical lines and transmission line poles will be designed and installed in accordance with the Avian Power Line Interaction Committee's (APLIC) Suggested Practices for Avian Protection on Power Lines (APLIC 2006).

15. **Overnight lighting.** Lighting at the facilities will be minimal, focused downward, and activated by motion sensors (not continually on) to reduce impacts to wildlife.

16. **Temporary laydown area restoration.** After Project completion, temporary disturbance areas will be restored as outlined in a BLM-approved Restoration Plan. Restoration efforts will help to recontour work areas, scarify soils to reduce compaction, establish natural vegetation with reseeding, and reduce possible weed infestations in the area.

Operation and Maintenance Minimization Measures

The following minimization measures will be implemented during O&M of the proposed action to reduce effects on the desert tortoise and other species:

17. **Speed limits.** While conducting maintenance activities or traveling outside of the desert tortoise exclusionary fenced area during periods of high desert tortoise activity (i.e., April 1 – May 31 and September 1 – October 31), a speed limit of 15 miles per hour will be maintained. A limit of 25 miles per hour will be maintained during periods of low tortoise activity. This will reduce dust and allow for observation of tortoises in the road. Speed limit and caution signs will be installed along access roads.

18. **Weed and Pest Management.** To address ongoing weed control, the prepared Integrated Weed Management Plan will be implemented throughout O&M of the Project. BLM-approved herbicides, such as Roundup (glyphosate), will be used to control noxious weeds, if required. Pest control may also be required, including control of rodents and insects inside of the buildings and electrical equipment enclosures.

19. **Desert tortoise exclusion fencing inspections.** Permanent desert tortoise exclusion fencing will be inspected weekly and within 24 hours following storm events during the desert tortoise active seasons (April 1 – May 31 and September 1 – October 31). Fencing will be inspected quarterly and within 24 hours following storm events during the desert tortoise inactive season. Any compromised fencing will be repaired within 48 hours of detection and reported to the Service to determine whether additional measures are necessary. Shade structures will also be inspected for effectiveness and adjusted as needed.

Decommissioning Minimization Measures

The same minimization measures used for construction activities will be used for decommissioning activities.

20. **Decommissioning and Reclamation Plan.** If the solar facility is not renewed at the end of the 30-year ROW, the Applicant and the BLM will develop a Decommissioning and Reclamation Plan to remove all structures and revegetate the area.

Compensatory Mitigation

21. **Habitat compensation.** To further offset the adverse effects from the proposed Project,

the Applicant will pay a desert tortoise remuneration fee of $1,002 per acre to the BLM (subject to increase on March 1, 2023), for a total of $85,170 (85 acres * $1,002). These fees will be used to support desert tortoise recovery actions.

ANALYTICAL FRAMEWORK FOR THE JEOPARDY DETERMINATION

Jeopardy Determination

Section 7(a)(2) of the Act requires that Federal agencies ensure that any action they authorize, fund, or carry out is not likely to jeopardize the continued existence of listed species. "Jeopardize the continued existence of" means "to engage in an action that reasonably would be expected, directly or indirectly, to reduce appreciably the likelihood of both the survival and recovery of a listed species in the wild by reducing the reproduction, numbers, or distribution of that species" (50 CFR 402.02).

The jeopardy analysis in this biological opinion relies on four components: (1) the Status of the Species, which describes the current range-wide condition of the Mojave desert tortoise, the factors responsible for that condition, and its survival and recovery needs; (2) the Environmental Baseline, which analyzes the condition of the desert tortoise in the action area, the factors responsible for that condition, and the relationship of the action area to the survival and recovery of the desert tortoise; (3) the Effects of the Action, which determines all consequences to the desert tortoise caused by the proposed action that are reasonably certain to occur in the action area; and (4) the Cumulative Effects, which evaluates the effects of future, non-Federal activities, that are reasonably certain to occur in the action area, on the desert tortoise.

In accordance with policy and regulation, the jeopardy determination is made by evaluating the effects of the proposed Federal action in the context of the current status of the Mojave desert tortoise, taking into account any cumulative effects, to determine if implementation of the proposed action is likely to reduce appreciably the likelihood of both the survival and recovery of the desert tortoise in the wild by reducing the reproduction, numbers, and distribution of that species. Desert tortoise designated critical habitat does not occur in the action area so no adverse modification to desert tortoise designated critical habitat is expected to occur with implementation of the proposed Project.

STATUS OF THE SPECIES

Listing History

The Service listed the Mojave population of desert tortoise (all desert tortoises north and west of the Colorado River in Arizona, Utah, Nevada, and California) as threatened on April 2, 1990 [55 Federal Register 12178].

Recovery Plan

In the revised recovery plan for the desert tortoise, the Service (2011) identified the need for "conservation areas" to protect existing desert tortoise populations and habitat. Box 2 and Figure 2 in the recovery plan (Service 2011) describe and depict these areas in a generalized manner,

respectively.

The revised recovery plan lists three objectives and associated criteria to achieve delisting. The first objective is to maintain self-sustaining populations of desert tortoises within each recovery unit into the future. The criterion is that the rates of population change for desert tortoises are increasing over at least 25 years (i.e., a single generation), as measured by extensive, range-wide monitoring across conservation areas within each recovery unit and by direct monitoring and estimation of vital rates (recruitment, survival) from demographic study areas within each recovery unit.

The second objective addresses the distribution of desert tortoises. The goal is to maintain well-distributed populations of desert tortoises throughout each recovery unit; the criterion is that the distribution of desert tortoises throughout each conservation area increase over at least 25 years.

The final objective is to ensure that habitat within each recovery unit is protected and managed to support long-term viability of desert tortoise populations. The criterion is that the quantity of desert tortoise habitat within each conservation area be maintained with no net loss until population viability is ensured.

The revised recovery plan (Service 2011) also recommends connecting blocks of desert tortoise habitat, such as critical habitat units and other important areas, to maintain gene flow between populations. Linkages defined using least-cost path analysis (Averill-Murray et al. 2013) illustrate a minimum connection of habitat for desert tortoises between blocks of habitat and represent priority areas for conservation of population connectivity.

Threats

The threats described in the listing rule and both recovery plans (Service 1994, 2011) continue to affect the species. The most apparent threats to the desert tortoise are those that result in mortality and permanent habitat loss across large areas, such as urbanization and large-scale renewable energy projects and those that fragment and degrade habitats, such as proliferation of roads and highways, off-highway vehicle activity, wildfire, and habitat invasion by non-native invasive plant species.

We remain unable to precisely quantify how particular threats affect desert tortoise populations relative to other threats. The assessment of the original recovery plan emphasized the need for a better understanding of the implications of multiple, simultaneous threats facing desert tortoise populations and of the relative contribution of multiple threats on demographic factors (i.e., birth rate, survivorship, fecundity, and death rate; Tracy et al. 2004).

For example, we have long known that the construction of a transmission line can result in the death of desert tortoises and loss of habitat. We have also known that common ravens, known predators of desert tortoises, use transmission line pylons for nesting, roosting, and perching and that the access routes associated with transmission lines provide a vector for the introduction and spread of invasive weeds and facilitate increased human access into an area. Increased human access can accelerate illegal collection and release of desert tortoises and their deliberate

maiming and killing, as well as facilitate the spread of other threats associated with human presence, such as vehicle use, garbage and dumping, and invasive plants (Service 2011). Changes in the abundance of native plants, because of invasive weeds, can compromise the physiological health of desert tortoises, making them more vulnerable to drought, disease, and predation.

Five-Year Reviews

Section 4(c)(2) of the Act requires the Service to conduct a status review of each listed species once every 5 years. The purpose of a 5-year review is to evaluate whether the species' status has changed since listing (or since the most recent 5-year review); these reviews, at the time of their completion, provide the most up-to-date information on the range-wide status of the species.

The Service's (2022a) second 5-year review of the status of the desert tortoise summarizes the information from its initial 5-year review (Service 2010) and describes substantive new information since 2011 (from the release of the updated recovery plan) relative to changes in threats, conservation measures, and regulatory mechanisms that pertain to the five listing factors outlined in section 4(a)(1) of the Act. For this reason, we are incorporating the 5-year review of the status of the desert tortoise (Service 2022a) by reference to provide most of the information needed for this section of the biological opinion; because it contains background information that is not in the recent document, we have also incorporated the 2010 5-year review by reference. The following paragraphs provide a summary of the relevant information in the most recent 5-year review. All references to "the 5-year review" in this section of the biological opinion are to the most recent document (Service 2022a), unless otherwise noted.

The 5-year review is replete with references to numerous studies and reports. We have not included references to those studies and reports in the following summary; the full citations are available in the 5-year review.

The 5-year review notes that while the Mojave distinct population segment of the desert tortoise was elevated to species status in 2011 as *Gopherus agassizii*, with most desert tortoises east of the Colorado River recognized as *G. morafkai*, "nine local populations that include *G. agassizii* or hybrids with *G. morafkai* have been genetically identified east of the Colorado River in Arizona." The 5-year review recommends evaluating the federal listing status of the Mojave desert tortoise relative to its current taxonomy and distribution.

In the revised 5-year review, the Service concluded that the "condition of most threats is similar to that described in the previous (Service 2010) status review" and summarized the new information within the context of the five listing factors outlined in section 4(a)(1) of the Act. We summarize that information below.

<u>Factor A: Present or Threatened Destruction, Modification, or Curtailment of Habitat or Range</u>

Various types of anthropogenic impacts continue to cause the loss of desert tortoise habitat. The Service has issued biological opinions or incidental take permits for approximately 74,000 acres of utility-scale solar energy development in occupied desert tortoise habitat. Solar development

has largely occurred outside of desert tortoise conservation areas, as described in the recovery plan (Service 2011).

The 5-year review also describes the Marine Corps' expansion of training onto approximately 167,982 acres of public and private land the Department of Army's plans to expand activities onto approximately 62,045 acres of its western training area in the near future. These activities are in the Western Mojave Recovery Unit.

Legal and illegal cannabis cultivation is causing smaller scale, more widely distributed losses of habitat, particularly in the Western Mojave Recovery Unit; illegal operations are likely to indirectly affect additional habitat because of various types of waste.

Wildfires fueled by invasive grasses have burned extensive areas of desert tortoise habitat. For example, fires in 2020 occurred in desert tortoise habitat in the Mojave National Preserve (Dome Fire, 43,273 acres), Nevada (Meadow Valley Fire, 23,500 acres), and the Red Cliffs Desert Reserve in the Upper Virgin River Recovery Unit (11,000 acres in several fires). The latter fire killed at least 25 desert tortoises.

The 5-year review notes that desert tortoises are "essentially absent" from habitat within 1 kilometer of areas with greater than 10 percent development; "development" includes urban areas, cultivated agriculture, energy facilities, mines and quarries, pipelines, transmission lines, roads and railroads. Approximately 5 percent of modelled desert tortoise habitat within conservation areas had development levels that exceeded this threshold. See Table 3 and Figure 7 in the 5-year review. Desert tortoise populations declined in conservation areas where the density of paved and unpaved roads exceeded 0.75 kilometer per square kilometer; population trends varied at lower density of routes. See Figure 8 in the 5-year review.

Factor B: Overutilization for Commercial, Recreational, Scientific, or Educational Purposes

The 5-year review notes that the Service continues to have little information on threats under Factor B. However, the potential for negative impacts to desert tortoise populations exists from collection and deliberate maiming/killing as a result of human access, vehicles on paved/unpaved roads, and non-motorized recreation. The 5-year review also notes that the effects of research activities permitted for the purposes of enhancing the recovery and conservation of the desert tortoise have been minor while providing valuable information that can be used to recover and improve management of the desert tortoise

Factor C: Disease or Predation

The 5-year review notes that "current research suggests that direct disease management of wild (desert) tortoise populations is less important … than managing factors that affect their habitat and its capacity to support healthy (desert) tortoises. However, management of disease when translocating desert tortoises between populations remains important. As an example of managing habitat, red brome (*Bromus rubens*), which is a non-native invasive plant, negatively affects the health and survival of juvenile desert tortoises.

Badgers (*Taxidea taxus*), coyotes (*Canis latrans*), kit foxes (V*ulpes macrotis*), dogs (*Canis*

familiaris), common ravens, and red-tailed hawks (*Buteo jamaicensis*) prey on desert tortoises. Badgers can have severe effects on desert tortoise populations at the local level; DNA analysis of scats suggest that badgers, coyotes, kit foxes, dogs, and red-tailed hawks may prey on desert tortoises more frequently than previously thought.

Common ravens, because their populations have greatly increased through human subsidies, severely affect the recruitment of desert tortoises into the breeding population through predation on small individuals. In California, management includes the broad-scale removal of common ravens from critical habitat of the desert tortoise.

Factor D: Inadequacy of Existing Regulatory Mechanisms

The BLM continues to face challenges in managing compliance with use of its off-highway vehicle network in the Western Mojave Recovery Unit. As of 2019, the BLM documented 24,518 kilometers of ground transportation linear features in this area, which is more than 2.5 times the 9,651 kilometers designated as open or limited. The BLM has an active program of restoring unauthorized routes and signing open routes.

Unauthorized cattle grazing continues within the Gold Butte National Monument in Nevada. We discussed cannabis farming in California previously in this section.

Factor E: Other Natural or Manmade Factors Affecting its Continued Existence

The 5-year review notes that, in the southwestern United States, 2000 through 2021 was the driest 22-year period in over 1,200 years; drought is likely to continue beyond 2022. Drought reduces the amount of annual plant forage for desert tortoises and, over longer times, will kill shrubs that desert tortoises rely on for cover.

Increased temperatures may affect hatchling sex ratios. Changes in climate may shift the timing of egg production and extend the egg-laying period. This change in egg production may not compensate for changes in the environment, such as the length of time eggs spend above their critical thermal maximum temperature and whether forage is available to support the production of eggs and forage for hatchlings. If climate change results in an overall decrease in reproduction, human-subsidized predation on young desert tortoises, particularly by common ravens, would exacerbate issues with the recruitment of desert tortoises into the breeding population.

Synthesis

Given the reproductive ecology of the desert tortoise, measurable increases in the size of populations will require years.

The Management Oversight Group for the desert tortoise “has taken steps to prioritize and implement actions that would be most effective at facilitating recovery across the range.” The Departments of Defense and the Interior have initiated a Recovery and Sustainment Partnership with the goal of implementing actions that would accelerate recovery of the desert tortoise while reducing the regulatory burden on military installations. The action plan focuses on identifying

ways to accelerate habitat restoration, fencing conservation areas and roadways, and addressing unauthorized routes in the Western Mojave Desert Recovery Unit.

In California, the BLM's Desert Renewable Energy Conservation Plan Land Use Plan Amendment to the California Desert Conservation Act Plan of 1980 included numerous conservation and management actions that addressed issues relevant to the desert tortoise. As part of the land use plan amendment, the BLM established new limits on ground-disturbing activities of 0.1–1.0 percent relative to its lands within desert tortoise conservation areas and mapped linkages between these areas. The land-use plan amendment also increased the amount of land that the BLM manages for conservation in California (e.g., ACECs, California Desert National Conservation Lands, etc.) from 6,118,135 to 8,689,669 acres. All of these areas are not within desert tortoise habitat; however, management as conservation areas will likely benefit desert tortoises indirectly because conservation management would limit subsidies to common ravens and other indirect effects.

The threats that led to the listing of the desert tortoise (i.e., the five-factor analysis required by section 4(a)(1) of the Act) continue. The status of the desert tortoise has continued to decline and most of the previously identified threats continue to affect populations.

In the 5-year review, the Service concluded by recommending that the status of the desert tortoise as a threatened species be maintained because of the large extent of its range and a total number in the "hundreds of thousands of individuals (all size classes) at last estimate."

Recommendations for Future Actions

The 5-year review provided eight recommendations for the highest priority actions over the next 5 years. These recommendations are from the revised recovery plan (Service 2011); their full text is in the 5-year review.

1. More aggressive implementation of habitat restoration, targeted predator control and limitation of subsidies, fencing priority stretches of highways, fire management planning and implementation, and environmental education;

2. Maintaining landscape connectivity and the resilience of desert tortoise conservation areas by managing all desert tortoise habitat for persistence and connectivity, limiting landscape-level disturbance across habitat managed for the desert tortoise by extending surface-disturbance caps similar to those enacted by the BLM in California to the rest of the Mojave desert tortoise's range, maximizing passage under roads, and adapting management based on information from research on: the effects of climate change on desert tortoise habitat, distribution, and population connectivity; the effects of large-scale fires, especially within repeatedly burned habitat, on desert tortoise distribution and population connectivity; the ability of solar energy facilities or similar developments to support desert tortoise movement and presence by leaving washes and native vegetation intact; and the design and frequency of underpasses necessary to maintain functional demographic and genetic connectivity across roads and highways;

3. Increasing law enforcement efforts across the range of the desert tortoise, especially within conservation areas to minimize impacts of habitat destruction and degradation as a result of unauthorized off-highway vehicle use, unpermitted cannabis farms, and trespass grazing;

4. Using population augmentation to help achieve recovery criteria in each of the five recovery units according to the Service's population augmentation strategy;

5. Updating the taxonomy, distribution, and listed status of the species, which we discussed previously in this section;

6. Incorporating updated population trend analysis and climate change and land-use modeling into the next 5-year review to inform management strategies under a framework for ecological adaptation;

7. Sustaining and more fully implementing range-wide monitoring efforts; and

8. Developing a revised spatial decision support system to improve models of threats, recovery actions, and demographics, using up-to-date underlying geospatial data, evaluation of prior conceptual models, and improved operationalization of recovery action terminology.

Core Criteria for the Jeopardy Determination

When determining whether a proposed action is likely to jeopardize the continued existence of a species, we are required to consider whether the action would "reasonably be expected, directly or indirectly, to reduce appreciably the likelihood of both the survival and recovery of a listed species in the wild by reducing the reproduction, numbers, or distribution of that species" (50 Code of Federal Regulations 402.02). We have used the best available information to summarize the status of the desert tortoise with respect to its reproduction, numbers, and distribution.

Reproduction

In the previous 5-year review, the Service (2010) notes that desert tortoises increase their reproduction in high rainfall years; more rain provides desert tortoises with more high quality food (i.e., plants that are higher in water and protein), which, in turn, allows them to lay more eggs. Conversely, the physiological stress associated with foraging on food plants with insufficient water and nitrogen may leave desert tortoises vulnerable to disease and the reproductive rate of diseased desert tortoises is likely lower than that of healthy animals. Young desert tortoises also rely upon high-quality, low-fiber plants (e.g., native annual plants) with nutrient levels not found in the invasive weeds that have increased in abundance across its range. Compromised nutrition of young desert tortoises likely represents an effective reduction in reproduction by reducing the number of animals that reaches adulthood. Consequently, although we do not have quantitative data that show a direct relationship, the abundance of weedy species within the range of the desert tortoise has the potential to affect the reproduction of desert tortoises and recruitment into the adult population in a negative manner.

Various human activities have introduced numerous species of non-native invasive plants into habitat of the desert tortoise. Routes that humans use to travel through the desert (paved and unpaved roads, railroads, motorcycle trails, etc.) serve as pathways for new species to enter habitat of the desert tortoise and for species that currently occur there to spread. Other disturbances of the desert substrate also provide invasive species with entry points into the desert. The abundance and distribution of invasive weeds may compromise, at least to some degree in localized areas across its range, the reproductive capacity of the desert tortoise; the continued increase in human access across the desert likely continues to facilitate the spread of weeds and further affect the reproductive capacity of the species.

Numbers

In the previous 5-year review, the Service (2010) discussed various means by which researchers have attempted to determine the abundance of desert tortoises and the strengths and weaknesses of those methods. Due to differences in area covered and especially to the non-representative nature of earlier study sites, data gathered by the Service's current range-wide monitoring program cannot be reliably compared to information gathered through other means at this time.

Range-wide monitoring from any single year samples a portion of the desert tortoise conservation areas; the conservation areas comprise only a portion of the recovery units. Additionally, any single-year estimate of the number of desert tortoises should be viewed as a snapshot that several variables likely influence. Consequently, considering trends derived from years of range-wide monitoring provides a more accurate view of the status of desert tortoise populations.

Allison and McLuckie (2018) used annual density estimates obtained from range-wide monitoring from 2004 through 2014 to evaluate range-wide trends in the density of desert tortoises over time. Allison and McLuckie (2018) extrapolated the densities of large desert tortoises derived by range-wide monitoring in the conservation areas to all modeled habitat in the recovery unit; the abundance columns in Table 2 contain these extrapolated numbers, which overestimate the number of desert tortoises.

Table 2. Change in desert tortoise abundance in recovery units (Allison and McLuckie 2018)*.

Recovery Units	Modeled Habitat (km^2)	Conservation Area (km^2)	2004 Abundance	2014 Abundance	Annual Trend (percent)
Western Mojave	23,139	6,873	131,540	64,871	-7.1
Colorado Desert	18,024	13,530	103,675	66,097	-4.5
Northeastern Mojave	10,664	4,889	12,610	46,701	-13.1
Eastern Mojave	16,061	3,720	75,342	24,664	-11.2
Upper Virgin River	613	115	13,226	10,010	-3.2

Recovery Units	Modeled Habitat (km^2)	Conservation Area (km^2)	2004 Abundance	2014 Abundance	Annual Trend (percent)
Total	68,501	29,127	336,393	212,343	

* Allison and McLuckie (2018) used modeled habitat within the entire range of the desert tortoise for this estimate. In other discussions in this biological opinion, we used information only from areas of monitored habitat within desert tortoise conservation areas to estimate the number of desert tortoises in the recovery unit.

Distribution

We discussed specific activities that have resulted or will result in the loss of desert tortoise habitat in the Factor A portion of this section of the biological opinion. Here, we summarize their overall effect on the distribution of the desert tortoise.

The 5-year review notes that the absolute amount of desert tortoise range-wide decreased by approximately 163,700 acres between 2005 and 2017, based on sudden changes in LandSat imagery in the trend of the normalized difference vegetation index at image pixels over time. However, several utility-scale solar energy development have been approved or constructed since 2017; additionally, LandSat imagery would not detect areas from which desert tortoises have been or will be translocated that have not undergone changes in vegetation to date.

Attempting to quantify the amount of habitat lost is difficult because of the varying methods used in studies. Also, models depicting desert tortoise habitat cannot differentiate between areas where desert tortoise populations maintain the ability to recruit young animals to breeding age and areas where recruitment has likely not occurred for years.

In summary, human activities have continued to reduce the distribution of the desert tortoise. Most of the losses of habitat have occurred outside of desert tortoise conservation areas, with the exception of those associated with Fort Irwin. The large size of the potential range of the desert tortoise and difficulties associated with determining areas that it actually occupies within that area (i.e., not including areas from which it has been extirpated or that are unsuitable habitat) precludes quantifying its distribution with precision.

ENVIRONMENTAL BASELINE

The regulations implementing section 7(a)(2) define the environmental baseline as the condition of the listed species or its designated critical habitat in the action area, without the consequences to the listed species or designated critical habitat caused by the proposed action. The environmental baseline includes the past and present impacts of all Federal, State, or private actions and other human activities in the action area, the anticipated impacts of all proposed Federal projects in the action area that have already undergone formal or early section 7 consultation, and the impact of State or private actions which are contemporaneous with the consultation in process. The consequences to listed species or designated critical habitat from ongoing agency activities or existing agency facilities that are not within the agency's discretion to modify are part of the environmental baseline (50 CFR 402.02).

Description of the Action Area

The implementing regulations for section 7(a)(2) of the Act (50 CFR 402.02) define the "action area" as all areas to be affected directly or indirectly by the Federal action and not merely the immediate area involved in the action.

The action area for this Project includes (1) the area of direct effects resulting from the Townsite Solar 2 facility (76-acre area that will contain the solar field, the 5-acre temporary laydown and staging area that will be reclaimed after construction is complete, and the 4-acre ROW with access roadway and transmission lines) and (2) the area of indirect effects on tortoise habitat and connectivity around the Project area. A translocation recipient site for desert tortoises was not included in the action area because no desert tortoises were found on two separate surveys conducted by APP and the BLM. Further information on translocation in the event of discovery during Project activities is outlined in the Effects of the Action section of this biological opinion.

The Project is located in the western portion of Eldorado Valley in the piedmont area of the McCullough Range. The Project is located within the Mojave Basin and Range ecoregion and is outside the boundaries of any ACEC, BLM wilderness area, or designated critical habitat for desert tortoise. The River Mountains ACEC is located 3 miles northeast and across I-11, the Boulder City Conservation Easement and Paiute-Eldorado desert tortoise critical habitat complex are located 3 miles southeast, the Paiute-Eldorado Valley ACEC is located 20 miles southwest, and the Sloan Canyon National Conservation Area (NCA) is located 5 miles west. The Project is bordered by the Townsite Solar 1 facility to the south, quarry operations to the west, and mining operations and a gun range to the north. The area immediately to the east of the Project site is open; however, US-95, which has tortoise exclusion fencing, and the edge of the Townsite Solar 1 facility are less than one mile away.

Previous Consultations in the Action Area

Programmatic Biological Opinions for Projects in the Action Area

Several Programmatic Biological Opinions (PBOs) have been issued to the BLM that include land in the action area. The first one was issued on November 25, 1997 (1-5-97-F-251; Service 1997), for implementation of various land management programs within the Las Vegas District planning area excluding desert tortoise critical habitat, ACECs, and the Las Vegas Valley. Activities proposed that may affect the desert tortoise in the action area include issuance of ROWs, Recreation and Public Purposes Act leases, mineral material sales and leases, and mining plans of operation. The programmatic consultation was limited to activities that could affect up to 240 acres per project and a cumulative total of 10,000 acres, excluding land exchanges and sales. Only land disposals by sale or exchange in Clark County, but outside the Las Vegas Valley, were covered under the consultation up to a total of 14,637 acres.

On June 18, 1998, the Service issued a PBO (1-5-98-F-053; Service 1998) to the BLM for implementation of various land management programs within desert tortoise habitat and the Las Vegas planning area, including desert tortoise critical habitat and ACECs. Activities that were proposed that may have affected the desert tortoise in the action area included recreation,

designation of utility corridors and mineral material extraction areas, and designation of the desert tortoise ACECs.

On June 17, 2010, the BLM submitted a programmatic biological assessment to the Service to request consultation for program-level and project-level actions that may affect and are likely to adversely affect 19 threatened and endangered species, including desert tortoise, of which 13 have designated critical habitat within the action area for the consultation. On January 2, 2013, the Service issued a non-jeopardy PBO to the BLM based on review of these activities (84320-2010-F-0365; Service 2013a). While the BLM's 1998 resource management plan remains in effect, the 2013 PBO replaces the Service's 1998 document, which covered a 10-year period. The PBO has been reinitiated six times to include additional acres or activity changes. The BLM requested reinitiation of the PBO in November 2017. The Service issued a new PBO (08ENVS00-2019-F-0153; Service 2019a) which was signed on January 14, 2020 and replaces the 2013 document. Several actions have been appended to the PBO that occur in the vicinity of the action area, including the Southern Nevada Intertie Project (84320-2015-F-0034.R002; Service 2019b) and the Sloan to Mead Rebuild Project (08ENVS00-2020-F-0028; Service 2019c).

On September 21, 2015, the Service issued a PBO (84320-2015-F-0120; Service 2015) to the Western Area Power Administration for proposed O&M activities to the West Area Transmission System infrastructure. This transmission system extends from southern Nevada through southern California and Arizona within an existing utility corridor ROW. Because the O&M activities are mostly within existing ROWs, the Service estimated that no more than 50 acres of new disturbance to desert tortoise habitat would occur during the actions completed under the PBO.

On April 18, 2022, the Service issued a PBO (2022-0030693-S7; Service 2022b) to the Federal Highway Administration (FHWA) for funding road and highway projects and use of mineral and material sites for these projects over a 10-year period. The Nevada Department of Transportation (NDOT) is the primary non-Federal Applicant of projects and activities under the PBO. The FHWA and the Service anticipate direct impacts to desert tortoise from implementation of activities covered under all programs in this PBO (including material sites) are estimated to be no more than 20,396 acres of suitable habitat and 768 acres of critical habitat.

Other Biological Opinions for Projects in the Action Area

On July 24, 2014, the Service issued a biological opinion (84320-2014-F-0224; Service 2014) to the WAPA for their proposed approval of an easement to Korean Western Power Company America, LLC to construct, operate, and maintain a 230 kV transmission line in the Eldorado Valley near Boulder City. The transmission line would cover 3.2 miles and would disturb a relatively small amount (20.4 acres) of desert tortoise habitat. The Service exempted incidental take of four tortoises captured and moved out of harm's way and one tortoise killed or injured. This transmission line transports power from the Townsite Solar 1 facility to the power grid and would also serve as transmission infrastructure for the proposed Townsite Solar 2 Project.

On December 28, 2012, the Service issued a biological opinion (84320-2012-F-0380; Service

2012a) to the BLM for issuance of a ROW to construct, operate, and maintain a 4.6-mile 230 kV gen-tie power line on 4.3 acres of BLM-managed land. The gen-tie line would connect a proposed 300 MW photovoltaic solar energy generating facility to the Eldorado Substation and McCullough Switching Station. Because the solar facility would not be constructed but for the proposed gen-tie lines, it was considered an interdependent action. Therefore, the biological opinion also addresses the proposal to construct, operate, and maintain a 300 MW solar facility on 2,200 acres of Boulder City-owned land by Techren Solar, LLC. The facilities were completed and expanded in 2020, to a total of 400 MW on 2,300 acres. The Techren Solar project is adjacent to the Eldorado Dry Lake Bed and is co-located with several other solar facilities on Boulder City land. An amendment to the biological opinion was issued on December 21, 2021 (08ENVS00-2012-F-0380-AMD1; Service 2021) to include additional communications and transmission lines within the existing ROW.

Habitat Conservation Plans

On November 19, 2000, the Service issued a biological opinion (1-5-00-FW-575; Service 2000) for the Clark County Multiple Species Habitat Conservation Plan (MSHCP) and a 30-year permit was issued to Clark County, the Cities of Las Vegas, North Las Vegas, Henderson, Boulder City, and Mesquite, and the NDOT. The MSHCP authorized the take of desert tortoise and 77 additional species resulting from the loss of up to 145,000 acres of land from urban development activities. Additionally, a $550 per acre mitigation fee was imposed to fund conservation programs and establish and manage the Boulder City Conservation Easement. The Service concluded that the MSHCP was not likely to jeopardize the desert tortoise or southwestern willow flycatcher or result in the adverse modification of critical habitat for either species. As of the 2021 Biennium Report (Desert Conservation Plan 2021), approximately 99,963 acres subject to mitigation fees had been disturbed and 15,000 acres of municipal development, exempt from mitigation fees, had been disturbed.

Under the MSHCP, several solar facilities have been constructed on Boulder City land in the vicinity of the proposed Project. These include Townsite Solar 1 (1,033 acres, 180 MW), which is adjacent to the Project area, and Copper Mountain Solar Facility 1 – 5 (4,000 acres, 802 MW), Nevada Solar One (400 acres, 64 MW), and Boulder Solar 1 and 2 (900 acres, 150 MW), which are co-located around and to the south of the Eldorado Dry Lake Bed, three miles south of the Project.

Habitat Characteristics of the Action Area

The U.S. Geological Survey (USGS) habitat model for desert tortoise shows a habitat suitability value of 0.9 out of 1.0 within the action area, indicating that the area has a high probability of being suitable for tortoises (Nussear et al. 2009). Soil survey data from the Natural Resources Conservation Service (NRCS) indicate that the Project site is underlain by loamy sand soils with high drainage and low runoff (NRCS 2022). Alluvial fans and valley wash features are also present in the action area. Despite some variation in foliar cover among these geological features, vegetation in the action area is composed entirely of Sonora-Mojave Cresosotebush-White Bursage Desert Scrub (USGS 2005). This vegetation community is dominated by creosote bush (*Larrea tridentata*) and white bursage (*Ambrosia Dumosa*) and has an open to intermittent

canopy. Vegetation in the Project area has been somewhat disturbed by off-highway vehicle recreation.

Status of the Species in the Action Area

Recovery Unit

The Project occurs within the southeastern portion of the Eastern Mojave Recovery Unit as described in the revised desert tortoise recovery plan (Service 2011). This recovery unit spans the Nevada-California border and is adjacent to the Northeastern Mojave Recovery Unit to the northeast, the Colorado Desert Recovery Unit to the southeast, and the Western Mojave Recovery Unit to the west.

Recent microsatellite data indicate that much of the Eastern Mojave Recovery Unit is relatively isolated from other recovery units (Hagerty and Tracy 2010). The Spring Mountains, which provided much of the separation between the former Northeastern Mojave and Eastern Mojave Recovery Units, narrowly channel gene flow through habitat corridors to the north and south, connecting this recovery unit to the Northeastern Mojave Recovery Unit (Hagerty 2008; Hagerty et al. 2011). The Sheep Mountains appear to form a barrier to tortoise movement between the eastern side of the recovery unit and the Northeastern Mojave Recovery Unit. The New York and Providence Mountains isolate Ivanpah/Shadow Valleys from Eldorado/Fenner Valleys in the Colorado Desert Recovery Unit to the east. Saline Valley and Death Valley extending south into Silurian Valley and Soda Dry Lake act as a barrier between this recovery unit and the Western Mojave Recovery Unit. Although gene flow likely occurred intermittently during favorable conditions across this western edge of the recovery unit, this area contains a portion of the Baker Sink, a low-elevation, extremely hot and arid strip that extends from Death Valley to Bristol Dry Lake. This area is generally inhospitable for desert tortoises.

As summarized in the Status of the Species section, the number of adult desert tortoises in conservation areas in the Eastern Mojave Recovery Unit is declining annually and decreased by 67.3 percent from 2004 to 2014 (Allison and McLuckie 2018). Furthermore, the proportion of juveniles decreased by 23 percent in the same time period. In the Eldorado Valley Tortoise Conservation Area specifically, tortoise density declined from 2.6 tortoises per square kilometer to 0.9 tortoises per square kilometer.

Project Surveys

Desert tortoise presence/absence surveys were conducted in compliance with Service protocol (Service 2017). The entire 85-acre Project area was surveyed by APP biologists as part of a broader survey effort on 458 acres of suitable desert tortoise habitat on BLM-managed land and land owned by Boulder City (Figure 2; BLM 2022). Biologists used 10-meter-wide transects with east/west orientation to conduct 100% coverage surveys. Surveys were conducted during the desert tortoise active season on October 27-30, 2018. During surveys, no tortoises or signs that indicate their presence, such as scat, burrows, carcasses, pallets, etc., were observed throughout the 458-acre survey area.

Additionally, to validate the results of the initial desert tortoise pre-construction survey, BLM biologists conducted a ground-truthing survey on 30 acres of the Project site in accordance with Service protocols. Three class-3 (i.e., deteriorated condition, definitely desert tortoise, including collapsed burrows) desert tortoise burrows were observed during the ground-truthing survey effort. No live tortoises were found, nor were any observations of recent sign documented. However, observations of inactive burrows within the project area confirm that the site has been and may currently be occupied by some individuals of the species. Although neither tortoises nor tortoise sign were found within the action area during surveys, the area may be part of an individual's home range. The home range of a female desert tortoise averages 0.15 to 0.16 km^2 (35 to 40 acres), which is about one third the size of male home ranges (0.39 to 0.47 km^2 or 96 to 116 acres; Harless et al. 2009). Therefore, projects that are less than 0.8 km^2 (200 acres) may overlap only part of a tortoise's home range.

Pre-construction field surveys conducted for other projects near the proposed Project area also found very few desert tortoises. In 2019, tortoises were encountered and handled during construction on two projects that were appended to the BLM's PBO (Service 2019a). Tortoises were encountered at the Southern Nevada Intertie project (Service 2019b) and the Sloan to Mead Rebuild 230 kV Transmission Line Replacement project (Service 2019c), approximately 3 and 11 miles away from the proposed Project site, respectively. In all cases, the observed tortoises were separated from the proposed Project site by other existing solar facilities, quarry operations, or the McCullough Range. No live tortoises were found within the action area for the Townsite Transmission Project (Service 2014) that the proposed Project will also utilize.

Surveys generally do not detect all desert tortoises in an area because they spend so much time underground. Because no tortoises were observed during either survey attempt, we are unable to provide an accurate estimate for the number of tortoises in the action area by using the equation in the Service's protocol (Service 2009). However, based on the location of the proposed Project, its proximity to existing development, and the lack of recent tortoise sign during two sets of surveys, we expect tortoise density to be no more than the average density reported for the Eastern Mojave Recovery Unit (1.5 tortoises per square kilometer; Allison and McLuckie 2018). Thus, we estimated that the 85-acre action area supports no more than 1 large adult desert tortoise (Table 3).

Previous solar projects have found more tortoises during clearance surveys than were originally estimated. Because tortoises are mobile, there may be more within the action area than were originally estimated during tortoise surveys. The K Road solar project found 13.6 percent more tortoises during clearance surveys than estimated in their biological opinion (Service 2012b), and the Silver State South solar project found 23.6 percent more tortoises than estimated (Service 2013b). Because such higher percentages have been found compared to estimated numbers, we allow for a 25 percent buffer for additional tortoises to be captured and moved. However, due to the low density of desert tortoises in the Project area, adding 25 percent to the estimated 1 adult tortoise within the Project area does not result in an increase to the total estimate of adult tortoises.

Turner et al. (1987) developed a life table for female desert tortoises based on studies conducted at Goffs, California in 1983. They estimated that 13.2 percent of the desert tortoises in that

population were larger than 180 mm in length. To estimate the number of all desert tortoises within the Project area, we used the methodology and calculations in Table 3. We estimate that 1 adult tortoise and 7 juvenile tortoises may be present in the Project area.

Table 3. Number of desert tortoises estimated to occur within the Project area.

Tortoise Calculation	**Tortoise Estimate (Rounded)**
Estimated number of adult desert tortoises (i.e., larger than 180 mm) (calculated by 85 acres * 0.004 acres/km^2 * 1.5 tortoises/km^2)	0.52 (1)
Project limit for translocation of adults (tortoise estimate + 25%)	1.25 (1)
Percentage of desert tortoises in size classes larger than 180 mm (Turner et al. 1987)	13.2
Total number of desert tortoises (calculated by 1 / 0.132)	7.6 (8)
Estimated number of juvenile desert tortoises (calculated by 8 - 1)	7

We are unable to provide estimates for the number of desert tortoise hatchlings or eggs that could occur in the Project area. Estimating the number of tortoise hatchlings and eggs is extremely difficult because the eggs are buried beneath the soil surface and numbers vary depending on the season (i.e., eggs are present during one time of the year but become hatchlings later in the year). Applying any assumptions has an unknown and high level of uncertainty. Therefore, we cannot calculate a precise estimate for the number of hatchlings or eggs that may be impacted by the proposed Project.

Population Connectivity

The Revised Desert Tortoise Recovery Plan (Service 2011) recommends connecting blocks of desert tortoise habitat, such as Critical Habitat Units and other important areas, to maintain gene flow between populations. Desert tortoise connectivity in the region is addressed in the Solar PEIS (BLM and DOE 2012) and the Supplement to the Solar PEIS. The Desert Tortoise Recovery Office performed a landscape-scale modeling exercise to identify priority habitat linkages between and among desert tortoise conservation areas (as defined in Service 2011) and to define other large blocks of habitat with important value to recovery of the desert tortoise. Based on our current understanding, the combination of linkages and existing desert tortoise conservation areas represents the basis for a conservation network for the desert tortoise. Priority 1 areas are designated where least-cost corridor modeling identified potential habitat linkages between existing conservation areas that have the best change of sustaining connectivity for desert tortoise populations. To identify these linkages, the Service began with the USGS desert tortoise habitat potential model (Nussear et al. 2009) and developed a cost surface where higher habitat potential equaled a lower cost to the desert tortoise. The linkages of least-cost to the desert tortoise between pairs of conservation areas (Beier et al. 2008) represent priority areas for conservation of desert tortoise population connectivity. Priority 2 areas are other blocks of habitat with the greatest potential to support populations of desert tortoises, outside least-cost corridors, may also have important value to recovery. Based on the USGS model, the Service identified areas of contiguous, high-value desert tortoise habitat as Priority 2 lands for conservation of desert tortoise within the context of the Solar PEIS.

While the Solar PEIS does not apply to the proposed Townsite Solar 2 Project because it is not a utility-scale solar development (i.e., generating less than 20 MW), the majority of the proposed Project area (80 acres) is within Priority 1 desert tortoise connectivity habitat (Figure 3). Connectivity for tortoises in the action area likely extends south into Eldorado Valley, through the Sloan Canyon NCA to the west, and through the Paiute-Eldorado critical habitat complex to the Colorado Recovery Unit in the south. However, potential movement of desert tortoises from the action area to the south is likely restricted by the recent construction of several solar facilities. The Project area may have limited connectivity north to the River Mountains ACEC and east due to I-11 and US-95 acting as barriers. US-95 is fenced with tortoise exclusion fencing and does not have culverts, resulting in an impermeable barrier to the east. I-11 is not fenced but likely deters tortoise movement. The habitat in and around the action area is likely already somewhat isolated from other desert tortoise habitat.

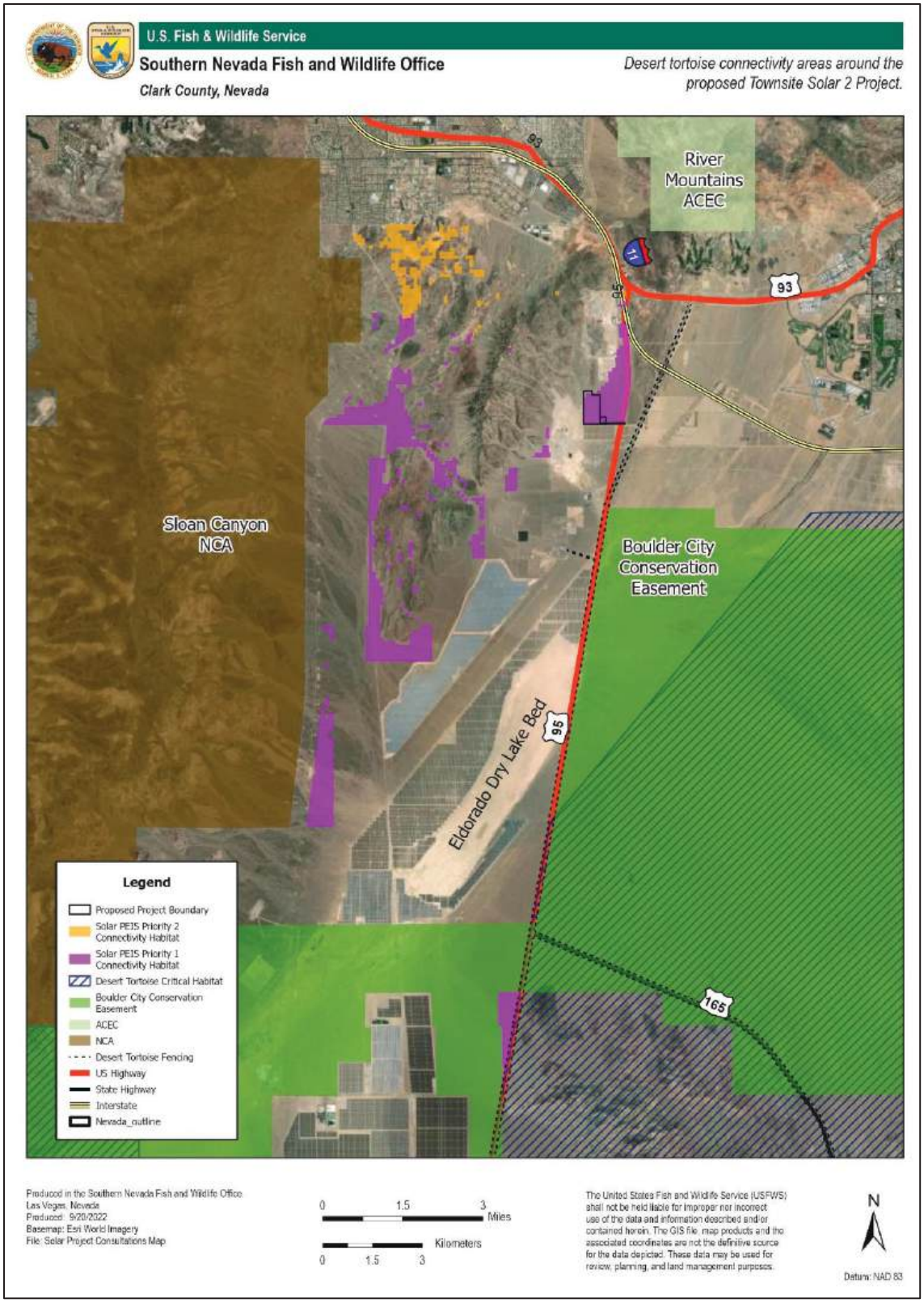


Figure 3. Desert tortoise connectivity areas around the Townsite Solar 2 Project.

EFFECTS OF THE ACTION

Effects of the action are all consequences to listed species or critical habitat that are caused by the proposed action, including the consequences of other activities that are caused by the proposed action. A consequence is caused by the proposed action if it would not occur but for the proposed action and it is reasonably certain to occur. Effects of the action may occur later in time and may include consequences occurring outside the immediate area involved in the action.

We conducted the analysis in the following sections based on the current conditions in the action area as we described in the Environmental Baseline section of this biological opinion. Several aspects of the proposed action may affect desert tortoises within the action area including the installation of the desert tortoise exclusion fence, killing or injuring of individuals and crushing of their burrows and eggs during construction and O&M activities, translocating tortoises from the Project site, loss of habitat, and other miscellaneous effects. We conclude this section by integrating our analysis with the best available information with regard to the numbers of desert tortoises and amount of habitat in the action area and recovery unit to determine whether the proposed action is likely to jeopardize the continued existence of the desert tortoise.

Effects of the Proposed Action

The construction, O&M, and decommissioning of the Project could result in death or injury to desert tortoises from the following activities: (1) project equipment usage and construction activities, (2) increased traffic and road access, (3) removal of tortoises from the Project site, (4) loss of habitat, and (5) an increase in habitat degradation and predator populations. To minimize incidental death and injury of desert tortoises residing in or entering the Project site, the Applicant would implement the actions specified in the Minimization Measures (MM) section of this biological opinion as part of the proposed action. The remainder of this section outlines risks to desert tortoises throughout all phases of the Project along with specific MM the Applicant will use to minimize adverse effects to desert tortoise.

Construction

To prevent tortoises from entering the solar facility during construction and O&M, the Applicant will install permanent Service-approved desert tortoise exclusion fencing (MM 4). The ROW corridor and temporary laydown area will not be permanently fenced but may receive temporary tortoise exclusion fencing during the construction phase of the Project. Once the exclusion fencing is installed, authorized biologists will conduct clearance surveys to translocate any tortoises found within the Project site (MM 6). We expect few, if any, desert tortoises will require translocation because the Project site and surrounding area support very few desert tortoises. While there is some potential for tortoises to be missed during clearance surveys, any undetected tortoises may be located by authorized biologists during routine inspections during construction activities. For these reasons, we anticipate that construction will kill few, if any, adult desert tortoises.

Construction of the perimeter fencing may adversely affect desert tortoises. Tortoises could attempt to gain access to areas of their home range that have been fenced and could follow fence

lines for long periods. Tortoises could die when exposed to harsh conditions (i.e., cold or hot temperatures) while pacing fences, however, incorporating shade structures along the fence and performing fence checks at regular intervals will reduce these effects (MM 4). The installation of fencing may also reduce the home range of tortoises that inhabit the areas immediately adjacent to the Project site or that overlap the area. This reduction could result in future injury or mortality of these individuals as they attempt to expand their home range into other areas where unknown threats may occur, such as vehicle traffic or adverse social interactions with other neighboring tortoises. Few, if any, desert tortoises will experience adverse effects from the presence of exclusion fencing for the same reasons we discussed in the previous paragraph.

The ROW corridor and temporary laydown area could have different effects on desert tortoises than the solar field if the Applicant does not install temporary tortoise exclusion fencing around these areas. Equipment used to clear vegetation, construct the powerlines, and stage materials could crush desert tortoises. Additionally, uncovered trenches or holes could trap desert tortoises, resulting in death or injury due to exposure or predation. During construction, the Applicant proposed several measures to reduce adverse effects to desert tortoises in unfenced areas. The Applicant will use ADTBs and authorized biologists to monitor work activities in unfenced areas, including checking trenches, to protect desert tortoises and move them from harm's way (MM 2, 5, 8).

Increased human activity and vehicle traffic in and around the action area could result in tortoise mortality or injury. Tortoises may take shelter under parked vehicles and be killed or injured. All personnel over the life of the Project will participate in a WEAP that includes practices to avoid impacts to desert tortoise (MM 1). Parked vehicles will be inspected prior to being moved each day, and if a tortoise is discovered, it will be moved out of harm's way by authorized biologists. Access to the construction site will be restricted to existing roads and speed limits will be posted and enforced (MM 11). We are unable to separate the potential effects of the use of paved access roads by Project-associated vehicles from those of the general public. However, we do not expect vehicles using this road to kill or injure many tortoises due to the low density of tortoises in and around the Project area.

Construction activities and vehicle traffic could also generate dust and increase noise in and around the action area. The buildup of dust on plant leaves could affect photosynthetic productivity, resulting in the loss of foraging plants and cover for desert tortoises. Increased noise levels may affect desert tortoise foraging and sheltering behavior, potentially resulting in injury or mortality. We expect these effects to be minimal and short-term because Project construction is expected to last between 9 and 12 months. Effects from dust will be minimized through implementation of a Dust Control Plan (MM 13). Additionally, the Applicant will translocate all individuals it finds during construction and the area surrounding the action area supports very few desert tortoises.

As discussed above in the Environmental Baseline section, we estimate that 7 juvenile desert tortoises may be present in the action area; however, we cannot provide an estimate for the number of hatchlings or eggs that may occur in the action area. Juvenile tortoises and hatchings are more difficult to detect, and while we anticipate that ADTBs or authorized biologists may detect some juvenile tortoises, hatchlings, or eggs during pre-construction clearance surveys or

other monitoring activities, it is unlikely that all individuals will be detected and moved from harm's way. Consequently, we expect that most of the juvenile tortoises, hatchlings, and eggs will remain in the action area and be killed or injured during construction. Given the lack of observations of live tortoises and recent sign during pre-construction surveys, we expect that few juvenile tortoises, hatchlings, and eggs will be present on the Project site.

Operations and Maintenance

Desert tortoises could access fenced Project areas during O&M if exclusion fencing is compromised due to flooding or storms. Once inside the fenced area, tortoises would be at risk of being injured or killed by O&M activities. Tortoise exclusion fencing would be inspected at regular intervals and following rain events throughout the life of the Project and damage would be repaired immediately (MM 19). As long as tortoise exclusion fencing remains intact, we expect that O&M activities within the fenced area are likely to injure or kill few desert tortoises. However, if fences are poorly maintained, risk to tortoises would likely increase.

Over the life of the Project, the Applicant would conduct routine visits to the solar facility to inspect equipment or perform maintenance. Tortoises could be injured or killed outside of fenced areas by vehicles traveling to and from the work site or by heavy machinery conducting ground-disturbing maintenance. Posting and enforcing speed limits on access roads is expected to minimize these effects (MM 17). Few, if any, desert tortoises are expected to be killed or injured by O&M due to the translocation of any tortoises discovered during construction and the continuing development and isolation of habitat in this area.

Decommission

Work activities associated with decommissioning the Project are unlikely to negatively affect desert tortoises. Tortoises would not be present within the fenced area and very few individuals are expected to be outside the fenced area due to disturbance and increasing isolation from other desert tortoise habitat. In the event that some individuals remain, the effects would be similar to those associated with construction and O&M that were previously described.

If the Project is decommissioned at the end of the 30-year ROW, the action area would likely be highly compacted and disturbed. Recovery time to pre-Project conditions would vary depending on the actions specified in the Decommissioning and Reclamation Plan that will be developed with the BLM (MM 20). Other variables, such as rainfall, could also affect recovery time. In the absence of restoration, recovery can take several decades (Abella 2010).

Translocation

The death and injury of desert tortoises associated with Project activities (i.e., construction, O&M, decommissioning) would be minimized through translocation, which recent studies have shown to be an effective conservation tool (Field et al. 2007, Esque et al. 2010, Nussear et al. 2012, Hinderle et al. 2015). All desert tortoises found on the Project site will be captured and relocated under specific direction from the Service and the BLM at the time of discovery (MM 7). We recommended this course of action because tortoise density is low in the action area and

the area of remaining habitat outside the Project perimeter is small and isolated from other suitable desert tortoise habitat due to development. Thus, short-distance relocations outside the Project perimeter are not possible. Effects from translocation would occur to both the translocated tortoises and to the resident tortoises where translocatees are moved.

We expect that few desert tortoises will require translocation from the Project site because no live individuals or recent sign were observed in the area on two pre-construction surveys. For the purposes of this biological opinion, we used the tortoise density reported for the Eastern Mojave Recovery Unit to estimate that 1 large adult tortoise and 7 juvenile tortoises could be translocated from the Project site.

Accidental death or injury of desert tortoises could result from capturing, handling, and moving tortoises for the purpose of translocation if these methods are performed improperly. Specifically, death or injury could result from (1) excavating tortoises from their burrows, (2) disease transmission from handling desert tortoises, (3) stress associated with handling and relocating tortoises outside of their home range, (4) stress associated with artificially increasing the density of desert tortoises in an area and thereby increasing competition for resources, and (5) disease transmission between translocated and resident tortoises. Because the Applicant will employ desert tortoise biologists approved by the Service, adhere to the most recent translocation guidance (Service 2019d), and implement the minimization measures outlined in the proposed action, we expect that mortality or injury to desert tortoises from activities associated with translocation is unlikely.

Following Service translocation guidance (Service 2019d), any desert tortoises that are found will be captured. Individuals within burrows would first be allowed time to leave on their own volition before being excavated by ADTBs. We expect that excavating tortoises from their burrows is likely more stressful for them than being captured on the surface of the ground. All tortoises captured at the Project site will undergo health screening, including a blood draw, to determine risk of disease. Any tortoises that have clinical signs of disease or blood tests positive for pathogens would not be translocated without approval by the Service and the BLM. Upper respiratory tract disease and other pathogens spread by direct contact between desert tortoises, which can increase disease prevalence in the population of resident tortoises at the recipient site that would not have been exposed otherwise.

Capture and handling of desert tortoises for the purposes of conducting health assessments, which includes visual inspection relative to body condition, clinical signs of disease, and collection of biological samples for disease screening (i.e., blood samples to test for antibodies to pathogens), could result in accidental death or injury. Desert tortoises store water in their bladders to be absorbed as necessary during times of low rainfall, and stress can cause tortoises to void their bladders. Averill-Murray (2002) determined that desert tortoises that voided their bladders during handling had lower overall survival rates than those that did not. Consequently, tortoises that void their bladders are at an increased risk of dying after their translocation. To reduce these effects, any tortoises that void their bladders during translocation activities will be rehydrated per Service guidelines. The capture, handling, and release of desert tortoises would also follow temperature restrictions to minimize the potential for stress, injury, or death resulting from hyperthermia or hypothermia.

Boarman (2002), in a review of literature on threats to the desert tortoise, stated that the adverse effects of translocation include increased risk of mortality, spread of disease, and reduced reproductive success. The tendency for translocated individuals to spend more time above ground, moving through their environment, than individuals within their home ranges exacerbates these threats. However, translocated animals seem to reduce movement distances following their first post-translocation hibernation to a level that is not significantly different from resident populations (Field et al. 2007, Nussear 2004, Nussear et al. 2012). As time increases from the date of translocation, most desert tortoises change their movement patterns from dispersed, random patterns to more constrained patterns, which indicate an adoption of a new home range (Nussear 2004).

As with other prior translocations, we anticipate that predation is likely to be the primary source of post-translocation mortality, particularly for small tortoises (Nussear 2004, Field et al. 2007). However, predation rates are similar between translocated and resident tortoises (Field et al. 2007, Esque et al. 2010). To minimize the risk of predation, the Service and BLM will select release sites preferentially located away from known areas of concentrated predator sign if any are identified. The level of winter rainfall can also affect predation rates. Tortoise mortality increased during periods of low winter precipitation at Fort Irwin, likely due to predators switching to available prey sources (Drake et al. 2010). Drought conditions appear to affect translocated and resident desert tortoises similarly and we do not anticipate that translocated tortoises will experience greater risk from drought than resident tortoises.

Depending on the specific goal of translocating desert tortoises, translocating individuals either short or long distances is likely to have different effects. Hinderle et al. (2015) found that desert tortoises translocated less than 2 kilometers are likely to attempt to return to their point of capture. If these individuals are unable to regain access to their home ranges, they are likely to wander more extensively or pace fence lines. These behaviors could expose tortoises to elevated risks of predation and extreme temperatures that they would otherwise avoid. Therefore, when tortoises will not be able to regain access to their point of capture, translocating them to suitable habitat more than 2 kilometers away is likely to prevent them from returning. In this case, the Applicant will contact the Service and the BLM immediately upon discovery of a tortoise within the Project site for disposition instructions. Depending on weather conditions, time of year relative to the desert tortoise active seasons (April 1 – May 31 and September 1 – October 31), and tortoise health screening, the Applicant may be instructed to translocate individuals to a recipient site immediately or transfer them to the Desert Tortoise Center first until conditions improve. Due to the Project's location relative to existing development and the need for a distant recipient site, the expected recipient site for translocated tortoises is the Paiute-Eldorado Valley ACEC, which is 20 miles (32.2 kilometers) southwest from the Project site.

Translocation could also affect resident desert tortoises within the recipient site due to local increases in population densities that may result in increased incidence of aggressive interactions between individuals and increased competition for forage, shelter sites, and other limited resources (Saethre et al. 2003). Therefore, recipient sites must be sufficiently large to accommodate and maintain the resident and translocated desert tortoises (Service 2019d). The Eldorado Valley tortoise conservation area, which encompasses the potential 329,440-acre Paiute-Eldorado Valley ACEC recipient site, currently supports a tortoise density of 2.3 tortoises

per square kilometer (Service 2020). The maximum recommended pre-translocation density within the Eastern Mojave Recovery Unit is 3.9 tortoises per square kilometer, making it eligible to receive tortoises via translocation (Service 2019d). Because the Project site appears to support few desert tortoises, we do not expect to surpass the recommended post-translocation density for the Eastern Mojave Unit (4.0 tortoises per square kilometer) and we do not expect any adverse effects related to increased density.

Habitat Loss

The Project would result in the permanent loss of 85 acres of desert tortoise habitat. Tortoises would be permanently excluded from the 76-acre fenced solar field and tortoise habitat would be modified due to activities in the temporary 5-acre laydown area and the 80-foot ROW corridor. Restoration of temporary disturbance areas will be completed per the BLM-approved Restoration Plan (MM 16). However, because full recovery of desert vegetation can take several decades (Abella 2010), we consider all ground-disturbing impacts associated with the proposed Project to be effectively permanent for mitigation compensation. To offset the permanent loss of desert tortoise habitat, the Applicant would pay a remuneration fee to the BLM for the 85-acre Project area (MM 21).

The proposed Project would not impede movement of desert tortoises in the Eastern Mojave Recovery Unit in general. The action area is bordered on three sides by existing disturbance or development, including the Townsite Solar 1 facility to the south, quarry operations to the west, and a gun range and mining operations to the north. US-95, which has desert tortoise fencing, is less than a mile to the east. The construction and use of the neighboring facilities and infrastructure have likely contributed to the current low density of desert tortoises in this area. Because the Project is isolated from adjacent desert tortoise habitat and any tortoises found during construction will be translocated elsewhere, we expect the loss of habitat will have minimal adverse effects on desert tortoises.

Habitat Degradation and Increased Predator Populations

Ground disturbing activities during construction could result in an increase of non-native invasive plant species in the area. Construction machinery may transport or spread non-native invasive plants in the action area, where they could become established. Invasive species could displace native species that provide forage for tortoises, resulting in reduced habitat quality and increased fire frequency (Brooks and Esque 2002). Potential degradation of habitat due to spread of invasive species would be avoided and minimized by the implementation of the Project's Integrated Weed Management Plan and other minimization measures described in this biological opinion (MM 12 and 18).

Construction of the solar facility and access roads could also result in adverse effects to adjacent vegetation communities through alteration of microclimate. Site grading and construction activities can alter water movement across the landscape and may lead to a decline in soil moisture (Grippo et al. 2015). Additionally, solar facilities often generate extra heat that is dispersed into adjacent lands along a gradient (Devitt et al. 2022). These alterations in microclimate could result in increased stressful conditions for plants, a shift in vegetation

communities, and ultimately reduced habitat suitability for desert tortoises. Impacts to habitat could also be exacerbated by climate change, which is likely to influence the amount of precipitation and lead to increased temperatures in the region (Barrows 2011, Christensen et al. 2007). We expect that habitat degradation in the area surrounding the proposed Project will affect few tortoises because much of this area is already developed. The small area of undeveloped habitat adjacent to the Project site is also becoming more isolated from contiguous desert tortoise habitat and pre-construction surveys indicated few tortoises are present in the area.

Human activities and construction also facilitate the expansion of predator populations into areas where they were previously absent or in low abundance. Predators such as common ravens, coyotes, or raptors may be attracted to the Project site due to a variety of resource subsidies provided by human activities and development. These subsidies include (1) increased food opportunities from litter, roadkill, or voluntary feeding from construction staff, (2) an increased number of perching substrates due to new distribution lines and fences, (3) increased cover from weather elements due to new infrastructure, and (4) increased water sources due to dust control protocols or water retention ponds. An increased presence of predators could lead to increased predation on tortoise eggs, juvenile tortoises, and adult tortoises. We do not expect predator subsidization will have a measurable effect on desert tortoises in the Project area because most tortoises will be removed from the site during pre-construction clearance surveys.

Common ravens were previously scarce in the Mojave Desert prior to 1940 but populations have increased rapidly due to expanding human use of the desert (Boarman 2002). While common ravens could travel beyond the Project area and prey on desert tortoises outside the action area, we expect raven subsidization to affect few tortoises outside the action area because this area is already highly developed and appears to support few desert tortoises. Minimization measures, including the WEAP, litter control program, and the BLM's *Common Raven Management Plan for Energy Development within the BLM Southern Nevada District*, would help reduce these adverse effects (MM 1, 9, 10).

Effects on Recovery

Reproduction

Disturbance associated with the construction of the Project would not have a measurable long-term effect on reproduction of individual desert tortoises that live within or adjacent to the Project site because construction activity would occur over a relatively brief period of time (approximately 9-12 months) relative to the reproductive life of female desert tortoises. Additionally, desert tortoises are well adapted to highly variable and harsh environments and their longevity helps compensate for their variable annual reproductive success (Service 1994).

We expect that few desert tortoises live within or adjacent to the Project site based on pre-construction surveys. However, if tortoises are present in the action area, we expect that most adult tortoises would be found and translocated either during pre-construction clearance surveys or routine monitoring by ADTBs and qualified biologists during construction. We expect few, if any, adult desert tortoises would die during construction. Juvenile tortoises may be killed because they are more difficult to find; however, the reproductive biology of the desert tortoise is

such that reproductive individuals (i.e., large, adult individuals) play a more important role in maintaining populations than those that are not yet able to reproduce (i.e., small, juvenile individuals), in large part because of the higher mortality rates of eggs and juvenile tortoises. The loss of eggs and juvenile desert tortoises should not have a measurable effect on the reproductive capacity of desert tortoises in the area.

If desert tortoises are found and translocated out of the Project site, they could exhibit reduced reproduction in the first year following translocation. However, reproductive rates of translocated tortoises are likely to match those of resident individuals in subsequent years (Nussear et al. 2012). Due to the Project's proximity to existing development and the small area of tortoise habitat remaining in the area, it is possible that any tortoises within the Project area are isolated from contiguous habitat in the recovery unit. The translocation of tortoises out of the Project area may provide a minor benefit to desert tortoise reproduction by relocating these isolated individuals and allowing them to reproduce. We do not expect the increased density of desert tortoises that would result from translocation to affect the reproduction of resident individuals (Saethre et al. 2003).

Disturbance associated with O&M and decommissioning of the Project would not have a measurable effect on reproduction of individual desert tortoises for the reasons discussed above. If desert tortoises are present within the action area, most individuals would be translocated and would continue to survive and reproduce at the location where they are moved.

Numbers

We expect that the construction of the Project is likely to injure or kill few adult desert tortoises. The proposed MMs, including the installation of permanent and temporary exclusion fencing around the perimeter of the Project site and clearance surveys and monitoring conducted by qualified biologists, would detect and translocate most adult desert tortoises within the action area during construction. The permanent exclusion fencing around the solar field would reduce the likelihood of injury or mortality to tortoises that may enter the Project site from adjacent habitat during O&M. Disturbance associated with decommissioning of the Project would not have an appreciable effect on desert tortoises because the Applicant would follow MMs specified in the Decommissioning and Reclamation Plan that will be developed with the BLM.

Desert tortoises are cryptic and difficult to detect, and some potential exists that the Applicant will not detect every desert tortoise during clearance surveys or construction monitoring. We anticipate that a small number of large desert tortoises could be injured or killed during construction, O&M, and decommissioning; however, we expect that most desert tortoises would be translocated. We do not anticipate that injury or mortality would result from the handling and moving of these individuals because the Applicant would work with the Service and the BLM to determine the best course of action based on current weather conditions and the health conditions of any individuals found. We also expect that a relatively small but unquantifiable number of eggs and hatchlings would be moved or destroyed during construction, O&M, and decommissioning.

Overall, the small number of adult desert tortoises that could be injured or killed as a result of the

Project equates to a small fraction of the estimated population of the Eastern Mojave Recovery Unit. While we cannot provide an accurate estimate of adult desert tortoises within the action area since no live tortoises or recent sign were observed on pre-construction surveys, we expect the Project area to support no more than 1 large adult desert tortoise, which if injured or killed during Project activities, would equate to approximately 0.004 percent of the 24,664 adult tortoises reported within conservation areas in the Eastern Mojave Recovery Unit in 2014. While the population has likely continued to decline since 2014, the losses from the proposed Project still would not appreciably reduce the number of desert tortoises within the Eastern Mojave Recovery Unit. The loss of 7 juvenile desert tortoises would not have an appreciable effect on the tortoise population in the area.

Distribution

The permanent loss of 85 acres of desert tortoise habitat would not appreciably reduce the distribution of the desert tortoise. Based on the Nussear et al. (2009) model and estimations of impervious surfaces (Fry et al. 2011), we estimate that approximately 3,937,849 acres of desert tortoise habitat remain in the Eastern Mojave Recovery Unit (Darst 2014). The proposed Project would result in the loss of 0.002 percent of the amount of desert tortoise habitat in the Eastern Mojave Recovery Unit and 0.0005 percent of the total amount of habitat range-wide (16,745,848 acres). Additionally, we do not expect the loss of habitat from this Project to appreciably impact regional population connectivity based on a qualitative evaluation of existing priority habitat and linkages for desert tortoise (Averill-Murray et al. 2013).

Due to the Project's proximity to existing disturbance, development, and US-95, which has tortoise exclusion fencing, very little desert tortoise habitat remains within and outside the Project boundary. Based on pre-construction surveys, we do not expect this area supports many desert tortoises, and any desert tortoises that are present may already be isolated from other habitat in the Eastern Mojave Recovery Unit. Translocation out of this small area of habitat may benefit individuals who were previously isolated.

Summary of Effects to Recovery

To achieve recovery, each recovery unit must contain well-distributed and self-sustaining populations across a sufficient amount of protected habitat to maintain long-term population viability and persistence (Service 2011). The loss of habitat from the proposed Project will not impede the recovery of the desert tortoise because the Project is not located within any designated critical habitat or protected conservation areas for tortoises. Additionally, the habitat in the action area is already somewhat disturbed by off-highway vehicle recreation, is surrounded by existing disturbance and development, and is likely to become increasingly isolated from other desert tortoise habitat due to additional development in the future. The translocation of desert tortoises from the Project area to a recipient site may provide a minor benefit to the recovery of desert tortoises. If tortoises are translocated to the Paiute-Eldorado Valley ACEC, they will be less vulnerable to encroaching development and will be within a larger population of desert tortoises with which they can interact and breed.

Based on the reasons discussed above, the proposed construction, O&M, and decommissioning

of the Townsite Solar 2 Project will not adversely affect recovery of the desert tortoise. The translocation of desert tortoises to a conservation area and the MMs proposed by the Applicant are commensurate to the expected Project impacts, considering the desert tortoise status and its threats.

CUMULATIVE EFFECTS

Cumulative effects include the effects of future State, tribal, local or private actions that are reasonably certain to occur in the action area considered in this biological opinion. We do not consider future Federal actions that are unrelated to the proposed action in this section because they require separate consultation pursuant to section 7 of the Act. Past and present impacts of non-federal actions are considered part of the environmental baseline conditions. All of the action area is federally owned, and any future projects on these lands would be subject to separate section 7 consultation.

Projects that may result in adverse effects to the desert tortoise on private and non-federal land are anticipated to fall under purview of existing Habitat Conservation Plans and associated incidental take permits. Much of the land surrounding the action area in Eldorado Valley is owned by Boulder City and is designated for energy production. There are many solar energy generation facilities near the proposed Project site and several electrical substations are also in the area to facilitate energy transmission. Additional planned solar facilities on surrounding private lands, including the Silver Peak Solar and Boulder Solar III projects, are reasonably certain to occur in the future and are anticipated to fall under purview of the Clark County MSHCP.

Increased development would cause continued habitat loss, degradation, and fragmentation for the local desert tortoise population, as well as increased harm of individual tortoises, contributing to the cumulative degradation of the area. Planned future actions such as future transmission line and road corridors, electrical power substations, and industrial solar power plants would likely continue this trend. However, as described above, much of the area surrounding the Project site has already been developed and the infrastructure in place. We are not aware of any actions that are reasonably certain to occur on non-federal lands that would result from implementation of the proposed Project.

CONCLUSION

Jeopardy Conclusion

The regulatory definition of “to jeopardize the continued existence of the species” focuses on assessing the effects of the proposed action on the reproduction, numbers, and distribution, and their effect on the survival and recovery of the species being considered in the biological opinion. For that reason, we have used those aspects of the desert tortoise status as the basis to assess the overall effect of the proposed action on the species.

After reviewing the current status of the desert tortoise, the environmental baseline for the action area, the effects of the proposed Townsite Solar 2 Project, and the cumulative effects, it is the

Service's biological opinion that the project, as proposed, is not likely to jeopardize the continued existence of the Mojave desert tortoise. We base this decision on the following:

1. The Applicant will implement numerous measures (i.e., exclusion fencing, clearance surveys, construction monitoring, translocation, and employing authorized desert tortoise biologists) to ensure that most desert tortoises are located and moved out of the Project site. Potential injury and death of desert tortoises will be minimized.

2. The Project would not considerably affect the survival or reproduction of the desert tortoise. Translocation of desert tortoises to the recipient site would minimize adverse effects by allowing those desert tortoises to remain in the population and contribute towards recovery of the species.

3. The Project would cause a minor decrease in the number of individuals. Few, if any, desert tortoises are likely to be injured or killed as a result of translocation out of the Project site.

4. The Project would not reduce the species' distribution range-wide. The amount of non-critical habitat proposed to be permanently and temporarily disturbed is small relative to the amount available in the Eastern Mojave Recovery Unit and range-wide.

5. Genetic and demographic connectivity will be minimally reduced and will continue to function range-wide. The Project area is already isolated from remaining desert tortoise habitat due to existing development and fenced highways and loss of the Project area will not considerably affect desert tortoise habitat connectivity overall.

6. There will be no impacts to designated critical habitat.

7. Mitigation and remuneration fees, based on acres disturbed, will contribute to funding desert tortoise recovery actions.

8. The project would not cause any effects that would delay or preclude the recovery of this species.

INCIDENTAL TAKE STATEMENT

Section 9 of the Act and Federal regulation pursuant to section 4(d) of the Act prohibit the take of endangered and threatened wildlife species, respectively, without special exemption. Take is defined as to harass, harm, pursue, hunt, shoot, wound, kill, trap, capture or collect, or to attempt to engage in any such conduct. Harm in the definition of "take" in the Act means an act which actually kills or injures wildlife. Such [an] act may include significant habitat modification or degradation where it actually kills or injures wildlife by significantly impairing essential behavioral patterns, including breeding, feeding, or sheltering (50 CFR 17.3). Incidental take is defined as take that is incidental to, and not the purpose of, the carrying out of an otherwise lawful activity. Under the terms of section 7(b)(4) and section 7(o)(2), taking that is incidental to and not the purpose of the agency action is not considered to be prohibited taking under the Act provided that such taking is in compliance with the terms and conditions of this incidental take

statement.

The measures described below are non-discretionary and must be undertaken by the BLM as binding conditions of any grant or permit issued to the Applicant, as appropriate, for the exemption in section 7(o)(2) to apply. The BLM has a continuing duty to regulate the activity covered by this incidental take statement. If the BLM (1) fails to assume and implement the terms and conditions or (2) fails to require the Applicant to adhere to the terms and conditions of the incidental take statement through enforceable terms that are added to the permit or grant document, the protective coverage of section 7(o)(2) may lapse. To monitor the impact of incidental take, the BLM must ensure that the Applicant report the progress of the action and its impact on the species to the Service as specified in the incidental take statement [50 CFR 402.14(i)(3)].

AMOUNT OR EXTENT OF TAKE

We anticipate the take of all desert tortoises and nests with eggs in the Project area. We expect that most desert tortoises in the Project area will be captured, as the form of take, and translocated. Because the Project site will be graded and disked, we expect that most desert tortoises and nests with eggs that are not detected during clearance surveys may be killed or injured; however, the MMs proposed by the BLM are likely to prevent mortality or injury of most individuals. We acknowledge that we cannot quantify the precise number of desert tortoises that may be taken because the species is cryptically colored to avoid predation and spends the majority of its life inhabiting burrows to avoid environmental extremes or predation, making the detection of live, dead, or injured individuals difficult. Additionally, population numbers fluctuate in response to weather patterns and other biotic and abiotic factors, and population levels are anticipated to fluctuate over the 30-year life of the Project.

While we are unable to reasonably anticipate the actual number of desert tortoises that would be taken by the proposed action, we must provide a level at which formal consultation would have to be reinitiated. The Environmental Baseline and Effects Analysis sections of this biological opinion indicate that adverse effects to desert tortoise would likely be low given the low density of desert tortoises in the action area and the nature of the proposed activities, and we, therefore, anticipate that take of desert tortoise would also be low. We also recognize that for every desert tortoise found dead or injured, other individuals may be killed or injured that are not detected, so when we determine an appropriate take level, we are anticipating that the actual take would be higher, and we set the number below that level.

For the purposes of this analysis, we estimate that 1 adult desert tortoise and 7 juvenile tortoises reside in the Project area. Our estimate is derived from the average density of the Eastern Mojave Recovery Unit, and we expect that we have overestimated the number of individuals that are present because no tortoises or recent sign were observed on two pre-construction surveys. We anticipate that the number of individuals killed or wounded resulting from construction of the Project would be a subset of the number of desert tortoises in the Project area; however, because the number of desert tortoises in the Project area is so small, we are exempting take in the form of mortality or injury for all individuals. For these reasons, we will consider that the Applicant exceeded the amount or extent of take if more than 1 adult desert tortoise is killed or injured

during construction. We used adult desert tortoises to establish this amount of take because juvenile desert tortoises are difficult to find and locating the carcasses of small tortoises is unlikely. Detecting more than 1 killed or injured adult desert tortoise on the Project site would indicate that a larger number of juveniles may be killed or injured during construction and would require reinitiation. All incidental take is outlined in Table 4.

The number of desert tortoise hatchlings and eggs taken as a result of the proposed action is unknown, but we exempt the incidental take of all eggs. In the effects analysis, we explained that we cannot estimate the number of eggs that may be present because they are underground and vary with the season. Although we cannot estimate a level of incidental take for these life stages, we expect it would be low.

We are not establishing a Reinitiation criterion for the number of desert tortoises translocated from the Project area because the Applicant has agreed to translocate all desert tortoises it finds. We estimate that up to 8 tortoises (1 adult and 7 juveniles) may be captured and translocated during construction activities. In the effects analysis, we explained there may be some mortality of desert tortoises that are translocated; however, we expect this number to be low due to the proposed MMs.

We anticipate that activities during the O&M and decommissioning phases of the Project are likely to kill or injure few, if any, desert tortoises because of the low number of desert tortoises in the action area and the proposed MMs to protect tortoises during these activities. O&M activities are generally low throughout the year and decommissioning activities would mostly occur within the fenced area. All desert tortoises that are found during construction would be translocated away from the Project site and we expect the small area of remaining habitat will support few, if any, desert tortoises in the future due to increasing development and isolation from other desert tortoise habitat. We cannot predict how many desert tortoises the Applicant is likely to encounter during O&M and decommissioning activities because these individuals would have moved into the area from offsite habitat.

Table 4. Desert tortoise incidental take thresholds.

Project Activity	**Exempted Mortality or Injury**		**Exempted Capture and Relocation (all age classes)**	**Anticipated Habitat Loss (acres)**	
	Adults and sub-adults (≥180 mm	**Juveniles (<180 mm**		Critical	Non-critical
Construction	1	7	All desert tortoises in harm's way may be captured or moved; we estimate that 8 tortoises may be moved during the project.	0	85

REASONABLE AND PRUDENT MEASURES WITH TERMS AND CONDITIONS

The BLM and Applicant will implement numerous MMs for this project as part of the proposed action to minimize the incidental take of desert tortoise. Our evaluation of the proposed action is based on the assumption that the actions as set forth in the "Proposed Minimization Measures" section of this biological opinion will be implemented. The service believes these measures are adequate and appropriate to minimize the incidental take of desert tortoise. Therefore, we are not including any reasonable and prudent measures with terms and conditions in this incidental take statement. Any proposed changes to the MMs or in the conditions under which these activities will occur may constitute a modification of the proposed action and may warrant reinitiation of formal consultation, as specified at 50 CFR § 402.16.

REPORTING REQUIREMENTS

Pursuant to 50 CFR 402.14(i)(3), the BLM must report the progress of the action and its impact on the species to the Service as specified in this incidental take statement. The BLM will ensure that a report documenting desert tortoise encounters, incidental take (including capture and relocation), and effectiveness and compliance with the desert tortoise protection measures is prepared and submitted to the Service's Southern Nevada Fish and Wildlife Office in Las Vegas.

Reports are required quarterly during the duration of construction and annually during O&M for the life of the facilities. The BLM may delegate this responsibility to the Applicant. Additionally, final construction reports will be submitted to the Service within 60 days of completion of construction of the Project. All quarterly reports are due by the 10th of each of the following months (January, April, July, and October), and annual reports are due February 1 of each year. Annual status updates shall be provided to the Service during O&M activities for the life of the facility.

Specifically, all reports must include information on any instances when desert tortoises were killed, injured, or handled; the circumstances of such incidents; and any actions undertaken to prevent similar incidents from reoccurring. Additionally, the reports should provide details regarding each desert tortoise handled or observed, the names of all monitors involved in the project, and the ADTB who supervised their actions. Information will include the following: date and time of observation, whether the desert tortoise was handled, general health, whether it voided its bladder, location that the desert tortoise was moved from and location moved to, unique physical characteristics of each tortoise, and effectiveness and compliance with the desert tortoise protection measures. Any incident occurring during project activities that was considered by the ADTB or biological monitor to be in non-compliance with this biological opinion will be documented immediately by the ADTB.

DISPOSITION OF DEAD OR INJURED SPECIMENS

To ensure that the protective measures are effective and are being properly implemented, the BLM shall contact the Service immediately if a desert tortoise is killed or injured as a result of any activity covered under this biological opinion. Upon locating a dead or injured desert tortoise within the action area, notification must be made by phone to the Southern Nevada Fish and Wildlife Office (702-515-5230). At that time, the Service and the BLM shall review the circumstances surrounding the incident to determine whether additional protective measures are

required. Care should be taken when handling sick or injured individuals to ensure effective treatment and care, and in handling dead specimens to preserve biological material in the best possible state for later analysis of cause of death.

In conjunction with the care of sick or injured desert tortoises or preservation of biological materials from a dead animal, the finder has the responsibility to carry out instructions provided by the Service to ensure that evidence intrinsic to the specimen is not unnecessarily disturbed.

Injured desert tortoises shall be delivered to a qualified veterinarian for appropriate treatment or disposal. Dead desert tortoises suitable for preparation as museum specimens shall be frozen immediately and provided to an institution holding appropriate Federal and State permits per their instructions. Should no institutions want the desert tortoise specimens, or if it is determined that they are too damaged (crushed, spoiled, etc.) for preparation as a museum specimen, then they may be buried away from the project area or cremated, upon authorization by the Service. The Applicant shall bear the cost of any required treatment of injured desert tortoises, euthanasia of sick desert tortoises, or cremation of dead desert tortoises. Should sick or injured desert tortoises be treated by a veterinarian and survive, they may be transferred as directed by the Service.

CONSERVATION RECOMMENDATIONS

Section 7(a)(1) of the Act directs Federal agencies to use their authorities to further the purposes of the Act by carrying out conservation programs for the benefit of endangered and threatened species. Conservation recommendations are discretionary agency activities to minimize or avoid adverse effects of a proposed action on listed species or critical habitat, to help implement recovery plans, or to develop information.

1. We recommend that the Applicant elect to install temporary desert tortoise exclusion fencing around the laydown area and 80-foot ROW corridor.

2. We recommend that the BLM salvage plants on the solar project site for use in habitat enhancement or restoration. If the BLM chooses to salvage plants from permanent disturbance on the project site, these plants may be held in a nursery or other temporary holding location until needed. No monitoring or other requirements would be required for these plants.

3. We recommend that the BLM mark any desert tortoise that it translocates in accordance with the Desert Tortoise Recovery Office’s protocol (Service 2009). This marking will allow the Service to identify the translocated desert tortoises in the future, if we encounter them during range-wide sampling.

The Service requests notification of the implementation of any conservation recommendations so we may be kept informed of actions minimizing or avoiding adverse effects or benefitting listed species or their habitats.

REINITIATION NOTICE

This concludes formal consultation on the actions outlined in the request. As provided in 50 CFR § 402.16, reinitiation of formal consultation is required where discretionary Federal agency involvement or control over the action has been retained (or is authorized by law) and if: (1) the amount or extent of incidental take is exceeded; (2) new information reveals effects of the agency action that may affect listed species or critical habitat in a manner or to an extent not considered in this opinion; (3) the agency action is subsequently modified in a manner that causes an effect to the listed species or critical habitat not considered in this opinion; or (4) a new species is listed or critical habitat designated that may be affected by the action. In instances where the amount or extent of incidental take is exceeded, the exemption issued pursuant to section 7(o)(2) may have lapsed and any further take could be a violation of section 4(d) or 9. Consequently, we recommend that any operations causing such take cease pending reinitiation.

If you have any questions about this biological opinion, please contact Rebecca Windsor of my staff at (702) 515-5254, or by electronic mail at rebecca_windsor@fws.gov.

LITERATURE CITED

Abella, S.R. 2010. Disturbance and Plant Succession in the Mojave and Sonoran Deserts of the American Southwest. International Journal of Environmental Research and Public Health 7:1248-1284. doi:10.3390/ijerph7041248

Allison, L.J. and A.M. McLuckie. 2018. Population trends in Mojave desert tortoises (*Gopherus agassizii*). Herpetological Conservation and Biology 13(2):433-452.

[APLIC] Avian Power Line Interaction Committee. 2006. *Suggested Practices for Avian Protection On Power Lines: The State of the Art in 2006.* Available Online At: http://www.aplic.org/uploads/files/2643/SuggestedPractices2006%28LR-2%29.pdf (Accessed: April 26, 2022).

Averill-Murray, R.C. 2002. Effects on Survival of Desert Tortoises (*Gopherus agassizii*) Urinating During Handling. Chelonian Conservation and Biology 4(2):430-435.

Averill-Murray, R., C.R. Darst, N. Strout, and M. Wong. 2013. Conserving population linkages for the Mojave desert tortoise. Herpetological Conservation Biology 8(1):1-15.

Barrows, C.W. 2011. Sensitivity to climate change for two reptiles at the Mojave–Sonoran Desert interface. Journal of Arid Environments 75:629–635.

Beier, P., Majka, D., and Spencer, W. 2008. Forks in the road: Choices in procedures for designing wildland linkages. Conservation Biology 22:863-851.

[BLM] Bureau of Land Management. 1998. Las Vegas Resource Management Plan and Final Environmental Impact Statement. Volumes I and II. Dated May 1998. Las Vegas, NV.

[BLM] Bureau of Land Management. 2006. Noxious Weed Plan. Las Vegas Field Office, Bureau of Land Management. Dated December 2006. Las Vegas, Nevada.

[BLM] Bureau of Land Management. 2007. Vegetation Treatments on Bureau of Land Management Lands in 17 Western States. Programmatic Environmental Impact Statement. Bureau of Land Management, Washington, D.C.

[BLM] Bureau of Land Management. 2018. Common Raven Management Plan for Energy Development Within the BLM Southern Nevada District. Dated 2018. Las Vegas, Nevada.

[BLM] Bureau of Land Management. 2022. Biological Assessment for the Townsite Solar 2 Project. Bureau of Land Management, Southern Nevada District Office, Las Vegas, Nevada. Prepared for the BLM by Allied Pacific Partners, LLC, San Diego, California. Dated June 2022.

[BLM and DOE] Bureau of Land Management and U.S. Department of Energy. 2012. Final Programmatic Environmental Impact Statement (PEIS) for Solar Energy Development in

Six Southwestern States. Volumes 1 – 7. Dated July 2012. Washington, D.C.

Boarman, W.I. 2002. Threats to Desert Tortoise Populations: A Critical Review of the Literature. Unpublished report prepared for the West Mojave Planning Team, Bureau of Land Management. Western Ecological Research Center, U.S. Geological Survey, Sacramento, California. August 9, 2002.

Brooks, M.L., and T.C. Esque. 2002. Alien Plants and Fire in Desert Tortoise (*Gopherus agassizii*) Habitat of the Mojave and Colorado Deserts. Chelonian Conservation and Biology 4(2):330-340.

Christensen, J H., B. Hewitson, A. Busuioc, A. Chen, X. Gao, R. Held, R. Jones, R.K. Kolli, W.K.won, R. Laprise, V. Magana Rueda, L. Mearns, C.G. Menendez, J. Räisänen, A. Rinke, A. Sarr, P. Whetton, R. Arritt, R. Benestad, M. Beniston, D. Bromwich, D. Caya, J. Comiso, R. de Elia, and K.ethloff. 2007. Regional climate projections, Climate Change, 2007: The Physical Science Basis. Contribution of Working group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change, University Press, Cambridge, Chapter 11, ISBN: 978-0-521-88009-1.

Darst, C. 2014. Impervious surface calculations. Excel spreadsheet. Dated May 6. Fish and Wildlife Biologist, Desert Tortoise Recovery Office. Ventura, California.

Desert Conservation Plan. 2021. 2019-2021 Biennium Progress Report. December 2021. Accessed August 4, 2022. https://www.clarkcountynv.gov/government/departments/environment_and_sustainability/desert_conservation_program/biennial_progress_reports.php.

Devitt, D.A., L. Apodaca, B. Bird, J.P. Dawyot, Jr., L. Fenstermaker, and M.D. Petrie. 2022. Accessing the Impact of a Utility Scale Solar Photovoltaic Facility on a Down Gradient Mojave Desert Ecosystem. Land 11:1315.

Drake, K.K., T.C. Esque, K.E. Nussear, C. Aiello, P. Emblidge, and P.A. Medica. 2010. An Annual Report for Desert Tortoise Translocation Research at the Fort Irwin Southern Expansion Translocation Area. Las Vegas Field Station, Western Ecological Research Center, U.S. Geological Survey. Henderson, Nevada.

Esque, T.C., K.E. Nussear, K.K. Drake, A.D. Walde, K.H. Berry, R.C. Averill-Murray, A.P. Woodman, W.I. Boarman, P.A. Medica, J. Mack, and J.S. Heaton. 2010. Effects of subsidized predators, resource availability, and human population density on desert tortoise populations in the Mojave Desert, USA. Endangered Species Research 12:167-177.

Field, K.J., C.R. Tracy, P.A. Medica, R.W. Marlow, and P.S. Corn. 2007. Return to the wild: Translocation as a tool in conservation of the Desert Tortoise (*Gopherus agassizii*). Biological Conservation 136:232-245.

Fry, J.A., G. Xian, S. Jin, J.A. Dewitz, C.G. Homer, L. Yang, C.A. Barnes, N.D. Herold, and

J.D. Wickham. 2011. National Land Cover Database for the Conterminous United States. Analysis of land cover change in the continental United States from 2001 to 2006 using Landsat ETM+ and TM imagery.

Grippo, M., J.W. Hayse, and B.L. O'Conner. 2015. Solar Energy Development and Aquatic Ecosystems in the Southwestern United States: Potential Impacts, Mitigation, and Research Needs. Environmental Management 55:244-256.

Hagerty, B.E. 2008. Ecological Genetics of the Mojave Desert Tortoise. Dissertation, University of Nevada, Reno. 244 pp.

Hagerty, B.E., K.E. Nussear, T.C. Esque, and C.R. Tracy. 2011. Making molehills out of mountains: Landscape genetics of the Mojave desert tortoise. Landscape Ecology 26:267-280.

Hagerty, B.E. and C.R. Tracy. Defining population structure for the Mojave desert tortoise. 2010. Conservation Genetics 11:1795-1807.

Harless, M.L., A.D. Walde, D.K. Delaney, L.L. Pater, and W.K. Hayes. 2009. Home Range, Spatial Overlap, and Burrow Use of the Desert Tortoise in the West Mojave Desert. Copeia 2:378-389.

Hinderle, D., R.L. Lewison, A.D. Walde, D. Deutschman, and W.I. Boarman. 2015. The Effects of Homing and Movement Behaviors on Translocation: Desert Tortoises in the Western Mojave Desert. The Journal of Wildlife Management 79(1):137-147.

[NRCS] Natural Resources Conservation Service. 2022. Web Soil Survey. Available at: https://websoilsurvey.sc.egov.usda.gov/app/WebSoilSurvey.aspx. Accessed August 9, 2022.

Nussear, K.E. 2004. Mechanistic investigation of the distributional limits of the desert tortoise, *Gopherus agassizii*. Dissertation, University of Nevada, Reno. 213 pp.

Nussear, K.E., T.C. Esque, R.D. Inman, L. Gass, K.A. Thomas, C.S.A. Wallace, J.B. Blainey, D.M. Miller, and R.H. Webb. 2009. Modeling habitat of the desert tortoise (*Gopherus agassizii*) in the Mojave and parts of the Sonoran Deserts of California, Nevada, Utah, and Arizona. U.S. Geological Survey Open-File Report 2009-1102.

Nussear, K.E., C.R. Tracy, P.A. Medica, D.S. Wilson, R.W. Marlow, and P.S. Corn. 2012. Translocation as a Conservation Tool for Agassiz's Desert Tortoises: Survivorship, Reproduction, and Movements. The Journal of Wildlife Management 76(7):1341-1353. DOI: 10.1002/jwmg.390

Saethre, M.B., T.C. Esque, P.A. Medica, R. Marlow, and C.R. Tracy. 2003. Determining carrying capacity of desert tortoises. Pages 149-150 in Proceedings of the 2003 Symposia of the Desert Tortoise Council, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 1994. Desert tortoise (Mojave population) recovery plan. Portland, Oregon.

[Service] U.S. Fish and Wildlife Service. 1997. Programmatic Biological Opinion for Implementation of Multiple Use Activities Within the Las Vegas Field Office. Dated November 25. Memorandum to District Manager, Las Vegas District, Bureau of Land Management, Las Vegas, Nevada. From Field Supervisor, Reno Fish and Wildlife Office, Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 1998. Biological Opinion for Implementation of Proposed Actions in the Las Vegas District's Proposed Resource Management Plan/Final Environmental Impact Statement. Dated June 18. Memorandum to District Manager, Las Vegas District, Bureau of Land Management, Las Vegas, Nevada. From Field Supervisor, Reno Fish and Wildlife Office, Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2000. Intra-Service Biological and Conference Opinion on Issuance of an Incidental Take Permit to Clark County, Nevada for a Multiple Species Habitat Conservation Plan. Dated November 19. Memorandum to Manager, California Nevada Operations Office, Sacramento, California. From Field Supervisor, Nevada Fish and Wildlife Office, Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2009. Desert Tortoise (Mojave Population) Field Manual (*Gopherus agassizii*). Region 8, Sacramento, California.

[Service] U.S. Fish and Wildlife Service. 2010. Mojave population of the desert tortoise (*Gopherus agassizii*) 5-year review: summary and evaluation. Desert Tortoise Recovery Office. Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2011. Revised recovery plan for the Mojave population of the desert tortoise (*Gopherus agassizii*). Sacramento, California.

[Service] U.S. Fish and Wildlife Service. 2012a. Biological opinion for the Techren Boulder City Solar Project, Boulder City, Clark County, Nevada. Dated December 28. Memorandumto Field Manager, Las Vegas Field Office, Bureau of Land Management, Las Vegas, Nevada. From State Supervisor, Nevada Fish and Wildlife Office. Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2012b. Biological opinion for the K Road Moapa Solar Project, Moapa River Indian Reservation, Clark County, Nevada. Dated March 7. Memorandum to Superintendent, Southern Paiute Agency, Bureau of Indian Affairs. St. George, Utah. From State Supervisor, Nevada Fish and Wildlife Office. Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2013a. Formal Programmatic Consultation under Section 7 of the Endangered Species Act for Effects to Threatened and Endangered species and their Critical Habitat that May Occur as a Result of Actions Proposed by the Southern Nevada District Office, Bureau of Land Management. Dated January 2. Memorandum to Assistant Field Manager, Division of Renewable Resources, Las Vegas

Field Office, Bureau of Land Management, Las Vegas, Nevada. From State Supervisor, Nevada Fish and Wildlife Office, Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2013b. Biological opinion for the Stateline Solar and Silver State Solar South Projects, San Bernardino County, California, and Clark County, Nevada. Dated September 30. Memorandum to Field Manager, Needles Field Office, Bureau of LandManagement, Needles California, and Assistant Field Manager, Las Vegas Field Office, Bureau of Land Management, Las Vegas, Nevada. From Acting Field Supervisor, Ventura Fish and Wildlife Office. Ventura, California.

[Service] U.S. Fish and Wildlife Service. 2014. Biological opinion for the Townsite Solar Transmission Project. Dated July 24. Memorandum to Environmental Manager, Western Area Power Administration, U.S. Department of Energy, Phoenix, Arizona. From Assistant Field Supervisor, Nevada Fish and Wildlife Office. Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2015. Programmatic Biological Opinion and Informal Consultation for Western Area Power Administration (Western) West Area Operations and Maintenance Programs, Nevada, Arizona, and California. Dated September 21. Memorandum to Environmental Manager, Western Area Power Administration, U.S. Department of Energy, Phoenix, Arizona. From Field Supervisor, Southern Nevada Fish and Wildlife Office, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2017. Preparing for any action that may occur within the range of the Mojave desert tortoise (*Gopherus agassizii*). Region 8, Sacramento, California.

[Service] U.S. Fish and Wildlife Service. 2019a. Formal Programmatic Consultation under Section 7 of the Endangered Species Act for Effects to Threatened and Endangered Species and their Critical Habitat that May Occur as a Result of Actions Proposed by the Southern Nevada District Office, Bureau of Land Management. Dated January 14, 2020. Memorandum to Field Manager, Division of Renewable Resources, Bureau of Land Management, Las Vegas Field Office, Las Vegas, Nevada. From Field Supervisor, Southern Nevada Fish and Wildlife Office, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2019b. Second Reinitiation of Formal Consultation for the Southern Nevada Intertie Project (DesertLink) Transmission Line Biological Opinion for the Bureau of Land Management Southern Nevada District Office, Nevada (File No. 84320-2015-F-0034). Dated November 14. Memorandum to Assistant Field Manager, Bureau of Land Management, Las Vegas, Nevada. From Field Supervisor, Southern Nevada Fish and Wildlife Office, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2019c. Request to Append the Sloan to Mead Rebuild 230 kV Transmission Line Replacement to the Bureau of Land Management's Southern Nevada District Office's Programmatic Biological Opinion. Dated January 9. Memorandum to Assistant Field Manager, Las Vegas Field Office, Bureau of Land Management, Las Vegas, Nevada. From Field Supervisor, Southern Nevada Fish and

Wildlife Office, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2019d. Translocation of Mojave Desert Tortoises from Project Sites: Plan Development Guidance. Dated December 2019. U.S. Fish and Wildlife Service, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2020. Range-wide Monitoring of the Mojave Desert Tortoise (*Gopherus agassizii*): 2019 Annual Reporting. Report by the Desert Tortoise Recovery Office, U.S. Fish and Wildlife Service, Reno, Nevada.

[Service] U.S. Fish and Wildlife Service. 2021. Amendment to Biological Opinion for Techren Boulder City Solar Project, Clark County, Nevada. Dated December 21. Memorandum to Assistant Field Manager, Bureau of Land Management, Division of Resources, Las Vegas, Nevada. From Field Supervisor, Southern Nevada Fish and Wildlife Office, Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2022a. Mojave desert tortoise (*Gopherus agassizii*) 5-year review: Summary and evaluation. Dated May. Desert Tortoise Recovery Office, Southern Nevada Fish and Wildlife Office. Las Vegas, Nevada.

[Service] U.S. Fish and Wildlife Service. 2022b. Programmatic Biological Opinion for Federal Highway Administration activities implemented by the Nevada Department of Transportation (2022-003069-S7). Dated April 18. Letter to FHWA, Carson City, Nevada. From Field Supervisor, Nevada Fish and Wildlife Office, Reno, Nevada.

Tracy, C.R., R. Averill-Murray, W.I. Boarman, D. Delehanty, J. Heaton, E. McCoy, D. Morafka, K. Nussear, B. Hagerty, and P. Medica. 2004. Desert tortoise recovery plan assessment. Prepared for the U.S. Fish and Wildlife Service. Reno, Nevada.

Turner, F.B., K.H. Berry, D.C. Randall, and G.C. White. 1987. Population Ecology of the Desert Tortoise at Goffs, California, 1983-1986. Annual Report to the Southern California Edison Company, Rosemead, California.

[USGS] U.S. Geological Survey. 2005. National Gap Analysis Program. Southwest Regional GAP Analysis Project-Land Cover Descriptions. RS/GIS Laboratory, College of Natural Resources, Utah State University.

APPENDIX F

Air Quality and Greenhouse Gas Support Data and Calculations

Townsite Solar - NAAQS and Nevada AAQS

Pollutant	Averaging Period	Primary NAAQS	Secondary NAAQS	Nevada State AAQS
NO2	1-hour	0.100 ppm*	N/A	0.100 ppm*
	Annual	0.053 ppm†	0.053 ppm†	0.053 ppm†
SO2	1-hour	0.075 ppm‡	N/A	N/A
	3-hour	N/A	0.50 ppm§	0.5 ppm¶
	24-hour	N/A	N/A	0.14 ppm¶
	Annual	N/A	N/A	0.03 ppm†
PM10	24-hour	150 µg/m#	150 µg/m#	150 µg/m#
PM2.5	24-hour	35 µg/m**	35 µg/m**	35 µg/m**
	Annual	12 µg/m††	15 µg/m††	12 µg/m††
CO	1-hour	35 ppm§	N/A	35 ppm§
	8-hour	9 ppm§	N/A	6 ppm¶, ‡‡
Pb	3-month	0.15 µg/m¶	0.15 µg/m ¶	0.15 µg/m ¶
O3	8-hour	0.070 ppm§§	0.070 ppm§§	0.070 ppm§§
H2S	1-hour	N/A	N/A	0.08 ppm¶

Sources: EPA (2022a)

Notes: N/A = not applicable; H2S = hydrogen sulfide; ppm = parts per million; µg/m = micrograms per cubic meter

EPA has proposed a final rule to reduce the PM2.5 annual standard to between 9 and 10 µg/m3

*The standard is based on the 3-year average of the 98th percentile of the daily maximum 1-hour average.

†Annual mean value.

‡The standard is based on the 3-year average of the 99th percentile of the daily maximum 1-hour average.

§Not to be exceeded more than once per calendar year.

¶Not to be exceeded.

Not to be exceeded more than once per calendar year on average over 3 years.

**The standard is based on the 3-year average of the 98th percentile of the 24-hour average.

†† Annual mean value. The standard is based on the 3-year average of the weighted annual mean.

‡‡The standard is 9 ppm for areas with an elevation less than 5,000 feet above mean sea level.

§§The standard is based on the annual fourth-highest daily maximum 8-hour concentration averaged over 3 years.

Townsite Solar - NAAQS and Nevada AAQS

EPA Air Quality Standards:

Ozone: 0.12 ppm (1-hour), 0.070 ppm (8-hour)

Location	2021					
1200 Ave G. Boulder City, Nv	8-hour			1-hour		
	1st High	2nd High	Days Exceeding Standard	1st High	2nd High	Days Exceeding Standard
	0.072	0.070	1	0.077	0.076	0
	2022					
	8-hour			1-hour		
	1st High	2nd High	Days Exceeding Standard	1st High	2nd High	Days Exceeding Standard
	0.072	0.069	1	0.076	0.073	0

*The 1200 Ave G. Boulder City, Nv monitoring station started data collection in 2021

Source: EPA. Air Monitor Reports. Available at: https://www.epa.gov/outdoor-air-quality-data/monitor-values-report

Townsite Solar - NAAQS and Nevada AAQS

Summary of Commuting Emissions

Activity	Emission Totals (Tons)							
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs
Townsite Solar Annual Operation	28.01	0.12	0.09	0.01	0.004	0.000	0.02	0.007

Summary of Fugitive Dust Emissions

Activity	Emission Totals (Tons)	
	PM_{10}	$PM_{2.5}$
Townsite Solar Annual Operation	0.26	0.039

Operation Emissions Summary

Activity	Annual Emission Totals (Tons)							
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs
Non-Road and On-Road Commute Vehicles	28.007	0.121	0.095	0.006	0.004	0.000	0.018	0.007
Fugitive Dust	0.0	0.0	0.0	0.26	0.039	0.0	0.0	0.0
TOTAL	28.01	0.12	0.09	0.26	0.044	0.00	0.02	0.007

Emission Totals (Metric Tons)			
CO_2e	CO_2	CH_4	N_2O
25.41	25.35	0.0010	0.0001

Townsite Solar - NAAQS and Nevada AAQS

Site-Specific Information

Parameter	Units	Value
Average Weight of Vehicle (W)[2]	tons	3
Precip > 0.01 in (p)[1]	days/yr	102

Emission Factor Equation Parameters

Parameter	Units	PM-10	PM-2.5
Surface Material Silt Loading (sL)[3]	g/m2	0.6	
Particle Size Multiplier (k)[4]	lb/VMT	0.0022	0.00054

Calculated Emission Factors

	Units	PM-10	PM-2.5
E (Short-Term Emission Factor)	lb/VMT	0.0042	0.00104
Eext (Annual Emission Factor)	lb/VMT	0.0039	0.00097

Facility	County	Paved Road Vehicle Miles Traveled (VMT) [5]	Total PM Emissions (tons)	
			Annual	
			PM10	PM2.5
Townsite Solar Annual Operation	N/A	42,924	0.085	0.021
	Total	42,924	0.085	0.021

Calculation Sources:

(1) EPA's AP 42 Fifth Edition Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources Chapter 13.0 – Introduction to Miscellaneous Sources

Section 13.2 – Introduction to Fugitive Dust Sources Final Section of 13.2.1 Paved Roads (January 2011)

Emission Factor Equations:

Equation 1 from Calculation Source (1) - Hourly Emission Factor (lb/VMT): E = k*[sL^0.91]*(W^1.02)

Equation 2 from Calculation Source (1) - Annual Emission Factor (lb/VMT): Eext = E*[1-P/(4*N)]

Where:

E = particulate emission factor (having units matching the units of k);

k = particle size multiplier for particle size range and units of interest;

sL = road surface silt loading (grams per square meter) (g/m2); and

W = average weight (tons) of the vehicles traveling the road.

Notes:

1 30-year average of days > 0.1 inch precipitation (1981-2010)

2 Based on estimate of the average vehicle weight for onroad vehicles, including worker commutes

3 Based on Table 13.2.1-2 of Calculation Source (1); worst-case baseline

4 Based on Table 13.2.1-1 of Calculation Source (1) for paved roads

5 Based on the number of worker round trips times a round trip distance of forty miles. It is assumed that 98% of travel takes place on paved roads due to fugitive dust plan requirement to limit unnecessary unpaved road travel and because most of the roads in the project area are paved.

Townsite Solar - NAAQS and Nevada AAQS

Site-Specific Information

Parameter	Units	Value
Average Weight of Vehicle (W)[1]	tons	3
Precip > 0.01 in (p)[2]	days/yr	102

Emission Factor Equation Parameters

Parameter	Units	PM-10	PM-2.5
Surface Material Silt Content (s)3	%	8.5	
Particle Size Multiplier (k)[4]	lb/VMT	1.5	0.15
Emperical Constant (a)4	--	0.9	0.7
Emperical Constant (b)4	--	0.45	0.45
Water Spray Control Efficiency	%	50	

Calculated Emission Factors

	Units	PM-10	PM-2.5
E	lb/VMT	0.550	0.059
Eext	lb/VMT	0.396	0.042

Facility	County	Unpaved Road Vehicle Miles Traveled (VMT) [5]	Total PM Emissions (tons)	
			Annual	
			PM10	PM2.5
Townsite Solar Annual Operations	N/A	876	0.1735	0.0186
	Total	876	0.1735	0.0186

Calculation Sources:

(1) EPA's AP 42 Fifth Edition Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources Chapter 13.0 – Introduction to Miscellaneous Sources,

Section 13.2 – Introduction to Fugitive Dust Sources Final Section of 13.2.2 Unpaved Roads (November 2006)

Emission Factor Equations:

Equation 1a from Calculation Source (1) - Hourly Emission Factor (lb/VMT): E = k*[(s/12)^a]*(W/3)^b

Equation 2 from Calculation Source (1) - Annual Emission Factor (lb/VMT): Eext = E*[(365-P)/365]

Where:

E = particulate emission factor (having units matching the units of k);

k = particle size multiplier for particle size range and units of interest;

sL = road surface silt loading (grams per square meter) (g/m2); and

W = average weight (tons) of the vehicles traveling the road.

Water spray control efficiency is based on fugitive dust plan measures and WRAP Fugitive Dust Handbook control efficiency values.

Notes:

1 30-year average of days > 0.1 inch precipitation (1981-2010)

2 Based on estimate of the average vehicle weight for onroad vehicles, including worker commutes and material removal/delivery vehicles

3 Based on Table 13.2.1-2 of Calculation Source (1); worst-case baseline

4 Based on Table 13.2.1-1 of Calculation Source (1) for paved roads

5 Based on the number of workers and the round trip distance of forty miles. It is assumed that 2% of travel takes place on unpaved roads due to fugitive dust plan requirement to limit unnecessary unpaved road travel and because most of the roads in the project area are paved.

Townsite Solar - NAAQS and Nevada AAQS

Operations/Maintenance Workers		Round Trip Distance	Number of Vehicle Miles Traveled per year	MOVES Emission Factors (g/VMT)											Annual Emission Totals (Tons)									
		(miles)		CO2	CH4	NO2	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	NO2	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Number of Operations/Maintenance Workers (3 workers daily assumed)	3	40	43,800.0	578.7421	0.0229	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	27.9424	0.0011	0.0001	28.0070	0.1208	0.0946	0.0061	0.0044	0.0001	0.0182	0.0073
									Total Onroad Exhaust Emissions (direct and indirect)						27.9424	0.0011	0.0001	28.0070	0.1208	0.0946	0.0061	0.0044	0.0001	0.0182	0.0073

Townsite Solar - NAAQS and Nevada AAQS

Summary of Construction Activity & Worker Mobile Emissions

Year	Emission Totals (Tons)							
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs
Townsite Solar 2023	4,574.69	7.94	11.99	0.62	0.52	0.01	1.25	0.50

Summary of Fugitive Dust Emissions

Year	Emission Totals (Tons)	
	PM_{10}	$PM_{2.5}$
Townsite Solar 2023	50.48	5.96

Emission Totals (Metric Tons) Amortized			
CO_2e	CO_2	CH_4	N_2O
138.34	137.56	0.00	0.00

Year	Emission Totals - Tons per Year							
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs
Townsite Solar 2023	4,574.69	7.94	11.99	51.10	6.48	0.01	1.25	0.50

Year	Emission Totals (Metric Tons)			
	CO_2e	CO_2	CH_4	N_2O
Townsite Solar 2023	4,150.09	4,126.76	0.07	0.08

Townsite Solar - NAAQS and Nevada AAQS

County	Construction Year	Acreage Disturbed	Months of Construction	Exposed Soils (acre-months)	Total PM Emissions (tons)	
					PM10	PM2.5
N/A	2023	50.0	12	600	37.752	4.014

Fugitive Dust Emission Factors (Construction)

PM10[1] 5.50E-02 ton/acre-month

PM2.5[1,2] 5.50E-03 ton/acre-month

[1]WRAP Fugitive Dust Handbook, Table 3-2, level 1, average conditions. Fifty percent control efficiency is applied to account for watering control measures.

[2]WRAP Fugitive Dust Handbook, September 2006, Section 3.4.1. PM2.5 to PM10 ratio for fugitive dust from construction is assumed to be 0.1.

Fugitive Dust Emission Factors (Wind Erosion)

PM10[3,5] 7.92E-03 ton/acre-month

PM2.5[3,4,5] 1.19E-03 ton/acre-month

[3]Wind erosion of exposed areas (seeded land, stripped or graded overburden) = 0.38 ton TSP/acre/yr (WRAP Fugitive Dust Handbook, Table 11-6)

[4]PM10/TSP = 0.5, PM2.5/PM10 = 0.15, (WRAP Fugitive Dust Handbook, Section 7-2)

[5]Water and other approved dust suppressants would be used at construction sites (50% minumum control applied per WRAP Fugitive Dust Handbook).

Summary of Construction Activity Mobile Emissions per Year

Activity	Emission Totals (Tons per year)								Emission Totals (Tons per year)				Emission Totals (metric tons per year)			
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs	CO_2e	CO_2	CH_4	N_2O	CO_2e	CO_2	CH_4	N_2O
Geotechnical Exploration	27.19	0.01	0.08	0.00	0.00	0.000	0.00	0.00	27.19	27.00	0.0002	0.0007	24.67	24.49	0.00	0.00
Site Preparation - Install Best Management Practices	13.03	0.01	0.03	0.00	0.00	0.000	0.00	0.00	13.03	12.95	0.0002	0.0003	11.82	11.75	0.00	0.00
Site Preparation - Solar Arrays	583.62	0.22	1.16	0.04	0.04	0.002	0.04	0.02	583.62	579.53	0.00	0.01	529.45	525.74	0.00	0.01
Roads & Fencing	665.67	0.28	1.68	0.05	0.05	0.002	0.06	0.02	665.67	660.99	0.00	0.02	603.89	599.64	0.00	0.02
Site Preparation - Post Installation	312.09	0.37	0.99	0.06	0.06	0.001	0.06	0.02	312.09	309.86	0.00	0.01	283.13	281.10	0.00	0.01
Install Support Structures	108.85	0.11	0.27	0.02	0.02	0.000	0.02	0.01	108.85	108.08	0.00	0.00	98.75	98.05	0.00	0.00
Install Inverters, Switchgear, and Support Equipment	378.22	0.24	0.84	0.04	0.04	0.001	0.05	0.02	378.22	375.55	0.00	0.01	343.12	340.70	0.00	0.01
Install DC and AC Electrical	1,104.84	0.72	2.27	0.12	0.11	0.003	0.12	0.05	1,104.84	1,097.03	0.01	0.03	1,002.29	995.21	0.01	0.03
TOTAL	3,193.53	1.98	7.32	0.32	0.31	0.01	0.35	0.14	3,193.53	3,170.99	0.02	0.08	2,897.12	2,876.67	0.02	0.07

Summary of Construction Worker Mobile Emissions per Year

Year	Emission Totals (Tons)								Emission Totals (Tons)				Emission Totals (metric Tons)			
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs	CO_2e	CO_2	CH_4	N_2O	CO_2e	CO_2	CH_4	N_2O
Townsite Solar 2023	1,381.17	5.96	4.67	0.30	0.22	0.00	0.90	0.36	1,381.17	1,377.98	0.05	0.01	1,252.97	1,250.08	0.05	0.01

Summary of Construction Activity Mobile Emissions per Year

Year	Emission Totals (Tons)								Emission Totals (Tons)				Emission Totals (metric Tons)			
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs	CO_2e	CO_2	CH_4	N_2O	CO_2e	CO_2	CH_4	N_2O
Townsite Solar 2023	3,193.53	1.98	7.32	0.32	0.31	0.01	0.35	0.14	3,193.53	3,170.99	0.02	0.08	2,897.12	2,876.67	0.02	0.07

Year	Emission Totals (Tons)								Emission Totals (Tons)				Emission Totals (metric tons)			
	CO_2e	CO	NO_X	PM_{10}	$PM_{2.5}$	SO_2	VOC	HAPs	CO_2e	CO_2	CH_4	N_2O	CO_2e	CO_2	CH_4	N_2O
TOTAL - 2023	4,574.69	7.94	11.99	0.62	0.52	0.01	1.25	0.50	4,574.69	4,548.97	0.08	0.09	4,150.09	4,126.76	0.07	0.08

Construction Workers			Round Trip Distance	Annual Number of Vehicle Miles Traveled	MOVES Emission Factors (g/VMT)												Annual Emission Totals (Tons)										
			(miles)		CO2	CH4	BSFS	NO2	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	NO2	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Number of Construction Workers	200		30	2,160,000.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	1377.9805	0.0545	0.0057	1381.1654	5.9581	4.6663	0.3019	0.2181	0.0047	0.8994	0.3607
											Total Onroad Exhaust Emissions (direct and indirect)						1377.981	0.055	0.006	1381.165	5.958	4.666	0.302	0.218	0.005	0.899	0.361

Townsite Solar - NAAQS and Nevada AAQS

Site-Specific Information

Parameter	Units	Value
Average Weight of Vehicle (W)[2]	tons	3
Precip > 0.01 in (p)[1]	days/yr	102

Emission Factor Equation Parameters

Parameter	Units	PM-10	PM-2.5
Surface Material Silt Loading (sL)[3]	g/m2	0.6	
Particle Size Multiplier (k)[4]	lb/VMT	0.0022	0.00054

Calculated Emission Factors

	Units	PM-10	PM-2.5
E (Short-Term Emission Factor)	lb/VMT	0.0042	0.00104
Eext (Annual Emission Factor)	lb/VMT	0.0039	0.00097

Facility	County	Paved Road Vehicle Miles Traveled (VMT) [5]	Total PM Emissions (tons)	
			Annual	
			PM10	PM2.5
Townsite Solar 2023	N/A	2,116,800	4.173	1.024

Calculation Sources:

(1) EPA's AP 42 Fifth Edition Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources Chapter 13.0 – Introduction to Miscellaneous Sources

Section 13.2 – Introduction to Fugitive Dust Sources Final Section of 13.2.1 Paved Roads (January 2011)

Emission Factor Equations:

Equation 1 from Calculation Source (1) - Hourly Emission Factor (lb/VMT): E = k*[sL^0.91]*(W^1.02)

Equation 2 from Calculation Source (1) - Annual Emission Factor (lb/VMT): Eext = E*[1-P/(4*N)]

Where:

E = particulate emission factor (having units matching the units of k);

k = particle size multiplier for particle size range and units of interest;

sL = road surface silt loading (grams per square meter) (g/m2); and

W = average weight (tons) of the vehicles traveling the road.

Notes:

1 30-year average of days > 0.1 inch precipitation (1981-2010)

2 Based on estimate of the average vehicle weight for onroad vehicles, including worker commutes

3 Based on Table 13.2.1-2 of Calculation Source (1); worst-case baseline

4 Based on Table 13.2.1-1 of Calculation Source (1) for paved roads

5 Based on the number of worker round trips times a round trip distance of forty miles. It is assumed that 98% of travel takes place on paved roads due to fugitive dust plan requirement to limit unnecessary unpaved road travel and because most of the roads in the project area are paved.

Townsite Solar - NAAQS and Nevada AAQS

Site-Specific Information

Parameter	Units	Value
Average Weight of Vehicle (W)[1]	tons	3
Precip > 0.01 in (p)[2]	days/yr	102

Emission Factor Equation Parameters

Parameter	Units	PM-10	PM-2.5
Surface Material Silt Content (s)3	%	8.5	
Particle Size Multiplier (k)[4]	lb/VMT	1.5	0.15
Emperical Constant (a)4	--	0.9	0.7
Emperical Constant (b)4	--	0.45	0.45
Water Spray Control Efficiency	%	50	

Calculated Emission Factors

	Units	PM-10	PM-2.5
E	lb/VMT	0.550	0.059
Eext	lb/VMT	0.396	0.042

Facility	County	Unpaved Road Vehicle Miles Traveled (VMT) [5]	Total PM Emissions (tons)	
			Annual	
			PM10	PM2.5
Townsite Solar 2023	N/A	43,200	8.5584	0.9169

Calculation Sources:

(1) EPA's AP 42 Fifth Edition Compilation of Air Pollutant Emission Factors, Volume 1: Stationary Point and Area Sources Chapter 13.0 – Introduction to Miscellaneous Sources,

Section 13.2 – Introduction to Fugitive Dust Sources Final Section of 13.2.2 Unpaved Roads (November 2006)

Emission Factor Equations:

Equation 1a from Calculation Source (1) - Hourly Emission Factor (lb/VMT): E = k*[(s/12)^a]*(W/3)^b

Equation 2 from Calculation Source (1) - Annual Emission Factor (lb/VMT): Eext = E*[(365-P)/365]

Where:

E = particulate emission factor (having units matching the units of k);

k = particle size multiplier for particle size range and units of interest;

sL = road surface silt loading (grams per square meter) (g/m2); and

W = average weight (tons) of the vehicles traveling the road.

Water spray control efficiency is based on fugitive dust plan measures and WRAP Fugitive Dust Handbook control efficiency values.

Notes:

1 30-year average of days > 0.1 inch precipitation (1981-2010)

2 Based on estimate of the average vehicle weight for onroad vehicles, including worker commutes and material removal/delivery vehicles

3 Based on Table 13.2.1-2 of Calculation Source (1); worst-case baseline

4 Based on Table 13.2.1-1 of Calculation Source (1) for paved roads

5 Based on the number of workers and the round trip distance of forty miles. It is assumed that 2% of travel takes place on unpaved roads due to fugitive dust plan requirement to limit unnecessary unpaved road travel and because most of the roads in the project area are paved.

Townsite Solar - NAAQS and Nevada AAQS

Site Preparation - Install Best Management Practices

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)										
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	**Nonroad construction equipment**																										
Forklift	Rough Terrain Forklifts	4	10	300.0	32524.5564	0.3258	10199.7084	0.8234	32759.0585	31.8977	80.4851	4.8605	4.7147	0.0920	4.5494	1.9335	10.7557	0.0001	0.0003	10.8332	0.0105	0.0266	0.0016	0.0016	0.0000	0.0015	0.0006
**Specific horsepowers were not available so emission factors are based on g/hr																											
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																						
Delivery/Work Trucks	Construction Pick-up diesel vehicles	2	20	1,200.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	0.5742	0.0000	0.0000	0.5755	0.0025	0.0019	0.0001	0.0001	0.0000	0.0004	0.0002
Delivery/Work Trucks	Heavy duty diesel vehicles (>6,000 lb GVW)	4	10	1,200.0	1633.0408	0.0200	-	0.0021	1634.2034	1.9802	4.0794	0.2051	0.0969	0.0055	0.1350	0.0355	1.6201	0.0000	0.0000	1.6213	0.0020	0.0040	0.0002	0.0001	0.0000	0.0001	0.0000
																Total Construction Equipment Emissions	12.950	0.000	0.000	13.030	0.015	0.033	0.002	0.002	0.000	0.002	0.001

Site Preparation - Solar Arrays

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)										
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	**Nonroad construction equipment**																									
Water Truck	Off Highway Truck	4	10	300.0	248212.2308	1.3901	77822.9137	6.2826	249968.8077	60.4068	686.6881	11.2143	10.8778	0.6675	16.2729	7.2830	229.8300	0.0013	0.0058	231.4565	0.0559	0.6358	0.0104	0.0101	0.0006	0.0151	0.0067
Compactor	Other Construction Equipment	4	10	300.0	104149.6637	1.0917	32666.9412	2.6372	104902.1478	124.3094	339.3837	19.6859	19.0954	0.3175	20.7478	7.8014	34.4416	0.0004	0.0009	34.6904	0.0411	0.1122	0.0065	0.0063	0.0001	0.0069	0.0026
Graders	Graders	6	10	300.0	64841.2264	0.2423	20328.7132	1.6411	65296.4763	16.5738	51.7574	3.0811	2.9886	0.1756	2.9071	1.1692	90.0588	0.0003	0.0023	90.6911	0.0230	0.0719	0.0043	0.0042	0.0002	0.0040	0.0016
Scrapers	Graders	8	10	300.0	64841.2264	0.2423	20328.7132	1.6411	65296.4763	16.5738	51.7574	3.0811	2.9886	0.1756	2.9071	1.1692	68.6162	0.0003	0.0017	69.0980	0.0175	0.0548	0.0033	0.0032	0.0002	0.0031	0.0012
Tractor/Disc	Crawler Tractor/Dozers	6	10	300.0	82758.1508	0.4236	25948.0144	2.0948	83342.6467	37.4722	138.1369	5.8424	5.6671	0.2288	6.0237	2.4734	114.9438	0.0006	0.0029	115.7556	0.0520	0.1919	0.0081	0.0079	0.0003	0.0084	0.0034
BackHoe/Front Loader	Rubber Tire Loaders	4	10	300.0	77245.0705	0.5996	24222.5435	1.9555	77796.7825	53.3570	177.0397	8.3987	8.1467	0.2199	9.1142	3.7619	40.8711	0.0003	0.0010	41.1630	0.0282	0.0937	0.0044	0.0043	0.0001	0.0048	0.0020
**Specific horsepowers were not available so emission factors are based on g/hr	**Specific horsepowers were not available so emission factors are based on g/hr																									
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																					
Truck, Pick up (Survey Crew)	Construction Pick-up diesel vehicles	4	20	2,400.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	0.7655	0.0000	0.0000	0.7673	0.0033	0.0026	0.0002	0.0001	0.0000	0.0005	0.0002
															Total Construction Equipment Emissions		579.527	0.003	0.015	583.622	0.221	1.163	0.037	0.036	0.002	0.043	0.018

Roads & Fencing

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)											
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																										
Water Truck	Off Highway Truck	4	10	300.0	248212.2308	1.3901	77822.9137	6.2826	249968.8077	60.4068	686.6881	11.2143	10.8778	0.6675	16.2729	7.2830	229.8300	0.0013	0.0058	231.4565	0.0559	0.6358	0.0104	0.0101	0.0006	0.0151	0.0067
Dump Truck	Off Highway Truck	8	10	300.0	248212.2308	1.3901	77822.9137	6.2826	249968.8077	60.4068	686.6881	11.2143	10.8778	0.6675	16.2729	7.2830	196.9971	0.0011	0.0050	198.3913	0.0479	0.5450	0.0089	0.0086	0.0005	0.0129	0.0058
Compactor	Other Construction Equipment	4	10	300.0	104149.6637	1.0917	32666.9412	2.6372	104902.1478	124.3094	339.3837	19.6859	19.0954	0.3175	20.7478	7.8014	96.4365	0.0010	0.0024	97.1333	0.1151	0.3142	0.0182	0.0177	0.0003	0.0192	0.0072
Graders	Graders	6	10	300.0	64841.2264	0.2423	20328.7132	1.6411	65296.4763	16.5738	51.7574	3.0811	2.9886	0.1756	2.9071	1.1692	90.0588	0.0003	0.0023	90.6911	0.0230	0.0719	0.0043	0.0042	0.0002	0.0040	0.0016
Rough Terrain Forklift	Rough Terrain Forklifts	4	10	300.0	32524.5564	0.3258	10199.7084	0.8234	32759.0585	31.8977	80.4851	4.8605	4.7147	0.0920	4.5494	1.9335	10.7557	0.0001	0.0003	10.8332	0.0105	0.0266	0.0016	0.0016	0.0000	0.0015	0.0006
Backhoe/ Front Loader	Rubber Tire Loaders	2	10	300.0	77245.0705	0.5996	24222.5435	1.9555	77796.7825	53.3570	177.0397	8.3987	8.1467	0.2199	9.1142	3.7619	35.7622	0.0003	0.0009	36.0176	0.0247	0.0820	0.0039	0.0038	0.0001	0.0042	0.0017
**Specific horsepowers were not available so emission factors are based on g/hr	**Specific horsepowers were not available so emission factors are based on g/hr																										
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																						
Delivery/Work Trucks	Construction Pick-up diesel vehicles	6	20	3,600.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	1.1483	0.0000	0.0000	1.1510	0.0050	0.0039	0.0003	0.0002	0.0000	0.0007	0.0003
																Total Construction Equipment Emissions	660.989	0.004	0.017	665.674	0.282	1.679	0.048	0.046	0.002	0.058	0.024

Site Preparation - Post Installation

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)										
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																									
Post Machine	Other Construction Equipment	10	10	300.0	104149.6637	1.0917	32666.9412	2.6372	104902.1478	124.3094	339.3837	19.6859	19.0954	0.3175	20.7478	7.8014	278.9770	0.0029	0.0071	280.9926	0.3330	0.9091	0.0527	0.0511	0.0009	0.0556	0.0209
Forklift	Rough Terrain Forklifts	4	10	300.0	32524.5564	0.3258	10199.7084	0.8234	32759.0585	31.8977	80.4851	4.8605	4.7147	0.0920	4.5494	1.9335	30.1158	0.0003	0.0008	30.3330	0.0295	0.0745	0.0045	0.0044	0.0001	0.0042	0.0018
**Specific horsepowers were not available so emission factors are based on g/hr	**Specific horsepowers were not available so emission factors are based on g/hr																									
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																					
Delivery/Work Trucks	Construction Pick-up diesel vehicles	4	20	2,400.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	0.7655	0.0000	0.0000	0.7673	0.0033	0.0026	0.0002	0.0001	0.0000	0.0005	0.0002
																Total Construction Equipment Emissions	309.858	0.003	0.008	312.093	0.366	0.986	0.057	0.056	0.001	0.060	0.023

Geotechnical Exploration

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)											
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																										
Drill Rigs	Bore/Drill Rigs	1	10	50.0	40786.9646	0.4120	12795.1885	1.0329	41081.2363	44.9329	158.5446	7.7587	7.5259	0.1192	10.3614	3.8689	1.6860	0.0000	0.0000	1.6982	0.0019	0.0066	0.0003	0.0003	0.0000	0.0004	0.0002
Water Truck	Off Highway Truck	1	10	100.0	248212.2308	1.3901	77822.9137	6.2826	249968.8077	60.4068	686.6881	11.2143	10.8778	0.6675	16.2729	7.2830	19.1525	0.0001	0.0005	19.2880	0.0047	0.0530	0.0009	0.0008	0.0001	0.0013	0.0006
Pile Driver	Bore/Drill Rigs	1	10	50.0	40786.9646	0.4120	12795.1885	1.0329	41081.2363	44.9329	158.5446	7.7587	7.5259	0.1192	10.3614	3.8689	1.6860	0.0000	0.0000	1.6982	0.0019	0.0066	0.0003	0.0003	0.0000	0.0004	0.0002
Rubber-tired Backhoe	Rubber Tire Loaders	1	10	20.0	77245.0705	0.5996	24222.5435	1.9555	77796.7825	53.3570	177.0397	8.3987	8.1467	0.2199	9.1142	3.7619	1.2772	0.0000	0.0000	1.2863	0.0009	0.0029	0.0001	0.0001	0.0000	0.0002	0.0001
Forklift	Rough Terrain Forklifts	2	10	50.0	32524.5564	0.3258	10199.7084	0.8234	32759.0585	31.8977	80.4851	4.8605	4.7147	0.0920	4.5494	1.9335	2.6889	0.0000	0.0001	2.7083	0.0026	0.0067	0.0004	0.0004	0.0000	0.0004	0.0002
**Specific horsepowers were not available so emission factors are based on g/hr	**Specific horsepowers were not available so emission factors are based on g/hr																										
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																						
Delivery/Work Trucks	Construction Pick-up diesel vehicles	8	20	1,600.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	0.5104	0.0000	0.0000	0.5115	0.0022	0.0017	0.0001	0.0001	0.0000	0.0003	0.0001
																Total Construction Equipment Emissions	27.001	0.000	0.001	27.191	0.014	0.077	0.002	0.002	0.000	0.003	0.001

Install Support Structures

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)										
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																									
Forklift	Rough Terrain Forklifts	4	10	1,050.0	32524.5564	0.3258	10199.7084	0.8234	32759.0585	31.8977	80.4851	4.8605	4.7147	0.0920	4.5494	1.9335	105.4054	0.0011	0.0027	106.1654	0.1034	0.2608	0.0158	0.0153	0.0003	0.0147	0.0063
	**Specific horsepowers were not available so emission factors are based on g/hr																									
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																					
Delivery/Work Trucks	Construction Pick-up diesel vehicles	4	20	8,400.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	2.6794	0.0001	0.0000	2.6856	0.0116	0.0091	0.0006	0.0004	0.0000	0.0017	0.0007
																Total Construction Equipment Emissions	108.085	0.001	0.003	108.851	0.115	0.270	0.016	0.016	0.000	0.016	0.007

Install Inverters, Switchgear, and Support Equipment

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)										
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																									
Cranes	Cranes	4	10	1,050.0	52935.3846	0.4428	16599.4561	1.3401	53314.4181	25.9232	110.4818	4.4319	4.2989	0.1515	6.2161	2.6276	122.5377	0.0010	0.0031	123.4151	0.0600	0.2557	0.0103	0.0100	0.0004	0.0144	0.0061
Backhoe/Front Loader	Rubber Tire Loaders	4	10	1,050.0	77245.0705	0.5996	24222.5435	1.9555	77796.7825	53.3570	177.0397	8.3987	8.1467	0.2199	9.1142	3.7619	250.3355	0.0019	0.0063	252.1234	0.1729	0.5737	0.0272	0.0264	0.0007	0.0295	0.0122
	**Specific horsepowers were not available so emission factors are based on g/hr																									
On-road construction vehicles	**On-road construction vehicles**		**Round Trip Distance per day (mi)**	**Number of Vehicle Miles Traveled**	**MOVES Emission Factors (g/VMT)**																					
Delivery/Work Trucks	Construction Pick-up diesel vehicles	4	20	8,400.0	578.7421	0.0229	-	0.0024	580.0797	2.5024	1.9598	0.1268	0.0916	0.0020	0.3778	0.1515	2.6794	0.0001	0.0000	2.6856	0.0116	0.0091	0.0006	0.0004	0.0000	0.0017	0.0007
																Total Construction Equipment Emissions	375.553	0.003	0.009	378.224	0.245	0.839	0.038	0.037	0.001	0.046	0.019

Install DC and AC Electrical

On-site Road and Nonroad Construction Equipment	On-site Road and Nonroad Construction Equipment	Quantity	Schedule	Operating Hours	MOVES Emission Factor (g/hr)												Monthly Emission Totals (Tons)											
			hours/day		CO2	CH4	BSFS	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP	CO2	CH4	N2O	CO2e	CO	NOx	PM10	PM2.5	SO2	VOC	HAP
Nonroad construction equipment	Nonroad construction equipment																										
Mini-Excavators	Excavators	8	10	1,050.0	54811.3390	0.2158	17184.1492	1.3873	55196.4931	13.2261	52.3487	2.3193	2.2497	0.1476	2.4032	0.9889	355.2647	0.0014	0.0090	357.7611	0.0857	0.3393	0.0150	0.0146	0.0010	0.0156	0.0064
Crawling Trencher	Other Construction Equipment	4	10	1,050.0	104149.6637	1.0917	32666.9412	2.6372	104902.1478	124.3094	339.3837	19.6859	19.0954	0.3175	20.7478	7.8014	241.0912	0.0025	0.0061	242.8331	0.2878	0.7856	0.0456	0.0442	0.0007	0.0480	0.0181
Backhoe/Front Loader	Rubber Tire Loaders	8	10	1,050.0	77245.0705	0.5996	24222.5435	1.9555	77796.7825	53.3570	177.0397	8.3987	8.1467	0.2199	9.1142	3.7619	500.6709	0.0039	0.0127	504.2469	0.3458	1.1475	0.0544	0.0528	0.0014	0.0591	0.0244
	**Specific horsepowers were not available so emission factors are based on g/hr																										
															Total Construction Equipment Emissions		1097.027	0.008	0.028	1104.841	0.719	2.272	0.115	0.112	0.003	0.123	0.049

"For Nevada:

Under a higher emissions pathway (RCP8.5), historically unprecedented warming is projected to continue through this century. Even under a lower emissions pathway (RCP2.6), annual average temperatures are projected to most likely exceed historical record levels by the middle of this century. However, a large range of temperature increases is projected under both pathways, and under the lower pathway, a few projections are only slightly warmer than historical records.

Extreme high temperatures are projected to increase substantially, with potentially large impacts in the very hot southern deserts, particularly the Las Vegas metro area, where 70% of the state's population resides. Extreme heat, combined with the urban heat island effect, would result in poor air quality and an increased risk of chronic respiratory conditions and heat stress.

Projected rising temperatures in Nevada would raise the snow line (the average lowest elevation at which snow falls). This would increase the likelihood that precipitation would fall as rain rather than snow, reducing water storage in the snowpack, particularly at those lower mountain elevations that are now on the margins of reliable snowpack accumulation. Higher spring temperatures would also result in earlier melting of the snowpack, further decreasing water availability during the already dry summer months. "

"Warmer temperatures are likely to decrease the amount of water in the mountain snowpack and increase the demand for water. Higher temperatures would also increase the evaporation rate, which would reduce streamflow and soil moisture. Thus, the intensity of future droughts is likely to increase, as would the risk of wildfires in some ecosystems. Increases in population and potentially decreased water flow from the Colorado River may lead to future water security issues across the state."

Source: BLM 2022b

Social Cost of CO_2 by Discount Rate and Statistic, 2020 to 2050 (in 2020 dollars per metric ton of CO2)

Emission Year	Discount Rate and Statistic			
	5% – Average	3% – Average	2.5% – Average	3% – 95th Percentile
2020	$14	$51	$76	$152
2025	$17	$56	$83	$169
2030	$19	$62	$89	$187
2035	$22	$67	$96	$206
2040	$25	$73	$103	$225
2045	$28	$79	$110	$242
2050	$32	$85	$116	$260

Source: IWG (2021)

Social Cost of CH_4, by Discount Rate and Statistic, 2020 to 2050 (in 2020 dollars per metric ton of CH4)

Emission Year	Discount Rate and Statistic			
	5% – Average	3% – Average	2.5% – Average	3% – 95th Percentile
2020	$670	$1,500	$2,000	$3,900

Emission Year	Discount Rate and Statistic			
	5% – Average	3% – Average	2.5% – Average	3% – 95th Percentile
2025	$800	$1,700	$2,200	$4,500
2030	$940	$2,000	$2,500	$5,200
2035	$1,100	$2,200	$2,800	$6,000
2040	$1,300	$2,500	$3,100	$6,700
2045	$1,500	$2,800	$3,500	$7,500
2050	$1,700	$3,100	$3,800	$8,200

Source: IWG (2021)

Social Cost of N_2O, 2020 to 2050 by Discount Rate and Statistic (in 2020 dollars per metric ton of N2O)

Emission Year	Discount Rate and Statistic			
	5% – Average	3% – Average	2.5% – Average	3% – 95th Percentile
2020	$5,800	$18,000	$27,000	$48,000
2025	$6,800	$21,000	$30,000	$54,000
2030	$7,800	$23,000	$33,000	$60,000
2035	$9,000	$25,000	$36,000	$67,000
2040	$10,000	$28,000	$39,000	$74,000
2045	$12,000	$30,000	$42,000	$81,000
2050	$13,000	$33,000	$45,000	$88,000

GHG Emission Factors for Comparison of Electrical System Energy Sources

Units	Coal	Natural Gas	Biopower	Geothermal	Wind	Solar	Hydro	Nuclear
grams CO_2e/kWh	980.00	465.00	40.00	11 - 47*	11.00	27 - 44	7.00	12.00
lbs CO_2e/kWh	2.16	1.03	0.09	0.02-0.10	0.02	0.06-0.10	0.02	0.03

Project No Action Coal & Natural Gas CO2e Estimates

Project kWh/month	Coal	Natural Gas
	MT CO_2e/month	
4,083,333	4,001.66	1,898.75

Project No Action Coal & Natural Gas CO2e Estimates

Project kWh/yr	Coal	Natural Gas
	MT CO_2e/yr	
49,000,000	48,019.94	22,784.97

APPENDIX G

Data Responses
Nevada Department of Wildlife
U.S. Fish and Wildlife Service

STATE OF NEVADA

DEPARTMENT OF WILDLIFE

6980 Sierra Center Parkway, Suite 120
Reno, Nevada 89511
Phone (775) 688-1500 • Fax (775) 688-1595

STEVE SISOLAK
Governor

TONY WASLEY
Director

BONNIE LONG
Deputy Director

JACK ROBB
Deputy Director

Khushali Desai
Natural Resources
SWCA
7210 Placid Street
Las Vegas, Nevada 89119

May 17, 2022

Re: Townsite Solar Project

Dear Khushali Desai:

I am responding to your request for information from the Nevada Department of Wildlife (NDOW) on the known or potential occurrence of wildlife resources in the vicinity of the Townsite Solar Project located in Clark County, Nevada. In order to fulfill your request, a query was performed using the best available data from the NDOW's wildlife occurrences, raptor nest sites and ranges, greater sage-grouse leks and habitat, and big game distributions databases. No warranty is made by the NDOW as to the accuracy, reliability, or completeness of the data for individual use or aggregate use with other data. The absence of data does not imply the absence of species or associated habitat, and additional coordination may be necessary to adequately address species or habitat presence or limits to existing data. These data should be considered **sensitive** and may contain information regarding the location of sensitive wildlife species or resources. All appropriate measures should be taken to ensure that the use of this data is strictly limited to serve the needs of the project described on your GIS Data Request Form. Abuse of this information has the potential to adversely affect the existing ecological status of Nevada's wildlife resources and could be cause for the denial of future data requests.

To adequately provide wildlife resource information in the vicinity of the proposed project the NDOW delineated an area of interest that included a four-mile buffer around the project area provided by you on Monday, May 16, 2022. Wildlife resource data was queried from the NDOW databases based on this area of interest. The results of this analysis are summarized below.

Big Game - Occupied bighorn sheep distribution exists outside of the project area within portions of the four-mile buffer area. No known occupied elk, mule deer, or pronghorn antelope distributions exist in the vicinity of the project area. Please refer to the attached maps for details regarding big game distributions relative to the proposed project area.

Greater Sage-Grouse - There is no known greater sage-grouse habitat in the vicinity of the project area.

Raptors - Various species of raptors, which use diverse habitat types, may reside in the vicinity of the project area. American kestrel, bald eagle, barn owl, burrowing owl, Cooper's hawk, ferruginous hawk, flammulated owl, golden eagle, great horned owl, long-eared owl, merlin, northern goshawk, northern harrier, northern saw-whet owl, osprey, peregrine falcon, red-tailed hawk, rough-legged hawk, sharp-shinned hawk, short-eared owl, turkey vulture, and western screech owl have distribution ranges that include the project area and four-mile buffer area. Furthermore, golden eagle has been directly observed in the vicinity of the project area.

Raptor species are protected by State and Federal laws. In addition, bald eagle, burrowing owl, California spotted owl, ferruginous hawk, flammulated owl, golden eagle, northern goshawk, peregrine falcon, prairie falcon, and short-eared owl are NDOW species of special concern and are target species for conservation as outlined by the Nevada Wildlife Action Plan. Per the *Interim Golden Eagle Technical*

Guidance: Inventory and Monitoring Protocols; and Other Recommendations in Support of Golden Eagle Management and Permit Issuance (United States Fish and Wildlife Service 2010) we have queried our raptor nest database to include raptor nest sites within ten miles of the proposed project area. There are 53 known or suspected raptor nest sites within ten miles of the project area. Please refer to the appendix for details regarding these raptor nest sites.

Other Wildlife Resources

There are no water developments in the vicinity of the project area. Additional species have also been observed in the vicinity of the project area. Please refer to the appendix for details regarding these species.

The proposed project area may also be in the vicinity of abandoned mine workings, which often provide habitat for state and federally protected wildlife, especially bat species, many of which are protected under NAC 503.030. To request data regarding known abandoned mine workings in the vicinity of the project area please contact the Nevada Division of Minerals (http://minerals.state.nv.us/).

The above information is based on data stored at our Reno Headquarters Office and does not necessarily incorporate the most up to date wildlife resource information collected in the field. Please contact the Habitat Division Supervising Biologist at our to discuss the current environmental conditions for your project area and the interpretation of our analysis. Furthermore, it should be noted that the information detailed above is preliminary in nature and not necessarily an identification of every wildlife resource concern associated with the proposed project. Consultation with the Supervising Habitat biologist will facilitate the development of appropriate survey protocols and avoidance or mitigation measures that may be required to address potential impacts to wildlife resources.

Brad Hardenbrook - Southern Region Supervising Habitat Biologist (702.668.3960)

Federally listed Threatened and Endangered species are also under the jurisdiction of the United States Fish and Wildlife Service. Please contact them for more information regarding these species.

If you have any question regarding the results or methodology of this analysis, please do not hesitate to contact Jinna Larkin at (755) 688-1580.

Appendix A: Raptor Nest Sites Table

Active Date	Check Date	Township/Range/Section	Probable Use
	4/29/2011		buteo
	4/29/2011		buteo
	4/30/2012		buteo
	5/1/2015		buteo
	5/1/2015		buteo
	5/1/2015		corvid
	5/1/2015		corvid
	5/1/2015		corvid
	5/6/2008		eagle
	4/29/2011		eagle
	4/29/2011		eagle
	4/29/2011		eagle
	4/29/2011		eagle
	4/29/2011		eagle
	4/29/2011		eagle
	4/29/2011		eagle/buteo
	4/30/2012		eagle/buteo
	5/1/2015		eagle/buteo
	5/26/1996		falcon
	6/10/1998		falcon
	6/17/1998		falcon
	6/11/1999		falcon
	5/4/2001		falcon
	6/8/2001		falcon
	6/25/2001		falcon
	6/12/2002		falcon
	1/1/2003		falcon
	5/19/2003		falcon
	4/30/2004		falcon
	6/2/2004		falcon
	6/25/2004		falcon
	4/24/2005		falcon
	6/3/2005		falcon
	6/30/2005		falcon
	4/27/2006		falcon
	6/5/2006		falcon
	7/6/2006		falcon
	7/6/2006		falcon
	6/1/2007		falcon
	6/1/2008		falcon
	5/1/2009		falcon
	3/13/2010		falcon
	5/1/2010		falcon
	5/1/2010		falcon
	6/29/2010		falcon
	5/27/2012		falcon

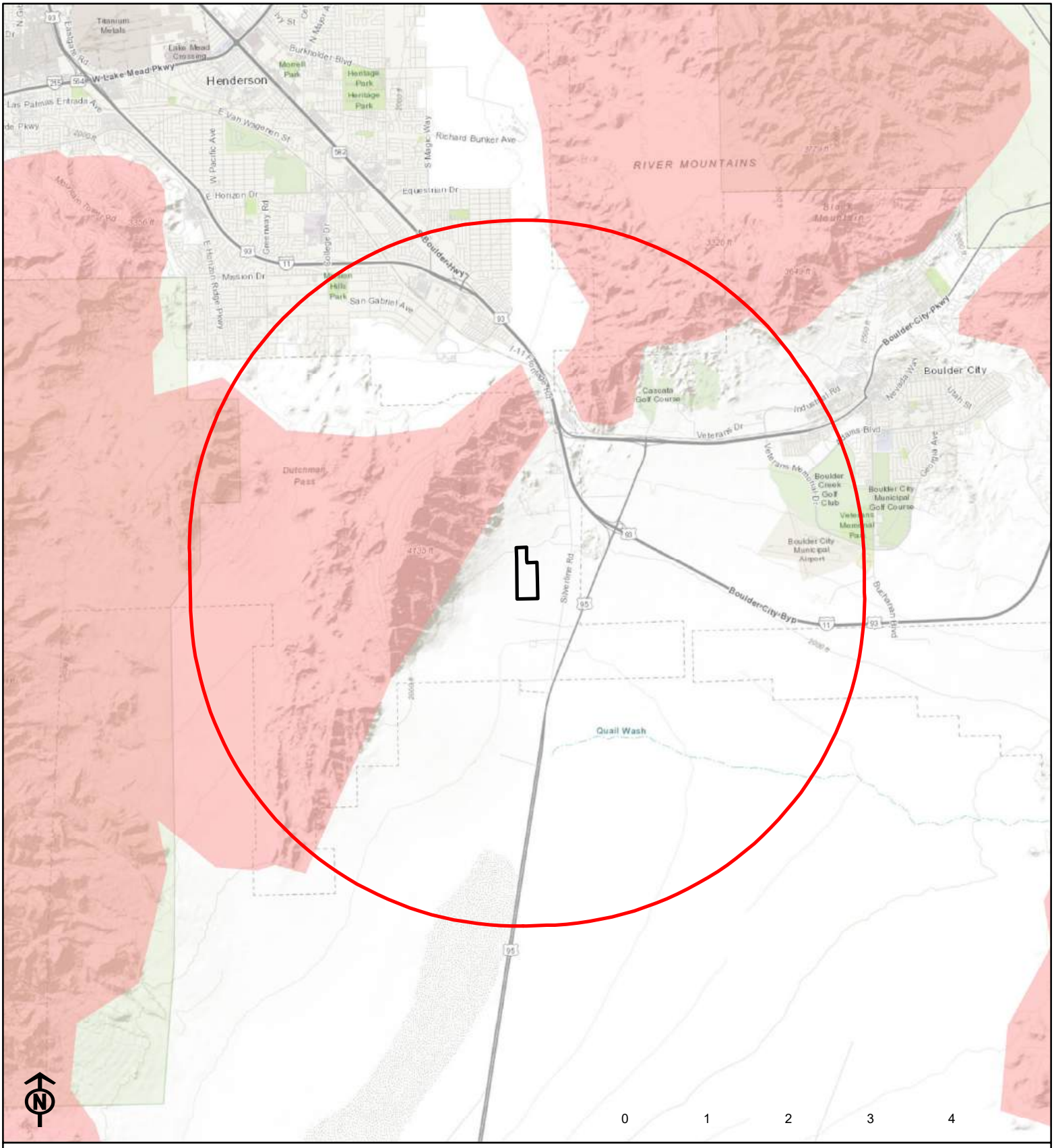
6/3/2012	falcon
6/6/2012	falcon
6/13/2012	falcon
6/18/2012	falcon
5/1/2015	falcon
5/6/2008	unknown
4/2/2019	

Appendix B: Other Wildlife Species Table

Common Name	ESA	State	SWAP SoCP
American avocet		Protected	Yes
black-necked stilt		Protected	
black-tailed gnatcatcher		Protected	
brown pelican		Protected	
brush mouse			
coachwhip			
common chuckwalla			Yes
common kingsnake			
common raven		Protected	
common side-blotched lizard			
desert banded gecko			Yes
desert glossy snake			
desert horned lizard			Yes
desert night lizard			Yes
desert tortoise	Threatened	Threatened	Yes
desert woodrat			
glossy snake			
gophersnake			
Great Basin collared lizard			Yes
Great Basin gophersnake			
Great Basin rattlesnake			
Great Basin whiptail			
greater sandhill crane			Yes
long-nosed leopard lizard			Yes
long-nosed snake			
long-tailed pocket mouse			
Merriam's kangaroo rat			
Mojave Desert sidewinder			Yes
Mojave patch-nosed snake			
Mojave rattlesnake			
Mojave shovel-nosed snake			Yes
Nevada shovel-nosed snake			Yes
Nevada side-blotched lizard			
North American racer			
northern desert iguana			Yes
northern desert nightsnake			
Panamint rattlesnake			
red racer			
Sierra gartersnake			
southwestern speckled rattlesnake			
spotted leaf-nosed snake			Yes
tiger whiptail			
variable groundsnake			
western banded gecko			Yes
western fence lizard			
western harvest mouse			

western long-tailed brush lizard	Yes
western shovel-nosed snake	Yes
yellow-backed spiny lizard	
zebra-tailed lizard	

ESA: Endangered Species Act Status
State: State of Nevada Special Status
SWAP SoCP: Nevada State Wildlife Action Plan (2012) Species of Conservation Priority



Project Area

Four Mile Buffer Area Boundary

Bighorn Sheep Distribution

Townsite Solar Project
Bighorn Sheep Distribution

May 17, 2022

Projection: UTM Zone 11 North, NAD83

No warranty is made by the Nevada Department of Wildlife as to the accuracy, reliability, or completeness of the data for individual use or aggregate use with other data.

V:\ActiveProjects\DataRequests\Template\Data Request - Response Template.mxd

IPaC **U.S. Fish & Wildlife Service**

IPaC resource list

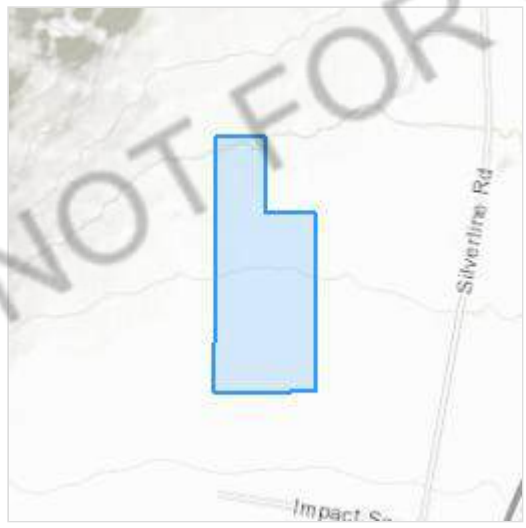
This report is an automatically generated list of species and other resources such as critical habitat (collectively referred to as *trust resources*) under the U.S. Fish and Wildlife Service's (USFWS) jurisdiction that are known or expected to be on or near the project area referenced below. The list may also include trust resources that occur outside of the project area, but that could potentially be directly or indirectly affected by activities in the project area. However, determining the likelihood and extent of effects a project may have on trust resources typically requires gathering additional site-specific (e.g., vegetation/species surveys) and project-specific (e.g., magnitude and timing of proposed activities) information.

Below is a summary of the project information you provided and contact information for the USFWS office(s) with jurisdiction in the defined project area. Please read the introduction to each section that follows (Endangered Species, Migratory Birds, USFWS Facilities, and NWI Wetlands) for additional information applicable to the trust resources addressed in that section.

NOT FOR CONSULTATION

Location

Clark County, Nevada



Local Office

Southern Nevada Fish And Wildlife Office

(702) 515-5230
(702) 515-5231

4701 N. Torrey Pines Drive
Las Vegas, NV 89130-2301

NOT FOR CONSULTATION

Endangered species

This resource list is for informational purposes only and does not constitute an analysis of project level impacts.

The primary information used to generate this list is the known or expected range of each species. Additional areas of influence (AOI) for species are also considered. An AOI includes areas outside of the species range if the species could be indirectly affected by activities in that area (e.g., placing a dam upstream of a fish population even if that fish does not occur at the dam site, may indirectly impact the species by reducing or eliminating water flow downstream). Because species can move, and site conditions can change, the species on this list are not guaranteed to be found on or near the project area. To fully determine any potential effects to species, additional site-specific and project-specific information is often required.

Section 7 of the Endangered Species Act **requires** Federal agencies to "request of the Secretary information whether any species which is listed or proposed to be listed may be present in the area of such proposed action" for any project that is conducted, permitted, funded, or licensed by any Federal agency. A letter from the local office and a species list which fulfills this requirement can **only** be obtained by requesting an official species list from either the Regulatory Review section in IPaC (see directions below) or from the local field office directly.

For project evaluations that require USFWS concurrence/review, please return to the IPaC website and request an official species list by doing the following:

1. Draw the project location and click CONTINUE.
2. Click DEFINE PROJECT.
3. Log in (if directed to do so).
4. Provide a name and description for your project.
5. Click REQUEST SPECIES LIST.

Listed species[1] and their critical habitats are managed by the Ecological Services Program of the U.S. Fish and Wildlife Service (USFWS) and the fisheries division of the National Oceanic and Atmospheric Administration (NOAA Fisheries[2]).

Species and critical habitats under the sole responsibility of NOAA Fisheries are **not** shown on this list. Please contact NOAA Fisheries for species under their jurisdiction.

---

1. Species listed under the Endangered Species Act are threatened or endangered; IPaC also shows species that are candidates, or proposed, for listing. See the listing status page for more information. IPaC only shows species that are regulated by USFWS (see FAQ).

NOT FOR CONSULTATION

2. NOAA Fisheries, also known as the National Marine Fisheries Service (NMFS), is an office of the National Oceanic and Atmospheric Administration within the Department of Commerce.

The following species are potentially affected by activities in this location:

Birds

NAME	STATUS
Southwestern Willow Flycatcher Empidonax traillii extimus Wherever found There is **final** critical habitat for this species. The location of the critical habitat is not available. https://ecos.fws.gov/ecp/species/6749	Endangered
Yellow-billed Cuckoo Coccyzus americanus There is **final** critical habitat for this species. The location of the critical habitat is not available. https://ecos.fws.gov/ecp/species/3911	Threatened
Yuma Ridgway''s Rail Rallus obsoletus yumanensis Wherever found No critical habitat has been designated for this species. https://ecos.fws.gov/ecp/species/3505	Endangered

Reptiles

NAME	STATUS
Desert Tortoise Gopherus agassizii There is **final** critical habitat for this species. The location of the critical habitat is not available. https://ecos.fws.gov/ecp/species/4481	Threatened

Insects

NAME	STATUS
Monarch Butterfly Danaus plexippus Wherever found No critical habitat has been designated for this species. https://ecos.fws.gov/ecp/species/9743	Candidate

Critical habitats

NOT FOR CONSULTATION

Potential effects to critical habitat(s) in this location must be analyzed along with the endangered species themselves.

THERE ARE NO CRITICAL HABITATS AT THIS LOCATION.

Migratory birds

Certain birds are protected under the Migratory Bird Treaty Act[1] and the Bald and Golden Eagle Protection Act[2].

Any person or organization who plans or conducts activities that may result in impacts to migratory birds, eagles, and their habitats should follow appropriate regulations and consider implementing appropriate conservation measures, as described below.

1. The Migratory Birds Treaty Act of 1918.
2. The Bald and Golden Eagle Protection Act of 1940.

Additional information can be found using the following links:

- Birds of Conservation Concern https://www.fws.gov/program/migratory-birds/species
- Measures for avoiding and minimizing impacts to birds https://www.fws.gov/library/collections/avoiding-and-minimizing-incidental-take-migratory-birds
- Nationwide conservation measures for birds https://www.fws.gov/sites/default/files/documents/nationwide-standard-conservation-measures.pdf

The birds listed below are birds of particular concern either because they occur on the USFWS Birds of Conservation Concern (BCC) list or warrant special attention in your project location. To learn more about the levels of concern for birds on your list and how this list is generated, see the FAQ below. This is not a list of every bird you may find in this location, nor a guarantee that every bird on this list will be found in your project area. To see exact locations of where birders and the general public have sighted birds in and around your project area, visit the E-bird data mapping tool (Tip: enter your location, desired date range and a species on your list). For projects that occur off the Atlantic Coast, additional maps and models detailing the relative occurrence and abundance of bird species on your list are available. Links to additional information about Atlantic Coast birds, and other important information about your migratory bird list, including how to properly interpret and use your migratory bird report, can be found below.

NOT FOR CONSULTATION

For guidance on when to schedule activities or implement avoidance and minimization measures to reduce impacts to migratory birds on your list, click on the PROBABILITY OF PRESENCE SUMMARY at the top of your list to see when these birds are most likely to be present and breeding in your project area.

NAME	BREEDING SEASON (IF A BREEDING SEASON IS INDICATED FOR A BIRD ON YOUR LIST, THE BIRD MAY BREED IN YOUR PROJECT AREA SOMETIME WITHIN THE TIMEFRAME SPECIFIED, WHICH IS A VERY LIBERAL ESTIMATE OF THE DATES INSIDE WHICH THE BIRD BREEDS ACROSS ITS ENTIRE RANGE. "BREEDS ELSEWHERE" INDICATES THAT THE BIRD DOES NOT LIKELY BREED IN YOUR PROJECT AREA.)
Black-chinned Sparrow *Spizella atrogularis* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9447	Breeds Apr 15 to Jul 31
Clark's Grebe *Aechmophorus clarkii* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.	Breeds Jun 1 to Aug 31
Costa's Hummingbird *Calypte costae* This is a Bird of Conservation Concern (BCC) only in particular Bird Conservation Regions (BCRs) in the continental USA https://ecos.fws.gov/ecp/species/9470	Breeds Jan 15 to Jun 10
Golden Eagle *Aquila chrysaetos* This is not a Bird of Conservation Concern (BCC) in this area, but warrants attention because of the Eagle Act or for potential susceptibilities in offshore areas from certain types of development or activities. https://ecos.fws.gov/ecp/species/1680	Breeds Dec 1 to Aug 31

NOT FOR CONSULTATION

Long-eared Owl asio otus — Breeds Mar 1 to Jul 15

This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.
https://ecos.fws.gov/ecp/species/3631

Marbled Godwit Limosa fedoa — Breeds elsewhere

This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.
https://ecos.fws.gov/ecp/species/9481

Willet Tringa semipalmata — Breeds elsewhere

This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.

Probability of Presence Summary

The graphs below provide our best understanding of when birds of concern are most likely to be present in your project area. This information can be used to tailor and schedule your project activities to avoid or minimize impacts to birds. Please make sure you read and understand the FAQ "Proper Interpretation and Use of Your Migratory Bird Report" before using or attempting to interpret this report.

Probability of Presence (■)

Each green bar represents the bird's relative probability of presence in the 10km grid cell(s) your project overlaps during a particular week of the year. (A year is represented as 12 4-week months.) A taller bar indicates a higher probability of species presence. The survey effort (see below) can be used to establish a level of confidence in the presence score. One can have higher confidence in the presence score if the corresponding survey effort is also high.

How is the probability of presence score calculated? The calculation is done in three steps:

1. The probability of presence for each week is calculated as the number of survey events in the week where the species was detected divided by the total number of survey events for that week. For example, if in week 12 there were 20 survey events and the Spotted Towhee was found in 5 of them, the probability of presence of the Spotted Towhee in week 12 is 0.25.
2. To properly present the pattern of presence across the year, the relative probability of presence is calculated. This is the probability of presence divided by the maximum probability of presence across all weeks. For example, imagine the probability of presence in week 20 for the Spotted Towhee is 0.05, and that the probability of presence at week 12 (0.25) is the maximum of any week of the year. The relative probability of presence on week 12 is 0.25/0.25 = 1; at week 20 it is 0.05/0.25 = 0.2.

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3. The relative probability of presence calculated in the previous step undergoes a statistical conversion so that all possible values fall between 0 and 10, inclusive. This is the probability of presence score.

To see a bar's probability of presence score, simply hover your mouse cursor over the bar.

Breeding Season (▪)
Yellow bars denote a very liberal estimate of the time-frame inside which the bird breeds across its entire range. If there are no yellow bars shown for a bird, it does not breed in your project area.

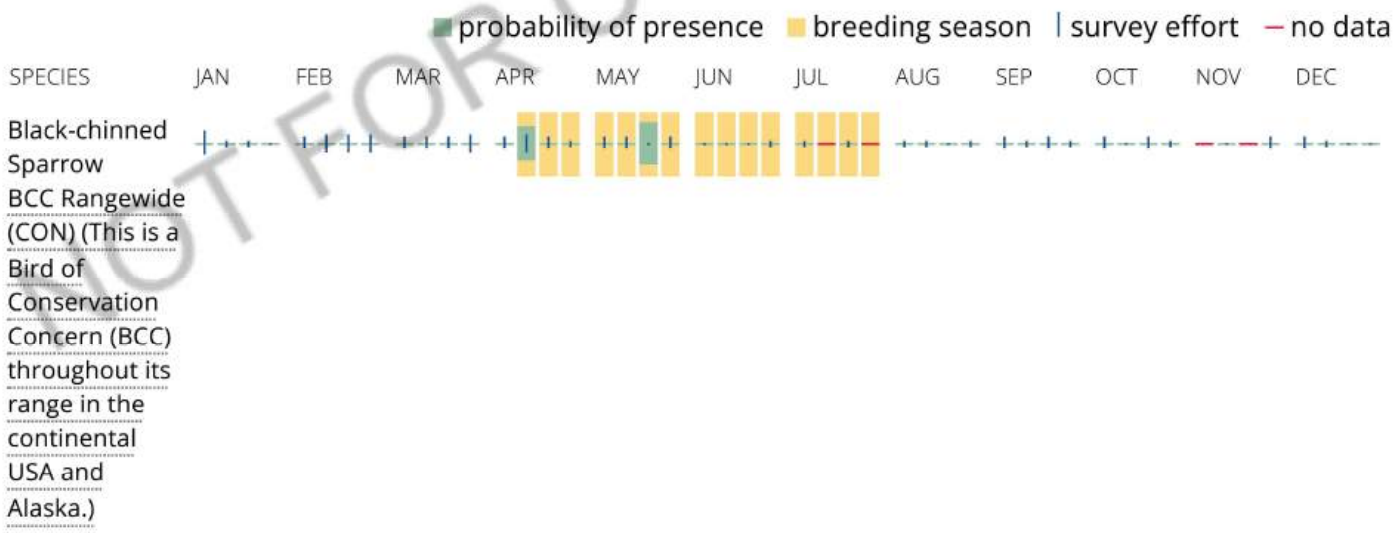
Survey Effort (|)
Vertical black lines superimposed on probability of presence bars indicate the number of surveys performed for that species in the 10km grid cell(s) your project area overlaps. The number of surveys is expressed as a range, for example, 33 to 64 surveys.

To see a bar's survey effort range, simply hover your mouse cursor over the bar.

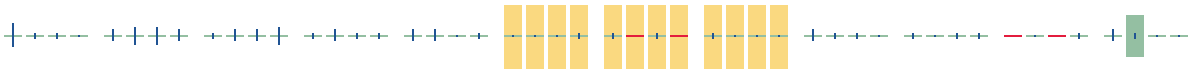
No Data (–)
A week is marked as having no data if there were no survey events for that week.

Survey Timeframe
Surveys from only the last 10 years are used in order to ensure delivery of currently relevant information. The exception to this is areas off the Atlantic coast, where bird returns are based on all years of available data, since data in these areas is currently much more sparse.

NOT FOR CONSULTATION



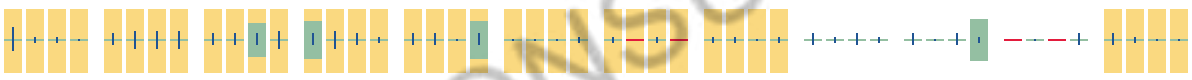
Clark's Grebe BCC Rangewide (CON) (This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.)



Costa's Hummingbird BCC - BCR (This is a Bird of Conservation Concern (BCC) only in particular Bird Conservation Regions (BCRs) in the continental USA)



Golden Eagle Non-BCC Vulnerable (This is not a Bird of Conservation Concern (BCC) in this area, but warrants attention because of the Eagle Act or for potential susceptibilities in offshore areas from certain types of development or activities.)



NOT FOR CONSULTATION

Long-eared Owl BCC Rangewide (CON) (This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.)



Marbled Godwit BCC Rangewide (CON) (This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.)



Willet BCC Rangewide (CON) (This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska.)



NOT FOR CONSULTATION

Tell me more about conservation measures I can implement to avoid or minimize impacts to migratory birds.

Nationwide Conservation Measures describes measures that can help avoid and minimize impacts to all birds at any location year round. Implementation of these measures is particularly important when birds are most likely to occur in the project area. When birds may be breeding in the area, identifying the locations of any active nests and avoiding their destruction is a very helpful impact minimization measure. To see when birds are most likely to occur and be breeding in your project area, view the Probability of Presence Summary. Additional measures or permits may be advisable depending on the type of activity you are conducting and the type of infrastructure or bird species present on your project site.

What does IPaC use to generate the migratory birds potentially occurring in my specified location?

The Migratory Bird Resource List is comprised of USFWS Birds of Conservation Concern (BCC) and other species that may warrant special attention in your project location.

The migratory bird list generated for your project is derived from data provided by the Avian Knowledge Network (AKN). The AKN data is based on a growing collection of survey, banding, and citizen science datasets and is queried and filtered to return a list of those birds reported as occurring in the 10km grid cell(s) which your project intersects, and that have been identified as warranting special attention because they are a BCC species in that area, an eagle (Eagle Act requirements may apply), or a species that has a particular vulnerability to offshore activities or development.

Again, the Migratory Bird Resource list includes only a subset of birds that may occur in your project area. It is not representative of all birds that may occur in your project area. To get a list of all birds potentially present in your project area, please visit the AKN Phenology Tool.

What does IPaC use to generate the probability of presence graphs for the migratory birds potentially occurring in my specified location?

The probability of presence graphs associated with your migratory bird list are based on data provided by the Avian Knowledge Network (AKN). This data is derived from a growing collection of survey, banding, and citizen science datasets .

Probability of presence data is continuously being updated as new and better information becomes available. To learn more about how the probability of presence graphs are produced and how to interpret them, go the Probability of Presence Summary and then click on the "Tell me about these graphs" link.

How do I know if a bird is breeding, wintering, migrating or present year-round in my project area?

To see what part of a particular bird's range your project area falls within (i.e. breeding, wintering, migrating or year-round), you may refer to the following resources: The Cornell Lab of Ornithology All About Birds Bird Guide, or (if you are unsuccessful in locating the bird of interest there), the Cornell Lab of Ornithology Neotropical Birds guide. If a bird on your migratory bird species list has a breeding season associated with it, if that bird does occur in your project area, there may be nests present at some point within the timeframe specified. If "Breeds elsewhere" is indicated, then the bird likely does not breed in your project area.

What are the levels of concern for migratory birds?

Migratory birds delivered through IPaC fall into the following distinct categories of concern:

1. "BCC Rangewide" birds are Birds of Conservation Concern (BCC) that are of concern throughout their range anywhere within the USA (including Hawaii, the Pacific Islands, Puerto Rico, and the Virgin Islands);
2. "BCC - BCR" birds are BCCs that are of concern only in particular Bird Conservation Regions (BCRs) in the continental USA; and
3. "Non-BCC - Vulnerable" birds are not BCC species in your project area, but appear on your list either because of the Eagle Act requirements (for eagles) or (for non-eagles) potential susceptibilities in offshore areas from certain types of development or activities (e.g. offshore energy development or longline fishing).

Although it is important to try to avoid and minimize impacts to all birds, efforts should be made, in particular, to avoid and minimize impacts to the birds on this list, especially eagles and BCC species of rangewide concern. For more information on conservation measures you can implement to help avoid and minimize migratory bird impacts and requirements for eagles, please see the FAQs for these topics.

NOT FOR CONSULTATION

Details about birds that are potentially affected by offshore projects

For additional details about the relative occurrence and abundance of both individual bird species and groups of bird species within your project area off the Atlantic Coast, please visit the Northeast Ocean Data Portal. The Portal also offers data and information about other taxa besides birds that may be helpful to you in your project review. Alternately, you may download the bird model results files underlying the portal maps through the NOAA NCCOS Integrative Statistical Modeling and Predictive Mapping of Marine Bird Distributions and Abundance on the Atlantic Outer Continental Shelf project webpage.

Bird tracking data can also provide additional details about occurrence and habitat use throughout the year, including migration. Models relying on survey data may not include this information. For additional information on marine bird tracking data, see the Diving Bird Study and the nanotag studies or contact Caleb Spiegel or Pam Loring.

What if I have eagles on my list?

If your project has the potential to disturb or kill eagles, you may need to obtain a permit to avoid violating the Eagle Act should such impacts occur.

Proper Interpretation and Use of Your Migratory Bird Report

The migratory bird list generated is not a list of all birds in your project area, only a subset of birds of priority concern. To learn more about how your list is generated, and see options for identifying what other birds may be in your project area, please see the FAQ "What does IPaC use to generate the migratory birds potentially occurring in my specified location". Please be aware this report provides the "probability of presence" of birds within the 10 km grid cell(s) that overlap your project; not your exact project footprint. On the graphs provided, please also look carefully at the survey effort (indicated by the black vertical bar) and for the existence of the "no data" indicator (a red horizontal bar). A high survey effort is the key component. If the survey effort is high, then the probability of presence score can be viewed as more dependable. In contrast, a low survey effort bar or no data bar means a lack of data and, therefore, a lack of certainty about presence of the species. This list is not perfect; it is simply a starting point for identifying what birds of concern have the potential to be in your project area, when they might be there, and if they might be breeding (which means nests might be present). The list helps you know what to look for to confirm presence, and helps guide you in knowing when to implement conservation measures to avoid or minimize potential impacts from your project activities, should presence be confirmed. To learn more about conservation measures, visit the FAQ "Tell me about conservation measures I can implement to avoid or minimize impacts to migratory birds" at the bottom of your migratory bird trust resources page.

NOT FOR CONSULTATION

Coastal Barrier Resources System

Projects within the John H. Chafee Coastal Barrier Resources System (CBRS) may be subject to the restrictions on federal expenditures and financial assistance and the consultation requirements of the Coastal Barrier Resources Act (CBRA) (16 U.S.C. 3501 et seq.). For more information, please contact the local Ecological Services Field Office or visit the CBRA

Consultations website. The CBRA website provides tools such as a flow chart to help determine whether consultation is required and a template to facilitate the consultation process.

THERE ARE NO KNOWN COASTAL BARRIERS AT THIS LOCATION.

Data limitations

The CBRS boundaries used in IPaC are representations of the controlling boundaries, which are depicted on the official CBRS maps. The boundaries depicted in this layer are not to be considered authoritative for in/out determinations close to a CBRS boundary (i.e., within the "CBRS Buffer Zone" that appears as a hatched area on either side of the boundary). For projects that are very close to a CBRS boundary but do not clearly intersect a unit, you may contact the Service for an official determination by following the instructions here: https://www.fws.gov/service/coastal-barrier-resources-system-property-documentation

Data exclusions

CBRS units extend seaward out to either the 20- or 30-foot bathymetric contour (depending on the location of the unit). The true seaward extent of the units is not shown in the CBRS data, therefore projects in the offshore areas of units (e.g., dredging, breakwaters, offshore wind energy or oil and gas projects) may be subject to CBRA even if they do not intersect the CBRS data. For additional information, please contact CBRA@fws.gov.

Facilities

National Wildlife Refuge lands

Any activity proposed on lands managed by the National Wildlife Refuge system must undergo a 'Compatibility Determination' conducted by the Refuge. Please contact the individual Refuges to discuss any questions or concerns.

THERE ARE NO REFUGE LANDS AT THIS LOCATION.

Fish hatcheries

THERE ARE NO FISH HATCHERIES AT THIS LOCATION.

NOT FOR CONSULTATION

Wetlands in the National Wetlands Inventory

Impacts to NWI wetlands and other aquatic habitats may be subject to regulation under Section 404 of the Clean Water Act, or other State/Federal statutes.

For more information please contact the Regulatory Program of the local U.S. Army Corps of Engineers District.

WETLAND INFORMATION IS NOT AVAILABLE AT THIS TIME

This can happen when the National Wetlands Inventory (NWI) map service is unavailable, or for very large projects that intersect many wetland areas. Try again, or visit the NWI map to view wetlands at this location.

NOT FOR CONSULTATION

Data limitations

The Service's objective of mapping wetlands and deepwater habitats is to produce reconnaissance level information on the location, type and size of these resources. The maps are prepared from the analysis of high altitude imagery. Wetlands are identified based on vegetation, visible hydrology and geography. A margin of error is inherent in the use of imagery; thus, detailed on-the-ground inspection of any particular site may result in revision of the wetland boundaries or classification established through image analysis.

The accuracy of image interpretation depends on the quality of the imagery, the experience of the image analysts, the amount and quality of the collateral data and the amount of ground truth verification work conducted. Metadata should be consulted to determine the date of the source imagery used and any mapping problems.

Wetlands or other mapped features may have changed since the date of the imagery or field work. There may be occasional differences in polygon boundaries or classifications between the information depicted on the map and the actual conditions on site.

Data exclusions

Certain wetland habitats are excluded from the National mapping program because of the limitations of aerial imagery as the primary data source used to detect wetlands. These habitats include seagrasses or submerged aquatic vegetation that are found in the intertidal and subtidal zones of estuaries and nearshore coastal waters. Some deepwater reef communities (coral or tuberficid worm reefs) have also been excluded from the inventory. These habitats, because of their depth, go undetected by aerial imagery.

Data precautions

Federal, state, and local regulatory agencies with jurisdiction over wetlands may define and describe wetlands in a different manner than that used in this inventory. There is no attempt, in either the design or products of this inventory, to define the limits of proprietary jurisdiction of any Federal, state, or local government or to establish the geographical scope of the regulatory programs of government agencies. Persons intending to engage in activities involving modifications within or adjacent to wetland areas should

seek the advice of appropriate federal, state, or local agencies concerning specified agency regulatory programs and proprietary jurisdictions that may affect such activities.

NOT FOR CONSULTATION

APPENDIX H

Visual Resources Mitigation Measures and Best Management Practices

Mitigation Measures

To limit the visual impact of the Project, the footprint and soil disturbance should be minimized during construction, O&M, and decommissioning after the anticipated 30-year commercial operation date. Where visual disturbance is inevitable, mitigation measures should be employed.

The solar arrays, battery storage, fencing, and roadways should be designed to be the least visually obtrusive by blending in with the existing surrounding landscape. This Project's structures would be viewed against the Eldorado Valley and surrounding mountains; therefore, colors, lighting, and surface treatments should reduce contrast with the existing landscape following BMPs identified in the 2013 publication *Best Management Practices for Reducing Visual Impacts of Renewable Energy Facilities on BLM-Administered Lands* (BLM 2013). Colors and finishes should be selected using the BLM Standard Environmental Color Chart and selection instructions. Recommended colors include covert green and/or shadow gray.

The following are recommended BMPs that can reduce visual impacts:

- Locate and operate solar collectors to avoid off-site glare.
- Use nonreflective materials, finishes, surface treatments, or color-treated solar collectors and support structures.
- Maintain color-treated surfaces of solar collectors.
- Maintain and use natural vegetation barriers for mitigation of solar glare where possible.
- Maintain and preserve the existing vegetation within the siting area to the greatest extent possible to reduce overall Project impact.
- If additional vegetation is used to enhance mitigation of Project components, it should be natural and native to the surrounding area to complement existing vegetation.
- Color treat structures to reduce contrasts with existing landscape.
- Color treat grouped structures using the same color.
- Maintain painted, treated, stained, or coated surfaces properly.
- Direct lights properly to eliminate light spill.
- Use timers or motion sensors on all building lighting to minimize unnecessary lighting.
- Minimize lighting usage during construction and operations.

After approximately 30 years, when the commercial operation of the Project would cease, the Project Area should be restored to a landscape that once again blends into the surrounding valley's forms and textures. The decommissioning of the site would create new visual impacts, which would be similar to construction, including the removal of all aboveground structures, fencing, and debris. This should be accomplished by restoring the site to its original contours while minimizing soil disturbance. Soils that were disturbed through the removal of the Project materials should be stabilized by seeding and planting native vegetation. Due to the arid climate of this Project location, planting and reseeding may need to occur during several seasons to ensure that native species are able to succeed.

APPENDIX I

Public Review Comments and Responses

Submitter	Comment Number	Public Comment	BLM Response
Basin and Range Watch	1	It is very unreasonable for BLM to throw a 15 day comment period at the public. That is a streamlining trick to reduce public feedback for the solar industry, At least give us 30 days.	The BLM emailed the following response to the Basin and Range Watch comment on August 9, 2023: "Thank you for your July 28 email requesting an extension of the Townsite Solar II public comment period. Throughout this process, including the August 3 public meeting, the public and interest groups have exhibited extremely low interest in this project. Therefore, a time extension of the public comment period will not be granted."
Clark County Department of Environment and Sustainability, Division of Air Quality	1	DAQ determines that this action should have no significant impact to ambient air quality if the project complies with air quality rules. The proposed project is located within Hydrographic Area 167 (Eldorado Valley), which is designated as a maintenance area for the 1997 ozone National Ambient Air Quality Standard (NAAQS) and attainment/unclassifiable for all other NAAQS pollutants. Therefore, Section 94 of the Air Quality Regulations (AQRs) apply. This rule requires a Dust Control Operating Permit (DCOP) be obtained prior to: (i) soil disturbing or construction activities impacting 0.25 acres or more in overall area, (ii) mechanized trenching of 100 feet or more in length, or (iii) mechanical demolition of any structure 1,000 square feet or more in area. Construction activities include, but are not limited to, the following practices: (i) land clearing, (ii) soil and rock excavation or removal, (iii) soil or rock hauling, (iv) soil or rock crushing or screening, (v) initial landscaping, (vi) establishing and/or using staging areas, parking areas, material storage areas, or access routes to or from a construction site.	Table A-1 lists the Clark County Dust Control Operating Permit as likely to be required. The applicant will acquire the permit prior to construction and will follow all permit requirements.
Clark County Department of Environment and Sustainability, Division of Air Quality	2	A construction project of ten (10) or more acres, trenching activities of one mile or greater, or structure demolition using implosive or explosive blasting techniques must include a detailed supplement to the DCOP Dust Mitigation Plan. The supplement must be in the form of a written report and must, at minimum, provide a project description, the area and schedule of the phases of land disturbance, and the control measures and the contingency measures to be used for all construction activities. The supplement will become part of the DCOP as an enforceable permit condition.	Table A-1 lists the Clark County Dust Control Operating Permit as likely to be required. The applicant will acquire the permit prior to construction and will follow all permit requirements.
Clark County Department of Environment and Sustainability, Division of Air Quality	3	Any construction project of fifty (50) or more actively disturbed acres must have in place an individual designated as the Dust Control Monitor to ensure that dust control measures are implemented, pursuant to the provisions of Section 94.8.	The following mitigation measure was added to Section 3.2.4 of the EA: "To mitigate the effects of fugitive dust, the applicant will adopt the following mitigation measure: a designated dust control monitor will be on-site to ensure dust control measures are implemented pursuant to the provisions of Section 94.8 of the Clark County Air Quality Regulations."

Submitter	Comment Number	Public Comment	BLM Response
Clark County Department of Environment and Sustainability, Division of Air Quality	4	In addition, Section 12 of the AQRs requires the issuance of a stationary source permit for any applicable source located in Clark County that has a potential to emit a regulated pollutant that is equal to or greater than the thresholds listed in that section. Any mechanical equipment (e.g., backup generators, boilers, cooling towers) may trigger air quality “stationary source” permitting in accordance with AQR Section 12.1. Therefore, stationary source permits should be obtained before commencing construction of any emissions unit.	The stationary source permit has been added to Table A-1 as a possible permit required by the applicant. In addition, the following language has been added to Section 3.2.3.1 of the EA: "... including the use of backup generators. These generators would be used for emergency backup service with limited operations for readiness testing, but not for prime power or peak shaving. The diesel generator engines would be maintained in accordance with the manufacturer’s written emission-related instructions and would also be subject to the ongoing maintenance and inspection provisions of 40 CFR 63 Subpart ZZZZ." Additionally, the following language has been added to Section 3.2.4: "Backup generator(s) will be present, and the applicant will obtain a stationary source permit prior to construction, as required (Table A-1 in Appendix A)."
Concerned American	1	Please protect threatened Mojave desert tortoises and their habitats. Tortoises sadly continue to decline in population despite ESA protection for over thirty years. BLM and FWS are responsible for much of this decline. Many decisions are improperly driven by political expediency instead of the laws and best science. This corrupt pattern must end. With better planning, there should be no need to sacrifice tortoises for solar developments.	As noted in Section 3.12.1 of the EA, pre-project desert tortoise surveys were conducted by Allied Pacific Partners in the fall of 2018. In total, 458 acres were surveyed, including the 85-acre Project Area. Although the 458-acre survey area contains habitat suitable for desert tortoise, no individuals or signs were found (Allied Pacific Partners, LLC 2022). Construction minimization measures outlined in the Project Biological Opinion will be implemented to reduce impacts to desert tortoise.
National Park Service	1	As currently written the document is unclear about whether or not the Old Spanish National Historic Trail was analyzed correctly as a Special Designation. To add clarity to the EA we suggest that you add a brief section under Special Designations that addresses the National Historic Trail (NHT) directly and comprehensively, I suspect you already have all the data on hand, it just needs to be slotted into the right place. Given the setting of this project and of the Old Spanish National Historic Trail we would suggest something as simple as the bullet points below. - Current setting description (ideally including pictures to back up the text) - Whether the proposed project will change the setting (as seen from the trail) - Whether there are any archaeological components of the NHT within the project area - Whether there are any high potential historic sites or high potential route segments within the project area or in the viewshed of the project area (note, this is not the same thing as national register listed/eligible sites).	A new section was added to the EA, Section 3.11 Special Designations, to address the National Park Service comments related to how the Proposed Action would affect the Old Spanish National Historic Trail (OSNHT). Please see Section 3.11. Also, the OSNHT was kept in Table A-6 in Appendix A with the following updates to the text: a reference was added to the first sentence of the first paragraph and the first sentence of the second paragraph was revised to "The segment of the OSNHT within the Project's analysis area is located in..."

Submitter	Comment Number	Public Comment	BLM Response
Concerned Citizen	1	I hope that this project won't kill tortoises or other wildlife. Too much habitat destruction has already occurred. Past mitigation has not been adequate. Federal agencies put political expediency ahead of following conservation laws. It is no wonder that the climate change and biodiversity loss crises are escalating in severity. When will this dangerous federal pattern be reformed to become much more sustainable?	As noted in Section 3.12.1 of the EA, pre-project desert tortoise surveys were conducted by Allied Pacific Partners in the fall of 2018. In total, 458 acres were surveyed, including the 85-acre Project Area. Although the 458-acre survey area contains habitat suitable for desert tortoise, no individuals or signs were found. Construction minimization measures outlined in the Project Biological Opinion will be implemented to reduce impacts to desert tortoise.
Desert Tortoise Council	1	We appreciate that the Bureau of Land Management (BLM) contacted us directly via email on July 27, 2023, which allows the Council this opportunity to provide comments on the above referenced project. Given the location of the proposed project in habitats known to be occupied by Mojave desert tortoise (Gopherus agassizii) (synonymous with Agassiz's desert tortoise), our comments pertain to enhancing protection of this species during activities funded, authorized, or carried out by the BLM, which we assume will be added to the Decision Record for this project as needed. Please accept, carefully review, and include in the relevant project file the Council's following comments for the proposed project. Unless otherwise noted, the referenced page numbers are from the BLM's Public Draft Environmental Assessment, DOI-BLM-NV-S010-2023-0016-EA, entitled "Townsite Solar 2 Project, Case File Number NVN-099875," dated August 2023. The Council provided scoping comments on this project on September 13, 2022, which are still considered to be pertinent, are incorporated by reference, and provided in the footnote given below. We appreciate that our scoping comments have been addressed in the Draft EA on page 5 with regards to perimeter fencing and that protective measures have been greatly expanded as demonstrated on page 8.	Comment noted. No changes have been made to the EA to address the comment.
Desert Tortoise Council	2	We note on page 47 that "Pre-project desert tortoise surveys were conducted by Allied Pacific Partners, LLC, in the fall of 2018 (October 27–30) for a broader survey effort on 458 acres that included the entire approximately 85-acre Project Area in accordance with the 2018 USFWS protocols Preparing [sic] for any action that may occur within the range of the Mojave Desert Tortoise (Gopherus agassizii [sic]) (Allied Pacific Partners, LLC 2022; USFWS 2018). No desert tortoise or sign was [sic] observed in either the Project Area or within the broader survey area. Although neither desert tortoise nor sign were found within the Project Area during surveys, the area may be part of an individual's home range (USFWS 2022d). Thus, USFWS estimates that the 85-acre Project Area supports no more than 1 large adult desert tortoise (USFWS 2022d)" [see Draft EA for cited references]. Again, we applaud BLM for requiring protective measures and requiring remuneration fees even though no desert tortoise sign was found on the site.	Comment noted. No changes have been made to the EA to address the comment.

Submitter	Comment Number	Public Comment	BLM Response
Desert Tortoise Council	3	We note that the project-specific biological opinion (USFWS 2022) identifies numerous protective measures that are applicable to this and other subsections of the Draft EA. We suggest that the main bullet at the top of page 8 be augmented to indicate that the seven bullets given below are only a partial list of protective measures (e.g., a speed limit of 25 mph is an additional measure given on page 11 of the biological opinion that is applicable), and that ALL measures, including those given in the biological opinion are required.	In the EA, Section 2.2.3.3, the fifth bullet was updated to include "In addition to the proposed minimization measures required in the Biological Opinion, the following best management practices will be followed in the interest of minimizing potential effects to desert tortoises."
Desert Tortoise Council	4	We appreciate this opportunity to provide comments on this project and trust they will help protect tortoises during any resulting authorized activities. Herein, we reiterate that the Desert Tortoise Council wants to be identified as an Affected Interest for this and all other projects funded, authorized, or carried out by the BLM that may affect species of desert tortoises, and that any subsequent environmental documentation for this project is provided to us at the contact information listed above. Additionally, we ask that you respond in an email that you have received this comment letter so we can be sure our concerns have been registered with the appropriate personnel and office for this project.	The BLM emailed the following response to the Desert Tortoise Council comment on August 18, 2023: "Per your request, I am confirming the BLM's receipt of your letter and formal comments."

Submitter	Comment Number	Public Comment	BLM Response
City of Henderson	1	On July 27, 2023, the City of Henderson received notice of the public comment period for the proposed Townsite Solar 2 Project in Clark County. A portion of the proposed project falls within the City of Henderson jurisdiction. The City of Henderson (the "City") has a critical need to reserve public right-of-way along the west and north boundaries of the proposed project, and requests that this land be removed from the proposed project or addressed in the Plan of Development. On May 2, 2023, the Henderson City Council adopted Ordinance No. 3956 annexing approximately 290.3 acres of land into the City. The annexation included the North fifty (50) feet and the west fifty (50) feet of the proposed project area. Concurrently, on May 9, 2023, the City of Boulder City adopted Ordinance No. 1723 annexing approximately 2554.88 acres, including the land abutting the Henderson annexation area. The proposed project crosses the boundary of the two jurisdictions. Prior to commencing the annexation, the City received a letter of no objection to the annexation, dated March 23,2022. Due to forthcoming industrial development on private land in the surrounding area, the City worked with Boulder City to identify needed road infrastructure. It was determined that it was in the public's best interest for safety and accessibility that the needed roads fall within the City for operation and maintenance of public roadways. These roads provide an expanded traffic network, provide recreational trails for City residents, provide customer access to utilities, and manage storm water runoff to protect private and public land. The City requests that the Plan of Development be revised to reflect the need for public roads along the west and north boundaries of the proposed project. The following are attached: 1. Exhibit A Showing the City Annexation Area 2. Legal Description of the City Annexation Area 3. City of Henderson Ordinance No. 3956 4. City of Boulder City Ordinance No. 1723 5. BLM Letter of No Objection dated March 23, 2023	The applicant reduced the Project Area to approximately 76 acres to remove all land within the limits of the City of Henderson from the Project Area. The applicant updated the plan of development (POD) and figures to reflect this revision to the Project Area.

Submitter	Comment Number	Public Comment	BLM Response
Los Angles Department of Water and Power (LADWP)	1	On March 30, 2021, a letter of non-objection from LADWP was previously provided to Mr. Josh Coon from Arevon Asset Management for the Townsite Solar Project (Townsite – Mead 230kV Gentie Transmission Line Project) crossing within the LADWP Mead-McCullough lines 1, 2, and Mead-Victorville Transmission Lines. The Right-of-Way Engineering Group, on behalf of the LADWP Power System, reviewed the request and submits the following comments and conditions: Comments: 1) The Townsites Solar 2 Project, herein referred to as "Applicant", shall pertain to its employees, agents, consultants, contractors, officers, patrons, or invitees of the Applicant, or by any other of the Applicant's affiliated entities. 2) The Applicant provided for our comments does not impact LADWP Transmission Lines. However, LADWP will need the enclosed document signed by the Project Owner to formalize the crossings within LADWP's transmission line right-of-way for the Townsite Solar 1 Project associated with this request. Please note, if any additional solar facility improvements are proposed within LADWP's transmission line corridors, formal approval by LADWP shall be required. See contact information below for submittal. 3) Plans shall be submitted for review to the LADWP Real Estate Services Office via the following email: RE.Office@ladwp.com and copy LADWP's Environmental Director at Katherine.Rubin@ladwp.com and Environmental Specialist Matthew Kerby at Matthew.Kerby@ladwp.com.	There is no Los Angeles Department of Water and Power (LADWP) line going across Section 14 of Townsite Solar 2. No changes have been made to the EA to address the comment.
U.S. Environmental Protection Agency (EPA)	1	While we appreciate the BLM's responses to our comments to date, the EPA continues to have concerns about potential direct, indirect and cumulative impacts to site hydrology, air quality, groundwater and communities with environmental justice concerns. Many of these concerns are based on the description of the proposed site preparation and vegetation management methods which will result in grading and disk and rolling the entire 85-acre project site. Replacing harmful site preparation techniques like 'grading' and 'disk and roll' with less intrusive measures such as 'overland travel' within solar panel array areas can reduce fugitive dust, potential exposures to Valley Fever, and impacts to site hydrology, and can help protect local communities with environmental justice concerns. Please see EPA's attached detailed comments identifying additional analysis that may be required to better assess and quantify project impacts and design appropriate mitigation measures in support of a Finding of No Significant Impact (FONSI).	Please see the responses to EPA comments 2 through 27.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	2	Site Preparation, Fugitive Dust and Environmental Justice The EPA continues to encourage the BLM and the Applicant to reconsider and utilize alternative (nontraditional) construction methods that can significantly reduce intensive project impacts on site. We note that the BLM has eliminated consideration of such a low impact alternative due to the relatively small size of the Townsite 2 Solar Project (compared to other operational and proposed utility-scale solar projects in the region) as well as the topography of project site and the amount of grading necessary to make the project feasible (pg. 12). As discussed in the Draft EA, removal of vegetation causes soil disturbance, which in turn would increase the amount of particulate matter in the Eldorado Valley's air (pg. 62). Surface disturbance of up to 85 acres would contribute to particulate matter and ozone pollution and possible airborne Valley Fever fungal spores that could adversely impact air quality and the health of City of Boulder City residents (p. 16). Past dust control measures from neighboring solar projects have proven ineffective resulting in air quality violations and concerns from local communities.1 Short-term cumulative effects on air quality from fugitive dust and ozone precursors (NOx and VOCs) emissions could be exacerbated by overlapping construction schedules from future projects identified in the Eldorado Valley Transmission and Utility Corridor Programmatic EA as well as upcoming projects from the Southern Nevada Water Authority and Southwest Gas (pg. 62). Furthermore, the Draft EA identifies a low-income EJ population present in the study area within 3 miles of the project site (pgs. 19 and 63). Under Executive Order 14096, while "each agency shouldmake achieving environmental justice part of its mission," each agency shall, "identify, analyze, and address disproportionate and adverse human health and environmental effects (including risks) and hazards of Federal activities, including those related to climate change and cumulative impacts of environmental and other burdens on communities with environmental justice concerns." In light of the above circumstances, we believe it is imperative to consider less invasive methods, regardless of project size, to limit soil and vegetation disturbance, fugitive dust, and cumulative impacts. Current best management practices are being developed in the BLM Southern Nevada District that should be considered at all proposed projects, including the Townsite Solar 2 Project. These BMPs, if utilized, will reduce significant project impacts by preserving soil structure, seed banks, and a set percentage of vegetation (including cacti and yucca), maintaining hydrologic flow patterns on site, and conserving small wildlife habitat. Using construction techniques that have less impact will also result in the retention of biological soil crusts, reduce air quality impacts from dust, reduce water use during construction and protect identified, existing ephemeral drainages (pg. 43). Such techniques have been used to date without significantly impeding project development or impacting worker safety.	Please see the responses to EPA comments 3 through 14.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	3	Utilize 'Overland Travel' as much as possible instead of high-impact methods like disk and roll or grading.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	4	Assemble as much of the racking material as possible in laydown areas, which minimizes travel along panel rows. Designate primary travel routes between panel arrays – every 3rd row – to minimize disturbance in other rows. Keep disturbance to one primary travel path to avoid zigzagging, which in the long run reduces other impacts.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	5	Ensure that there are well-trained construction monitors on site focused on ensuring that construction/vehicle trips impacts are minimized.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	6	Limit grading to specific areas – roads, substation, O&M facilities, laydown areas, some equipment pads, and in discrete areas within the arrays due to structural design limitations.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	7	Utilize smaller rubber-wheeled vehicles, lightweight skid steers, small cranes, tractors, and rubber-tired forklifts to minimize soil disturbance.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.
U.S. Environmental Protection Agency	8	Require the installation of real-time PM10 dust monitoring equipment to monitor during both the construction and operational phases of the project. Such real-time monitors can provide the data necessary to inform mitigation and adaptation measures for the proposed project.	As stated in Section 2.2.8 of the EA, the applicant will prepare a dust abatement plan and will obtain a Clark County Dust Control Operating Permit as outlined in the applicant-committed EPMs listed in Appendix C. The need for real-time PM10 dust monitoring could be determined as needed by Clark County, and if required, would be installed.
U.S. Environmental Protection Agency	9	Keep soils out of drainages, preserve protective buffers alongside washes, and maintain hydrologic flow patterns within the site.	As stated in Section 2.2.8 of the EA, the applicant will prepare a site drainage plan and a stormwater pollution prevention plan prior to construction to reduce potential risks to water quality from sediment inputs to on- and off-site drainages and washes. Topography and design constraints will inform the amount of grading required to build the Project, and the EA analyzes the effects of permanent disturbance of up to 80 acres and up to 5 acres of temporary disturbance associated with the Proposed Action.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	10	Mount the Battery Energy Storage System (BESS), transformers, and inverters on elevated platforms to allow soils underneath to remain pervious and adapt to any potential flooding.	The POD Section 2.5 states concrete footings would be installed for inverters/transformers and possible BESS equipment.
U.S. Environmental Protection Agency	11	Bend and pin tortoise fencing instead of trenching, to minimize disturbance along the fence line.	Bend and pin tortoise fencing requires a greater amount of maintenance and is used for temporary uses where desert tortoises will be reintroduced to the site. The proposed Project will use desert tortoise exclusion fencing for the life of the Project; therefore, bend and pin fencing would not be appropriate.
U.S. Environmental Protection Agency	12	Incorporate propagule islands, patches of intact vegetation and soils that provide seeds and soil microbial propagules, to facilitate revegetation or recolonization of adjacent disturbed areas.	The following language has been added to Section 2.4.2: "An alternative was evaluated that considered maintaining vegetation, soils, and surface water flow within the solar field area and was eliminated due to the relatively small size of the Project (compared to other operational and proposed utility scale solar projects in the region), the topography of Project Area, and the amount of grading necessary to make the Project feasible. The topography including surface sediments composed of cobbles, gravel, and sand in combination with design constrains will inform the amount of grading required to build the Project. Preliminary design work has indicated that grading would be required." The proposed Project is also within BLM's Henderson Community Pit for mineral materials. Once the operational life of the proposed Project concludes, it would be decommissioned, and project components would be removed. At that time, the entire Project Area would again be available for the sale and excavation of mineral materials to support community development in the future. As a result, the EA evaluates the effects of the Proposed Action on up to 80 acres of permanent disturbance and up to 5 acres of temporary disturbance.
U.S. Environmental Protection Agency	13	Construct the project in phases, which reduces dust and allows areas to begin recovery sooner.	The Project is of such a small size that constructing in phases is not technically or financially feasible.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	14	Monitor vegetation recovery on site after construction by developing a Restoration Plan. Use benchmarks and required restoration measures (if much disturbance has taken place) to ensure adherence to Biological Opinion and to ensure sufficient plant growth after construction.	The POD Section 6.3 identifies that a Facility Decommissioning and Site Reclamation Plan will be developed at the conclusion of the life of the Project. In the EA, Section 2.2.8 also identifies that a restoration plan will be prepared for the temporary laydown area. Restoration plans will be developed to follow the BLM restoration plan template requirements at the time of preparation.
U.S. Environmental Protection Agency	15	According to the Draft EA, although the existing soils have low susceptibilities to wind and water erosion, soils disturbed by grading and excavation are likely to become more susceptible to erosion once vegetation is removed. Water erosion could also lead to increased sedimentation into nearby drainages, and wind erosion could damage nearby plants (pg. 44).	Please see the response to EPA comment 20.
U.S. Environmental Protection Agency	16	The Draft EA also presents soil flooding frequency. According to the Natural Resources Conservation Service, the Arizo unit is rated with an occasional flood frequency, or a chance of flooding of 5% to 50% in any given year – a notably high frequency rate (pg. 43). The soil map unit with the highest percentage of land cover within the analysis area is the Arizo unit at 57%. The Caliza unit follows at 22%, and the Canutio-Akela complex is at 12% (pg 42). Flooding for the Caliza and Canutio-Akela complexes are expected less than once every 500 years.	Please see the responses to EPA comments 18 and 19.
U.S. Environmental Protection Agency	17	Unfortunately, no mitigation measures for potential flooding are identified (pg. 45) nor was a drainage plan or Stormwater Pollution Prevention Plan (SWPPP) included as an appendix to the Draft EA. Without such plans, we are unable to comment on their potential effectiveness or sufficiency.	Please see the response to EPA comment 20.
U.S. Environmental Protection Agency	18	Quantify the portion of the project site that falls within the Arizo unit.	Table A-17 in Appendix A of the EA provides soil map unit information. The Arizo unit covers approximately 46 acres (57%) of the Project Area. Figure 4 on page 10 of the Botanical Resource Survey Report for the Townsite Solar 2 Project provides a figure with the soil map units for the Project. The report is on the BLM National NEPA Register for the Project.
U.S. Environmental Protection Agency	19	Consider the impacts of changing precipitation patterns on the proposed project. For example, discuss the anticipated extent and depth of overland flows throughout the development areas given a 100-year flood event as compared to a 500-year event, including where critical infrastructure would be located, so that early consideration may be given to improving the resiliency of the project. Critical infrastructure should not be located in the floodplain. Mount batteries BESS, transformers, and inverters on elevated platforms. Doing this will improve the resiliency of the project.	According to Table A-6 in Appendix A of the EA, the Project Area is not within a floodplain, and the Project Area is subject to minimal flooding (Zone X) according to the Federal Emergency Management Agency (FEMA) Flood Map Service Center (FEMA 2011). All components of the Proposed Action are considered to be outside of a floodplain. As described in Section 2.5 of the POD, concrete footings will be used for inverters, transformers, and possible BESS equipment.

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	20	Include, as appendices to the Final EA, any drainage plan, SWPPP and technical reports that lead to conclusions regarding environmental consequences and ensure all documents are available on the BLM National NEPA Register Eplanning website. Providing such plans in the appendices as well as requisite summary information helps to ensure a comprehensive report easily accessible to reviewers, the public and the decision maker.	The BLM is requiring a site drainage plan and a stormwater pollution prevention plan for the Notice to Proceed (construction). The BLM can include EPA in the review of those plans. The following text was added to Section 2.2.8 of the EA: "As appropriate, the BLM will include other agencies to review these plans prior to the issuance of the notice to proceed."
U.S. Environmental Protection Agency	21	According to the Draft EA, non-potable reclaimed water may be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored onsite in aboveground water tanks at the temporary laydown area. If water from the City of Boulder City Wastewater Treatment Plant is not available, water would be obtained from commercial sources (pg. 7). No further discussion is provided on the availability of water from the City of Boulder nor potential commercial sources. During the Project construction period, construction-related activities, including dust control, would require an estimated 25 acre-feet of water (pg. 9). We note that the Moapa Band of Paiutes has inquired about water resources (pg. 69) as there is an ongoing concern about adequate water sources for the myriad of projects proposed in the region. The EPA has concerns regarding the potential cumulative impacts to groundwater basins should multiple projects draw from the underlying basins and it is not clear where ultimately the water for construction and operations will come from for this project.	Please see the responses to the EPA comments 22 through 27.
U.S. Environmental Protection Agency	22	Describe whether the project can secure water sourced from the local wastewater treatment plant.	As described in Section 1.2 of the POD, non-potable reclaimed water will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities. The following text has been added to Section 1.2 of the POD and Section 2 of the EA: “Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available."

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	23	If an identified source of water cannot be secured, discuss, in the Final EA potential commercial sources, what party would be responsible for providing water, any anticipated limitations regarding quantity or timing of water allowances, and whether potentially groundwater will be used.	As described in Section 1.2 Project Elements of the POD, non-potable reclaimed water would be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. The following text has been added to Section 1.2 of the POD and Section 2 of the EA: "Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. Water transport on public roads would adhere to all federal and state transportation rules and regulations. The applicant does not intend to develop or purchase water rights."
U.S. Environmental Protection Agency	24	Identify the potentially affected groundwater basin(s) and discuss whether the basin is overappropriated or has been classified as a 'designated groundwater basin.'	The following text has been added to Table A-6 of Appendix A of the EA: "The Project is located in Nevada Hydrographic Area number 167 – Eldorado Valley. The perennial yield of the Eldorado Valley is 500 acre-feet per year, and there are 2,256 acre-feet per year of committed water rights, making the basin over allocated. This is a designated groundwater basin, meaning the permitted groundwater rights are approaching or exceeding the estimated average annual recharge and the water resources are being depleted or require additional administration (Nevada Division of Water Resources 2023)...No groundwater withdrawal is anticipated..."
U.S. Environmental Protection Agency	25	Discuss water availability given the influx of renewable energy projects and mining projects in the general area and clearly indicate if the project will contribute to cumulative impacts to groundwater availability in the area.	The estimated water usage is included in the EA and will require up to 25 acre-feet of water during construction and 2 to 3 acre-feet per year during operation. As described in Section 1.2 of the POD, non-potable reclaimed water will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. The following text has been added to Section 1.2 of the POD and Section 2 of the EA: "Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. The applicant does not intend to develop or purchase water rights."

Submitter	Comment Number	Public Comment	BLM Response
U.S. Environmental Protection Agency	26	If water will be supplied from an off-site source, analyze environmental impacts associated with the transport and storage of such an alternative water supply.	As described in Section 1.2 of the POD, non-potable reclaimed water will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities. The following additional text has been added to Section 1.2 of the POD and Section 2 of the EA: "Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. Water transport on public roads would adhere to all federal and state transportation rules and regulations."
U.S. Environmental Protection Agency	27	Identify available technologies to minimize or recycle water.	The BLM-approved dust palliatives as well as non-potable reclaimed water are proposed by the applicant.

Submitter	Comment Number	Public Comment	BLM Response
Fort Independence Indian Reservation, Tribal Historic Preservation Office	1	Non substantive	BLM will send a repsonse letter to the tribe and include a copy in the project record.
International Brotherhood of Electrical Workers - 1	1	Non substantive	Comment noted. No changes have been made to the EA to address the comment.
International Brotherhood of Electrical Workers - 2	1	Non substantive	Comment noted. No changes have been made to the EA to address the comment.

TOWNSITE SOLAR 2 LLC
TOWNSITE DATA CENTER
PLAN OF DEVELOPMENT

Prepared for

Department of the Interior
Bureau of Land Management
Southern Nevada District
Las Vegas Field Office
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MAY 2026

CONTENTS

1 Project Description 4
1.1 Introduction 4
1.1.1 Applicant's Schedule for the Proposed Project 9
1.1.2 Need for the Proposed Project 9
1.1.3 Applicant's Purpose and Need 10
1.1.4 General Facility Description, Design and Operation 10
1.1.5 Total Acreage and Dimensions of Facilities 12
1.2 Project Elements 13
1.2.1 Laydown Area 19
1.2.2 Site Security and Fencing 19
1.2.3 Lighting 19
1.2.4 BESS Inverters, Transformers, and Medium Voltage Switchgear 19
1.2.5 34.5-230-kV Collection System 20
1.2.6 Interior Data Center Access Way 20
1.2.7 Electric Distribution Line 20
1.2.8 Water 21
1.2.9 Wastewater 21
1.2.10 Waste and Hazardous Materials Management 21
1.2.11 Nonhazardous Wastes 21
1.2.12 Hazardous Materials and Hazardous Waste 22
1.2.13 Fire Protection 22
1.2.14 Health and Safety Program 24
1.2.15 Stormwater Management 24
1.2.16 Vegetation Management 25
1.2.17 Weed and Pest Control 25
1.3 Alternatives 25
1.3.1 No Action Alternative 25
1.3.2 Alternatives Considered but Eliminated from Further Analysis 25
1.3.3 Resource conflicts 26
1.4 Financial and Technical Capability of the Applicant 27
2 Construction of the Facilities 28
2.1 Geotechnical Investigation 28
2.2 Site Preparation for Data Center and Related Facilities 28
2.2.1 Surveying and Staking 28
2.2.2 Fencing 28
2.2.3 Vegetation Removal and Treatment 29
2.2.4 Site Clearing, Grading, and Excavation 29
2.2.5 Gravel, Aggregate, and Concrete Needs and Sources 29
2.3 Data Center Assembly and Construction 29
2.4 Electrical Collection and Transmission System Construction 30
2.5 Road System Construction 31
2.6 Workforce, Schedule, Equipment, and Materials 31
2.7 Construction Traffic 32
2.8 Construction Power 33
3 Related Facilities and Systems 34
3.1 Transmission System Interconnect 34

3.1.1 Proposed Transmission System 34
3.1.2 Ancillary Facilities 34
3.1.3 Status of Interconnection Agreement 34
3.2 Gas Supply Systems 34
3.3 Other Related Systems 34
3.3.1 Communication System Requirements 34
4 Operation and Maintenance (O&M) 35
5 Decommissioning and Site Reclamation 38
6 Environmental Considerations 39
6.1 Site Characteristics and Environmental Issues 39
6.2 Other Uses on the Project Area 39
6.2.1 Grazing and Mining 39
6.2.2 Recreation and Off-Highway Vehicle Conflicts 39
6.2.3 Project Military Aviation 40
6.2.4 Other Existing Uses and Access 40
6.3 Design Features 40
6.4 Worker Education and Awareness Plan (WEAP) Environmental Protection Measures 41
6.4.1 Air Quality 41
6.4.2 Cultural Resources 42
6.4.3 Desert Tortoise 43
6.4.4 Fire Protection 43
6.4.5 Reclamation 43
6.5 Mitigation Methods 43
6.5.1 Desert Tortoise 43
7 References 44

FiguresFigure 1-1. Proposed Project location map. 5
Figure 1-2. Project Area and Temporary Laydown Area. 6
Figure 1-3. Legal description for Project Area (separate ROW application) 7
Figure 1-4. Proposed Project and land ownership 8
Figure 1-5. Proposed Project elements 16
Figure 1-6. Detailed view of the linear facilities segment. 17
Figure 1-7. Interconnection pathways 18

Tables

Table 1-1. Proposed Project Schedule 9
Table 1-2. Legal Land Description of the Data Center Facility Footprint ...**Error! Bookmark not defined.**
Table 1-3. Estimated Permanent and Temporary Disturbance Acreages Associated with the Project Facilities 12
Table 1-4. Hazardous Materials That May be Used During Construction and O&M Activities 22
Table 1-5. Potential Permits and Authorizations 26
Table 2-1. Estimated On-Site Equipment for Data Center and Collection System Construction 32
Table 4-1. Routine Maintenance Protocol 36

1 PROJECT DESCRIPTION

1.1 Introduction

The Townsite Solar 2 Project is currently entitled as 19 MW solar project plus a 35 MW Battery Storage System (BESS) pursuant to Bureau of Land Management ("BLM") Right of Way ("ROW") grant N-99875 issued to Townsite Solar 2 LLC ("Applicant") on September 22,2023. The Applicant proposes to modify the ROW as outlined in this updated plan of development and in the Amended SF 299 submitted on November 4, 2025, as further updated on January 8, 2026.

Townsite Solar 2 LLC, a wholly owned subsidiary of Townsite Solar 2 Parent, LLC, a joint venture between TS2 Investments, LLC and Arevon Energy Devco 1, LLC, proposes to construct, own, operate, maintain, and decommission a data center on a 76-acres of the requested 85 acre ROW, with an additional 4 acres to l be used for an 80 × 2,000–foot linear facilities segment ("Linear Facilities").The project will be referred to as the Townsite Data Center (the "Proposed Project").

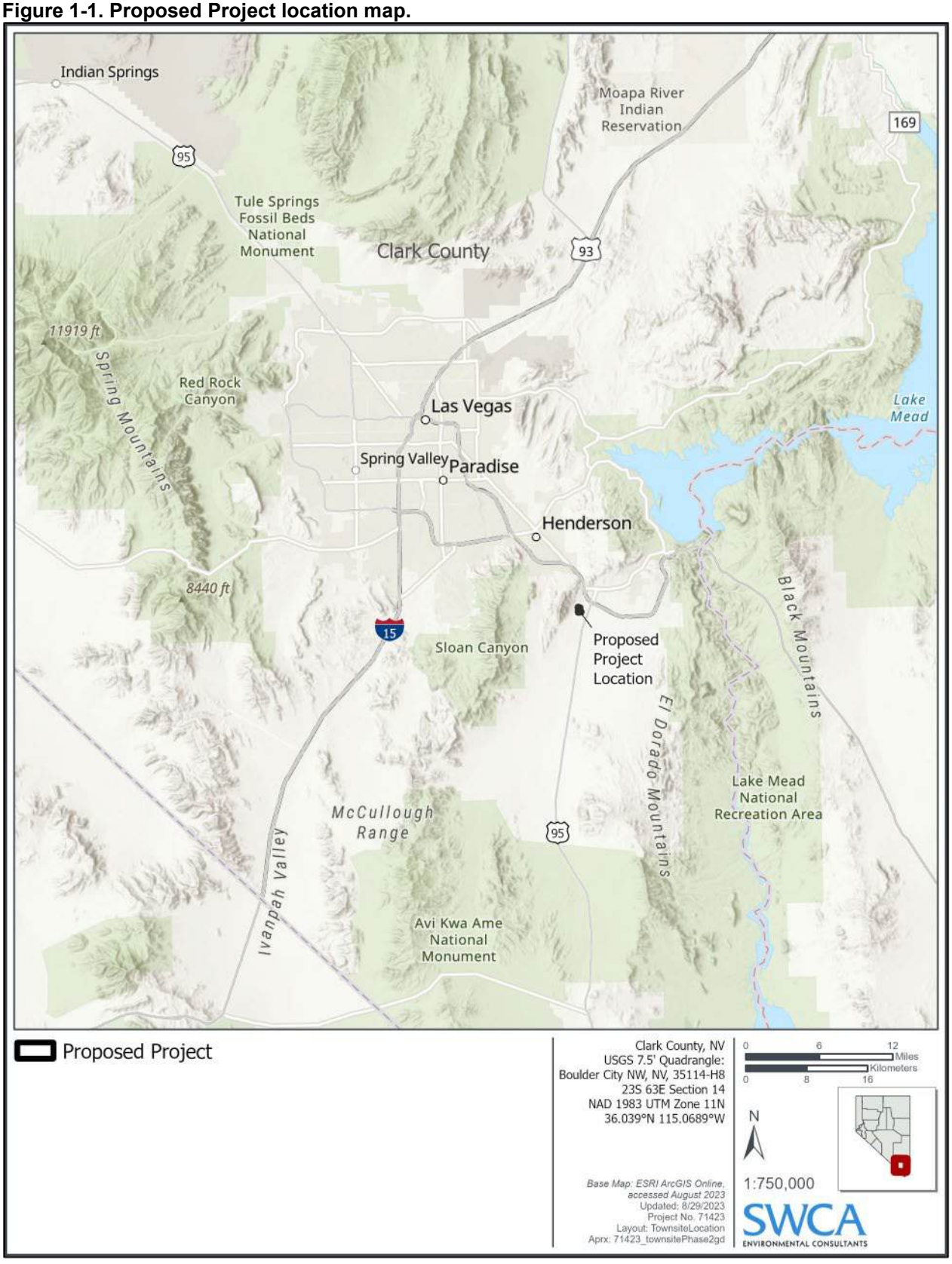
The Proposed Project consists of the data center building which is a building that houses infrastructure including computer servers, storage, and network equipment, to store, manage, process, and disseminate large amounts of data and digital information and administrative facilities. Contiguous to the data center building will be surface parking, a 70 MW BESS, a substation, and emergency backup generators fueled by biodiesel or natural gas.

In addition, the Linear Facilities portion of the Proposed Project will contain a road to access the data center site during construction (and temporary laydown area) and operation, installation of an electric permanent 12- to 13.8-kilovolt (kV) distribution power line (the "distribution line"), and for an underground 34.5-kV to 230 kV power collection system("Power Collection Line) used to interconnect the Proposed Project into a new substation proposed to support the expansion of the Southern Nevada Water Authority (SNWA) 230-kV transmission line (N-100382). The Linear Facilities segment will commence on the southeast corner of the data center and run due east for approximately 2,000 feet to the western border of the project being developed by the Applicant on municipal land owned by the City of Boulder City(("BCNV Property").

Site access will use two separate locations for site access on U.S. Route 95. The first point of site access is located approximately 1.24 miles south of the Interstate 11 (I-11)/U.S. Route 93 and U.S. Route 95 interchange, then west onto non-BLM lands using an unnamed road (same road to access Customers Team Members Community Crushing Boulder Ranch Quarry ("CTC BRQ Pit")) for 0.25 mile, then north on Old U.S. Highway 95 (Old Hwy 95) for 0.35 mile to the Proposed Project's linear facilities segment on BLM land (see Figure 1-4). The second point of site access is a right in/right out only point of access located approximately 0.9 miles south of the Interstate 11 (I-11)/U.S. Route 93 and U.S. Route 95 interchange that was recently approved by the Nevada Department of Transportation. If needed, the applicant will obtain authorization from the governing jurisdictions for use of non-BLM roads for access.

To minimize any environmental impacts, the Project Area contains a slightly larger area than required to allow for adjustments in the facility layout. Any land not needed for the Proposed Project will be relinquished back to the BLM at the appropriate time.

Figure 1-1. Proposed Project location map.



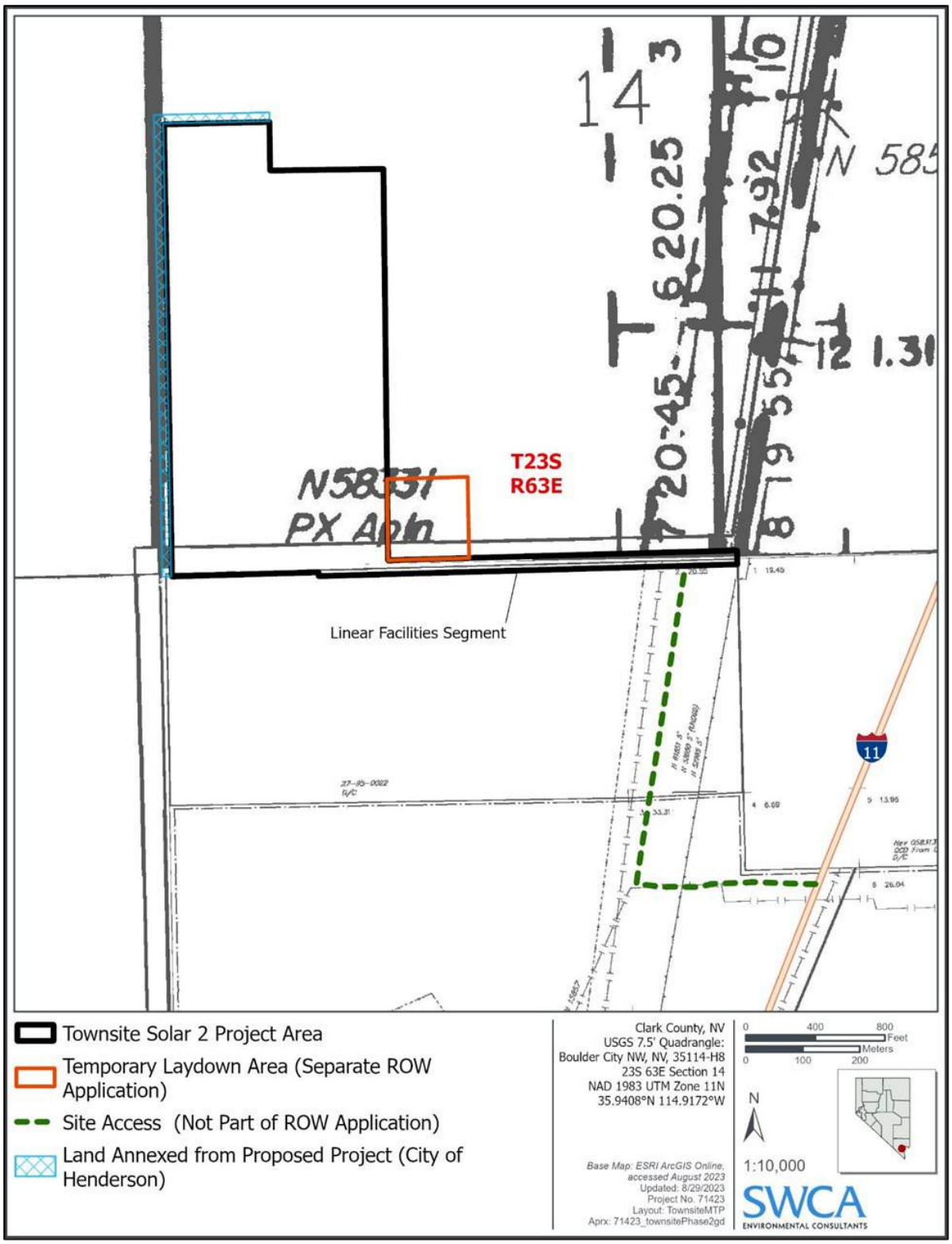


Figure 1-2. Project Area and Temporary Laydown Area.

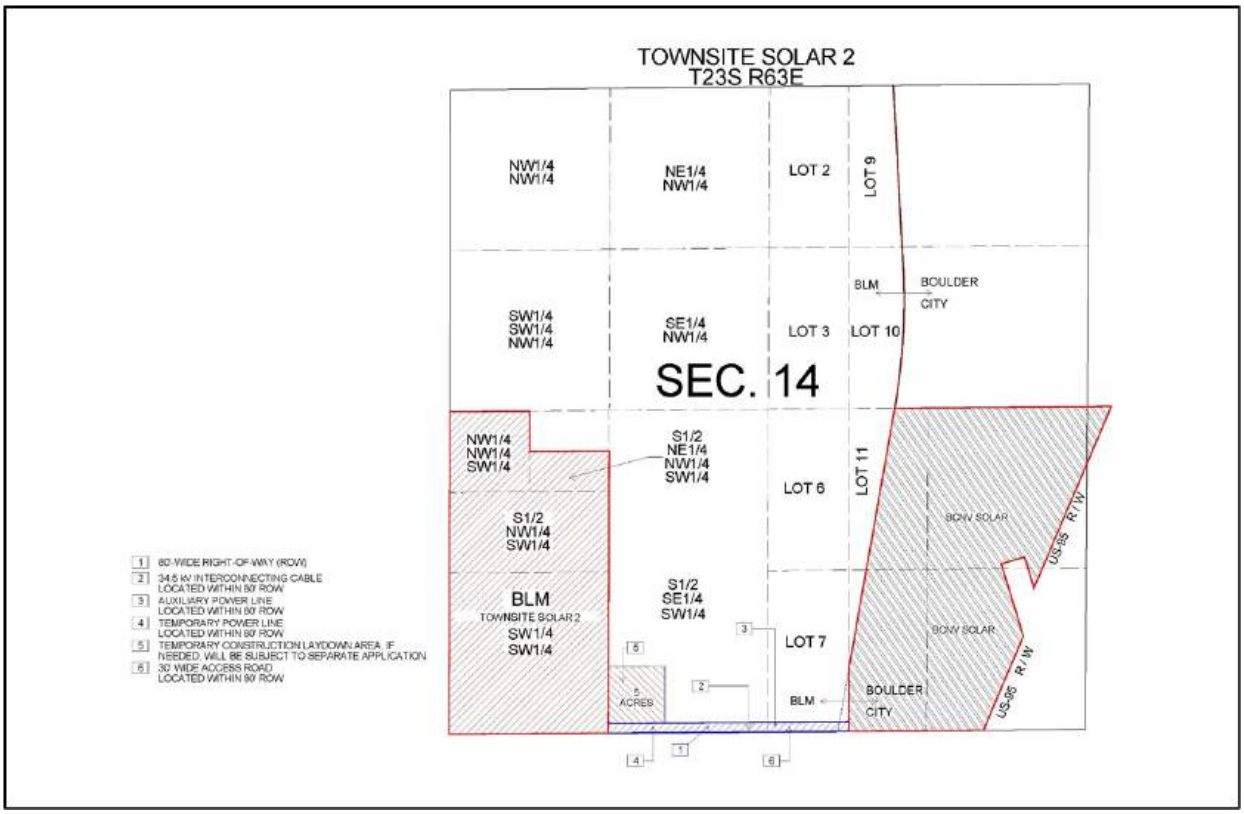


Figure 1-3. Legal description for Project Area (

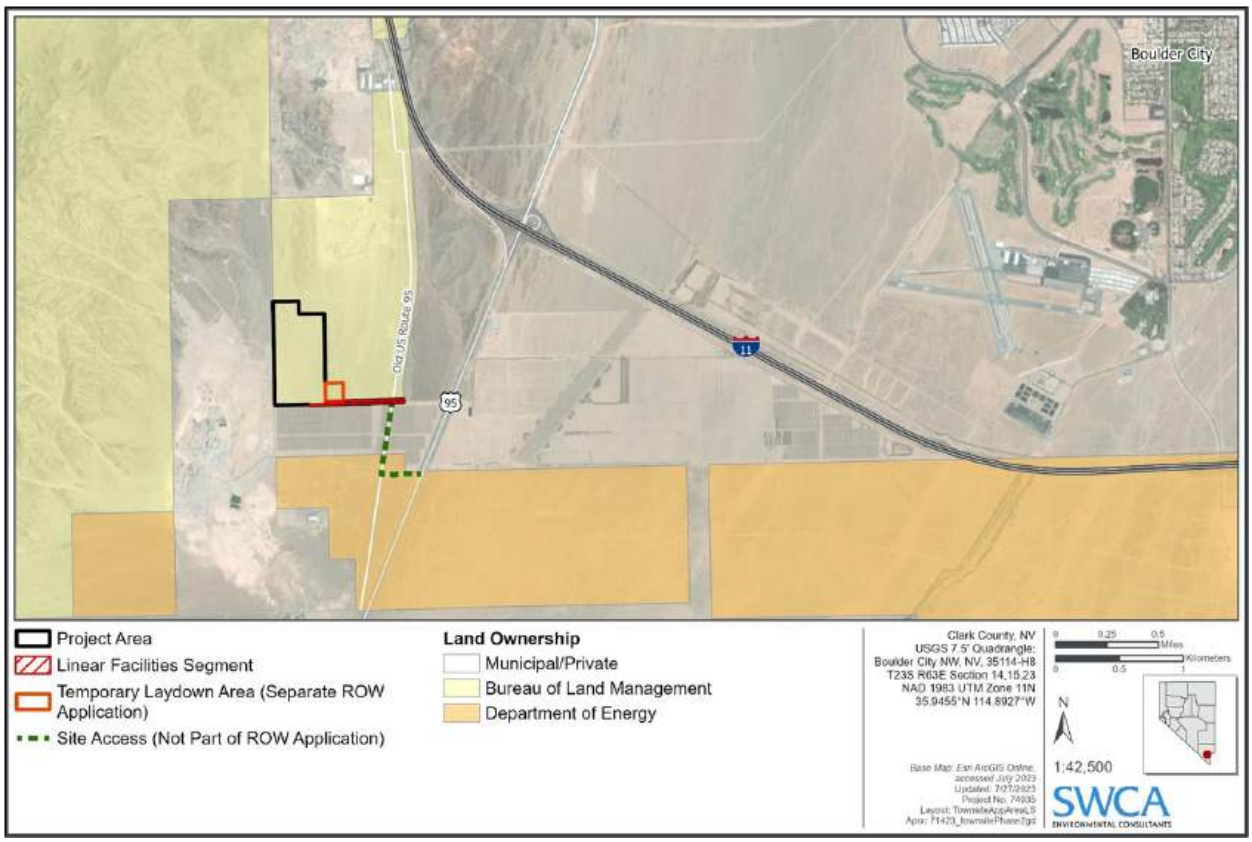


Figure 1-4. Proposed Project and land ownership

Construction of the Proposed Project will require approximately up to -12 months and will commence following completion of all necessary permits and authorizations are obtained.

1.1.1 Applicant's Schedule for the Proposed Project

After approval of the Proposed Project and prior to any activity on the site, design features including management plans (as listed in Section 6.3) will be developed and approved, and regulatory and permit conditions will be integrated into the final construction compliance documents. Construction will begin once all applicable approvals and permits have been obtained. Construction is expected to take up to 12 months and will include mobilization, construction grading and site preparation, installation of drainage and erosion controls, data center and BESS construction.

Table 1-1 outlines the Proposed Project's schedule with construction planned to commence in Q1-2027. Once construction is completed, the Proposed Project will be in operation for at least 50 years, with the possibility of a subsequent replacement of equipment for additional years of operation if grant renewal is requested and authorized at the end of the initial 50 years.

Table 1-1. Proposed Project Schedule

Activity	Date
BLM Permitting/NEPA Process Complete	Q2-2026
Construction Commencement	Q1-2027
Startup and Testing Phase	Q4-2027
Commercial Operation	Q2-2028

1.1.2 Need for the Proposed Project

Corporations such as Amazon Web Services (AWS), Advanced Micro Devices (AMD), Microsoft Azure, Google Cloud, Meta Platforms, Equinix, and Digital Realty are constructing data centers worldwide. These top data center companies are adding nearly 50 million square feet and over 1 gigawatt of capacity globally. The expansion of data centers is vital to support advancements in technologies like artificial intelligence (AI) and strengthen the U.S.'s resilient digital infrastructure. The United States has a greater need for artificial intelligence than any other industrialized nation. Development of data center and AI infrastructure is a top priority the United States Government. The importance of the development of data centers on federal land is demonstrated in America's AI Action Plan which was published by the Office of the President in July 2025. The AI Action Plan provides the following directives to promote the development of AI infrastructure in America:

- Establish new Categorical Exclusions under NEPA to cover data center-related actions.

Where possible, adopt Categorical Exclusions already established by other agencies so that each relevant agency can proceed with maximum efficiency.

- Expand the use of the FAST-41 process to cover all data center and data center energy projects eligible under the Fixing America's Surface Transportation Act of 2015.25
- Explore the need for a nationwide Clean Water Act Section 404 permit for data centers, and, if adopted, ensure that this permit does not require a Pre-Construction Notification and covers development sites consistent with the size of a modern AI data center.

- Expedite environmental permitting by streamlining or reducing regulations. promulgated under the Clean Air Act, the Clean Water Act, the Comprehensive Environmental Response, Compensation, and Liability Act, and other relevant related. laws.
- Make Federal lands available for data center construction and the construction of power. generation infrastructure for those data centers by directing agencies with significant. land portfolios to identify sites suited to large-scale development.
- Maintain security guardrails to prohibit adversaries from inserting sensitive inputs to this infrastructure. Ensure that the domestic AI computing stack is built on American products and the infrastructure that supports AI development such as energy and telecommunications are free from foreign adversary information and communications. technology and services (ICTS)including software and relevant hardware.
- Expand efforts to apply AI to accelerate and improve environmental reviews, such as through expanding the number of agencies participating in DOE's Permit

Development of AI infrastructure in the United States reduces reliance on foreign sources of fuel, promotes national security and diversifies power demand.

1.1.3 Applicant's Purpose and Need

The fundamental purpose of the Proposed Project is to provide an efficient, economic, and reliable data center to support local, state and federal objectives paving the way for expansion of other similar projects in the area. Considering the entire process, from raw material sourcing through end-of-life-cycle collection and recycling, the additional generating capacity of the Proposed Project will produce a small fraction of the greenhouse emissions of a similar capacity fossil fuel plant.

1.1.4 General Facility Description, Design and Operation

The Proposed Project Facilities consist of power and cooling systems, security infrastructure, and the physical IT components including servers, storage, and networking equipment. The power systems include uninterruptible power supplies (UPS), backup generators and energy storage, while cooling systems maintain optimal temperatures through air conditioning and other methods. Security is both physical (e.g., access controls, surveillance) and digital (e.g., firewalls).

Power and cooling infrastructure

- Power delivery: High-capacity, redundant power infrastructure, including (UPS)for immediate backup, up to 70 MW BESS and 170 MW of biodiesel generators for long outages and system emergencies.

- Cooling systems: Advanced systems like precision air conditioning, fans, and a state of the art air-cooled closed loop cooling to manage the heat generated by equipment and maintain optimal temperature and humidity. This system eliminates the need for evaporative or water-based cooling, and it does not release water or vapor onto the ground or through evaporation.

- Electrical distribution: Infrastructure like switchgears, transformers, and Power Distribution Units ("PDUs") to distribute power efficiently and reliably.

Physical IT components

- Servers: High-performance computers that run applications and process data.

- Storage systems: Hardware like hard drives, SSDs, and storage area networks ("SANs") to store and manage vast amounts of data.
- Networking equipment: Routers, switches, and firewalls that route data traffic and enable communication between servers, storage, and the outside world.

Security systems

- Physical security: Measures to prevent unauthorized access to the facility, including biometric scanners, security guards, surveillance cameras, and secured entry points.
- Fire suppression: Advanced systems like fire alarms with specific extinguishing agents (such as inert gas or microdroplets) to quickly put out fires without damaging equipment.
- Cybersecurity: Digital defenses like firewalls and intrusion detection systems to protect against cyber threats.

Other key facilities

- BESS: The Proposed Project design will include a BESS with capacity of up to 70 MW for 4 to 8 hours. The BESS will be used by the data center to provide backup power and help regulate seasonal and hourly variances in energy demand of the data Center, voltage and frequency.
- Biodiesel/Gas Fired Back up Generators: The Proposed Project design will include 100% back-up diesel generators fired with biodiesel or natural gas. Such generators will be permitted and operated in accordance with local, state and federal guidelines and standards.
- Maintenance and staging areas: Space for unpacking and preparing new equipment.

The Proposed Project site located on approximately 85 acres of public land managed by BLM. The Project will interconnect into the newly expanded SNWA 230-kV transmission line or into the existing Western Area Power Administration (WAPA) Mead 230-kV Substation from the existing Townsite Solar 1 facilities, both located on municipal land. The Proposed Project may share its BESS with the proposed BCNV Facility.

Prior to Project construction, geotechnical exploration including soil borings, soil resistivity testing, and pile load tests will be conducted to provide information for preliminary design necessary for the construction of the facilities. Authorization to conduct geotechnical investigation is requested in SF299 filed with the BLM for the Project. No reclamation is proposed for geotechnical investigation as the testing will occur within areas proposed as part of the permanent disturbance associated with the Project. The Applicant is planning to begin geotechnical exploration in Q1-2026 to minimize potential impacts to federally managed lands and environmental resources.

The Applicant reduced potential impacts of the Proposed Project this effort by selecting a site in proximity to the City of Boulder City's Eldorado Valley Energy Zone.

The Proposed Project is in the south half of Section 14, Township 23 South (T23S), Range 63 East (R63E), Mount Diablo Base and Meridian. Table 1-2 shows the complete legal land description for the Proposed Project.

Table 1-2. Legal Land Description of the Data Center Facility Footprint

Mount Diablo Meridian, Clark County, Nevada	
Township 23 South, Range 63 East	
Section 14	S½NE¼NW¼SW¼, NW¼NW¼SW¼, S½NW¼SW¼, SW¼SW¼, S½SE¼SW¼, and Lot 7

1.1.5 Total Acreage and Dimensions of Facilities

Table 1-3 lists the Proposed Project elements and the estimated permanent and temporary disturbance acreages. Once the buildings, roads, BESS, switching stations and parking lots are installed, at least 80% of the acreage will be covered with impervious surfaces.

Table 1-3. Estimated Permanent and Temporary Disturbance Acreages Associated with the Project Facilities

Facility Type	Acres of Disturbance	Notes
Permanent Disturbance		
Data Center facility	76*	The Proposed Project consisting of power and cooling systems, security infrastructure, and the physical IT components including servers, storage, and networking equipment. The power systems include uninterruptible power supplies (UPS) and backup generators, while cooling systems maintain optimal temperatures through air conditioning and other methods. Security is both physical (e.g., access controls, surveillance) and digital (e.g., firewalls).
BESS	(5)	Battery Storage, inverters and appurtenant systems.
Switching Station	5	Switching station including transformer and HV Breakers) within the construction laydown area that will become part of the permanent ROW following construction.
Drainage features	(TBD)	The drainage features are included within the data center footprint.
Berms	(TBD)	The berms are within the data center footprint.
Fencing (permanent)	(0.5)	Security perimeter fencing (hereinafter, perimeter fence) and entrance gates will be installed. Permanent Mojave desert tortoise (*Gopherus agassizii*; hereinafter "desert tortoise") fencing, up to 7,917 feet long × 3 feet wide (surface clearance area for installation) will be attached to the bottom of the perimeter fence and data center facility entrance gates.
Interior access way	(2)	The access way will be graded and compacted with site materials, will measure up to 20 feet wide × 6,000 feet long, and is anticipated to be placed around the center of the data center
Project interconnection	N/A	The Project will interconnect through either the new Boulder Flats substation through the newly expanded SNWA 230-kV transmission line; or the proposed BCNV Facility into the WAPA Mead 230-kV Substation via the Townsite Solar 1 Facility (on municipal No new surface disturbance associated with the interconnection is expected.

Facility Type	Acres of Disturbance	Notes
Linear facilities segment (site road, distribution line, and underground Power Collection System)	4	The Proposed Project intends to utilize the Linear Facilities approved in Right of Way Grant N-99875 issued to TS2 on September 29, 2023(TS2 ROW) The TS2 ROW includes a segment that is approximately 4 acres in size, measures 80 feet wide × 2,000 feet long, and will commence from the southeast corner of the site and extend eastward along the southern border of S½SE¼NW¼ Section 14, T23S, R63E of BLM land to end at the BCNV Property. The Proposed Project will use the newly constructed site road, measuring 30 feet wide × 2,000 feet long, co-located within the Linear Facilities segment) and will be used to access the Project during construction and operations. The site road will be graded and covered with gravel base and sited at least 25 feet from the distribution power line's centerline. The remaining 50-foot-wide × 2,000-foot-long portion of the Linear Facilities segment will be used for an overhead 12 to 13.8-kV auxiliary distribution line to provide power to the Proposed Project during construction and operations. The Applicant intends to assign the 50 × 2,000–foot portion of the linear facilities segment to the owner of the distribution line at a later date. The construction and operation of the interconnecting underground Power Collection line and a fiber optic cable (installed either underground with Power Collection Line as a separate conduit or on overhead lines along the distribution line) would be co-located in the Linear Facilities segment between the site road and the centerline of the distribution line (25-foot offset). Installation of the underground power collector line will require at least two pulling sites, up to 80 feet wide × 300 feet long to be located within a portion of the linear facilities segment (80 × 2,000–foot area) for a 2-month construction time frame.
Total Project Area	85	
Temporary Disturbance for Geotechnical Exploration (within data center facility footprint)		
Borehole/soil resistivity	(1.5)	Up to six boreholes at depths of approximately 30 and 50 feet. Geotechnical boring sites will be approximately 100 × 100 feet wide (to account for space for drilling and support vehicles), for a total disturbance of 0.23 acre per site. At each borehole, two soil resistivity surveys will be conducted along two perpendicular traverses. Resistivity testing will occur entirely within the disturbance area for boring geotechnical sites.
Pile Load Testing	(0.06)	At ten locations, a group of six piles (6 × 9 feet or 54 square feet) will be installed using a pile installation rig for a total disturbance of 0.06 acre. Target depth will be 7 and 13 feet. Pile load testing will occur at the boring geotechnical sites.
Site access	(0)	Access will be by existing trails/overland travel. All disturbance will be restricted to the Project Area.
Total	(1.56)	

Notes:

() Parentheses indicate the acreage is within the data center facility footprint and is part of the 76-acre total, or within the Linear Facilities segment and is part of the 4-acre total, or within the laydown/BESS area and is part of the 5-acre total.

N/A = not applicable

TBD = to be determined

† All acreages listed under permanent disturbance are included within the requested 76-acre data center facility and the acres listed under temporary disturbance (1.5 acres) will be included within the requested long term ROW.

1.2 Project Elements

Project elements are included in Figure 1-5 and Figure 1-6, and the proposed interconnection pathways are included in Figure 1-7. Below is a brief description of the Proposed Project elements:

- Geotechnical investigation work includes soil borings, soil resistivity testing, pile load testing, and access. Up to six exploratory test borings to depths of approximately 30 and 50 feet, two field soil resistivity tests (Wenner four-pin method) along two perpendicular traverses at each borehole, and pile load testing at four locations with a group of three piles installed at target

depths of 7 and 13 feet. Underground Service Alert will locate and mark the horizontal and vertical locations of exiting utilities prior to boring. Access will be by existing trails and overland travel using a truck-mounted drill rig and conventional pickup truck(s) or four-wheel drive truck(s). All disturbance will be restricted to the Project Area.

- Construction Laydown area : Located adjacent to and outside of the southeast corner of the Project Area, measuring approximately 467 × 464 and totaling 5-acres. This acreage will initially be used as the laydown area and will subsequently be the location of a 70 MW BESS and switching station.
- Data center facility footprint:
    - A data center building would likely cover several million square feet. For reference, a 100 MW facility is often cited as being at least 100,000 square feet and can be much larger. See Figure 5-1 for current planned area for the data center.
    - The exact dimensions are not fixed, as they depend on factors like server density, cooling systems, and overall building design. Some large-scale campuses could have multiple buildings to achieve the total power requirement. The data center building will not exceed 25 feet in height.
    - Permanent fencing will be installed around the perimeter of the data center. The perimeter fencing will include a 6- or 7-foot-tall chain-link fence with 1-foot-high barbed-wire security strands along the top. Mojave desert tortoise (*Gopherus agassizii*; hereinafter "desert tortoise") fencing (permanent) and installation will follow U.S. Fish and Wildlife Service (USFWS) guidelines and will be attached to the bottom of the perimeter fencing (USFWS 2009). Both will be installed concurrently.
    - One auxiliary voltage transformer, measuring 61 × 62 × 71 inches mounted on a concrete pad adjacent to and connected to the distribution line.
    - Installation of a BESS and appurtenant systems will provide up to 70MW, 4- to 8-hour capacity system for the electric grid
    - Installation of a electric switching station within the five acres initially used as construction laydown area during construction.
    - Installation of 167 MW of back-up biodiesel or gas fired generators
    - Underground Power Collection Line alternating current (AC) collection system will transfer power from the DC collection system and PCSs and transfer the power through the underground Power Collection Line to the Linear Facilities segment where the electricity will interconnect into the grid.
- The Linear Facilities segment extending along the southern border of the project footprint to the east, measuring 80 feet wide × 2,000 feet long, ending at the western border of the proposed BCNV Property y, on municipal land owned by the City of Boulder City, and will be used for the site road, auxiliary distribution line, underground Power Collection Line, and fiber optic cable:
    - Site road: The Proposed Project will use the 30-foot-wide × 2,000-foot-long portion of the Linear Facilities segment for the main access to the data center facility, which will provide site access during construction and operation of the Proposed Project. The site road will be graded.
    - Auxiliary overhead distribution line: The remaining 50-foot-wide × 2,000-foot-long portion of the Linear Facilities segment will be used for an overhead 12- to 13.8-kV auxiliary distribution line to provide power during construction and operations. The Applicant intends to assign the 50 × 2,000–foot portion of the Linear Facilities segment to the owner of the distribution line, at a later date.

  - Interconnection and communications: The 50-foot-wide × 2,000-foot-long portion of the Linear Facilities segment will also be shared for the construction and operation of the underground Power Collection Line and fiber optic cabling. The underground 34.5-230 kV collection line will deliver electricity into the on-site BESS or to the new SNWA substation in support of the newly expanded SNWA 230-kV transmission line. The Power Collection Line will require two pulling sites, up to 80 × 300 feet each, to be within a portion of the linear facilities segment (80 × 2,000–foot area) for an approximate 2-month construction time frame. Fiber optic cabling will be installed either underground with the Power Connection Line as a separate conduit or on overhead lines along the distribution line.

- Project security using a combination of perimeter fencing around the data center buildings, BESS and switching station, controlled access gates, on-site security patrols, lighting, electronic security systems and/or remote monitoring.
- Temporary fencing: Desert tortoise exclusion fencing (temporary) will be installed around the Linear Facilities segment (4 acres) to reduce the number of biological monitors required on-site. At least two desert tortoise gates will be installed within the Linear Facilities segment and one entrance gate to the data center facility (with permanent desert tortoise fencing attached). The desert tortoise fencing will be installed concurrently with the perimeter fencing.

Non-potable reclaimed water will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. Water would be obtained from commercial sources with existing water rights or availability within a 50-mile radius of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. This water, if needed, would be trucked to the Project area.

The Proposed Project will be designed in accordance with federal and industrial standards, including American Society of Mechanical Engineers standards, the National Electrical Code (National Fire Protection Association [NFPA] 2008), the International Building Code (International Code Council 2006a), the International Energy Conservation Code (International Code Council 2006b), the Uniform Plumbing Code (International Association of Plumbing and Mechanical Officials 2006a), the Uniform Mechanical Code (International Association of Plumbing and Mechanical Officials 2006b), NFPA standards, and Occupational Safety and Health Administration (OSHA) standards.

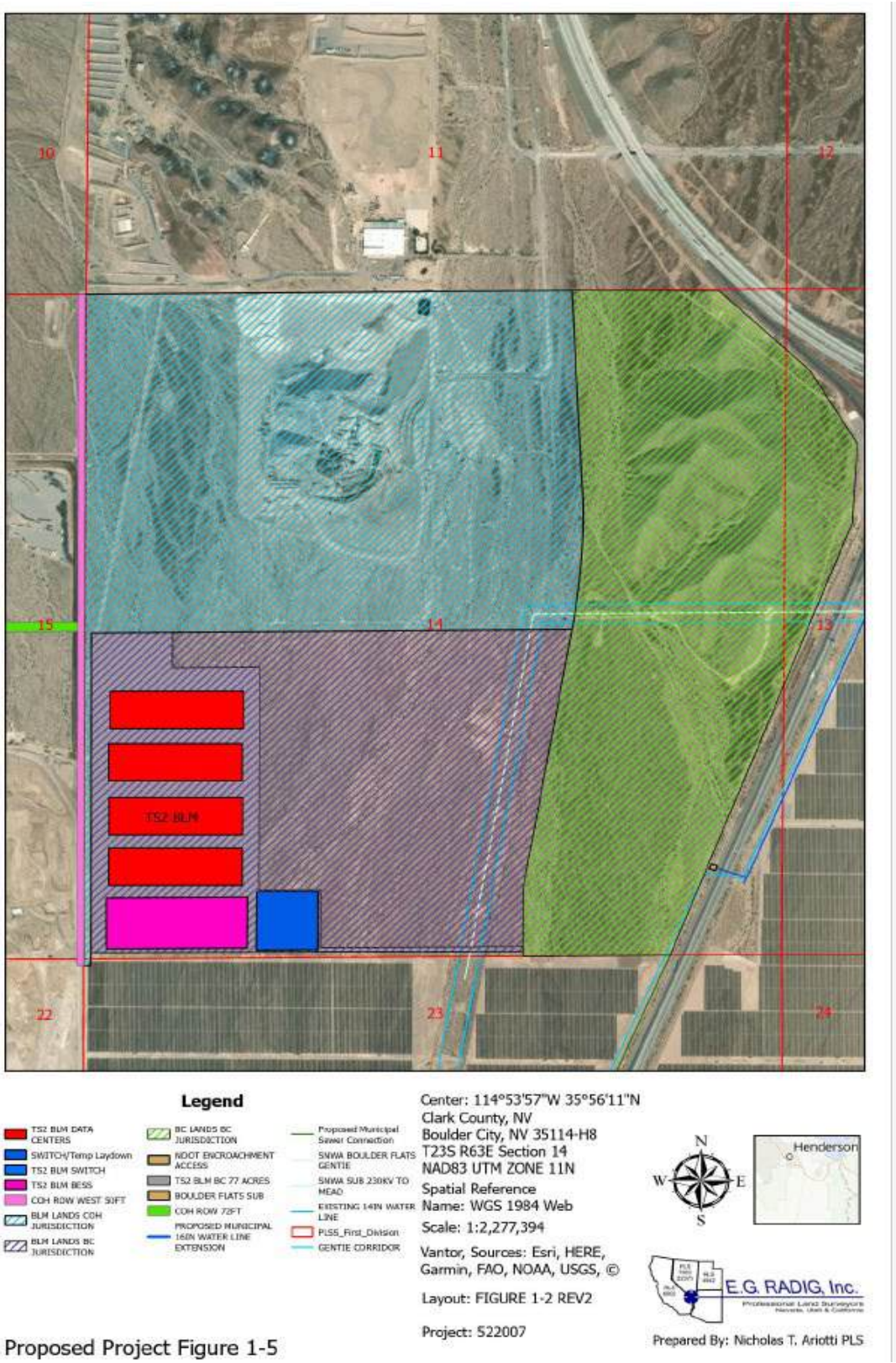


Figure 1-5. Proposed Project elements



Figure 1-6. Detailed view of the linear facilities segment.

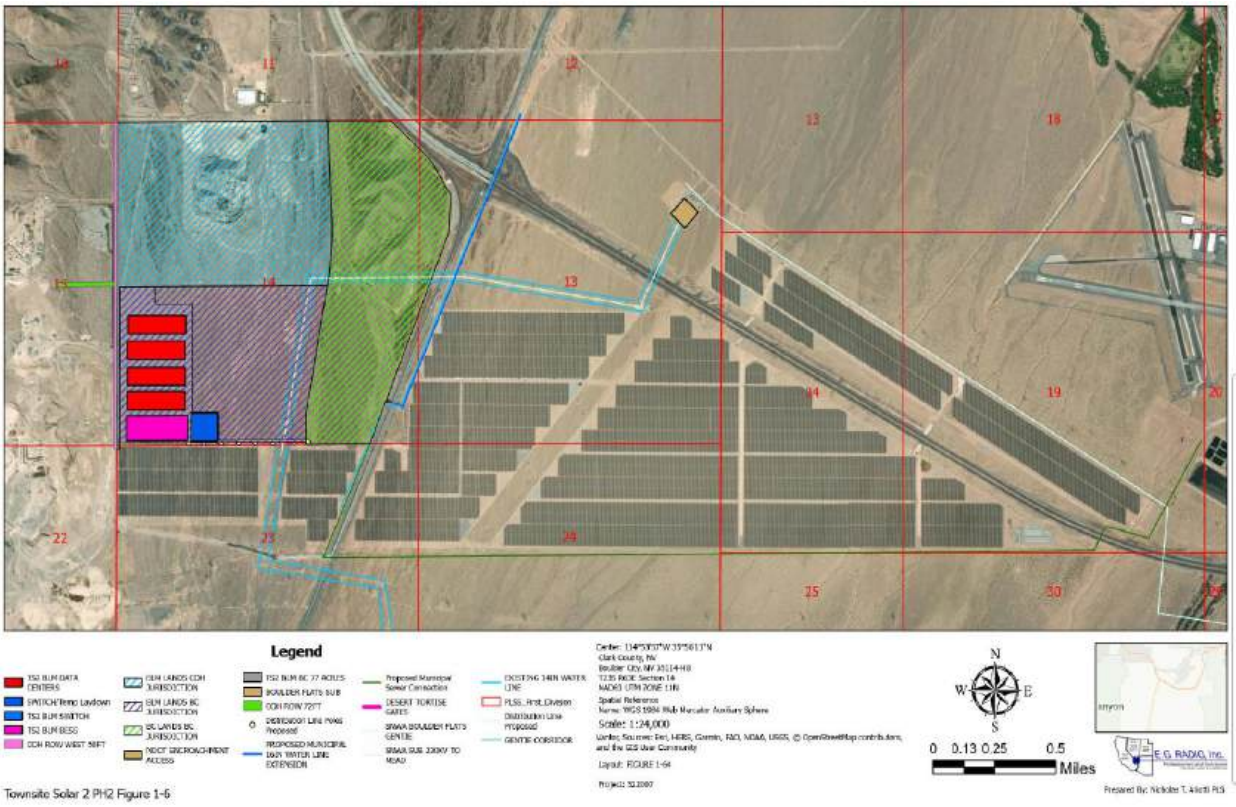


Figure 1-7. Interconnection pathways

Note: For illustration purposes only, not part of the Proposed Action.

1.2.1 Laydown Area

The 5-acre laydown area will be situated above the Linear Facilities segment. This area will initially be used as a laydown area for temporary installation of construction trailers, construction workforce parking, aboveground water tanks, receiving materials, and materials storage that will be graded/compacted earth. The area is subject of a Short Term ROW grant N-099875-01 (NVNV105860618) issued to the Applicant and will be used initially for the same purpose but will subsequently become the location of the switching station .

The laydown area will be enclosed to exclude desert tortoises prior to ground-disturbing work activity. The site road within the Linear Facilities segment will provide access to the laydown area. Aboveground vegetation will be cleared and graded, if necessary.

1.2.2 Site Security and Fencing

Security at the Proposed Project site will involve fencing around the data center, lighting, security patrols, and electronic security systems. The Proposed Project site will be monitored 24 hours per day, 7 days per week during all phases. Lighting will be provided at the Proposed Project entrance gate.

The data center perimeter will be secured with chain-link metal fabric security fencing. Controlled access gates will be at the site entrance. The perimeter fence will be a 6- to 7-foot-high chain-link fence with 1-foot-high barbed-wire security strands at the top (the fence in proximity to the distribution line will be properly grounded). Approved desert tortoise exclusion fencing will also be attached to the outside of the perimeter fence, up to 7,916 × 3 feet (surface clearance area for installation) and installed per USFWS guidelines (USFWS 2009).

Temporary desert tortoise exclusion fencing, up to 6,068 feet will be installed Linear Facilities segment (4 acres) to reduce the number of biological monitors required on-site. The temporary desert tortoise fencing will attach to the perimeter fence at the western boundary and the northwestern boundary of the linear facilities segment. If there are any delays installing the perimeter fencing, the temporary desert tortoise fencing will be extended in these locations to ensure exclusion as shown in Figure 1-6. At least two desert tortoise temporary gates will be installed within the linear facilities segment.

1.2.3 Lighting

Lighting will be provided at the Proposed Project entrance gate. Lighting for facilities and associated infrastructure will be down shielded to keep light within the boundaries of the Proposed Project site and the minimum amount and intensity necessary for the intended use. Nighttime activities will be performed with temporary lighting. Night lighting used during construction, operation, and maintenance of the Proposed Project will be controlled or reduced using directed lighting, shielding, and/or reduced lumen intensity.

The Applicant will prepare a Lighting Management Plan for construction and operation of the Proposed Project.

1.2.4 BESS Inverters, Transformers, and Medium Voltage Switchgear

The BESS includes a power conversion system containing inverters and medium voltage transformers, as well as other electrical equipment. Each PCS will contain communication equipment to wirelessly

communicate with the tracker units to control operation and detect anomalous conditions. Combining switchgear) will be along the underground Power Collection Line. All electrical equipment will be housed in protective enclosures on concrete pads.

1.2.5 34.5-kV-230 kV Power Collection Line and Switching Station

The underground Power Collection Line will be composed of underground cabling installed underground using 35-kV-230 kV rated voltage cables listed for direct buried applications. Underground electric cables will be installed to comply with the minimum burial depth in accordance with the National Electrical Code.

The Applicant will purchase the 167 MW maximum hourly electric power needed for data center operations. It will be sourced from the wholesale markets for delivery to the Mead Substation, generated on site, or from discharging the BESS that will be charged with power from the grid or power generated on site. The electric power will be transmitted from the Mead substation via the Applicant's firm transmission rights on SNWA's 230 kV line to the point of delivery on the project site where it will be transmitted through the underground Power Collection Line to the Project's Switching Station for use by the data center or to charge the on-site BESS.

The project site is within the electric service territory but does not use the Boulder City distribution network to deliver energy to the project power consumed by the Project. This arrangement will satisfy the state law requirement that Boulder City serves the project, while at the same time not utilizing power from the Boulder City electrical system. The Mead substation capacity exceeds 10,000 MW and is one of the most liquid trading hubs in the western United States. On average, 30-50% of the power delivered to the Mead Substation is from non-carbon renewable resources(hydropower, solar, wind, geothermal, etc.), The Applicant will seek to procure power from renewable resources when it is economically feasible. The Project load of 167 MW represents less than 2 % of the total capacity of the Mead Substation.

The Switching Station will consist of the main 34.5kV to 230kV transformer, a high voltage breaker and the appurtenant facilities. No structure within the BESS area will exceed twenty-five feet in height.

Interior Data Center Access Way

Within the data center, a new access way will be built to provide vehicle access to the buildings for operations and maintenance (O&M) activities. This access way will be approximately 20 feet wide × 6,000 feet long and is anticipated to be placed around the center of the data center (see Figure 1-5) to minimize disturbance. The access way will be graded and compacted using on-site materials to facilitate use of two-wheel-drive vehicles.

Similar to the disturbance from other project components (based on the assumption that all acreage within the fenced perimeter will be disturbed), the acreage identified for roads is also considered to be permanent disturbance.

1.2.6 Electric Distribution Line

A new 12- to 13.8-kV distribution line (up to approximately 2,000 feet in length) will be installed within the Linear Facilities segment. The distribution line will interconnect to NV Energy or some other electric utility or municipality to provide electric distribution service for auxiliary power to the Proposed Project during construction and operations. The centerline will be at least 25 feet from the site access

road. A portion of the Linear Facilities segment will be assigned to the owner of the distribution line once the electricity provider has been selected. In addition, a portion of the distribution line will be constructed on municipal-owned land by the City of Boulder City (BLM does not have authority to approve project components on municipal land; therefore, the portion of the distribution line planned for municipal land would require separate authorization from the City of Boulder City). Distribution line poles will be approximately 40 feet high from ground surface and spaced an average of 300 feet from one another. To the extent applicable, all overhead electrical lines will be designed and installed in accordance with the Avian Power Line Interaction Committee's (APLIC's) *Suggested Practices for Avian Protection on Power Lines* (APLIC 2006). Alternatively, generators may be used to provide temporary construction power. During operational daylight hours, the Proposed Project will generate its own power for equipment operation. During non-daylight hours, the Proposed Project will require auxiliary power to keep transformers energized, maintain communications to project equipment, and provide power for heating, ventilation, air conditioning, and lighting.

1.2.7 Water

Up to 500,000 gallons/day of water will be required during the Proposed Project's construction period for construction-related activities, including dust control. The Proposed Project expects to obtain the water for dust control from Boulder City from the waste effluent that is currently being discharged by the City in the desert. This same waste effluent was used for dust control during construction of the solar facilities in the area.

Water for the Proposed Project will be obtained from the City of Boulder City Wastewater Treatment Plant through an existing pipeline for dust suppression activities and stored on-site in aboveground water tanks at the temporary laydown area. Water would be obtained from commercial sources with existing water rights or availability within a reasonable distance of the Project Area if water from the City of Boulder City Wastewater Treatment Plant is not available. Water transport on public roads would adhere to all federal and state transportation rules and regulations. The applicant does not intend to develop or purchase water rights.

1.2.8 Wastewater

Wastewater generated during construction and O&M will include stormwater runoff, equipment washdown water, sewer for bathrooms and water from excavation dewatering during construction (if dewatering is required). Sewer service will be provided from the Boulder City. These wastewaters may be classified as hazardous or nonhazardous depending on their chemical quality and handled and disposed of in accordance with applicable law.

1.2.9 Waste and Hazardous Materials Management

The primary wastes generated at the Proposed Project during construction and O&M will be solid and liquid wastes, including sewage, scrap wood, steel, glass, plastic, paper and metals, empty hazardous waste containers, waste oil filters, oily rags and oil sorbent, spent lead acid batteries, waste oil and sanitary waste.

1.2.10 Nonhazardous Wastes

The Proposed Project will produce waste typically associated with construction and O&M activities. Such waste will include defective or broken electrical materials, computer materials, empty containers,

the typical refuse generated by workers, and other miscellaneous solid waste. All nonhazardous waste materials will be disposed of in accordance with applicable laws and regulations.

1.2.11 Hazardous Materials and Hazardous Waste

Limited quantities of hazardous materials will be used on-site for construction and O&M activities. Table 1-4 lists the hazardous materials anticipated to be used on-site. Refueling of equipment and vehicles during construction phase will take place in one centralized location in the construction laydown area. The Applicant will prepare a Spill Prevention and Emergency Response Plan, which will address waste and hazardous materials management, including best management practices (BMPs) related to spill response, transportation, and handling of materials and wastes. No hazardous materials will be stored on-site. Safety Data Sheets for each of these materials will be provided in the Spill Prevention and Emergency Response Plan.

Table 1-4. Hazardous Materials That May be Used During Construction and O&M Activities

Hazardous Material	Storage Description; Capacity	Storage Practices and Special Handling Precautions
Mineral insulating oil (or biodegradable oil)	Carbon steel transformers; total on-site inventory of 80,000 gallons.	Used only in transformers within PCSs; secondary containment for each transformer will be managed in accordance with the Spill Response and Emergency Response Plan.
Batteries, lead acid based and/or lithium ion	Battery-based emergency backup power at each of the PCS (which include both inverter and transformer equipment) plus BESS.	Enough cooling capacity to maintain ambient temperatures appropriate for the selected battery will be provided.
Propane	Generator-based emergency backup power at each of the PCS shelters (or one centralized generator); tanks at each PCS will be sized between 20 and 100 gallons (or 1,000 gallons if one centralized tank).	Will be managed in accordance with the Spill Response and Emergency Response Plan.
Herbicide Roundup® (glyphosate) or equivalent; pesticide	Brought on-site by licensed contractor, used immediately.	No mixing will occur on-site, and no herbicides will be stored on-site.

1.2.12 Fire Protection

The fire suppression system of the data center is a combination of advanced detection and waterless extinguishing methods designed to protect sensitive IT equipment and ensure business continuity. Unlike standard office spaces, data centers cannot use traditional water sprinkler systems due to the risk of irreparable damage to electronics.

A comprehensive data center fire suppression system includes these integrated components:

- Early warning detection: Uses aspirating smoke detection (ASD) systems like VESDA to actively pull air samples for analysis. This allows detection of combustion particles at their earliest, invisible stage, well before a fire starts.
- A pre-alarm and notification system: Alerts on-site personnel and facility monitoring teams, giving them a chance to investigate and abort the suppression release if it is a false alarm.
- Clean agent or inert gas suppression: Floods the protected area with a fire-extinguishing agent that leaves no residue and is safe for electronics.
- A backup sprinkler system (optional): A double-interlock pre-action sprinkler system may be installed in some data centers as a last resort. This system requires both a detection signal and a sprinkler head activation before releasing water, minimizing the risk of accidental discharge.

Common suppression system types include the following:

Inert gas systems
Inert gas systems use a blend of naturally occurring atmospheric gases, such as nitrogen and argon, to extinguish fires by reducing oxygen concentration to a level that can no longer support combustion, while remaining safe for occupants to breathe for a brief period.

- Agents: The most common inert gas agents include Proinert and Inergen.
- Pros: Environmentally friendly, zero ozone depletion or global warming potential, and safe for occupants.
- Cons: Require a large footprint for gas cylinder storage and create loud noise during discharge, which can potentially harm hard disk drives (HDDs) without proper acoustic nozzles.

Water mist systems
Water mist systems use highly pressurized, atomized water droplets to suppress fires by using a fine mist to increase the water's surface area, allowing it to absorb heat rapidly and turn into steam. This cools the fire and displaces oxygen.

- Pros: Uses significantly less water than conventional sprinklers, minimizing water damage.
- Cons: Less common as a primary system in data halls compared to gaseous agents.

In-rack and cabinet-level systems
These are supplementary systems designed to suppress fires at the source, preventing them from spreading by using either a clean agent or inert gas, The system is installed directly inside a server cabinet. Early-warning detection triggers a localized release of the suppression agent with extremely fast response time and effective in preventing a small fire from causing widespread damage, saving crucial equipment.

The specific design of a data center fire suppression system is determined by numerous factors and guided by industry standards:

- NFPA standards: The National Fire Protection Association (NFPA) sets the standards for data center fire protection. Relevant codes include NFPA 75 (Fire Protection of Information Technology Equipment) and NFPA 2001 (Clean Agent Fire Extinguishing Systems).
- Risk assessment: An expert risk assessment identifies potential fire hazards, the value of the equipment, and acceptable downtime.
- Space and layout: Factors like high airflow, raised floors, and hot/cold aisle containment influence the type of detection and suppression system used.
- Acoustic hazards: Gaseous systems can create significant noise upon discharge. Special acoustic nozzles can be used to protect sensitive equipment, especially hard disk drives from damage.

The Proposed Project also includes the possible installation of an up to 70MW BESS and appurtenances within the southern portion of the data center facility footprint. The Eldorado Valley has highlighted the potential for fire and explosion hazards associated with lithium-ion batteries, which are used in BESSs. The main factors that contribute to BESS fire incidents include thermal runaway, which is caused by lithium-ion battery defects or damage that generates excessive heat; failure of the BESS control system by one or more control components, which can cause overheating and fire; and the release of flammable gases from the heated battery cells (Cummings 2019; Spector 2019). To minimize potential fire risks

associated with thermal runaway and control system component failure, multiple levels of safety design are incorporated at the software and hardware levels of the battery management system. To address the risk of flammable gas release, the inclusion of a fire system suppression can include combustible gas sensor systems, auto ventilation systems, smoke detectors, and dry pipe sprinkler systems running to each battery container. Additionally, the proposed BESS will meet NFPA Standard 69: Standard on Explosion Prevention Systems (NFPA 2019) for the ventilation system and NFPA Standard 855: Standard for the Installation of Stationary Energy Storage Systems (NFPA 2022) for each BESS container, as well as UL 9540: the Standard for Energy Storage Systems and Equipment. These BESS design measures not only minimize the potential risk associated with fire but also provide containment so in case of a fire, the risk of spreading outside of the container is unlikely.

Fire services are provided by the BLM SND Office for wildland fire protection to public lands in southern Nevada, and the Clark County Fire Department maintains first responder responsibility within areas of unincorporated Clark County. Fire protection on municipal land within the Eldorado Valley is provided by the City of Boulder City Fire Department.

During construction, one electric and one diesel-fueled backup firewater pump will deliver water to the fire protection water-piping network. The electrical equipment enclosures that house the PCSs will be either metal or concrete structures. Any fire that could occur will be contained within the structures, which will be designed to meet National Electric Manufacturers Association (NEMA) 1 or NEMA 3R IP44 standards for electrical enclosures (heavy-duty sealed design to withstand harsh outdoor environmental conditions). The Applicant will prepare and implement a Fire Management Plan.

1.2.13 Health and Safety Program

The Applicant will require that all employees and contractors adhere to appropriate health and safety and emergency response plans. All construction and operations contractors will be required to operate under a Health and Safety Program that meets industry standards. All site personnel will be required to go through a new hire orientation and follow a Worker Education and Awareness Plan (WEAP), which will address Project-specific safety, health, and environmental concerns.

1.2.14 Stormwater Management

The existing landscape surrounding the Proposed Project has been modified over time by mining operations, major highways, transmission lines, and solar energy development. As a result, off-site flows to the Proposed Project come from the southern side of U.S. Route 11/U.S. Route 93. In most cases, the runoff generated from these off-site areas flows onto the site through culverts crossing under U.S. Route 11/U.S. Route 93 or by overtopping it. A lesser source of storm flows comes from the northwest. These lesser flows also cross U.S. Route 95 and some are directed south toward the Proposed Project site.

Potential stormwater management measures may include a series of channels constructed to convey water flows to existing washes or spreading the flows to allow them to leave the site as sheet flow. The channels will be reinforced with rip-rap and grouted as required to reduce erosion. One or more detention basins may be required above the channels or placed to intercept flows at the top of the site to manage stormwater entering the site. Spreader basins and/or rip-rapping may be required below each channel to reduce flow velocity before stormwater enters existing downstream washes. The goal is for runoff generated on-site to be conveyed as sheet flow leaving the site. These proposed drainage measures are examples that may be used to help minimize off-site runoff. A Site Drainage Plan and Stormwater Pollution Prevention Plan will be prepared.

1.2.15 Vegetation Management

The site will be allowed to revegetate following construction. Vegetation will typically be maintained no higher than commercially reasonable for site maintenance and fire-risk management; mechanical and chemical controls will be used, as needed. Proposed Project roads will remain free of vegetation. The Applicant will address post-construction vegetation management, including invasive and noxious weed control, as part of a BLM-approved Integrated Weed Management Plan for the Proposed Project.

1.2.16 Weed and Pest Control

The Applicant will prepare an Integrated Weed Management Plan for the Proposed Project that will follow the Las Vegas Resource Management Plan issued by the BLM LVFO (BLM 1998), Noxious Weed Plan (BLM 2006), and the interagency guidance Partners Against Weeds (BLM 2007) for an active integrated weed management program. BLM-approved herbicides such as Roundup (glyphosate) will be used to control noxious weeds, if required. Nonnative invasive species will also be reduced to pre-project levels or less. Pest control may also be required, including control of rodents and insects inside of the buildings and electrical equipment enclosures.

1.3 Alternatives

1.3.1 No Action Alternative

Under the No Action Alternative, the BLM would not approve the ROW for the Project. The data center facility would not be constructed, operated, maintained, or decommissioned. Therefore, there would be no ground disturbance or loss of habitat.

1.3.2 Alternatives Considered but Eliminated from Further Analysis

The original application requested 250 acres for development of the Proposed Project, but 135 acres were removed from the Proposed Project due to overlap with a community aggregate pit. Accordingly, the 250-acre alternative was considered but eliminated from further analysis.

The Applicant evaluated several locations in Southern Nevada in proximity to existing supporting infrastructure.
Alternative locations were eliminated for one or more of the following reasons:
• Existing land use applications
• Excessive slopes
• Wetland/flood zones
• Distance to interconnection facility

1.3.3 Resource conflicts

Federal lands constitute a significant percentage of lands available in Nevada for energy facilities. Development of energy facilities on federal lands is an objective of the federal government, with a particular emphasis on facilities with low energy density capacities. There are no other non-federal lands within practical connection distance to SNWA's Boulder Flats substation. Other Permits and Authorizations

Federal, state, and local permits, authorizations, or interagency consultations that may be required for the Proposed Project are listed in Table 1-5. The City Council of the City of Boulder City has voted to annex the Project Area; therefore, approvals through the City of Boulder City will be required and the approvals through Clark County will be limited.

Table 1-5. Potential Permits and Authorizations

I. Federal Permits, Authorizations, or Interagency Consultations
U.S. Department of the Interior, BLM
• ROW grant under Title V of Federal Land Policy and Management Act
• NEPA compliance
• National Trails System Act
U.S. Department of the Interior, BLM and State Historic Preservation Office (SHPO)
• BLM/SHPO, National Historic Preservation Act Section 106 Consultation (If required)
U.S. Department of the Interior, USFWS
• Endangered Species Act Section 7 Consultation and Biological Opinion/Incidental Take Statement
II. State of Nevada Permits or Authorizations
Nevada Department of Wildlife
• Special Purpose Permit
Nevada Division of Forestry
• Native Cacti and Yucca Commercial Salvaging and Transportation Permit
Nevada Department of Transportation (NDOT)
• Over Legal Size/Load Permit
Nevada Division of Environmental Protection
• Groundwater Discharge Permit
Nevada Division of Water Resources
• Water Reuse Permit
III. Local Permits*
Clark County Division of Air Quality
• Dust Control Construction Permit

* Various permits may also be required from local government(s) with jurisdiction over the Project Area and authority over certain components/actions of the Project.

1.4 Financial and Technical Capability of the Applicant

The Applicant has successfully completed the development, financing and construction of over 2,000 MW of energy projects throughout the world. In 2019, Skylar Opportunities LLC (Skylar), an affiliate of the Applicant, successfully completed the development of the $350 million Townsite Solar 180 MW energy project located in Boulder City, NV. Skylar currently has over 1,200 MW of energy projects under development. Skylar will continue to use its own resources to fund its development efforts and has many years of experience with first tier banks and investment banks in raising investment capital, tax equity and debt capital for similar energy projects.

The Applicant has sufficient financial resources to complete the development and construction of the Proposed Project. Prior to the start of construction of the Proposed Project the Applicant may supplement its own financial resources with tax equity, private equity, and/or bank financing. Personnel for the Applicant have successfully developed and raised financing for 3000 MW of energy projects.

2 CONSTRUCTION OF THE FACILITIES

2.1 Geotechnical Investigation

Prior to construction, geotechnical exploration investigations including soil borings, soil resistivity testing, and pile load tests will be conducted in the Project Area. The geotechnical activities are the subject of a separate application for a short term right of way.

Approximately 5 acres of the Proposed Project area will be used as the temporary location of construction trailers with administrative offices, construction worker parking, temporary water service and fire water supply holding tanks, temporary construction power services, and tool sheds and containers, as well as an area for construction equipment and material delivery and storage. The area will be utilized for the Switching station during operations(See Figure 5-1for details

Desert tortoise fencing (temporary) will be constructed and installed around the laydown area and Linear Facilities segment (see Section 2.4.2 for description).

Also, temporary construction areas in the Linear facilities segment will be required for each pole location for the overhead distribution line as well as temporary locations for conductor tension (stringing) and pulling operations to accommodate the construction of the underground Power Collection Line.
The tension and pulling sites each measure approximately 80 feet wide × 300 feet long and will be required for a 2-month time frame.

2.2 Site Preparation for Data Center and Related Facilities

A geotechnical investigation and environmental clearance surveys will be performed prior to commencement of construction activities. During the environmental clearance phase, the boundaries of the construction area will be delineated and marked. The perimeter fence installation will be sequenced at the beginning of construction. Permanent desert tortoise fencing will be attached to the bottom of the perimeter fence and will be in place prior to commencement of grading activities. The site will then be prepared for use; existing vegetation removal and grading will be minimized to the extent reasonably practicable. Site preparation techniques are described below.

2.2.1 Surveying and Staking

Prior to construction, the limits of construction disturbance areas will be determined by surveying and staking. Where necessary, the limits of the ROW will also be flagged. All construction activities will be confined to these areas to prevent unnecessary impacts affecting sensitive areas. These areas, which will include buffers established to protect biological resources, will also be staked and flagged. The locations of underground utilities will be located, staked, and flagged to guide construction activities.

2.2.2 Fencing

Permanent fencing will be installed around the perimeter of the site and will include a 6- or 7-foot-tall chain-link fence with 1-foot-high barbed-wire security strands along the top. Approved permanent desert tortoise fencing, up to 7,917 feet long × 3 feet wide (surface clearance area for installation) will be attached to the bottom of the perimeter fencing to prevent desert tortoises from moving onto the site from adjacent areas during construction and O&M. Both will be installed concurrently.

Prior to construction, up to 6,068 × 3 feet (surface clearance area for installation) of temporary desert tortoise fencing will be installed around the laydown area and linear facilities segment. The perimeter fence will be installed concurrently with the temporary desert tortoise fencing installation via overland travel. The temporary desert tortoise fencing may be attached to the perimeter fence along the western boundary of the laydown area and the northwestern boundary of the linear facilities segment, reducing the length of temporary desert tortoise fencing required, as shown in Figure 1-6. However, if the perimeter fence installation is delayed, temporary desert tortoise fencing can be extended in these areas to provide closure. All desert tortoise fencing will be installed per USFWS guidelines (USFWS 2009).

2.2.3 Vegetation Removal and Treatment

Within the data center areas that will be graded, existing vegetation will be worked into the underlying surface soils. Vegetation will be permanently cleared from the site road, access way, and where concrete foundations are used for the inverter/medium-voltage transformer equipment (PCSs), combiner boxes, optional BESS, and other ancillary equipment.

2.2.4 Site Clearing, Grading, and Excavation

All earthwork required to install drainage detention control basins and foundations will be coordinated as part of civil works or the Proposed Project. Trenching will be required for placement of collector lines. The data center will require a positive natural terrain slope of less than 5%. Where possible, the disk and roll technique will be used to prepare the surface of the data center buildings, BESS and switching station areas . The disk and roll technique will use conventional farming equipment to prepare the site for construction. Typical farming equipment will have rubber-tired tractors with disking equipment and drum rollers with limited use of scrapers to perform micro grading. In areas where the terrain is not suitable for disk and roll, conventional cut and fill grading will be used to prepare the area.

Within the data center, grading will be required for an interior access way and for electrical equipment pads. In general, the design standard for the interior access way within the data center will be consistent with the amount and type of use they will receive.

2.2.5 Gravel, Aggregate, and Concrete Needs and Sources

Concrete will be poured in place for equipment footings, building foundations, fence footings, and miscellaneous small pads. Aggregate material will be used for the trench backfill, temporary construction area, site road, and access way within the Proposed Project. Rip-rap material may be required for erosion control. Gravel will be sourced from a nearby gravel pit and it is not anticipated that any cut materials will be disposed of offsite. The Applicant will determine if alternate sources for these materials are needed and will be presented to BLM for review and approval, as necessary.

2.3 Data Center Assembly and Construction

Building a data center is the process of constructing a facility to house computer systems, storage, and IT equipment for data handling. It involves building the physical infrastructure, including building space, power, and cooling, to support the critical technology inside this specialized facility. Prior to any construction of the buildings the clearance and site preparation steps for those areas will be completed. Within each area designated data center buildings, the construction sequence will follow a consecutive order.

- Clear land.

- The initial focus is on constructing the building's core, shell, and substations. Each building will contain power and cooling systems, security infrastructure, and the physical IT components including servers, storage, and networking equipment. The data center's exterior walls are typically made of reinforced concrete or precast concrete panels to provide structural strength and security.
- Trenches for underground cable will be prepared.
- Install underground Power Collection Line.
- Backfill trenches.
- Install auxiliary transformer equipment.
- Introduction of BESS equipment
- Install Switching Station, if included as part of Proposed Project.
- Perform electrical terminations.
- Inspect, test, and commission equipment.

Cable trenches will be used to provide underground connection for Project equipment. Trenches will contain electrical conductors for power generation and fiber-optic cables for equipment communication. Trenches will vary between 2 to 3 feet wide and 2 to 3 feet deep depending on the number of conductors and voltage of equipment to comply with applicable electrical codes.

Trucks will be used to transport the equipment to the data center. A small mobile crane may be used to assist construction workers in setting up the cooling system and racking the computers. Final data center assembly will require small cranes, tractors, and forklifts.

2.4 Electrical Collection and Transmission System Construction

Electrical construction will consist primarily of the following elements:

- **Equipment**—Installation of all electrical equipment, including inverters and transformers), auxiliary transformers, circuit breakers, disconnect switches, switchgear and distribution panels, lighting, communication, control, and supervisory control and data acquisition (SCADA) equipment.
- **Cables**—Installation of all cables necessary to energize the Proposed Project equipment, including instrument control wiring. High-, medium-, and low-voltage cables will be routed via cable trays, above-grade conduits, below-grade conduit in duct bank, and overhead structures.
- **Grounding**—All equipment and structures will be grounded as necessary. Within the data center, an appropriate grounding system will be engineered and constructed to maintain personnel safety and equipment protection.
- **Telecommunications**—Multiple communication systems will be required for the Proposed Project to properly operate, including T-1 internet cables, fiber optic, and telephone. All communications will be installed during electrical construction.

Electrical Collection System. The Proposed Project will include an underground Power Collection Line. Standard transmission line construction techniques will be used to construct the Power Collection Line. The primary stages in underground collection line construction include trenching and cable placement.

Typical equipment expected to be used for construction includes backhoe, various pickup and flatbed trucks, conductor reel, bucket trucks, and truck-mounted tensioner and puller.

2.5 Road System Construction

Preconstruction activities for the Project-related roadways will include surveying desert tortoises, relocating desert tortoises if found during the surveys, and meeting any necessary cactus and yucca salvage requirements. The Proposed Project's site road will be graded sufficiently to facilitate travel to the Proposed Project site and will connect to the existing Old Hwy 95 road. Within the data center, grading will be required for the interior access way. All Project-related roadways are proposed to be native graded/compacted dirt; however, roads may alternatively use an aggregate base in some or all areas to meet Proposed Project dust and flood control requirements.

Any temporary or permanent crossings under existing transmission lines will be coordinated and approved with the respective owners of the transmission lines.

Site access will use two separate locations for site access on U.S. Route 95. The first point of site point of site access is located approximately 1.24 miles south of the Interstate 11 (I-11)/U.S. Route 93 and U.S. Route 95 interchange, then west onto non-BLM lands using an unnamed road (same road to access Customers Team Members Community Crushing Boulder Ranch Quarry [CTC BRQ Pit]) for 0.25 mile, then north on Old U.S. Highway 95 (Old Hwy 95) for 0.35 mile to the Proposed Project's linear facilities segment on BLM land (see Figure 1-4). The second point of site access is a right in/right out only point of access located approximately 0.9 miles south of the Interstate 11 (I-11)/U.S. Route 93 and U.S. Route 95 interchange that was recently approved by the Nevada Department of Transportation. The use of all existing roads on non-BLM land will be coordinated with the appropriate landowners and authorities.

2.6 Workforce, Schedule, Equipment, and Materials

The on-site construction workforce will consist of laborers, craftsmen, supervisory personnel, support personnel, and construction management personnel. The Proposed Project on-site construction workforce is anticipated to be an average of 250-350 construction workers, with a peak not expected to exceed 750 workers at any given time. Most construction staff and workers commute daily to the jobsite from within Clark County, primarily from the Las Vegas area. The Proposed Project is expected to take up to 18 months to complete.

Construction will take place between 5:00 a.m. and 5:00 p.m. and may occur 7 days per week. Additional hours may be necessary to make up schedule deficiencies, or to complete critical construction activities. For instance, during hot weather, it may be necessary to start work earlier (e.g., at 3:00 a.m.) to avoid work during high ambient temperatures. Further, construction requirements will require some nighttime activity for installation, service or electrical connection, inspection, and testing activities.

Construction activities will follow a generally consecutive order; however, most construction activities associated with each construction component will overlap to some degree and will include:

- Preparing the site, including surveying and staking and geotechnical activities
- Performing site clearance surveys and installing desert tortoise exclusion (temporary and permanent) and perimeter fencing
- Performing site grading activities and constructing stormwater management measures
- Constructing the site road and interior access way within the Project Area

- Trenching and installing the underground electrical system in the data center
- Assembling building foundation/installing building shell and ancillary facilities
- Constructing the electrical collection and transmission system
- Installing BESS, Switching Station and pertinent facilities
- Testing and commissioning
- Restoring temporary laydown area

Table 2-1 provides a description of the on-site equipment expected to be used for data center and BESS. Actual construction equipment details and durations of use may vary.

2.7 Construction Traffic

Typical construction traffic will consist of trucks transporting construction equipment and materials to and from the site and vehicles of management and construction employees during the construction period. Most construction staff will commute daily to the jobsite from within Clark County, primarily from the Las Vegas area. All Project-related traffic will use U.S. Route 95 to access the site from Old Hwy 95. The Applicant will prepare a traffic management plan to address Project-related traffic.

Table 2-1. Estimated On-Site Equipment for Data Center, Collection System and BESS Construction

Equipment Description	Daily Quantity	Horsepower	Fuel Type	Equivalent Full-Load Operating Time (hours/day)	Vehicle Miles (VMT) per Day on Unpaved Surface
Install BMP Measures (Part of Site Preparation)					
Rough Terrain Forklift	8	75	Diesel	1.7	10
Delivery/Work Trucks	12	200	Diesel	2.0	5
Site Preparation					
Truck, Pick-up (Survey Crew)	5	180	Gas	1.7	5
Grader	6	200	Diesel	6.8	20
Backhoe/Front Loader	4	120	Diesel	3.4	20
Tractor/Disc	6	210	Diesel	6.8	40
Scraper	8	265	Diesel	3.4	30
Compactor	6	120	Diesel	1.7	10
Water Truck	4	175	Diesel	6.8	N/A
Site Preparation – Roads					
Grader	6	200	Diesel	6.8	20
Backhoe/Front Loader	2	120	Diesel	6.8	10
Compactor	4	120	Diesel	6.8	20
Water Truck	4	175	Diesel	6.8	N/A
Dump Truck	8	235	Diesel	2.7	10
Install Fencing					

Equipment Description	Daily Quantity	Horsepower	Fuel Type	Equivalent Full-Load Operating Time (hours/day)	Vehicle Miles (VMT) per Day on Unpaved Surface
Rough Terrain Forklift	4	75	Diesel	1.7	10
Delivery/Work Trucks	6	200	Diesel	1.0	5
Post-Installation					
Delivery/Work Trucks	8	200	Diesel	1.0	5
Post Machine	10	45	Diesel	8.1	1
Rough Terrain Forklift	4	75	Diesel	6.8	10
Install Support Structure					
Rough Terrain Forklift	4	75	Diesel	6.8	10
Delivery/Work Trucks	4	200	Diesel	1.0	5
Install Inverters and Switchgear and Substructure					
Crane	4	125	Diesel	4.5	1
Backhoe/Front End Loader	4	120	Diesel	6.8	10
Delivery/Work Trucks	4	200	Diesel	1.0	5
DC and AC Wire Installation (UG)					
Backhoe/Front Loader	8	120	Diesel	6.8	10
Crawling Trencher	4	100	Diesel	4.1	1
Mini-Excavator	8	42	Diesel	6.8	10

2.8 Construction Power

A distribution line (up to approximately 2,000 feet in length) interconnecting to the existing utility distribution service will be installed along the Linear Facilities segment to provide auxiliary power during construction and operation. A portion of the distribution line will be constructed on municipal-owned land by the City of Boulder City (the BLM does not have authority to approve project components on municipal land; therefore, the portion of the distribution line planned for municipal land would require separate authorization from the City of Boulder City). Distribution line poles will be made of wood, steel, or concrete and will be 40 feet high from ground surface and spaced an average of 300 feet from one another. The distribution line will be installed during construction to provide power to the laydown area during construction and to data center facility to maintain communications to Project equipment, and provide power for heating, ventilation, air conditioning, and lighting at the PCSs BESS and Switching Station during operations. Alternatively, generators may be used to provide temporary construction power.

3 RELATED FACILITIES AND SYSTEMS

3.1 Transmission System Interconnect

3.1.1 Proposed Transmission System

The Proposed Project will deliver electricity to an on-site BESS or to the data center with energy purchased from the Mead Substation and transported via bidirectional firm transmission on SNWA's 230-kV transmission line or will interconnect into the existing WAPA Mead 230-kV Substation from the existing Townsite Solar 1 Facility (on municipal land).

3.1.2 Ancillary Facilities

The underground Power Collection Line will be constructed throughout the data center and will connect the buildings with the substation and the BESS s. The Power Collection Line will deliver electricity purchased at the Mead Substation into data center or to the on-site BESS or generate electricity with onsite bio diesel or natural gas fired generation. The Power Collection Line will be co-located within the Linear Facilities segment that includes the site road and the distribution line.

3.1.3 Status of Interconnection Agreement

The Applicants affiliate, TS2, has executed a large generator interconnection agreement dated September 20, 2024, with SNWA (the Interconnection Agreement). The Interconnection Agreement provides the Applicant with up to 167 MW of firm bilateral transmission capacity on the SNWA transmission line from the Mead Substation to the newly constructed Boulder Flats substation. The Applicant is seeking to increase its interconnection capacity under the Interconnection Agreement.

3.2 Gas Supply Systems

The Proposed Project includes require a biodiesel for emergency backup generators. The biodiesel will be stored in tanks connected to the generators.

Southwest Gas has indicated they could transport up to 3000Mcf/day at 300psi to the facility.

3.3 Other Related Systems

3.3.1 Communication System Requirements

Multiple communication systems will be used for construction and operation. These items will include telephone, fiber optics, and T1 internet. The Applicant expects to use existing wired or wireless telecommunications facilities. In addition, the Applicant will install hard-wired (landline) systems as part of the electrical construction activities or will supplement with small aperture (less than 1 meter) satellite communications gear.

4 OPERATION AND MAINTENANCE (O&M)

An operations and maintenance (O&M) plan for a data center is a holistic strategy that includes daily, weekly, monthly, and annual checklists for critical infrastructure and equipment. The goal is to maximize uptime, extend the life of equipment, and ensure continuous, efficient, and secure operations.

Environmental controls

- **Daily**: Monitor temperature and humidity levels in the data center to ensure they remain within the optimal range of 64.4°F to 80.6°F (18°C to 27°C) and 40% to 60% humidity.
- **Weekly:** Check and inspect HVAC and cooling units for leaks or other issues.
- **Quarterly:** Conduct comprehensive visual and technical inspections of all cooling systems. Test the cooling system and check for any material buildup.
- **Annually**: Perform preventive maintenance on HVAC systems, which includes changing air filters, cleaning coils, and checking refrigerant level

Power systems

- **Daily:** Inspect Uninterruptible Power Supply (UPS) systems and generators to confirm they are functioning correctly.
- **Quarterly:** Test backup generators under load to ensure they can sustain the data center during an extended power outage.
- **Annually:** Conduct comprehensive maintenance on all electrical components, including UPS units, batteries, Power Distribution Units (PDUs), and switchboards. Test the integrity of aging power cables.

Project operations will require a workforce of up to forty full-time-equivalent positions at an average salary of $100,000/year including benefits. This workforce will include administrative/management personnel, operator/ maintenance, and security personnel.

Dust during O&M will be controlled and minimized by applying water and/or BLM-approved palliatives (see Section 6.4.1, *Air Quality*).

Table 4-1. Routine Maintenance Protocol

Equipment	Maintenance Interval	Task
Data Center	Daily	• Monitor temperature and humidity levels in the data center to ensure they remain within the optimal range of 64.4°F to 80.6°F (18°C to 27°C) and 40% to 60% humidity Conduct comprehensive visual and technical inspections of all cooling systems. Test the cooling systems and check for any material buildup.
	Quarterly	• Test backup generators underload to ensure they can sustain the data center during an extended power outage. • Visually inspect modules for discoloration • Visually inspect wiring for connections and secure mounting • Visually inspect mounting structure for rust and erosion around foundations • Manually clean localized debris from bird droppings, etc.
	Annually	• Conduct comprehensive maintenance on all electrical components, including UPS units, batteries, Power Distribution Units (PDUs), and switchboards. Test the integrity of aging power cables. • Perform preventive maintenance on HVAC systems, which includes changing air filters, cleaning coils, and checking refrigerant level. • Audit physical security controls, update visitor management procedures, and review access authorizations.
Inverters/trans-formers and BESS	Semi-annually	• Perform temperature checks on breakers and electrical terminations. • Visually inspect all major components and wiring harnesses for discoloration or damage • Measure all low voltage power supply levels. • Inspect/remove any dust/debris inside cabinet. • Inspect door seals. • Check proper fan and HVAC operation. • Inspect and clean (replace if necessary) filters. • Check electrical termination torque. • Check the operation of all safety devices (e-stop, door switches, ground fault detection)
	Annually	• Check all nuts, bolts, and connections for torque and heat discoloration. • Calibrate control board and sensors. • Inspect air conditioning units for proper operation
Medium-voltage transformers	Semi-annually	• Perform temperature check. • Inspect door seals. • Record all gauge readings. • Clean any dirt/debris from low-voltage compartment
Breakers and switchgear	Semi-annually	• Inspect for discoloration of equipment and terminations. • Inspect door seals
	Annually	• Check open/close operation
Overhead transmission lines (if any)	Annually (and after heavy rains)	• Inspect guy wires and tower angle. • Visually inspect supports/insulators. • Visually inspect discoloration at terminations
Roadways	Annually (and after heavy rain)	• Inspect access way and site road that cross drainage paths for erosion

Equipment	Maintenance Interval	Task
Vegetation	Semi-annually	• Noxious weed inspections will be conducted in accordance with the BLM-approved Integrated Weed Management Plan • Inspect localized vegetation and control to no higher than what is commercially reasonable. • Apply herbicides as necessary to control noxious weeds
	As needed	• Mechanical and/or other approved treatment, as required to reduce vegetation height
Backup Power	Annually	• Visually inspect backup power system • Perform functional test of backup power system
Fencing	Quarterly (and after heavy rain)	• Inspect perimeter fence for vandalism and erosion at base. • Conduct desert tortoise fence inspections in accordance with the terms and conditions of the Project-specific Biological Opinion

5 DECOMMISSIONING AND SITE RECLAMATION

The Proposed Project has an anticipated operational life of 50 years, after which the Applicant may choose to apply to the BLM for renewal of the ROW authorization, or choose to decommission the Project, remove the systems and their components, and restore the Project Area. In order to renew, the Applicant will request a renewal from the BLM at least 120 calendar days before the grant or lease expires consistent with the renewal terms and conditions specified in the grant or lease.

A final Facility Decommissioning and Site Reclamation Plan will be created in accordance with BLM policy and objectives at the time and will be approved by the BLM once the operational life of the Proposed Project has been reached.

6 ENVIRONMENTAL CONSIDERATIONS

6.1 Site Characteristics and Environmental Issues

An Environmental Assessment ("EA"), including a detailed analysis of site characteristics and environmental considerations was performed for Townsite Solar 2 Phase 1, which is adjacent to the Project. The Phase 1 EA assessed the environmental impact of the Proposed Project in the following areas:

- Air Resources
- Cultural Resources
- Environmental Justice and Socioeconomics
- Fish and Wildlife, Excluding Federally Listed Species
- Forestry (analyzed under Vegetation)
- Fuels and Fire Management
- Hydrologic Conditions (analyzed under Soils)
- Invasive Species/Noxious Weeds
- Migratory Birds (and Eagles)
- Recreation/Travel/Wild and Scenic Rivers
- Soils
- Threatened, Endangered, and Candidate Animal Species (i.e., the desert tortoise)
- Vegetation, Including Special-Status Plants, Cacti, and Invasive Species and Noxious Weeds
- Visual Resources
- Wastes (Hazardous or Solid)

It is anticipated that the same resources as existed and were analyzed in Phase 1 EA are present in the Phase 2 Project area.

6.2 Other Uses on the Project Area

6.2.1 Grazing and Mining

There are no grazing allotments within the Project site.

6.2.2 Recreation and Off-Highway Vehicle Conflicts

In accordance with the existing Las Vegas Resource Management Plan, off-highway vehicle use in the surrounding area has been designated as "Limited to existing roads, trails, and dry washes" (BLM 1998). In the vicinity of the Project Area, there are no known casual recreation uses in the area, even though there are existing dirt roads and trails throughout the Project Area. There have not been any special recreation permits for events or commercial activities in the locale issued by BLM LVFO. There are no developed recreation facilities or opportunities in this portion of the Special Recreation Management Area

(BLM 2021). The data center facility portion of the Proposed Project will make this area unavailable for dispersed recreational activities for the operational life of the Proposed Project. The Proposed Project will limit recreational access through the site once site preparation has begun by blocking potential east-west recreational movement across the site.

6.2.3 Project Military Aviation

Although the Project is not under any military airspace or in a Department of Defense Consultation Area, it is approximately 40 miles south of Nellis Air Force Base. Given that distance, Project elements will not affect the approach or departure corridors for runways at the base. Because the Project will not construct facilities taller than 200 feet, it will not require Federal Aviation Administration evaluation of safety hazards. However, the military has indicated that structures higher than 50 feet within the vicinity of the base may present electromagnetic compatibility concerns for test missions at the Nevada Test and Training Range. The Proposed Project does not include components over 50 feet in height.

6.2.4 Other Existing Uses and Access

The Proposed Project will not adversely affect the existing ROWs within and adjacent to the site. The Proposed Project is within an undisturbed portion of the BLM's community pit (N-048712) and was reduced in size to accommodate possible future expansion of the pit, which has valid existing rights for the west half of Section 14, T23S, R63E (BLM 2021).

Existing development in the vicinity of the Proposed Project consists of an operating gravel quarry and private shooting range, approximately 0.5 mile to the north, which will not be impacted by Project construction or operations. Approximately 2,000 feet to the east is the proposed BCNV Facility on municipal land and directly south of the Proposed Project is the Townsite Solar 1 Project, also on municipal land. To the west is CTC BRQ Pit, an existing privately held sand and gravel pit. An evaluation of these and other uses on the Proposed Project site will be evaluated in the NEPA process.

6.3 Design Features

The Applicant will incorporate applicable design features from the 2024 PEIS ROD into the Proposed Project development process, including the following management plans, as required by the BLM:

- Dust Abatement Plan
- Facility Decommissioning and Site Reclamation Plan (at the conclusion of the life of the Project)
- Fire Management Plan
- Health and Safety Plan
- Integrated Weed Management Plan
- Lighting Management Plan
- Raven Management Plan
- Site Drainage Plan and Stormwater Pollution Prevention Plan
- Spill Prevention and Emergency Response Plan
- Traffic Management Plan

6.4 Worker Education and Awareness Plan (WEAP) Environmental Protection Measures

The following are preliminary Applicant-proposed environmental protection measures (EPMs). These measures are subject to change based on the findings of site-specific technical analyses and the analysis during the NEPA process.

6.4.1 Air Quality/Heat/Water Use/Noise/Visual Impacts

- Water will be applied to the ground during the construction and use of the Project Area, roads, and other disturbed areas as necessary to control dust.
- A Dust Control Operating Permit will be obtained from Clark County Department of Environment and Sustainability, Division of Air Quality prior to construction, and requisite dust control measures and BMPs will be implemented during the construction of the Proposed Project.
- During operations, air quality in and around Boulder City area will not be impacted by the data center. While proposed data center will have back-up generators fueled by biodiesel, they will only be used during testing and system emergencies. The back-up generators will be operated in accordance with local, state and federal air quality requirements
- The facility will use cooling towers that move warm air upward several hundred feet above the buildings and directs heat away from ground level. The nearest residence is 400 feet above the site and approximately 3 miles away. The projected wind-adjusted temperature increase is approximately 0.04°C under normal operating conditions, well below the average human perception threshold for temperature change of 0.92°C. It is a one-time steady increase that does not accumulate over time.
- Appropriate water erosion and dust control measures will be implemented to prevent an increased dust and sediment load to ephemeral washes around the construction site and to comply with Clark County dust control requirements. Dust during construction will be controlled and minimized by applying water and/or BLM-approved palliatives.
- The proposed data center would use an advanced closed-loop air-cooled (dry) cooling system with direct-to-chip liquid cooling and dry cooling towers to eliminate excessive water use and protect Southern Nevada water resources. Up to 400,000 gallons of a purified water/propylene glycol mix for a one-time initial fill of the closed-loop cooling system. The purified water/propylene glycol mix will be purchased from a vendor of that product. Up to 2,000 gallons/ yr. of the purified water/propylene glycol will be needed for periodic maintenance and will likewise be sourced from a vendor of that mixture. The cooling system will not be filled with any municipal water, as it is not appropriate for this type of cooling system. The initial fill of the cooling system and any topping up of the system must be with the purchased purified water/ propylene glycol mix. Any waste from the system associated with periodic maintenance will be trapped and properly disposed of offsite through contractors approved to handle and dispose of the material.
    - Potable Water Uses:
        - Ongoing use of building restrooms and kitchen (1,000-2,000 gallons/day) for employees.
        - The potable water will be stored in tanks on site or will be procured from Boulder City and transported to the site via a city owned pipeline.

        - Approximately 90% of potable water used on site will be returned to the Boulder City wastewater treatment plan through a new sewer line constructed and paid for by the project.
    - Wastewater Usage:
        - During construction, the Applicant will seek to purchase up to 500,000 gallons per day of wastewater for dust control from the City.
        - This effluent is currently being discharged into the desert by City's wastewater treatment plant.

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- The Proposed Project will follow the BLM Guidelines and standards and all colors will be selected from the approved BLM color palette. The project will use a lighting design that focuses on safety and complies with all local regulations, including requirements to reduce light pollution in accordance with the Dark Skies Project. The Applicant will work with local authorities to develop landscaping and building colors that fit with the surrounding environment. The height of the building will comply with all City ordinances, and the building's lower elevation will reduce its visibility by residents. It is not anticipated that there will be any significant reflectivity of any major component of the Proposed Project.

- The Applicant will employ BMPs to protect the soil surface by covering or binding soil particles. The Proposed Project will incorporate erosion control measures required by regulatory agency permits and contract documents, as well as other measures selected by the contractor. Project-specific BMPs will be designed by the contractor and included in the Project Stormwater Pollution Prevention Plan. BLM has allowed the use of several dust palliatives on other projects within the SND. If dust palliatives are used in place of water for the Proposed Project, the total amount of water needed during construction will be reduced. The Applicant may opt to use such palliatives, as authorized by the BLM for the Proposed Project. The soil binder/dust palliatives that are proposed for the Project, and that the BLM has previously allowed, are:
    - Road Bond 1000
    - For roads and heavy traffic areas: Soil Cement
    - For non-traffic areas on finer soils: Formulated Soil Binder FSB 1000
    - For non-traffic areas on sandier/rockier soils: Plas-Tex

- The Project will conduct a noise study before the commencement of construction and. The Project will have no audible impact on local residents due to its location nearly 3 miles from and 400 feet below the nearest residence. Sound levels drop with distance, from 76 dBA 10 feet from the building inside the facility fence line (similar to a hair dryer or dishwasher), to 40 dBA at half a mile (similar to a quiet room or refrigerator), and approximately 7 dBA at the nearest home. If emergency generators were running, the sound level at approximately 3 miles would be 24 dBA, similar to rustling leaves. These impacts will be further reduced through acoustically-treated generator enclosures, strategic equipment placement, and compliance with local noise ordinances. The Applicant will ensure that noise levels will comply with all local, state and federal guidelines

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6.4.2 Cultural Resources

- Cultural resources are defined as buildings, sites, structures, or objects, each of which has historical, architectural, archaeological, cultural, and/or scientific importance. Numerous laws,

regulations, and statutes, on both the federal and state levels, seek to protect and target the management of cultural resources.

- In consultation with the BLM and with SHPO concurrence, any areas that contain cultural resources of significance or whose eligibility for inclusion on the National Register of Historic Places (NRHP) is unevaluated, will be avoided, mitigated, or "treated" and recorded as appropriate.

6.4.3 Desert Tortoise

- The Applicant will implement the minimization measures and compensatory mitigation requirements as required and outlined in the applicable Biological Opinion.

6.4.4 Fire Protection

- All federal, state, and county laws, ordinances, rules, and regulations that pertain to prevention, pre-suppression, and suppression of fire will be strictly adhered to.

6.4.5 Reclamation

- For temporary disturbance areas that require clearing and/or grading work, restoration and reclamation procedures will follow the Restoration Plan and will be based on site-specific requirements and techniques commonly employed at the time the area is to be reclaimed and/or based on the Restoration Plan and would include regrading, top soiling, and revegetating all disturbed areas.
- After construction is complete, disturbed work areas will be graded to the approximate original contour, and the area will be revegetated with BLM-approved seed mixtures according to the Restoration Plan. Most postconstruction work will entail scarifying soils to reduce compaction and reseeding.
- A Facility Decommissioning and Site Reclamation Plan will be prepared in accordance with BLM policy and objectives at the time and approved by the BLM, once the operational life of the Proposed Project has been reached.

6.5 Mitigation Methods

6.5.1 Desert Tortoise

- Per the Project-specific Biological Opinion, the following compensatory mitigation is required to avoid, minimize, and offset potential adverse effects to the desert tortoise (USFWS 2022).
- **Habitat Compensation:** To further offset the adverse effects from the Project, the Applicant will pay the currently applicable desert tortoise remuneration fee to BLM.

7 REFERENCES

Avian Power Line Interaction Committee (APLIC). 2006. *Suggested Practices for Avian Protection on Power Lines*. Available at: http://www.aplic.org/uploads/files/2643/SuggestedPractices2006(LR-2).pdf.

Bureau of Land Management (BLM). 1998. *Record of Decision for the Approved Las Vegas Resource Management Plan and Final Environmental Impact Statement*. October 1998. Available at: http://www.blm.gov/style/medialib/blm/nv/field_offices/las_vegas_field_office/las_vegas_resource.Par.28808.File.dat/ROD_LV_RMP.pdf.

———. 2006. Noxious Weed Plan. Available at: https://www.blm.gov/sites/blm.gov/files/2006_Signed_LVFO_Weed_Plan.pdf. Accessed February 2023.

———. 2007. *Final Programmatic Environmental Impact Statement (PEIS) for Vegetation Treatments on Bureau of Land Management Lands in 17 Western States*. June 29, 2007. Available at: http://www.blm.gov/wo/st/en/prog/more/veg_eis.html.

———.

International Association of Plumbing and Mechanical Officials (IAPMO). 2006a. Uniform Plumbing Code. Available at: https://archive.org/details/gov.law.iapmo.upc.2006. Internet Archive. Accessed February 2023.

———. 2006b. Uniform Mechanical Code. Available at: https://archive.org/details/gov.law.iapmo.umc.2006. Accessed February 2023.

International Code Council. 2006a. International Building Code. Available at: https://archive.org/details/gov.law.icc.ibc.2006. Internet Archive. Accessed February 2023.

———. 2006b. International Energy Conservation Code. Available at: https://archive.org/details/gov.law.icc.iecc.2006. Accessed February 2023.

National Fire Protection Association (NFPA). 2008. National Electrical Code. Available at: https://archive.org/details/gov.law.nfpa.nec.2008/page/n1/mode/2up. Accessed February 2023.

U.S. Fish and Wildlife Service (USFWS). 2009. *Desert Tortoise (Mojave Population) Field Manual (*Gopherus agassizii*)*. Region 8, Sacramento, California. Available at: https://www.fws.gov/sites/default/files/documents/Desert-Tortoise-Field-Manual.pdf.

———. 2022. *Biological Opinion on the Townsite Solar 2 Project, Clark County, Nevada.* Southern Nevada Fish and Wildlife Office. File No. 2022-0035189-S7-001. October 18, 2022.

———. 2023. Adjustment for 2023 Mojave Desert Tortoise Remuneration Fees Collected under Endangered Species Act Section 7 Biological Opinion. Southern Nevada Fish and Wildlife Office. January 20, 2023.